

**From roots to trees:
Thematic roles and the lexicon-syntax interface**

A Dissertation Presented

by

Eva Neu

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Approved as to style and content by:

Rajesh Bhatt, Chair

Kyle Johnson, Member

Alejandro Pérez Carballo, External Member

Seth Cable, Department Head
Linguistics

How can we know the dancer from the dance?

—W. B. Yeats, ‘Among School Children’

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Abstract

From roots to trees: Thematic roles and the lexicon-syntax interface

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Eva Neu

B.A., Free University of Berlin

M.A., University College London

Ph.D., University of Massachusetts Amherst

Directed by: Professor Rajesh Bhatt

This study is about thematic roles (θ -roles). The research question I pursue is how speakers know what it means for an argument to be interpreted with a given θ -role in the context of a given event description. For instance, speakers who understand the sentence *Brutus stabbed Caesar* know not only that Caesar is a patient, but more specifically that Caesar is the patient *of* a stabbing event, and they know what this means: viz., that he was penetrated by a sharp weapon. I ask how this knowledge about what it means to be the agent, patient, etc., of a particular event is represented in speakers' mental grammars.

I begin by refuting a competing answer to this question, which I label the individual role theory of θ -roles. Under this theory, verbs are stored in the lexicon with a finite set of discrete verb-specific roles (individual roles). For instance, the verb *stab* would be stored with the individual roles of a stabber and a stabbee. The function of θ -roles would then be to categorize these individual roles into more abstract clusters, with a stabber being an instance of an agent, and a stabbee an instance of a patient. I highlight several theoretical problems for this theory, and I discuss a number of argument-structural alternations cross-linguistically for which I argue that it makes the wrong predictions. In these alternations, the individual roles of a verb would not stand in a one-to-one mapping relationship to θ -roles, which under the individual role theory is unexpected.

The alternative theory I propose takes as its starting point the assumption that arguments are introduced by functional heads in the syntax. I argue that what it means to be, say, the patient of a stabbing event is not already stored as part of the lexical entry of the verb, but generated compositionally in the syntax by merging the verb *stab* with a functional head introducing a patient argument. I show that this can be captured in the formal denotation of the sentence, and I discuss the consequences of this view for the question of what the meaning of a verb is before it combines with argument-introducing heads. Analogous to the

verb's first argument, the interpretation of its second argument – such as *Brutus* in *Brutus stabbed Caesar* – is equally generated compositionally in the syntax by merging an argument-introducing head with an event description. Due to syntactic constituency, this event description already contains the first argument. As a result, lower arguments are expected to affect the interpretation of higher arguments but not vice versa, a prediction that I show to be borne out. Lastly, I argue that the proposed analysis aligns with an exemplar-theoretic view on θ -roles, according to which abstract roles such as agent and patient are generalizations formed over agents and patients of particular events.

Copyright Note

A previous version of Chapter 4.2 has appeared in *Natural Language and Linguistic Theory*.

Contents

Acknowledgments	i
Abstract	ii
Copyright Note	iv
List of abbreviations	viii
1 Introduction	1
1.1 Why θ -roles?	1
1.2 The problem.	3
1.3 The solution, in brief	7
1.4 Outlook, or: A reader's manual	7
2 Some formalities	9
2.1 Proto-roles and the thematic inventory	9
2.2 The status of θ -roles in the grammar	11
2.3 How to diagnose a θ -role	11
3 A brief history of argument structure	14
3.1 Projectionism	14
3.2 Cognitive Construction Grammar	16
3.3 Distributed Morphology	22
3.4 Summary	23
4 Six variations on variable linking	25
4.1 Variable unaccusativity	26
4.1.1 Data and analysis	26
4.1.2 Evidence for the analysis	27
4.1.3 Competing analyses	29
4.1.3.1 Against syntactic unaccusativity: Van Valin (1990) and Dowty (1991) . .	29
4.2 Direct causatives of unergatives	30
4.2.1 Data and analysis	30
4.2.2 Evidence for the analysis	32
4.2.2.1 Background	32
4.2.2.2 Syntactic diagnostics	39
4.2.2.3 Thematic interpretation	44

4.2.3	Competing analyses	45
4.2.3.1	The low subject analysis: Massam (2009) and beyond	46
4.2.3.2	Covertly transitive unergatives: Ramchand (2008)	47
4.2.3.3	An ‘in-motion’ θ -role: Biggs (2019)	52
4.3	Direct causatives of ingestives	56
4.3.1	Data and analysis	56
4.3.2	Evidence for the analysis	58
4.3.2.1	Background	58
4.3.2.2	Syntactic diagnostics	60
4.3.2.3	Thematic interpretation	63
4.3.3	Competing analyses	64
4.3.3.1	Coindexation of agent and goal: Amberber (2002)	64
4.3.3.2	Antireflexivization: Krejci (2012)	66
4.4	The agent-affectee alternation	71
4.4.1	Data and analysis	71
4.4.2	Evidence for the analysis	72
4.4.2.1	Background	72
4.4.2.2	Syntactic diagnostics	77
4.4.2.3	Thematic interpretation	83
4.4.3	Competing analyses	87
4.4.3.1	Pragmatic inference: Rivero (2004)	88
4.4.3.2	A feature-based analysis: Kallulli (2008)	89
4.4.3.3	Affected vs. high applicatives: Cuervo (2003)	91
4.4.3.4	A remaining puzzle from German: Schäfer (2008)	97
4.5	Pseudo-symmetric predicates	101
4.5.1	Data and analysis	101
4.5.2	Evidence for the analysis	103
4.5.2.1	Background	103
4.5.2.2	Syntactic diagnostics	107
4.5.2.3	Thematic interpretation	111
4.5.3	Competing analyses	115
4.5.3.1	A threshold model: Kruitwagen (2024)	115
4.5.3.2	Covert reciprocals: Wehbe (2025)	117
4.5.3.3	Covert intransitives: Winter (2016, 2018) and van Craenenbroeck and Johnson (2026)	127

4.6	Statives of ingestives	130
4.6.1	Data and analysis	130
4.6.2	Evidence for the analysis	133
4.6.2.1	Background	133
4.6.2.2	Syntactic diagnostics	143
4.6.2.3	Thematic interpretation	152
4.6.3	Competing analyses	154
4.6.3.1	Agentivity via a meaning postulate: Paparounas (2025)	154
5	Analysis	157
5.1	Rethinking argument interpretation	157
5.1.1	Argument interpretations can be located in the tree	157
5.1.2	Variable linking and the nature of prethematic verb meaning.	160
5.1.3	Beyond intransitives	163
5.1.4	Proto-properties and exemplars	167
5.2	Revisiting the case studies.	169
5.2.1	Direct causatives of unergatives	169
5.2.2	Direct causatives of ingestives	172
5.2.3	The agent-affectee alternation	172
5.2.4	Pseudo-symmetric predicates	174
5.2.5	Statives of ingestives	175
6	Ramifications	177
6.1	Entailments	177
6.2	Verb distribution	182
7	Conclusion	186
	Bibliography	190

List of abbreviations

Abbreviations follow the Leipzig glossing conventions, with the following additions:

AOR	aorist
CAUS2	indirect causative
CL	clitic
PTCL	discourse particle
INTENS	intensive pronoun
MID	middle Voice
NACT	non-active
O	object agreement
VBZ	verbalizer

Chapter 1: Introduction

1.1 Why θ -roles?

This study is about thematic roles (θ -roles). Since Gruber (1965) and Fillmore (1968), θ -roles have been used to analyze a wide variety of issues at the syntax-semantics interface, from generalizations on argument positions, to the distinction between expletive and contentful arguments, and even to binding and control. Therefore, the purpose of this section is to clarify in very broad terms from what angle we will be concerned with θ -roles, or in other words, the reason why I am interested in θ -roles to begin with.

A general tenet of much linguistic theory is that speakers' knowledge of a language consists of two categorically distinct domains: lexical knowledge and grammatical knowledge, roots and trees. Knowledge about root meaning – for instance, what distinguishes a cup from a mug – is fuzzy, gradient and inextricably tied up with world knowledge. Knowledge about functional structure – for instance, the mechanisms of subject-verb agreement – is discrete, abstract and unaffected by the state of the world. Perhaps nobody has drawn the line between these two domains more sharply than Borer (2005a):

I believe that the proposed dividing line is a real one, and that it distinguishes between what is grammatically real – structures and formal properties of functional items, and what may be very real, but not grammatically so – properties of substantive vocabulary. The latter, I propose, are creatures born of perception and conceptualization, representing an intricate web of layers upon layers of a complex perceptual structure and emerging world knowledge, concepts which come to represent it, the reflection upon these concepts, and so on. Their properties, however characterized, are thus fundamentally not grammatical. (Borer, 2005a:10-11)

The strong claim that is made here is that lexical knowledge and grammatical knowledge are separate systems that are fundamentally unable to communicate with each other. Grammar is insensitive to root meaning because this is not information that the grammar trades in; lexical meaning is oblivious of grammatical rules and constraints. If the grammar and the lexicon were pieces of code, they would be written in different programming languages. No information can simply be transferred from one to the other.

This is an interesting claim, and I will interpret it as radically as possible. The challenge it faces, however, is how to explain cases where lexical meaning does seem to be able to affect grammatical structure after all. The primary area where this can be observed is in argument structure. In the following, I first lay out this challenge in more detail and then show how θ -roles come to the rescue. To anticipate the upshot of this discussion, the angle from which we will be concerned with θ -roles in this work is thus as a theoretical construct that mediates between lexical knowledge and grammatical knowledge.

Consider the sentence in (1):

(1) Brutus stabbed Caesar.

A simplified representation of the denotation of (1), glossing over tense and aspect, is given in (2):

(2) $\lambda e. \text{stab}(e) \wedge \text{Agent}(e)(\text{Brutus}) \wedge \text{Patient}(e)(\text{Caesar})$

The question I would like us to focus on is what meaning is expressed by two particular components of the denotation in (2): $\text{Agent}(e)(\text{Brutus})$ and $\text{Patient}(e)(\text{Caesar})$. Part of the meaning conveyed is of course that Brutus is acting in a generically agent-like way, and Caesar in a generically patient-like way. That is, the two arguments have abstract properties associated with the θ -roles of agent and patient, respectively. For instance, Brutus is acting intentionally and possesses causal power, whereas Caesar passively undergoes the action against his will. Whatever the properties are that generally lead to an argument being categorized as an agent, Brutus must have a non-negligible subset of them, and *mutatis mutandis* for Caesar.

But this is not all there is to it. A speaker who understands the sentence in (1) knows something much more specific about Brutus and Caesar: roughly, that the former wielded a sharp weapon and that the latter was penetrated by a sharp weapon. This is of course knowledge that is somehow associated with the root *stab*. But the root itself does not tell us which role in the event was played by Brutus and which one by Caesar; for all the root cares, Caesar could be the one doing the stabbing. Rather, it is the combination of a θ -role with a particular root that tells us which action concretely a participant in the event performed. We know that Caesar was penetrated by a sharp weapon because that is what it means to be the patient of a stabbing event.

Thus, a speaker who understands the sentence *Brutus stabbed Caesar* knows not only that Brutus is an agent and Caesar a patient, but that they are the agent and the patient, respectively, *of* a very particular event, and they know what this means: that Brutus wielded a sharp weapon and Caesar was penetrated by a sharp weapon. Compare (1) to a different sentence, (3):

(3) Brutus kicked Caesar.

The denotation, following the same format as (2), is in (4):

(4) $\lambda e. \text{kick}(e) \wedge \text{Agent}(e)(\text{Brutus}) \wedge \text{Patient}(e)(\text{Caesar})$

Here again, Brutus and Caesar are agent and patient, respectively, but they are agent and patient of a different event. A speaker who understands (4) knows that Brutus moved his leg in a particular way and that Caesar was hit by Brutus's leg. Being the patient of a stabbing event is different from being the patient of a kicking event, and neither reduces to being a patient *tout court*. The same holds for any other θ -role.

Returning to the dichotomy between lexical and grammatical knowledge, what is special about θ -roles is that they occupy a position at the interface between the two domains. On the one hand, agent, patient,

goal, etc. are *bona fide* grammatical concepts, represented in compositional semantics and mapped onto syntactic positions in some regular fashion. At the same time, when interpreted in the context of a given event description, θ -roles are associated with very particular pieces of lexical meaning, such as someone wielding a sharp weapon. In a sense, what it means to be an agent or patient is grammatical knowledge, but what it means to be the agent or patient *of* a particular event is lexical knowledge.

It is this position at the interface that makes θ -roles such a crucial concept in linguistic theory. Consider for a moment a world without θ -roles, in which speakers know that the verb *stab* comes with two arguments – someone who wields a sharp weapon and someone who is penetrated by a sharp weapon –, but where these arguments are not categorized as agent and patient, respectively. In this world, no grammatical principle could require one argument to be realized as the subject of the sentence and the other as the object, rather than vice versa. There is no specialized syntactic position for arguments that denote participants wielding a sharp weapon because wielding a sharp weapon is a matter of lexical knowledge that the grammar, under the strict dichotomy assumed here, does not have access to. The difference between wielding a sharp weapon and being penetrated by a sharp weapon is not in and of itself a grammatical difference. It is only by virtue of being categorized as agents and patients that arguments denoting participant who wield sharp weapons and are penetrated by sharp weapons, respectively, become things that the grammar can make reference to in the first place. Being an agent of *stab* is different from being an agent of *kick*, but the grammar does not see these differences. All it sees is that these arguments are agents, and it will treat them as such, and it will treat them equally.

This is what θ -roles do for us as a theoretical construct, regardless of how concretely we believe them to be represented in the grammar. They bridge the gap between lexical knowledge and grammatical knowledge; they get us from roots to trees. It is only by virtue of θ -roles that pieces of lexical meaning – someone stabbing, kicking, being stabbed, etc. – become items that the grammar can operate on, be it to assign them a particular argument position, to make them the object of binding rules, or any other business that the grammar deals in. The broader goal of this study is to better understand how θ -roles do their job.

1.2 The problem

This section outlines the specific research question that I pursue here. Recall that earlier, I introduced a distinction between being an agent or patient, and being an agent or patient *of* a particular event. The main question that the present work tries to answer is how the latter kind of knowledge is encoded in speakers' grammars. What exactly do we know when we know that Caesar is the patient of a stabbing event? How do we know it? Where in the architecture of the language faculty is this knowledge located? How does a θ -role such as that of a patient come to access or activate one particular part of the lexical knowledge associated with *stab* (being penetrated by a sharp weapon) rather than another (wielding a sharp weapon)? How must lexical knowledge be structured for this to be possible?

I readily admit that this is not an obvious question, in the sense that it might not be obvious that there is

a question to begin with. Part of what this study tries to achieve is to convince the reader that the question is real and non-trivial. In the following, I first outline what might be considered a commonsensical answer to it. For reasons that should become clear later, I label this view the individual role theory of θ -roles. This theory has been proposed explicitly in the framework of Cognitive Construction Grammar (Goldberg, 1995, 2006), but I believe that it is implicitly taken for granted in much work on argument structure of any theoretical flavor. After having introduced this view, I then briefly anticipate why I believe it to be wrong, an argument that will be developed later in much more detail in Chapters 3 and 4. By the end of this section, I hope to have made it clear that what it means for an argument to be interpreted with a given θ -role in the context of a given event description is a genuine question that is relevant for argument structure more broadly and has not yet been given an adequate answer.

The core idea underlying what I call the individual role theory of θ -roles is that the lexical meaning of a verb consists of a set of verb-specific roles. For instance, a verb like *stab* comes with two roles, what we might in purely heuristic terms call a stabber (someone who wields a sharp weapon) and a stabbee (someone who is penetrated by a sharp weapon). In *Brutus stabbed Caesar*, Brutus is the stabber and Caesar is the stabbee. Borrowing a term from Dowty (1989), I will refer to these verb-specific roles that would be stored as part of the lexical meaning of a verb as *individual roles*.

What θ -roles do, under this view, is organize these verb-specific individual roles into abstract clusters. For the verb *stab*, the stabber role is classified as an agent and the stabbee role as a patient. Similarly, for the verb *kick*, the kicker role is classified as an agent and the kickee role as a patient. We can thus imagine lexical knowledge about verbs taking the form of a table, with all θ -roles listed on the x -axis and all verbs listed on the y -axis, as seen in (5). Note that the nominalizations that populate these cells have no deeper theoretical import; they are simply a shorthand for the individual roles that are specified as part of the lexical meaning of the verb.

(5)

	agent	patient	goal	...
<i>stab</i>	stabber	stabbee		
<i>run</i>	runner			
<i>hand</i>	hander	handed	handee	
<i>fall</i>		faller		
...				

Under this view, knowing what it means to be the agent, patient, goal, etc. of a particular event means finding the relevant verb on the y -axis and then looking up the individual role specified for the relevant θ -role on the x -axis. Crucially, this means that the interpretation of an argument in the context of a given event always only depends on two variables: the verb and the θ -role, the y -axis and the x -axis. Overall, under the individual role theory, θ -roles interpreted in the context of a given event are individual roles, stored as part of the lexical meaning of a given verb.

This picture faces both theoretical and empirical problems. To begin with the former, the individual role theory of θ -roles presupposes that verb meaning takes the form of structured lexical entries. For instance, the verb *stab* would need to be stored with two roles as part of its lexical entry, a stabber and a stabbee, which can then in some way be mapped onto θ -roles. Structured lexical entries are assumed in such otherwise heterogeneous frameworks as projectionism (Levin and Rappaport Hovav, 1995, 2005) and Cognitive Construction Grammar (Goldberg, 1995, 2006), but have been rejected in more recent work in argument structure, especially in the tradition of Distributed Morphology (DM; Halle and Marantz, 1993, 1994). Instead, verb meaning is assumed to take the form of unstructured blobs of conceptual knowledge, with arguments being introduced by functional heads in the syntax (Borer, 2005a, 2005b; Kratzer, 1996; Marantz, 1984, a.m.o.). Reviewing these theoretical developments more in detail later in Chapter 3, I argue that the objections that DM has raised against structured lexical entries in general also undermine the concept of individual roles.

By way of example, recall that according to the individual role theory, the interpretation of an argument should only depend on the verb and the θ -role, the *y*-axis and the *x*-axis. However, Marantz (1984) has pointed out that in examples such as (6) and (7), the interpretation of the external argument depends crucially on the internal argument: what it means to be an agent of a taking-a-book event is very different from what it means to be an agent of a taking-a-break event.

- | | |
|---|--|
| (6) a. take a book from the shelf
b. take a bus to New York
c. take a nap
d. take an aspirin for a cold
e. take a letter in shorthand | (7) a. kill a cockroach
b. kill a conversation
c. kill an evening watching TV
d. kill a bottle (i.e., empty it)
e. kill an audience (i.e., wow it) |
|---|--|
- (Marantz, 1984:25)

The interpretation of an argument is thus not simply determined by an individual role – here, that of a taker or a killer – which is already stored on the verb, but more holistically by the broader sentential context. I later discuss a number of other, similar challenges for the idea that verbs come with a discrete set of contentful individual roles that determine the interpretation of an argument in the sentence.

Turning to the empirical problems with the individual role theory of θ -roles, Chapter 4 will survey a range of concrete argument-structural alternations for which I claim that the theory makes the wrong predictions. In a nutshell, in these alternations, the individual roles and the θ -roles of a verb would not stand in a one-to-one mapping relationship. I label this pattern variable linking, and I argue that it crucially undermines the individual role theory.

Here, I briefly introduce one very simple case of variable linking to give the reader a sense of the kind of data that we will be concerned with. As is widely known, intransitive verbs are assumed to allow for two different syntactic structures: unergatives that only have an external argument and unaccusatives with

only an internal argument. However, some verbs allow for both an unergative and an unaccusative use. An example of variable unaccusativity from Hindi-Urdu is given in (8):

- (8) a. **ur*-ii (huu-ii) ciṛyaa b. *ur*-ii (huu-ii) patang
 fly-PFV.FSG be-PFV.FSG bird.FSG fly-PFV.FSG be-PFV.FSG kite.FSG
 Intended: ‘the flown bird’ ‘the flown kite’ (Ahmed, 2010:8f.)

In both, the verb *ur* ‘fly’ is used in a reduced relative clause, a context that has been argued to only be compatible with unaccusative verbs. If the verb takes an animate argument, as in (8a), the reduced relative is unavailable, as expected for an unergative structure. However, with an inanimate argument as in (8b), the verb can form a reduced relative, indicating an unaccusative structure.

This switch in syntactic properties does not accidentally correlate with a change in the animacy status of the argument. For animates, flying is an intentional and effortful activity that is best expressed with an agent θ -role. For inanimates, on the other hand, flying is rather akin to passive floating and more compatible with a patient θ -role. Under the assumption that the external argument position is associated with an agent θ -role and the internal argument position with a patient θ -role, this semantic contrast matches the different syntactic properties of the verb in the two examples. In sum, the verb ‘fly’ can come either with an agent or with a patient θ -role depending on whether the flying is performed in an agent-like or patient-like manner.

I assume that under the individual role theory of θ -roles, both arguments in (8) would receive the same individual role, which we might label a flyer. However, this individual role would then be mapped onto two different θ -roles, namely, an agent role in (8a) and a patient role in (8b). This is the pattern that I call variable linking: an argument-structural alternation in which the individual roles and the θ -roles of a verb do not stand in a one-to-one mapping relationship. A flyer can be either an agent or a patient.

Chapter 4 will discuss several other cases of variable linking which are more interesting and sophisticated than variable unaccusativity. I believe that variable linking is ubiquitous and that any functioning theory of argument structure needs to be able to account for it. As should be evident, the individual role theory does not. If the meaning of a verb indeed consisted of a set of discrete individual roles which are then categorized into more abstract θ -roles, it is not clear how any flexibility or variation in the mapping between the two should arise. I argue that the solution to this conundrum consists not in modifying this mapping relation or in positing more fine-grained individual roles, but in tossing out the concept of individual roles altogether, alongside the understanding of verb meaning and θ -roles that it entails.

It bears highlighting that the individual role theory of θ -roles has rarely been explicitly defended in the literature, but it is not a straw man. It has been articulated explicitly within Cognitive Construction Grammar, as reviewed in Chapter 3, and to my knowledge, it remains the only theory on the market that directly addresses our research question. I suspect that implicitly, the concept of individual roles is taken for granted in much work on argument structure past and present, regardless of theoretical orientation. If I am wrong, and if every reader of these pages already recognizes from the outset that the individual role theory is flawed, the better.

To recap the main points of this section, the research question I pursue is how speakers know what it means for an argument to be interpreted with a given θ -role in the context of a given event description. Concretely, this study has three goals. The first goal, as already flagged earlier, is to convince the reader that the research question is a genuine question: it is not trivial, there is more than one possible answer to it, and the answer matters for argument-structural analysis and our understanding of the syntax-semantics interface and the nature of lexical meaning more broadly. My second goal is to show that the individual role theory of θ -roles fails even on its own terms. My third and loftiest goal is to contribute to developing an alternative account. The next section foreshadows the solution I will have to offer.

1.3 The solution, in brief

The proposal I develop in the following rests on the assumption that arguments are introduced by functional heads in the syntax. This idea is in no way novel, although I will take it to its extreme by applying it not only to the verb's external argument, as is commonly done, but to all arguments. The verb itself does not introduce any arguments.

Against this background, I will propose that the interpretation assigned to an argument with a given θ -role in the context of a given event – for example, what it means to be the patient of a stabbing event – is not already stored on the verb, but is generated compositionally by merging a functional head associated with a θ -role with an event description. This idea is anticipated in Marantz (1984) but has, to my knowledge, so far not been formulated explicitly as a comprehensive theory. Instead of taking ‘individual roles’ such as ‘stabbees’ and ‘kickees’ to be primary, which are then classified or categorized by more abstract θ -roles, I take θ -roles to be primary. What it means to be the patient of a stabbing event is not a primitive, but is derived by combining a patient-introducing head with a stabbing event.

This approach makes crucially different predictions from the individual role theory of θ -roles. First, I show that variable linking falls out naturally. Second, under my analysis, arguments are not arguments of verbs as such, but of whichever event description they combine with, and this is sensitive to syntactic constituency. For instance, in *Brutus stabbed Caesar*, the individual role theory would make Caesar the patient and Brutus the agent of a stabbing event, but under my analysis, Caesar is the patient of a stabbing event and Brutus the agent of a stabbing-Caesar event, which I argue is different from a stabbing event. I demonstrate that this asymmetry in the treatment of internal and external arguments is reflected in the data. Lastly, I argue that the analysis proposed naturally aligns with an exemplar-theoretic perspective on θ -roles, under which, say, an agent θ -role is a generalization formed over different agents *of* specific events, such as the agent of dancing, the agent of flying, and the agent of stabbing Caesar.

1.4 Outlook, or: A reader's manual

I end this introduction with an overview over the structure of this work and some advice on how to read it.

Chapter 2, ‘Some formalities,’ outlines the theoretical and methodological assumptions that I rely on throughout. All but the most superficial readers are advised to at least skim it.

Chapter 3, ‘A brief history of argument structure,’ discusses three main developments in the field of argument structure: projectionism, Cognitive Construction Grammar and Distributed Morphology (DM). In doing so, I focus on how these theories have approached the research question I pursue here. My goal in doing so is twofold. First, the individual role theory of θ -roles that I have sketched out above has been proposed explicitly by Goldberg (1995, 2006) within the framework of Cognitive Construction Grammar. Discussing Goldberg’s proposal in detail, I argue that it faces significant theoretical problems. My second goal is to show that in other approaches to argument structure such as projectionism and DM, no alternative account has been developed, and in fact, the research question has remained wholly unaddressed. For anyone who is interested in the broader argument that the present study makes, this chapter is recommended.

Chapter 4, ‘Six variations on variable linking,’ forms the empirical core of this work. It covers six argument-structural alternations that I argue qualify as cases of variable linking as defined above: variable unaccusativity, direct causatives of unergatives, direct causatives of ingestives, the agent-affectee alternation, pseudo-symmetric predicates and statives of ingestives. Each of these six case studies begins with a very short section entitled ‘Data and analysis,’ which introduces the relevant data, explains why they constitute an instance of variable linking, sketches out the cross-linguistic picture and points to relevant references. Readers who are mainly interested in the theoretical claims that this work makes can read these introductory sections only and skip the remainder of the case studies. Readers who are not invested in the general narrative but are interested in one specific case study should be able to read the relevant sections only, provided that they have skimmed Chapters 1 and 2.

Chapter 5, ‘Analysis,’ then presents the main theoretical proposal of this work. I begin by introducing the general idea that the interpretation of an argument bearing a given θ -role in the context of a given event description is not stored on the verb, but generated compositionally in the syntax. I then walk through the six case studies from Chapter 4 and show how they can be analyzed against this background. The first part of this chapter is key; the second is optional.

Chapter 6, ‘Ramifications,’ discusses some further questions and complications that arise for the proposal. Chapter 6.1 deals with entailments between different alternants of a verb, and Chapter 6.2 with the question of how knowledge about the syntactic distribution of verbs is encoded in speakers’ grammars. This chapter is optional.

Chapter 7, ‘Conclusion,’ summarizes the main claims of this work and outlines limitations and directions for future research. It is highly recommended.

Chapter 2: Some formalities

This chapter spells out the theoretical and methodological assumptions that the present work relies on, with a focus on θ -roles. I begin by sketching out an influential approach to θ -roles that I will in some ways rely on – Dowty’s (1991) proto-role theory – and introduce the thematic inventory we will work with. Next, I outline my assumptions about the status of θ -roles in the grammar, here diverging from Dowty. Lastly, I discuss how to diagnose θ -roles and to what extent syntactic evidence can be probative in this endeavour.

Before diving into these specifics, a few very general remarks on the kind of world that this study lives in. This is a generative approach. While I try to keep theoretical assumptions to a minimum, I will take for granted a standard constituency syntax with binary branching that largely obeys the principles of Minimalism (Chomsky, 1995) and a compositional semantics that can be expressed in predicate logic and typed lambda calculus (Heim and Kratzer, 1998). Verb meaning will be formalized in terms of events (Davidson, 1980; Parsons, 1990). Furthermore, I subscribe to two basic tenets of the DM architecture, namely, that there is no modular divide between the syntax and the morphology (‘syntax all the way down’) and that the distinction between lexical meaning and grammatical meaning is reflected in the syntax as the distinction between acategorical roots and functional heads (Halle and Marantz, 1993, 1994). This is not the only possible approach to the problem I tackle, and different architectures might make different contributions, but I believe that the present framework should allow us to make some progress in this matter.

2.1 Proto-roles and the thematic inventory

The function of θ -roles is to relate arguments to event descriptions. A θ -role specifies the way in which an argument is integrated into an event by providing a set of properties that characterize the argument in the event. In this study, I will adopt the idea going back to Dowty (1991) that θ -roles are not tidy categories defined by necessary and sufficient conditions, but prototypes in the sense of Rosch and Mervis (1975). Under this view, θ -roles have a center and a periphery. There is such a thing as a perfect, prototypical agent, at least in theory, which is defined by a set of properties diagnosable via entailments. But the arguments that actually occur in sentences typically possess only a subset of the relevant properties, resembling the agent prototype to varying degrees. In the case studies in Chapter 4, I will use proto-properties to detect θ -roles and to argue for or against particular thematic analyses of specific argument-structural alternations.

Dowty highlights that θ -roles can come at different levels of granularity and that nothing precludes different areas of the grammar – let alone different languages – being sensitive to different salient sets of properties. There can be a prototypical mammal at the same time as a prototypical dog, or a prototypical farm animal. I do not commit here to a comprehensive list of θ -roles that exist in the grammar, and the literature has never been able to produce such a list that has been met with anything resembling consensus. Rather, in

each case study, I will merely argue that two arguments which might be assumed to receive the same θ -role are in fact assigned different θ -roles. To this end, I will assume a minimal set of thematic distinctions that I believe at least some areas of the grammar can refer to, without thereby claiming that this exhausts the thematic inventory and without ruling out that what I refer to as a θ -role is perhaps better understood as a bundle or cluster of different roles. None of this affects the argument made.

Dowty himself posited only two broad proto-roles, agent and patient, with the proto-properties listed in (1) and (2):¹

- (1) *Contributing properties for the agent proto-role*
 - a. volitional involvement in the event or state
 - b. sentience (and/or perception)
 - c. causing an event or change of state in another participant
 - d. movement (relative to the position of another participant)
 - e. (exists independently of the event named by the verb)
- (2) *Contributing properties for the patient proto-role*
 - a. undergoes change of state
 - b. incremental theme
 - c. causally affected by another participant
 - d. stationary relative to movement of another participant
 - e. (does not exist independently of the event named by the verb) (Dowty, 1991:572)

While Dowty proposed that other θ -roles can simply be defined as various subsets or mixtures of the basic agent and patient proto-roles, for our purposes it will be helpful to have a few more concrete labels at our disposal. Chapter 4.3 will feature what I will refer to as a goal θ -role, defined by entailments such as being the endpoint of a path and receiving a transfer of possession, whether literal or metaphorical. Goals are commonly encountered as the indirect objects in standard ditransitives such as *give*. I remain agnostic as to whether goals, recipients and sources are best analyzed as distinct roles or should be lumped together (see, e.g., Beavers and Koontz-Garboden, 2020 for discussion). In Chapter 4.4, we will encounter a somewhat less canonical candidate, an affectee θ -role, defined by entailments such as being emotionally affected and suffering positive or negative consequences as a result of the event. The very minimal claim I make is simply that this θ -role is not identical to that of an agent. Lastly, Chapter 4.6 examines the θ -role featuring in stative passives, which I tentatively describe as the holder of a result state. Again, the only key claim is that this θ -role is not an agent.

The fact remains that θ -roles are hard to pin down and hard to tease apart. The semantic differences associated with them are real but also undeniably fuzzy, to an extent that can be physically uncomfortable for

¹Dowty parenthesizes the last item on each list, arguing that they could also be attributed to the discourse-related properties of subjecthood.

a formal linguist. If Dowty was right, then this fuzziness is not noise in the data; rather, θ -roles *are* inherently fuzzy concepts. Part of what the present study will achieve, I believe, is explain how this fuzziness arises from the grammar. I hope that this might also in future work allow us to gain a more robust understanding of the semantic profile of different θ -roles.

2.2 The status of θ -roles in the grammar

The prototype approach to θ -roles still leaves the question open how they are represented in speakers' grammars. Dowty himself regarded θ -roles merely as constraints on possible lexicalizations that play no active role in the grammar. In contrast, I assume that θ -roles are explicitly represented in the denotation of a sentence and correspond to functional heads in the syntax. In doing so, I follow much recent syntactically oriented work on argument structure, beginning with Marantz (1984) and Kratzer (1996).

Kratzer's original denotation for a Voice head introducing the agent argument is shown in (3a). This Voice head combines with the VP in its complement, e.g., (3b), via Event Identification, deriving (3c).

- (3) a. $\llbracket \text{Voice} \rrbracket = \lambda x. \lambda e. \text{Agent}(x)(e)$
 b. $\llbracket \text{VP} \rrbracket = \lambda e. \text{stab}(\text{Caesar})(e)$
 c. $\llbracket \text{Voice}' \rrbracket = \lambda x. \lambda e. \text{Agent}(x)(e) \wedge \text{stab}(\text{Caesar})(e)$

To anticipate, the formal analysis I will develop in Chapter 5 will diverge from Kratzer's original framework in two ways. First, I will change the type of argument-introducing heads so that they combine via Function Application. Second, while Kratzer argued that only external arguments are introduced by functional heads instead of by the verb itself, I will extend this treatment to all arguments. These two changes to Kratzer's original proposal are not novel or particularly radical.

As a final note on the semantics, it is widely acknowledged that verbal events, even those corresponding to a single verbal domain, can often be decomposed into several subeventualities. For instance, I will assume that direct causatives contain at least a result eventuality and a separate causing event (Schäfer, 2009, a.m.o.), as diagnosed most famously by *again* modification (Beck and Johnson, 2004; von Stechow, 1995, 1996). For the most part, I will sidestep this additional complexity, assuming that the subevents of each verbal domain together make up a superevent that constitutes the domain for thematic uniqueness (Carlson, 1984, 1998; Williams, 2015, a.o.). Once the basics of the analysis are spelled out, we will add a more complex event structure to the picture starting in Chapter 5.2.

2.3 How to diagnose a θ -role

With the grammatical status of θ -roles clarified, the question remains how to determine which θ -role a given argument receives. Unfortunately, the semantic intuitions that linguists have about θ -roles are notoriously slippery and contested. The point that the present work tries to make is precisely that these intuitions are

often mistaken in one particular respect, namely, in being guided by an argument's verb-specific entailments – 'individual roles' –, rather than by thematic entailments alone. In the absence of additional evidence, debating conflicting semantic intuitions could easily become an argumentative *cul-de-sac*.

In the case studies, I will thus buttress the thematic analyses I propose with syntactic evidence. However, while there is near-universal consensus that there is some sort of correlation between θ -roles and argument positions, the precise mapping relation between the two is, again, contested. In the remainder of this section, I outline the assumptions I make about this mapping and how I will use syntactic evidence in the case studies. Note that in doing so, I remain fully agnostic about the question why θ -roles would correlate with syntax in the first place; we are solely concerned with the descriptive generalizations.

In the literature, the perhaps strongest view that has been advanced on the θ -role-syntax interface is the UTAH (Uniformity of Theta-Assignment Hypothesis; Baker, 1988, see also Perlmutter and Postal, 1984), according to which each θ -role stands in a one-to-one mapping relationship with a particular argument position. The UTAH has been widely argued to be too strong, and more complex, many-to-many mapping algorithms have been proposed instead (see Levin and Rappaport Hovav, 2005 for an overview). Much hinges on the syntactic analysis assumed, such as whether deep unaccusativity is adopted. No clear consensus has emerged. At the same time, claiming that we know nothing about the relation between θ -roles and syntax would not be a valid stance either, nor one that is widely adopted; the correlations between the two are imperfect and murky, but undeniably real.

Thus, my pitch is as follows. Recall that I have posited different θ -roles for agents, patients, goals and affectees, and tentatively for holders of result states. Moreover, I posit that the verbal domain contains (at least) four different argument positions: external arguments, internal arguments, indirect objects (in the terminology of Pykkänen (2008), low applicatives) and high applicatives. In the spirit of rampant agnosticism, I do not commit to any particular syntax for any of these positions. My strategy will be to simply demonstrate that an argument in question patterns with what are widely believed to be external arguments, or internal arguments, etc., for the purposes of various diagnostics. There are competing proposals on the market on how these argument positions can be formalized in terms of the syntactic category of the head introducing the argument and the position of this head in the structure, but this work is not the place to adjudicate between them.

Furthermore, I take it to be well-established that agents are often associated with the external argument position, patients with the internal argument position, goals with the indirect object position, and affectees with the high applicative position. This is almost certainly not a strict one-to-one mapping. But if two arguments have syntactic properties associated with two different syntactic positions, and if the arguments also differ in their thematic properties, and if these thematic properties furthermore match those standardly associated with the respective syntactic positions of the arguments in question, then I believe that there is a strong case to be made that the two arguments are assigned different θ -roles. For instance, if one argument displays properties associated with the external and the other with the internal argument position, and if

the first argument receives a more agent-like, the second a more patient-like interpretation, then I will take this as evidence for two different θ -roles. Note that this is not a biconditional: if these requirements fail to be met, it does not follow that the arguments receive the same θ -role. The goal here is merely to establish a standard that allows us to diagnose with some degree of certainty when the θ -roles of two arguments differ, not when they are identical.

As for holders of result states – the arguments of statives, discussed in Chapter 4.6 –, the syntactic position canonically associated with this θ -role is particularly controversial. In the relevant case study, I will merely show that certain arguments, which I argue to be holders of result states rather than agents, do not occupy the external argument position.

To summarize, I adopt Dowty's (1991) view that θ -roles are prototypes, but unlike Dowty, I assume that θ -roles are explicitly represented in the formal denotation of a sentence and correspond to functional heads in the syntax. I do not commit to any comprehensive thematic inventory or a detailed picture of the syntax of the verbal domain. Rather, in each of the case studies, I merely argue that by relying on semantic and syntactic evidence in conjunction with each other, a strong case can be made that two arguments that might be assumed to receive the same θ -role are better analyzed as being assigned different roles. This way of operationalizing θ -roles and their mapping onto syntax is, I hope, sufficiently precise, coherent and plausible, but it is certainly not the only one. θ -roles are not things in the world that we could straightforwardly observe, but theoretical constructs whose validity depends on the work they do in explaining the world to us. Whether the view on θ -roles adopted here is a good one will need to be judged based on the overall success of this study as a whole.

Chapter 3: A brief history of argument structure

This chapter surveys previous approaches to the research question I pursue here, namely, how θ -roles are interpreted in the context of specific events. I focus on three main developments in the history of argument structure that have conceptualized the interface between lexical meaning and syntactic structure in very different ways: projectionism, Cognitive Construction Grammar and Distributed Morphology. This is a brief overview limited to only three frameworks, but taken together, I believe that they do a good job at defining the theoretical space in which previous accounts can be located. Moreover, even my overview over these three frameworks will not be exhaustive, glossing over many questions that are not directly relevant for our purposes. The goal of this chapter is not to evaluate the relative merit of different theories of argument structure in general, but to understand how they deal with a very particular problem.

3.1 Projectionism

I begin with a line of research on argument structure that places the main emphasis on structured lexical representations from which syntactic structures are projected, in accordance with the Projection Principle (Chomsky, 1981). Proponents of this view can be found both within the generative tradition, with Grimshaw (1979) and Pesetsky (1982) being some prominent early works, as well as outside of it in the context of LFG (Bresnan, 1982) and HPSG (Pollard and Sag, 1994). Here, I focus on the highly articulated and influential framework developed by Levin and Rappaport Hovav (Levin, 1993; Levin and Rappaport, 1986, 1988; Levin and Rappaport Hovav, 1995, 2005; Rappaport Hovav and Levin, 1988, 1998).

Unlike more recent theories, projectionism posits a generative lexicon as a grammatical module distinct from the syntax. Argument-structural knowledge – including knowledge that takes the form of productive rules – is located in this lexicon. A verbal lexical entry contains a structured semantic representation of a verb’s meaning together with a syntactic frame. In Levin and Rappaport Hovav’s work, the semantic representation is composed of the root itself and one or several abstract predicates such as BECOME, CAUSE and ACT. By way of example, the denotations of intransitive *break* and *open* and their respective transitive variants are given in (1) and (2):

- (1) a. [y BECOME <*BROKEN*>]
b. [y BECOME <*OPEN*>]
- (2) a. [[x ACT] CAUSE [y BECOME <*BROKEN*>]]
b. [[x ACT] CAUSE [y BECOME <*OPEN*>]] (Levin and Rappaport Hovav, 2005:71)

Note that although these denotations are not formulated in terms of standard θ -roles such as agent and

patient, an argument is still assigned a semantic role by virtue of being the argument of a particular abstract predicate. The resulting categories fulfill the same function as θ -roles in that they identify abstract properties that are shared by arguments of different verbs, such as having causal power (being the argument of CAUSE) or undergoing a change of state (being the argument of BECOME). For our purposes, whether abstract roles are represented in terms of θ -terminology or in terms of abstract predicates is secondary.

In addition to a semantic representation, each verb also comes with a syntactic frame labeled its argument structure; an example for ditransitive verbs such as *put* is given in (3). Levin and Rappaport Hovav assume that a verb's argument structure trivially maps onto D-structure.

- (3) $e, x \langle y, P_{loc}z \rangle$ (Levin and Rappaport Hovav, 1995:21)

Here, x , y and z stand in for the three arguments. The argument that is outside the brackets, x , is the external argument. Of the two internal arguments, y is a direct and z an indirect argument.

Finally, the mapping between the semantic and the syntactic representation of a given verb is governed by a number of linking rules that map positions in the semantic structure to positions in the syntactic structure. Two examples are given in (4); of course they do not exhaust the set of linking rules required.

- (4) a. *Immediate Cause Linking Rule*
 The argument of a verb that denotes the immediate cause of the eventuality described by that verb is its external argument. (Levin and Rappaport Hovav, 1995:135)
- b. *Directed Change Linking Rule*
 The argument of a verb that corresponds to the entity undergoing the directed change described by that verb is its internal argument. (Levin and Rappaport Hovav, 1995:146)

We can now turn to the status of verb-specific entailments in projectionism and the question of how speakers know what it means for an argument to be interpreted with a given θ -role in the context of a given event description. Recall that earlier, I have introduced the concept of individual roles such as stabbers and stabbees that form part of verb meaning and are mapped onto more abstract θ -roles. Projectionism does not explicitly represent such individual roles. In the denotations shown above, a verb like *break* does not come with a breaker and a breakee role, but with an agent and a patient role (or more precisely, an argument of CAUSE and an argument of BECOME). Arguments are described semantically only by their abstract role: the CAUSE arguments of *break* and *open* are both simply that, CAUSE arguments, and what this means in the context of a particular event description is not reflected anywhere in the semantics.

I believe that the implicit assumption that work in this school rests on is still the same as what I have sketched out earlier and what we will see spelled out explicitly in Cognitive Construction Grammar: that verbs come with verb-specific individual roles, and that θ -roles describe sets of these roles by identifying shared abstract components. The next section on Cognitive Construction Grammar will discuss in detail

where this view goes wrong. But for projectionism, the main problem is that what it means to be, say, an agent *of* a particular kind of event is simply not represented in the architecture. As a result, the theory does not offer an answer to our research question, or even a clear perspective on how to approach it in the first place.

3.2 Cognitive Construction Grammar

Construction grammar is an umbrella term for a broad range of theoretical approaches, originating with Fillmore and Kay (1993), Fillmore et al. (1988), Lakoff (1987), and Lambrecht (1994). A key premise underlying all of them is that the meaning of a complex construction is not reducible to the semantics of its parts – words or morphemes – composed in a predictable fashion. Rather, the construction itself is imbued with meaning, independently of the words or morphemes in the sentence. In the following, I focus on one prominent framework within this camp, Cognitive Construction Grammar, developed by Goldberg (1995, 2006; see also Boas, 2013 for an overview).

The central term ‘construction’ in this context denotes very broadly any pairing of form and meaning. Constructions exist at different levels, comprising individual words and idiomatic phrases such as *kick the bucket* as well as larger grammatical schemata such as passives or ditransitives. Construction grammar thus abandons the line between the syntax and the lexicon, instead positing a continuum between smaller and larger form-meaning pairings. Like words, larger constructions can be polysemous, being associated with a ‘core sense’ as well as peripheral interpretations. For instance, the ditransitive construction comes with the core sense of ‘agent successfully causes recipient to receive patient,’ but can also mean ‘agent enables recipient to receive patient,’ ‘agents intends to cause recipient to receive patient,’ ‘agent causes recipient not to receive patient,’ etc. (Goldberg, 1995:38). Constructions express basic types of human experience, being shaped through usage.

In consequence, Cognitive Construction Grammar rejects the projectionist claim that all relevant semantic components of an argument-structural configuration are contributed by the lexical entry of the verb. Rather, some information is encoded on the construction as a whole, whereas the verb itself comes with a more minimal semantic representation. Concretely, Goldberg proposes that the denotation of a verb contains a set of *participant roles*, corresponding to what I have labeled here individual roles, and the denotation of the construction contains a set of *argument roles*, corresponding to θ -roles.

Participant roles are part of the lexical semantics of the verb and describe the kinds of participants involved in the event described. For instance, the verb *hand* comes with three participant roles (5):

- (5) HAND < **hander handee handed** >

Goldberg further classifies participant roles as either profiled or not profiled. Syntactically, profiled participant roles are normally obligatorily expressed in the sentence. Semantically, they correspond to entities

that are ‘obligatorily accessed and function as focal points within the scene, achieving a special degree of prominence’ (Goldberg, 1995:44). For instance, the verbs *rob* and *steal* differ in that in the former, the goods are not profiled and can remain unexpressed, whereas in the latter, it is the target that is not profiled and thus does not need to surface (6)–(8). Profiled roles are boldfaced.

- (6) a. Jesse robbed the rich (of all their money).
b. *Jesse robbed a million dollars (from the rich).
- (7) a. Jesse stole money (from the rich).
b. *Jesse stole the rich (of money).
- (8) a. rob < **thief target** goods >
b. steal < **thief target goods** > (Goldberg, 1995:45)

In general, the function of the profiled/non-profiled distinction is to allow for, but also to constrain, argument-structural flexibility of verbs; that is, to account for the fact that verbs can often surface in more than one syntactic structure, but not in all. We will see further below how concretely profiling is leveraged to this end.

In addition to the participant roles (i.e., individual roles) on the verb, the construction itself is associated with a set of argument roles (i.e., θ -roles). For instance, the ditransitive CAUSE-RECEIVE construction comes with an agent, a recipient and a patient role (9).

- (9) a. X CAUSES Y to RECEIVE Z
b. CAUSE-RECEIVE < **agt rec pat** > (Goldberg, 1995:49)

As seen previously for participant roles, argument roles can be either profiled or not profiled; here, the distinction does not matter for our purposes.

The semantic representations corresponding to the verb and the construction, respectively, are then mapped onto each other; in particular, participant roles and argument roles need to be fused. This fusion is subject to two principles (10):

- (10) a. *The Semantic Coherence Principle*

Only roles that are semantically compatible can be fused. Two roles *r1* and *r2* are semantically compatible if either *r1* can be construed as an instance of *r2*, or *r2* can be construed as an instance of *r1*.¹ For example, the kicker participant of the *kick* frame may be fused with the agent role of the ditransitive construction because the kicker role can be construed as an instance of the agent role. Whether a role can be construed as an instance of another role is determined by general categorization principles.

¹Note that Goldberg’s original text says: ‘...or *r1* can be construed as an instance of *r1*.’ I believe that this is an error.

b. *The Correspondence Principle*

Each participant role that is lexically profiled and expressed must be fused with a profiled argument role of the construction. If a verb has three profiled participant roles, then one of them may be fused with a non-profiled argument role of a construction. (Goldberg, 1995:50)

I exemplify the fusion of roles with the verb *hand* and the ditransitive construction (11):

(11)	Sem	CAUSE-RECEIVE	< agt	rec	pat	>
	Root: instance, means	HAND	< hander	handee	handed	>
	Syn	V	SUBJ	OBJ	OBJ ₂	

(Goldberg, 1995:50)

We see three levels of representation: the semantics of the construction, the meaning of the verb merged in this construction, and the syntax. The event itself is a CAUSE-RECEIVE event at the level of the construction, with the root specifying further that the instance or means of this event is a handing. In the syntax, the event corresponds to a verb. As for the arguments, the *hander* participant is an agent realized as a subject, the *handee* participant is a recipient realized as an object, and the *handed* participant is a theme realized as a second object.

To briefly recap, Cognitive Construction Grammar posits two levels of roles – participant roles on the verb and argument roles on the construction – and formulates a mapping algorithm between the two. This is, to my knowledge, the most explicit formalization of the intuition that I have sketched out in the introduction and which I believe underlies a lot of work in argument structure from different theoretical backgrounds. The problems for Goldberg’s theory that I will discuss in the following should thus not be understood as being confined to a particular variant of Construction Grammar, but arise for this way of thinking about the lexicon-syntax interface in general.

The first problem concerns the mapping algorithm that Goldberg proposes. Leaving aside the fact that the ‘general categorization principles’ in (10a) would need to be fleshed out empirically, the architecture does not predict any variable linking, the crucial pattern introduced briefly in Chapter 1 and documented in much more detail in Chapter 4. It appears to be impossible for a single participant role to be mapped onto different argument roles in different alternants, or vice versa. The mapping depends exclusively on the fit between the intrinsic semantic content of the two sets of roles; it is not sensitive to context or otherwise expected to vary. Goldberg does allow for certain mismatches between participant roles and argument roles – one of which we will turn to in a moment –, but all of them involve cases where a profiled argument on one level corresponds to a non-profiled role or even no role on another, or where an argument fails to be realized at all. In short, variation in the number of arguments is predicted; variation in the mapping itself is not.

The problems go beyond the mapping algorithm. A general challenge in argument structure is how to account for the fact that verbs can productively appear in a wide range of syntactic structures, famously

demonstrated by Borer (2005b) for the English verb *siren* derived from the corresponding noun (12):

- (12) a. The factory horns sired throughout the raid.
 b. The factory horns sired midday and everyone broke for lunch.
 c. The police car sired the Porsche to a stop.
 d. The police car sired up to the accident site.
 e. The police car sired the daylight out of me. (Borer, 2005b:8)

Such data are hard to reconcile with the idea that verbs come with a limited number of roles stored as part of their lexical entry, and this has long been used as an argument against lexicalist frameworks like projectionism. By positing a finite set of participant roles per verb, Cognitive Construction Grammar appears to fall prey to the same criticism.

Goldberg does have an answer to this challenge. One way in which participant roles and argument roles are allowed to mismatch is that a profiled argument role can fail to be fused with any participant role. For instance, recall that the CAUSE-RECEIVE construction has three profiled argument roles. The verb *kick* can be merged in this construction despite the fact that it only has two participant roles (profiled or not profiled) (13)–(14):

- (13) Joe kicked Bill the ball.

(14)	Sem	CAUSE-RECEIVE	< agt	rec	pat	>
	Root: instance, means	KICK	< kicker		kicked	>
	Syn	V	SUBJ	OBJ	OBJ ₂	

(Goldberg, 1995:54)

The interpretation of *Bill* in (13) thus exclusively comes from the argument role and does not rest on a participant role; it is what is often called an unselected argument. This mismatch between the levels of participant roles and argument roles lets Cognitive Construction Grammar circumvent the problem that equipping verbs with a finite set of roles underpredicts their argument-structural flexibility.

However, the question remains how such unselected arguments are nonetheless integrated into the lexical semantics of the verb. In (14), what is predicated of Bill is that he receives the ball specifically by kicking, not via any other means of transfer, in the same way that Joe causes him to have the ball specifically by kicking. Hence, there is no clear reason why Joe but not Bill should receive a participant role from the verb. Both the agent and the recipient role must, as it were, be customized by the root. Arguments that are introduced only by the broader syntactic construction, not the lexical entry of the verb, will be the norm in DM territory as surveyed in the next section, and the challenges outlined for the latter framework will also apply to Goldberg's participant-role-free arguments. For now, I merely point out that nothing warrants the asymmetry in the treatment of the arguments in (14).

Moving on, a more fundamental problem is how to determine the number and type of participant roles that are specified as part of the verb's lexical entry. Goldberg proposes the 'No ____ing occurred' test to be used for this purpose (15). Any participant that is understood to obligatorily form part of the event thus described must correspond to a participant role encoded on the verb. For instance, (15a) denies that a kicking relation obtained between two participants. Goldberg does point out that the 'No ____ing occurred' test sometimes yields more than one possible interpretation, as in (15c), which she takes as evidence that the relevant verb is polysemous.

- (15) a. No kicking occurred. (two-participant interpretation)
 b. No sneezing occurred. (one-participant interpretation)
 c. No hammering occurred. (one-participant [sound emission] or two-participant [impact] interpretation)
 d. No giving occurred. (three-participant interpretation) (Goldberg, 1995:43f.)

The logic behind this diagnostic is that verbs themselves come with participant roles independent of the constructions in which they appear, but that in order to see these participant roles, we must observe the verbs in a construction-free environment. The question is why the 'No ____ing' context would constitute such an environment. Goldberg simply describes the diagnostic as a useful heuristic, but the concern is real. For one, at the risk of being petty, the verbs in (15) are not verbs, but nouns. More importantly, the 'No ____ing occurred' context *is* a construction in Goldberg's sense; it does not somehow allow the verbs to present themselves in a vacuum.

One might admit that the 'No ____ing occurred' test fails to achieve its objective, but still maintain that a verb comes with its own set of roles that can be determined by truly staring at the verb in isolation. Simply by contemplating *kick*, we could say, we just *know* that this verb comes with a kicker and a kickee argument. But the status of these intuitions is dubious. A different way of interpreting them is that what we are accessing when staring at *kick* by itself is a prototypical usage of the verb, perhaps as in *The boy kicked the ball*. It is not obvious that these prototypicality intuitions can be identified with the meaning of the verb as it enters any given derivation. In short, positing a level of participant roles as part of the verb's denotation, independent of the syntax, raises the question of how we ever have access to these roles, and the intuitions we might have in this respect cannot necessarily be taken at face value.

Lastly, Marantz (1984) has pointed out that the interpretation of an external argument can depend not only on the verb itself, but on the verb phrase as a whole. In (16)–(17), the subjects of the different verb phrases perform vastly different activities depending on the internal argument. Hence, it is not clear what the content of a taker or a killer participant role specified as part of the lexical meaning of *take* and *kill* would be, or, for that matter, what it would mean to be a takee or a killee.

- | | |
|--|---|
| (16) a. take a book from the shelf
b. take a bus to New York
c. take a nap
d. take an aspirin for a cold
e. take a letter in shorthand | (17) a. kill a cockroach
b. kill a conversation
c. kill an evening watching TV
d. kill a bottle (i.e., empty it)
e. kill an audience (i.e., wow it) |
|--|---|
- (Marantz, 1984:25)

It is tempting to dismiss cases like (16)–(17) as idiomatic uses of the verb that do not reflect its true denotation, but as Marantz argues, there is no clear line separating literal from idiomatic uses. What is more, Kratzer (1996) has pointed out that some of the examples in (16)–(17) can be extended productively, as in *kill an afternoon/a lovely morning/an entire day/....* This is incompatible with the view that these data are merely stored idioms. Thus, the fact remains that the interpretation of an argument is not reducible to a fixed participant role it receives from the verb, but is sensitive to context in a way that Goldberg’s architecture does not allow for.

Overall, individual roles (participant roles) and θ -roles (argument roles) are both explicitly represented in Cognitive Construction Grammar, and so is their mapping relationship. Unlike in projectionism, there are two levels of semantic representation – the lexical entry of the verb and the meaning of the construction –, which makes space for both verb-specific and abstract roles. However, I have pointed out four problems. First, since the mapping relation between participant roles and argument roles merely depends on their intrinsic semantic fit, variable linking is not predicted. Second, equipping verbs with a finite set of participant roles fails to account for their argument-structural flexibility, and allowing for some arguments to not be assigned a participant role merely introduces an unwarranted asymmetry. Third, it is an open question how we could know the number and kind of participant roles on a verb, and the status of the intuitions we have in this respect is dubious. Fourth, the concept of contentful participant roles does not square with the fact that the interpretation of arguments is determined not only by the verb itself, but sometimes by the verb phrase as a whole.

The first of these points of criticism – variable linking – will be the topic of the empirical work in Chapter 4. But the discussion of Cognitive Construction Grammar should have made it clear that variable linking notwithstanding, the idea that verbs come with individual roles that are then categorized by θ -roles faces serious theoretical challenges in and of itself. Thus, the strategy that I will pursue later in Chapter 5 is not to adopt Goldberg’s individual roles and only modify the mapping algorithm between individual roles and θ -roles or posit a more fine-grained set of individual roles, but to abandon the notion of individual roles altogether.

3.3 Distributed Morphology

The final theoretical development in argument structure that we need to discuss shares with Cognitive Construction Grammar the assumption that the semantics of an argument-structural configuration ought to be largely located in the structure itself, rather than – as proposed by projectionists – in the lexical entry of the verb. Unlike in Cognitive Construction Grammar, however, meaning is not holistically associated with a construction as such, but with functional morphemes that enter the syntactic derivation and combine in a predictable fashion. Such views have been espoused in a number of otherwise quite heterogeneous works (e.g., Alexiadou et al., 2015; Arad, 2003; Borer, 2005a, 2005b; Doron, 2003; Pytkäinen, 2008; Ramchand, 2008; Schäfer, 2008; see Marantz, 2013 for an overview). In the following, I focus on Distributed Morphology (DM; Halle and Marantz, 1993, 1994).

For our purposes, the crucial difference between the three theories of argument structure surveyed here concerns the lexical representation of verbs. While projectionism posited a richly structured semantic representation alongside a syntactic frame, and Cognitive Construction Grammar a more minimal entry only associated with participant roles (i.e., individual roles), DM reduces verbs even further to pure acategorical roots that have combined with a verbalizer. A verb like *kick*, under this view, simply denotes a kicking event; it has no further internal structure. Roots themselves are chunks of world knowledge that can be mapped onto a spell-out; they are linguistic objects only in the broadest sense. This architecture is motivated by DM's larger goal to place all productive, generative knowledge squarely in the syntax, in line with the Minimalist Program (Chomsky, 1995). Idiosyncratic lexical knowledge is reduced to inert conceptual blobs that are inserted into the syntactic machinery.

Accordingly, all the functional structure that projectionists placed within the lexical item, DM outsources to the syntax. The originally acategorical root is commonly assumed to first combine with a *v* head that introduces an eventuality. Additional events or states might be added by other heads to build a complex event meaning, similar to Levin and Rappaport Hovav's event decomposition in the lexicon. Arguments are introduced by, and receive a θ -role from, functional heads rather than the verb itself, a trend starting with Kratzer (1996) and Marantz (1984) for external arguments only and culminating in the highly modular and user-friendly framework of Pytkäinen (2008). Research on argument structure thus essentially becomes a matter of identifying the relevant functional heads involved in a particular configuration and of defining their denotation and their possible syntactic relations.

Having outlined the basics of the DM view on the lexicon-syntax interface, I now turn to the question of how the theory deals, or can deal, with our research question. The overall picture resembles what we saw for projectionism: what it means for a θ -role to be interpreted in the context of a given event is simply not addressed. For instance, it is not clear whether and to what extent DM predicts variable linking to be possible. Projectionism, I have suggested earlier, is arguably implicitly guided by the same idea that is spelled out explicitly in Cognitive Construction Grammar: that verbs come with verb-specific individual

roles which are then categorized by abstract θ -roles. In practice, this might be true for some work in the DM tradition as well. But leaving aside the various problems that the concept of individual roles raises in and of itself, I argue that it is also fundamentally at odds with the overall DM architecture.

Recall that a general disagreement between projectionism and Cognitive Construction Grammar on the one hand and DM on the other concerns the nature of lexical representations. Both projectionism and Cognitive Construction Grammar assume that verbs are represented with structured lexical entries, which could include individual roles such as stabbers and stabbees. In DM, however, lexical meaning takes the form of acategorical roots. One might claim that under this view, the acategorical root *stab* could then simply denote a relation between a stabber and a stabbee. But crucially, these individual roles would not be well-defined objects that the grammar can make reference to. If roots have internal semantic structure – e.g., if they contain a list of individual roles –, it cannot be structure that the grammar can operate on since all grammatical semantic structure in DM is syntactic structure, and roots are defined as whatever remains once all syntactic structure is factored out.

In other words, even if we could locate individual roles on the root, a mapping algorithm between individual roles and θ -roles could not be a well-defined component of the grammar because it strictly transcends the grammar’s boundaries. In DM, the fact that in *Brutus stabbed Caesar*, Brutus is the agent and Caesar the patient of the event is grammatical knowledge, but the fact that Brutus wields a sharp weapon and Caesar is penetrated by a sharp weapon is not. As far as the grammar is concerned, nothing enforces this interpretation rather than another because ‘wielding a sharp weapon’ or ‘being a stabber’ as components of lexical meaning would not be something that the grammar can make reference to in the first place.

Overall, what makes individual roles incompatible with DM is that they run counter to the fundamental assumption that lexical meaning takes the form not of structured lexical entries, but of acategorical roots whose internal semantic make-up is not accessible to the grammar. Making them accessible to the grammar would give us structured lexical entries, and it would give us projectionism, or perhaps Cognitive Construction Grammar, and this would bring us back to where we started.

In light of the challenges I have raised regarding the concept of individual roles, it is, in a sense, progress that DM abandons them altogether. However, this also leaves an explanatory gap. Like projectionism, DM does not provide a solution to our research problem, and unlike in projectionism, individual roles do not even make sense as an implicit assumption in the absence of an explicit theory. The task is thus to develop an alternative account.

3.4 Summary

This concludes our brief tour through three major theoretical developments in argument structure: projectionism, Cognitive Construction Grammar and DM. Only one of them, Cognitive Construction Grammar, explicitly addressed the question we are dealing with, proposing that the lexical entries of verbs come with individual roles which are then categorized by more abstract θ -roles. I have discussed problems with this view

not only for the mapping algorithm between individual roles and θ -roles, but for the concept of individual roles itself: data such as Borer's *siren* examples and Marantz's *take* and *kill* examples deal a very decisive blow to the idea that verbs come with a finite number of inherently contentful roles. Projectionism, while arguably animated by similar intuitions, does not have a separate level of representation for verb-specific roles; instead, θ -roles describe the arguments of verbs already in the lexical entry itself. Lastly, as for DM, I have argued that it is wholly incompatible with the notion of individual roles in the first place. If lexical meaning takes the form of unstructured acategorical roots, it cannot consist of a finite set of roles that are visible to the grammar and could be mapped onto θ -roles.

In sum, no ready-made answer to our research question exists. For moving forward, I believe that DM provides a better starting point for developing an answer to the puzzle than previous frameworks. In Chapter 5, I will develop an account of interpreting θ -roles in context that crucially relies on a DM-informed understanding of lexical meaning as acategorical roots. First, however, the next chapter will give us a better sense of the relevant data that any solution to the problem will need to be able to handle.

Chapter 4: Six variations on variable linking

This chapter surveys six argument-structural alternations that I analyze as instances of variable linking as defined earlier. That is, if the verbs involved in these alternations indeed came with a set of verb-specific individual roles, the latter would not stand in a one-to-one mapping relationship to θ -roles.

The purpose of these case studies is twofold. First, they demonstrate that variable linking is a real and pervasive phenomenon in argument structure, and they constitute a set of data that the theoretical proposal developed later will need to account for. A secondary motivation behind the case studies is that the alternations in question are interesting in their own right and, for the most part, contested. I will not attempt to present conclusive analyses for all of them and largely focus on the evidence they offer for the mapping between roots and structures. Nonetheless, my hope is that the perspective from variable linking will contribute to our understanding of these data more broadly.

All case studies are concerned with constructions that have been attested in a number of (often typologically unrelated) languages and that I take to be cross-linguistically common and widespread. However, to avoid sacrificing depth for breadth, I investigate each of them by focusing on data from two or three languages only. Naturally, I cannot make conclusive claims for languages I have not looked at in detail, and I hope that future research will fill those gaps.

That being said, only one of the case studies has any claim to originality, namely, the last one on statives of ingestives in Chapter 4.6. However, my analysis of these data also remains very preliminary. For all other case studies, the syntactic analysis I assume has been proposed before, and for some, the thematic analysis has too. While I do add some further diagnostics and data points to the picture, the contribution I aim to make mostly lies in assembling data from a range of seemingly disparate domains, showing that a common pattern emerges from them, and taking seriously the implications that this pattern has for the syntax-semantics interface.

The case studies are arranged in order of increasing complexity and all follow the same structure. A first section introduces the basic data and the analysis, explains why the alternation in question constitutes an instance of variable linking, and provides necessary references to cross-linguistic data and previous work. Readers who are mainly interested in the overarching plot of this study and do not wish to get too much into the empirical weeds can read these first sections only and skip the remainder of each case study. A second section then provides evidence for the analysis that makes the alternation a case of variable linking, and a third section discusses competing analyses.

4.1 Variable unaccusativity

4.1.1 Data and analysis

This first case study is concerned with variable behavior of intransitive verbs. As is well known, the Unaccusativity Hypothesis developed by Burzio (1981, 1986) and Perlmutter (1978) holds that intransitives fall into two syntactically distinct classes. In unergatives, the sole argument is base-generated in the same position as the subject argument of transitives, whereas in unaccusatives, it starts out in the same position as the object argument of transitives. In both cases, the argument subsequently raises to SpecTP. While unergatives and unaccusatives can thus be string-identical on the surface, a variety of diagnostics have been proposed to differentiate between the two. These include reduced relative clauses (Embick, 2004b), resultative secondary predicates (Levin and Rappaport Hovav, 1995), impersonal passives (Perlmutter, 1978), Hebrew possessor datives (Borer, 1994), Italian *ne*-cliticization (Burzio, 1986), Hindi-Urdu inabilitatives (Bhatt and Embick, 2017), Turkish adjectival participles on *-ık* (Acartürk, 2005; Acartürk and Zeyrek, 2010), and many more.

While the unergative/unaccusative distinction is sometimes treated as a contrast between two non-overlapping classes of verb types, it has long been noticed that many verbs sometimes pass as unergatives, sometimes as unaccusatives according to the same diagnostic (Borer, 2005b; Levin and Rappaport Hovav, 1995; Perlmutter and Postal, 1984; Sorace, 2000, a.m.o.). Example (1), already seen earlier, demonstrates that the Hindi-Urdu verb ‘fly’ variably qualifies as unergative (1a) and unaccusative (1b) for the purposes of reduced relatives depending on the choice of argument:

- | | | | | | | | | |
|-----|----|----------------------------|------------|----------|----|------------------------------------|------------|----------|
| (1) | a. | *uṛ-ii | (huu-ii) | ciṛyaa | b. | uṛ-ii | (huu-ii) | patang |
| | | fly-PFV.FSG | be-PFV.FSG | bird.FSG | | fly-PFV.FSG | be-PFV.FSG | kite.FSG |
| | | Intended: ‘the flown bird’ | | | | ‘the flown kite’ (Ahmed, 2010:8f.) | | |

Variable unaccusativity is highly pervasive cross-linguistically; I am not aware of any language in which it has been shown to be unattested. For Tsova-Tush, a Northeast Caucasian language from which we will see examples later, Holisky (1987:122–129) provides the following statistics:

- (2)
- a. 31 verbs behave exclusively as unaccusatives;
 - b. 27 verbs tend towards unaccusative behavior but can be coerced into an unergative use;
 - c. 61 verbs can behave either way, with a difference in meaning;
 - d. 36 verbs tend towards unergative behavior but can be coerced into an unaccusative use;
 - e. 78 verbs can only behave as unergatives.

As argued earlier, under a theory that posits individual roles, variable unaccusativity would constitute an instance of variable linking, defined as an argument-structural alternation in which the individual roles and the θ -roles of a given verb do not stand in a one-to-one mapping relationship. Example (1) would feature a single individual role, that of a flyer. However, I argue that the arguments are assigned two different θ -roles,

an agent role in (1a) and a patient role in (1b). Recall that I have proposed that positing two different θ -roles is licensed if a) the two arguments in question occupy different syntactic positions and b) they differ in their thematic properties in a way that is congruent with the θ -roles standardly associated with their respective positions. Assuming that unaccusativity diagnostics such as reduced relatives indeed diagnose a difference in syntactic position, the first condition is met: ‘the bird’ in (1a) occupies the external argument position, ‘the kite’ in (1b) the internal argument position. The second condition for positing distinct θ -roles is also met: in (1a), the participant is animate and moves of their own accord by expending energy, whereas in (1b), it is inanimate and merely passively floats in the air. Thus, the two arguments display abstract agent and patient properties, respectively.

Variable unaccusativity is a simple and robustly attested phenomenon and overall the least controversial case of variable linking that we will encounter in these pages. Accordingly, this case study is by far the shortest. To further corroborate the claim that the two syntactic variants of variable behavior verbs involve two different θ -roles, the next section surveys work by Sorace (2000, a.o.) on the semantic correlates of variable unaccusativity. A third section then briefly discusses a competing approach to the data, championed most prominently by Van Valin Jr (1990) and Dowty (1991), according to which unergative and unaccusative intransitives do not differ in their underlying syntax. I will argue that even from this perspective, variable linking persists.

4.1.2 Evidence for the analysis

In a series of papers, Sorace has accumulated evidence from various domains that syntactic diagnostics for unergativity/unaccusativity systematically correlate with semantic proto-agent and proto-patient properties, respectively (Bard et al., 2010; Keller and Sorace, 2003; Legendre and Sorace, 2003; Roehm et al., 2010; Sorace, 1993, 1995a, 1995b, 2000, 2004, 2011; Sorace and Cennamo, 2006; Sorace and Shomura, 2001). This finding holds on two levels. First, intransitives whose argument – by virtue of the lexical semantics of the verb – has neither clearly agent- nor clearly patient-like properties are more likely to display variable syntactic behavior, give rise to less clear-cut judgments in the context of unaccusativity diagnostics and are learned later both in L1 and in L2 acquisition. This includes, for example, existence of state verbs such as *last*, *seem*, *exist*, etc.

Secondly, and more crucially for our purposes, which syntax a variable behavior verb adopts in a given context depends on the thematic properties ascribed to the argument by the broader sentential context. As seen earlier in (1), animate arguments that act purposefully or volitionally are often realized as agents, inanimate arguments as patients. The same effect can be observed in the Italian example (3), where unaccusativity is diagnosed by the auxiliary ‘be,’ and unergativity by ‘have’:

- (3) a. Il pilota ha/?è atterrato sulla pista di emergenza.
 the pilot has/is landed on.the runway of emergency
 ‘The pilot landed on the emergency runway.’

- b. L'aereo è/?ha atterrato sulla pista di emergenza.
 the.plane is/has landed on.the runway of emergency
 'The plane landed on the emergency runway.' (Sorace, 2000:876)

By the same token, Closest Conjunct Agreement in Russian, only licensed with unaccusatives, is sensitive to the animacy of the argument (4) (Krejci, 2020):

- (4) a. *Na lestničnoj ploščadke stojal sosed i ego brat.
 on stairway landing stood.MSG neighbor.MSG.NOM and his brother.MSG.NOM
 Intended: 'My neighbor and his brother were standing on the stairway landing.'
 b. Na stole stojal stakan i kuvšin.
 on table stood.MSG glass.MSG.NOM and jug.MSG.NOM
 'On the table stood a glass and a jug.' (Krejci, 2020:126f.)

However, agentivity is not reducible to animacy. In Tsova-Tush, unergatives surface with ergative case marking, unaccusatives with nominative case (Holisky, 1987). In (5), both case marking patterns are licensed, but with differences in interpretation. If the argument is realized as an external argument with ergative case (5a), it is interpreted as performing the activity knowingly and purposefully, but if it is realized as an internal argument with nominative case (5b), it is interpreted as undergoing the activity against their will.

- (5) a. (as) vuiž-n-as.
 1SG.ERG fell-AOR.1SG-ERG
 'I fell down, on purpose.'
 b. so vož-en-sO.
 1SG.NOM fell-AOR.1SG-NOM
 'I fell down, by accident.' (Holisky, 1987:105)

In the examples seen so far, the decisive thematic properties were all associated with agenthood – animacy/sentience, purpose and volition. Among the properties associated with patienthood, variable unaccusativity has been shown to be sensitive to whether an argument undergoes a change of state. In the Dutch example (6), using again auxiliary selection as an unaccusativity diagnostic, the verb 'roll' in isolation takes the auxiliary 'have,' indicating an unergative syntax. However, once the verb combines with the PP *naar beneden* 'downstairs' that describes a result state, it surfaces with the auxiliary 'be,' indicating an unaccusative syntax. In (6b), the ball undergoes a change of state – from being upstairs to being downstairs – and is therefore assigned a patient θ -role. The same effect can be observed in the Italian example (7).

- (6) a. De bal heeft/*is gerold.
 the ball has/is rolled
 'The ball rolled.'
 b. De bal is/*heeft naar beneden gerold.
 the ball is/has to down rolled
 'The ball rolled downstairs.' (Sorace, 2000:876)

- (7) a. Ugo ha corso meglio ieri.
 Ugo has run better yesterday
 ‘Ugo ran better yesterday.’
 b. Ugo è corso a casa.
 Ugo is run to home
 ‘Ugo ran home.’ (Rosen, 1984:66f.)

To summarize, variable unaccusativity is not only a syntactic alternation but accompanied by systematic semantic differences. For a given variable behavior verb, its argument displays more agent properties in the unergative variant and more patient properties in the unaccusative variant. This clearly indicates that the argument is assigned two different θ -roles in the two alternants, making variable unaccusativity, for a theory that has individual roles, a case of variable linking.

4.1.3 Competing analyses

In the previous literature, the idea that variable unaccusativity signals the presence of two different θ -roles, while not always explicitly acknowledged, has to my knowledge never been explicitly denied. In the following, I briefly discuss only one competing approach, namely, how variable unaccusativity would be analyzed in a framework that denies syntactic unaccusativity.

4.1.3.1 Against syntactic unaccusativity: Van Valin (1990) and Dowty (1991)

The Unaccusativity Hypothesis holds that unergatives and unaccusatives differ in whether their sole argument is base-generated in the external or internal argument position, which are associated with different θ -roles. The difference in the underlying syntax is then reflected in unergativity/unaccusativity diagnostics. But an alternative perspective has always been that such diagnostics are rather directly sensitive to semantic factors without a difference in the underlying argument position. For instance, reduced relatives might not require an internal argument, but simply an argument with a patient interpretation, and impersonal passives not an external argument, but an argument with an agent interpretation. Defended perhaps most famously by Van Valin Jr (1990), this view has also been espoused by Dowty (1991). I will not endeavor here to defend a syntactic approach to unaccusativity, the evidence for which I believe to be strong, and I refer the reader to Levin and Rappaport Hovav (1995), *a.m.o.*, for further discussion. However, it is worth briefly considering for the sake of argument how variable unaccusativity would be analyzed under a purely semantic approach to unaccusativity.

Suppose that unaccusativity diagnostics are constructions that are directly sensitive to some sort of semantic role assigned to an argument, rather than to its syntactic position.¹ To return to example (1), repeated below as (8), reduced relatives in Hindi-Urdu would only be licensed with arguments that are assigned some role X. To account for variable unaccusativity, the sole argument of ‘fly’ would then have to bear role X in (8b), but not in (8a):

¹Note that Van Valin’s (1990) analysis of unaccusativity, couched in the framework of Role and Reference Grammar, distinguishes between semantic macroroles and thematic relations, and also argues that some unaccusativity diagnostics are sensitive to lexical aspect rather than properties ascribed to the argument. I here gloss over these implementational details.

- (8) a. *uṛ-ii (huu-ii) ciṛyaa
 fly-PFV.FSG be-PFV.FSG bird.FSG
 Intended: ‘the flown bird’
- b. uṛ-ii (huu-ii) patang
 fly-PFV.FSG be-PFV.FSG kite.FSG
 ‘the flown kite’ (Ahmed, 2010:8f.)

In consequence, variable linking persists, only in a slightly different guise. Whether θ -roles map onto argument positions which then determine the outcome of unaccusativity diagnostics, or whether θ -roles directly determine the outcome of unaccusativity diagnostics without a syntactic middleman, the fact remains that the flyer argument would be analyzed as being assigned different θ -roles in (8a) and (8b). The fact that variable unaccusativity constitutes an instance of variable linking is not contingent on endorsing the Unaccusativity Hypothesis, but holds under a purely semantic approach as well.

4.2 Direct causatives of unergatives

4.2.1 Data and analysis

The next case study of variable linking is concerned with a more complex case than variable unaccusativity: direct causatives of unergatives. In many languages, verbs that as intransitives pass unergativity diagnostics can also appear in direct causatives, shown here for Hindi-Urdu (9), Turkish (10) and Sason Arabic (11):

- (9) a. Rohan naach rahaa hai.
 Rohan.M dance PROG.MSG be.PRS.3MSG
 ‘Rohan is dancing.’
- b. Shama Rohan-ko nach-aa rahii hai.
 Shama.F Rohan-DOM dance-CAUS PROG.F be.PRS.3MSG
 ‘Shama is making Rohan dance/twirling him around (the dance floor).’
- (Bhatt and Embick, 2017:124)
- (10) a. Bebek uyu-du.
 baby sleep-PST
 ‘The baby slept.’
- b. (Ben) bebeğ-i uyu-t-tu-m.
 I baby-ACC sleep-CAUS-PST-1SG
 ‘I put the baby to sleep.’
- (11) a. i-zak.
 3M-laugh
 ‘He laughs.’
- b. a-zakkiy-u.
 1SG-laugh.CAUS-him
 ‘I make him laugh.’ (Akkuş, 2021a:175)

Note that in all three languages, the transitives surface with regular causative morphology. Cross-linguistically, direct causatives of unergatives have also been attested in Georgian (Harris, 1981; Nash, 2021), Eastern Armenian (Megerdumian, 2002), Niuean (Massam, 2009; Tollan and Massam, 2022), Acehnese (Legate, 2014), Samoan (Tollan, 2018; Tollan and Massam, 2022), Algonquian (Tollan and Oxford, 2018), Tagalog (Nie, 2020, Kipsigis Kouneli, 2021), Quechua (Myler, 2022) and Malayalam (Krishnan and Sarma, 2023).

I propose that direct causatives of unergatives in Hindi-Urdu, Turkish and Sason Arabic are syntactically and semantically identical to direct causatives of unaccusatives, with the subject merged as an external and

the object as an internal argument. Moreover, I argue that while the subject of the intransitive receives an agent θ -role, the object of the transitive is assigned a patient θ -role, with the agent θ -role instead going to the external causer.

For theories that posit individual roles, direct causatives of unergatives would thus constitute a case of variable linking. For instance, in the Hindi-Urdu example (9), I assume that the argument *Rohan* would be analyzed as receiving the individual role of a dancer. However, under the analysis proposed here, the argument is assigned an agent θ -role in the intransitive and a patient θ -role in the transitive, parallel to what we saw for variable unaccusativity. In addition, the same θ -role is mapped onto different individual roles in the two alternants: in the intransitive, the agent is a dancer, but in the transitive, it corresponds to whichever individual role is assigned to Shama.

The syntactic analysis I adopt here is not novel. It has previously been proposed by Harris (1981), Legate (2014), and Nie (2020) for Georgian, Acehnese and Tagalog, respectively;² both Legate and Nie have also pointed out the deagentivized interpretation of the object. In a similar vein, Levin and Rappaport Hovav (1995) have argued that English verbs of emission such as *beam*, *shine*, *buzz* and *flash* are ambiguous between an unergative and an unaccusative/causative use, with *the flashlight* in (12) being realized as an agent in the intransitive and as a patient in the transitive:

- (12) a. The flashlight beamed/shone.
 b. We beamed/shone the flashlight. (Levin and Rappaport Hovav, 1995:115)

However, Levin and Rappaport Hovav consider this alternation a spurious lexical idiosyncrasy not governed by a general rule. In contrast, part of the point that this chapter tries to make is that the phenomenon is cross-linguistically robust and ought to inform our understanding of θ -roles.³

Alternative approaches to direct causatives of unergatives have been developed, three of which are discussed at the end of this case study: Massam (2009) and subsequent work, Ramchand (2008) and Biggs (2019). But first, I provide evidence for the analysis proposed here: after confirming that the examples in question constitute true direct causatives of unergatives, I show that the objects of direct causatives of unergatives have syntactic properties associated with internal arguments and semantic properties associated with patients.

²Nie (2020) argues that only some causatives of unergatives in Tagalog have a transitive structure in which the object occupies the internal argument position, and that others, in which the object does not receive a deagentivized interpretation, have a Voice-over-Voice structure.

³The thematic analysis adopted here also appears to have been anticipated in Pāṇini's *kāraka* system (*Aṣṭ.* 1.4.52): 'Of the verbs having the sense of 'motion,' 'knowledge or information' and 'eating,' and of verbs that have some literary work for their object, and of intransitive verbs, that which was the agent of the verb in its primitive (non-ni or non-causal state), is called the object (karma) in its causative state [...].' *Karma* is a heterogeneous category but corresponds roughly to a patient/theme θ -role defined primarily as being affected by the action. For verbs of eating, knowing and reading, see the next case study on ingestives; Pāṇini analyzes the corresponding direct causatives as having two *karma* roles.

4.2.2 Evidence for the analysis

4.2.2.1 Background

I begin by establishing some basic facts about direct causatives of unergatives in Hindi-Urdu, Turkish and Sason Arabic: I show that the intransitives pass unergativity diagnostics, and that the transitives are direct rather than indirect causatives.

The intransitives are unergatives This section presents several diagnostics for each of the three languages according to which some causativizing verbs should be classified as unergatives.

For Hindi-Urdu, Bhatt and Embick (2017) provide three unergativity diagnostics. First, unergatives are not able to appear in reduced relative environments whereas unaccusatives are (13):

- (13) a. *hās-aa huaa laṛkaa
 laugh-PFV be.PFV boy
 Intended: ‘the laughed boy’
 b. khul-aa huaa darwaazaa
 open-PFV be.PFV door
 ‘the opened door’ (Bhatt and Embick, 2017:121)

Secondly, unergatives but not unaccusatives can form impersonal passives (14):

- (14) a. calo, daur-aa jaa-ye.
 come run-PFV PASS-SBJ
 ‘Come, let it be run.’ (i.e., come, let us run)
 b. *calo, kaṭ-aa jaa-ye.
 come cut.INTR-PFV PASS-SBJ
 Intended: ‘Come, let us get cut.’ (Bhatt and Embick, 2017:123)

Finally, unergatives and unaccusatives differ in their behavior in the so-called inabilitative construction, which is used to express that the subject is unable to perform a certain activity. Unergatives, patterning with transitives in this respect, can only appear in the inabilitative with passive (15a) but not with active syntax (15b). Unaccusatives, on the other hand, do not allow passive (16a) but only active syntax (16b):

- (15) a. Nina-se daur-aa nahī ga-yaa.
 Nina-INST run-PFV NEG PASS-PFV
 ‘Nina couldn’t run.’
 b. *Nina-se Mona nahī daur rahī hai.
 Nina-INST Mona.F NEG run PROG.F be.PRS.SG
 Intended: ‘Nina is unable to make Mona run.’
 (16) a. *dhabbō-se miṭ-aa nahī ga-yaa.
 stains-INST wipe.INTR-PFV NEG PASS-PFV
 Intended: ‘The stains weren’t able to bring themselves to erase themselves.’

- b. Nina-se dhabbe nahĩ miṭ-e.
 Nina-INST stains.M NEG wipe.INTR-PFV.MPL
 ‘Nina wasn’t able to wipe away the stains.’ (Bhatt and Embick, 2017:122)

The vast majority of Hindi-Urdu verbs which qualify as unergatives according to these diagnostics can form direct causatives (Bhatt and Embick, 2017).⁴

The following three diagnostics confirm the unergative status of some causativizing verbs in Turkish. First, unaccusatives but not unergatives can combine with the adjectival participle *-ık* (Acartürk, 2005; Acartürk and Zeyrek, 2010), as shown in (17):⁵

- (17) a. kır-ık bardak
 break-ADJ glass
 ‘broken glass’
 b. *uyu-k bebek
 sleep-ADJ baby
 Intended: ‘slept baby’

Secondly, (18) demonstrates that the agent nominalizer *-ıcı* can only surface with unergatives or transitives:

- (18) a. koş-ucu
 run-NMLZ
 ‘runner’
 b. sat-ıcı
 sell-NMLZ
 ‘seller’
 c. *düş-ücü
 fall-NMLZ
 Intended: ‘faller’

Finally, only unergatives can form impersonals in episodic contexts (19a) (Acartürk, 2005; Acartürk and Zeyrek, 2010; Akkuş, 2021a; Legate et al., 2020). Impersonals of unaccusatives, while felicitous under a habitual reading, cannot receive an episodic interpretation (19b):

- (19) a. Dün burada uyu-n-du.
 yesterday here sleep-IMPERS-PST
 ‘People/one slept here yesterday.’
 b. *Dün burada öl-ün-dü.
 yesterday here die-IMPERS-PST
 Intended: ‘People/one died here yesterday.’

⁴The only unergatives reported by Bhatt and Embick (2017) to not causativize are onomatopoeic denominatives such as *bilbilaa-naa*, ‘cry in pain.’ While I cannot discuss these cases in detail, it appears that such verbs already contain the morpheme *-aa*, suggesting that they are of a causative nature to begin with and thus cannot undergo further causativization.

⁵Interestingly, the addition of a causative morpheme does not improve the acceptability of unergatives with *-ık*:

- (i) *uyu-t-uk bebek
 sleep-CAUS-ADJ baby
 Intended: ‘slept baby’

This unacceptability can be attributed to morphological ill-formedness: Güler (2014) proposes that *-ık* attaches directly to the root, which suggests that the causative affix cannot intervene between the two.

Several Turkish verbs which pass these three unergativity diagnostics causativize, such as ‘sleep’ (10), ‘walk’ (54) and ‘sit’ (31).

Finally, unergatives can be detected in Sason Arabic using the following diagnostics. First, resultative secondary predicates require the presence of an internal argument and are thus only licensed with unaccusatives (20a). With unergatives (20b), the adjective can only have a depictive reading:

- (20) a. *sabi sar / var raxu.*
 boy became / fell sick
 ‘The boy became/fell sick.’
 b. *#sabi faqaz raxu.*
 boy ran sick
 Intended: ‘The boy ran himself sick, became sick as the result of running.’

Secondly, only unergatives are able to form impersonal passives. Unaccusatives in Sason Arabic can surface with the same affix, but the resulting constructions do not qualify as true passives. First, impersonals of unaccusatives are restricted to human referents: (21b) is infelicitous in a situation in which, for example, animals fall. True impersonal passives in Sason Arabic, on the other hand, can take non-human referents, as seen in (21a). Secondly, impersonals of unaccusatives do not license a *by*-phrase:

- (21) a. *in-nam nihane (mı zıyar / yorif).*
 PASS.IPFV-sleep.IPFV.3M here (by children / sheep)
 ‘It is slept here (by the children/sheep).’
 b. *in-vır nihane (*mı zıyar).*
 IMPERS-fall here (by children)
 ‘People fall here/one falls here.’

The last diagnostic comes from path arguments and cognate objects. Following Kuno and Takami (2004) and Nakajima (2006), I assume that the latter can have two distinct syntactic realization. If they are true arguments of the verb, they are located in its complement position, are thus only compatible with unergatives and can passivize. I remain agnostic as to whether such arguments are base-generated in this position, as well as if and how they are assigned a θ -role. Alternatively, path arguments and cognate objects can be realized as adjuncts, in which case they can also surface with unaccusatives and resist passivization. Thus, (22) demonstrates that ‘laugh’ has unergative status in Sason Arabic since it combines with a cognate object which can passivize. On the contrary, unaccusative ‘rot’ can only take an adjunct cognate object which fails to passivize (23):

- (22) a. *zake-ma kotti zak.*
 laugh-a bad laughed.3M
 ‘He laughed a bad laugh.’
 b. *zake-ma kotti in-zak (mı zıyar).*
 laugh-a bad PASS.PFV-laugh.PFV by children
 ‘A bad laugh was laughed (by the children).’
- (Akkuş and Öztürk, 2017:2)

- (23) a. badıncanad pat-ma gize kotti patto.
 tomatoes rotting-a such bad rotten.3PL
 ‘The tomatoes rotten such a bad rotting.’ (Akkuş and Öztürk, 2017:3)
- b. *pat-ma gize kotti in-pat (mı badıncanad).
 rotting-a such bad PASS.PFV-rot.PFV by tomatoes
 Intended: ‘Such a bad rotting was rotten (by the tomatoes).’

Like Hindi-Urdu and Turkish, Sason Arabic has several verbs which pass these unergativity diagnostics but also form direct causatives, including ‘laugh’ (11) ‘sleep’ (38), ‘jump’ (37) and ‘run’ (40). In sum, in all three languages, some causativizing verbs should be classified as unergatives based on standard diagnostics.

The transitives are direct causatives I now give an overview over the various causativization strategies in all three languages and show that unergatives can form direct as opposed to indirect causatives. To first briefly introduce a diagnostic I draw on, it is well-known that we can distinguish between direct and indirect causatives using adverbial modification (e.g., Martin and Schäfer, 2014), exemplified for English in (24):

- (24) a. John₁ awoke Bill₂ quickly_{1/*2}.
 b. John₁ made Bill₂ awake quickly_{1/2}.

In direct causatives, adverbials such as *quickly* can only describe the subject’s action, not the object’s: example (24a) is false if John is not acting quickly. Indirect causatives such as (24b), on the other hand, which contain a separate causing event, allow the adverbial to describe either the action of the causer or of the object, given that the latter is the subject of the embedded event.

Hindi-Urdu has three morphologically distinct causatives, derived via changes in vowel length of the root, the morpheme *-aa* and the morpheme *-vaa*, respectively (Bhatt and Embick, 2017). Since the first strategy does not apply to unergatives, I will focus on the latter two. Causatives formed with *-aa* receive a direct interpretation, those formed with *-vaa* an indirect interpretation: the former can only be used to describe a situation in which the causer directly and physically acts on the object whereas in the latter, the relation between causer and object is mediated in some way. Example (25) shows an unaccusative (25a), a direct *-aa* causative (25b) and an indirect *-vaa* causative (25c). An intermediate causer marked with *-se* is unavailable in the direct causative (25b).

- (25) a. makaan jal rahaa hai.
 house.M burn PROG.M be.PRES
 ‘The house is burning.’
- b. dakaitō-ne makaan jal-aa diyaa.
 bandits-ERG house.M burn-CAUS give.PERF.M
 ‘Bandits burned the house.’
- c. zamiindaar-ne (dakaitō-se) makaan jal-vaa diyaa.
 landlord-ERG bandits-INS house.M burn-CAUS2 give.PERF.M
 ‘The landlord had the house burned (by the bandits).’ (Bhatt and Embick, 2017:94f.)

Unergatives can form both *-aa* (26a) and *-vaa* (26b) causatives. Intermediate agents are only available with the latter, supporting the claim that the former are direct:⁶

- (26) a. Shama (*Mina-se) Rohan-ko nach-aa-egii.
 Shama Mina-INST Rohan-DOM dance-CAUS-FUT.F
 ‘Shama is making Rohan dance/twirling him around (the dance floor).’
 b. Shama Mina-se Rohan-ko nach-vaa-egii.
 Shama Mina-INST Rohan-DOM dance-CAUS2-FUT.F
 ‘Shama makes Mina make Rohan dance.’

Yet further evidence is provided by adverbial modification. In the *-aa* causative in (27a), the adverb ‘in a strange way’ can only target a single event, obligatorily giving rise to the interpretation that the way in which Shama is acting is strange. In *-vaa* causatives, on the other hand, the adverb can either modify the main clause event (27b), or – with a different word order – the event in the embedded clause (27c):

- (27) a. Shama Rohan-ko ajiib tarah-se nach-aa rahii hai.
 Shama Rohan-DOM strange way-INST dance-CAUS PROG.F be.PRS.3MSG
 ‘Shama, in a strange way, is making Rohan dance.’
 b. Shama-ne ajiib tarah-se Mina-se Rohan-ko nach-vaa-yaa.
 Shama-ERG strange way-INST Mina-INST Rohan-DOM dance-CAUS2-PFV
 ‘Shama, in a strange way, makes Mina make Rohan dance.’
 c. Shama-ne Mina-se Rohan-ko ajiib tarah-se nach-vaa-yaa.
 Shama-ERG Mina-INST Rohan-DOM strange way-INST dance-CAUS2-PFV
 ‘Shama makes Mina, in a strange way, make Rohan dance.’

In (27b), Shama herself is acting strangely, while in (27c), she asks Mina to behave in a strange way. Moreover, *-vaa* causatives can surface with two distinct adverbs, one targeting each event (28a), such that Shama is acting in a strange way but asks Mina to proceed in a good way to make Rohan dance. This is not possible with *-aa* causatives (28b):

- (28) a. Shama-ne ajiib tarah-se Mina-se Rohan-ko acchii tarah-se nach-vaa-yaa.
 Shama-ERG strange way-INST Mina-INST Rohan-DOM good way-INST dance-CAUS2-PFV
 ‘Shama, in a strange way, is making Mina, in a good way, make Rohan dance.’
 b. *Shama-ne ajiib tarah-se Rohan-ko acchii tarah-se nach-aa-yaa.
 Shama-ERG strange way-INST Rohan-DOM good way-INST dance-CAUS-PFV
 Intended: ‘Shama, in a strange way, is making Rohan dance in a good way.’

⁶A curious exception to this rule is the fact that *-aa* causatives formed from certain transitives receive an indirect reading, as demonstrated by their ability to combine with an intermediate agent:

- (i) Anjum-ne (Saddaf-se) paoda kaṭ-aa-yaa.
 Anjum-ERG Saddaf-INST plant cut-CAUS-PFV
 ‘Anjum had Saddaf cut the/a plant.’ (Ramchand, 2008:160)

Example (i) has the same meaning as the corresponding *-vaa* causative. I must leave it open why the *-aa* morpheme can be used in indirect causatives under these circumstances; however, as (26) shows, this confound does not affect the status of *-aa* causatives of unergatives and can thus be neglected for our purposes. Note also that my native speaker consultant rejects (i), but agrees with the broader point.

To summarize, *-vaa* causatives make two events available for modification, but *-aa* causatives only one, confirming that the latter are direct and do not involve a separate causing event.

Turkish has a single causativization strategy, realized with various allomorphs: *-Dir*, *-t*, *-Ir*, *-Ar*, *-It* and *-Art* (Akkuş, 2021a). The resulting causatives receive either a direct or an indirect reading depending on the verbal structure they combine with. Causatives of unaccusatives are obligatorily direct (29):⁷

- (29) a. Kalem masa-dan düş-tü.
pencil table-APL fall-PST
'The pencil fell from the table.'
- b. Leyla kalem-i masa-dan düş-ür-dü.
Leyla pencil-ACC table-APL fall-CAUS-PST
'Leyla dropped the pencil from the table.'
Not: 'Leyla caused someone to drop the pencil from the table.' (Akkuş, 2021a:216)

Causatives of transitives, in contrast, must be interpreted as indirect (30):

- (30) a. bütün misafir-ler araba-yı temizle-di-ler.
all guest-PL car-ACC clean-PST-3PL
'All the guests cleaned the car.'
- b. bütün misafir-ler-e araba-yı temizle-t-ti.
all guest-PL-DAT car-ACC clean-CAUS-3PL
'(S/he) made all the guests clean the car.' (Akkuş, 2021a:215)

Causatives of unergatives, finally, are ambiguous between a direct and an indirect reading (31):

- (31) a. Çocuk koltuğ-a otur-du.
child couch-DAT sit-PST
'The child sat on the couch.'
- b. (Ben) çocuğ-u koltuğ-a otur-t-tu-m.
I child-ACC couch-DAT sit-CAUS-PST-1SG
'I sat the child on the couch.'
'I made the child sit on the couch.'

If the causative is interpreted as direct, (31b) must mean that the speaker physically lifts up the child and places them on the couch, whereas under an indirect reading, the speaker might, for instance, order or persuade the child to sit on the couch or bring about this state of affairs in some other unspecified way. This ambiguity is confirmed by adverbial modification: if (32) receives a direct interpretation, no adverb can describe the action of the object, whereas this is possible under the indirect interpretation.

- (32) (Ben) sakince bebeğ-i koltuğ-a yavaşça otur-t-tu-m.
I calmly baby-ACC couch-DAT slowly sit-CAUS-PST-1SG
'Calmly and slowly, I sat the baby on the couch.'
'Calmly, I made the baby sit on the couch slowly.'

⁷Akkuş (2023) argues that what rules out indirect causatives of unaccusatives in Turkish is that the embedding *v* head obligatorily selects a VoiceP; see also Akkuş (2021b) for the same issue in Sason Arabic.

Again, this demonstrates that causatives of unergatives can have a direct interpretation without a separate causing event.

Finally, Sason Arabic has four distinct causatives, of which two are morphological and two periphrastic (Akkuş, 2021a). Since the latter are obligatorily indirect, I will only discuss morphological causatives, derived via ablaut and gemination, respectively. Ablaut causatives always receive a direct reading and are more restricted than geminates, applying only to a limited subset of unaccusatives and even fewer unergatives. Examples (33) and (34) show an ablaut causative of an unaccusative and an unergative, respectively, both obligatorily interpreted as direct:

- (33) a. *lāke tal-e.*
 stain came.out-3F
 ‘The stain came out.’
 b. *tel-tu lāke.*
 came.out.CAUS-1SG stain
 ‘I got the stain out.’
 Not: ‘I caused someone to get out the stain.’ (Akkuş, 2021a:91)
- (34) a. *nam-e.*
 sleep-3FSG
 ‘She slept.’
 b. *nem-tu-a.*
 sleep.CAUS-1SG-her
 ‘I put her to sleep.’
 Not: ‘I caused her to sleep.’

Geminate causatives, on the other hand, which are formed by geminating the second consonant of the root, pattern with Turkish causatives. When formed from an unaccusative, they are obligatorily direct (35):

- (35) a. *xaser xireb.*
 yoghurt spoiled.3M
 ‘The yoghurt spoiled.’
 b. *leyla xarrib-e xaser.*
 Leyla spoiled.CAUS-3F yoghurt
 ‘Leyla spoiled the yoghurt.’
 Not: ‘Leyla caused someone to spoil the yoghurt.’ (Akkuş, 2021a:91)

Geminate causatives formed from a transitive must receive an indirect reading. The object can be realized either as a bare DP (84b) or as a PP headed by *mışa* ‘to, for’ (36c):

- (36) a. *kemal ku i-qri lala kitab.*
 kemal be.3M 3M-read.IPFV this.M book
 ‘Kemal is reading this book.’ (Akkuş, 2021a:93)
- b. *oratman ki tı-qarri kemal lala kitab.*
 teacher be.3F 3F-read.CAUS Kemal this.M book
 ‘The teacher is making Kemal read this book.’ (Yakut, 2012:14)
- c. *oratman ki tı-qarri lala kitab mışa kemal.*
 teacher be.3F 3F-read.CAUS this.M book to Kemal
 ‘The teacher is making Kemal read this book.’ (Yakut, 2012:14)

Geminate causatives of unergatives are again ambiguous between a direct and indirect interpretation (37):

- (37) a. patk-e mī haydan.
 jumped-3F over wall
 ‘She jumped over the wall.’
 b. pattik-tu-a mī haydan.
 jumped.CAUS-1SG-her over wall
 ‘I jumped her over the wall.’
 ‘I made her jump over the wall.’

The direct reading of (37b) entails that the speaker lifts the object over the wall, whereas under the indirect interpretation, the speaker causes them to jump in some way, be it via physical coercion, persuasion, or any other measure. Moreover, in the latter case, it is possible for the causing and the jumping event to take place at different times, which is not conceivable under the direct reading.

I demonstrate the adverbial diagnostic for ablaut causatives which, being obligatorily direct, lend themselves most easily to this diagnostic, but note that the facts replicate for geminate causatives. In example (38a), the subject-oriented adverb can only modify the first person singular speaker, not the sleeper. This contrasts with the periphrastic indirect causative in (38b) formed with the causativizing verb ‘make’ in which two events can be targeted by modification: while the action of sleeping is peaceful, the action of causing someone to sleep is slow.

- (38) a. sakin nem-tu-a.
 peacefully slept-1SG-her
 ‘I put her to sleep peacefully.’
 b. hedi hedi si-te nom sakin.
 slow slow made-2SG.F sleep.INF peacefully
 ‘You.F slowly made someone sleep peacefully.’

As before, this confirms the direct nature of some causatives of unergatives in Sason Arabic.

With this background established, we can turn to more direct evidence for the analysis adopted here, namely, that direct causatives of unergatives in Hindi-Urdu, Turkish and Sason Arabic are regular transitives in which the external argument is assigned an agent θ -role and the internal argument a patient θ -role.

4.2.2.2 Syntactic diagnostics

This section presents four diagnostics according to which the object of direct causatives of unergatives behaves as an internal argument: reduced relatives, secondary predicates, telicity, and path arguments and cognate objects.

Reduced relatives We have seen in (13) that reduced relatives in Hindi-Urdu require the presence of an internal argument, thus being licensed with unaccusatives but not with unergatives. Causativized unerga-

tives, however, can form reduced relatives targeting the object, indicating that the latter is realized as an internal argument (39):

- (39) a. *daur-aa laṛkaa
 run-PFV.MSG boy
 Intended: ‘the run boy’
- b. [Ravi-dwaaraa daur-aa-yaa gayaa] laṛkaa
 Ravi-by run-CAUS-PFV PASS.PFV boy
 ‘the boy run by Ravi’ (i.e., the boy chased by Ravi) (Bhatt and Embick, 2017:124f.)

Resultatives In Sason Arabic, resultatives have been shown above in (20) to require the presence of an internal argument. Crucially, the object in causatives of unergatives licenses a resultative, as in (40):

- (40) faqqiz-tu-a raxu-e, yani cimd-e barra.
 ran.CAUS-1SG-her sick-F that.is got.cold-3F outside
 ‘I ran her sick, that is, she got a cold outside.’

Example (40) could be used in a situation in which the speaker made the object run outside in the cold for several hours, as a result of which she fell sick. Again, this is strong evidence for the internal argument status of the object.

Telicity A characteristic property of internal arguments is that they can often confer a telic interpretation on the verb phrase (Tenny, 1987, 1994). For instance, while the intransitive *Zeno ate* is atelic, *Zeno ate the apple* describes a telic event which comes to its natural endpoint when the apple is fully consumed. This is not to claim a one-to-one correspondence between internal arguments and telicity: not all internal arguments induce telic readings, and under certain circumstances, a telic reading can be generated in the absence of any internal argument (Ramchand, 2008). However, we can nevertheless use telic interpretations to detect the presence of an internal argument if other factors are carefully controlled for, as I will do in the following.

In various languages, the distinction between telic and atelic interpretation is reflected in the type of temporal modifier licensed in the sentence. For instance, in English, *in X hours* requires a telic but *for X hours* an atelic verb phrase (Vendler, 1957), shown in (41):

- (41) a. Zeno ate the apple in an hour.
 b. #Zeno ate the apple for an hour.
 c. Zeno ran for an hour.
 d. *Zeno ran in an hour.

Example (41d) is ungrammatical, supporting the claim that the absence of an internal argument correlates with an atelic reading. On the other hand, as (41b) shows, an atelic interpretation which allows *for X hours* to surface can often be enforced even in the presence of a definite internal argument if the event is construed

as having been interrupted. For instance, (41b) becomes more felicitous if the apple is not fully consumed. I will note this confound in the following where applicable but it will not be relevant for our purposes.

All these observations from English replicate in Turkish with the contrast between the postpositions *içinde* ‘in’ and *boyunca* ‘for.’ Unaccusatives, which have an internal argument and can receive a telic interpretation, preferably take *içinde* (42), whereas unergatives, being atelic, combine with *boyunca* (43):

- (42) a. Yağ üç dakika içinde eri-di.
butter three minutes in melt-PST
‘Butter melted in three minutes.’
b. *Yağ üç dakika boyunca eri-di.
butter three minutes for melt-PST
Intended: ‘Butter melted for three minutes.’
- (43) a. Kadın üç saat boyunca çalış-tı.
woman three hours for work-PST
‘The woman worked for three hours.’
b. *Kadın üç saat içinde çalış-tı.
woman three hours in work-PST
Intended: ‘The woman worked in three hours.’

(Nakipoğlu-Demiralp, 2002, cited and translated in Acartürk, 2005:45f.)

As in English, (42b) is acceptable if the melting process is interrupted after three minutes without the butter being fully melted yet. Note also that for my consultants, it is more natural to use a bare DP instead of a *boyunca* ‘for’ PP, and the locative suffix *-te* instead of an *içinde* ‘in’ PP, as seen in (44) and (45):

- (44) a. Yağ üç dakika-da eri-di.
butter three minutes-LOC melt-PST
‘Butter melted in three minutes.’
b. *Yağ üç dakika eri-di.
butter three minutes melt-PST
Intended: ‘Butter melted for three minutes.’
- (45) a. Kadın üç saat çalış-tı.
woman three hours work-PST
‘The woman worked for three hours.’
b. *Kadın üç saat-te çalış-tı.
woman three hours-LOC work-PST
Intended: ‘The woman worked in three hours.’

We can now employ this telicity contrast to diagnose internal arguments in direct causatives of unergatives. As a baseline, (46) demonstrates that the intransitive unergative ‘sleep’ can combine with *boyunca* or the bare DP (46a), as expected:

- (46) a. Bebek iki saat (boyunca) uyu-du.
 baby two hour (for) sleep-PST
 ‘The baby slept for two hours.’

However, to complicate matters, ‘sleep’ can also be used with *içinde/-te* but then receives the markedly different interpretation ‘fall asleep.’ I argue that in this case, the normally unergative verb is used as an unaccusative when appearing in a telic environment, as a result of which its sole argument is realized in the internal position. Acartürk (2005) has attested this effect of telicity extensively for Turkish. What matters for our current purposes is that ‘sleep’ allows for both an atelic and a telic use with different interpretations.

In causative environments, ‘sleep’ can take both *boyunca*/DP (47a) or *içinde/-te* (47b). Crucially, this correlates with a difference in the interpretation of the causative:

- (47) a. Bakıcı bebeğ-i sadece iki saat (boyunca) uyu-t-tu.
 caretaker baby-ACC only two hour (for) sleep-CAUS-PST
 ‘The caretaker let the baby sleep for only two hours.’
 b. Bakıcı bebeğ-i sadece {iki saat-te / iki saat içinde} uyu-t-tu.
 caretaker baby-ACC only {two hour-LOC / two hour in} sleep-CAUS-PST
 ‘The caretaker (was able to) put the baby to sleep in only two hours.’

In (47a), ‘sleep’ is used as an atelic predicate, accordingly taking a *boyunca*/bare DP modifier: the example features a cruel caretaker who only allows the baby two hours of sleep. In contrast, in (47b), ‘sleep’ is used as a telic predicate in the sense of ‘fall asleep’ as seen earlier, giving rise to the interpretation that the caretaker manages to make the extremely unruly baby fall asleep in merely two hours. This contrast in telicity predicts that (47b) but not (47a) should contain an internal argument.

This prediction aligns with the fact that (47a) is interpreted as an indirect and (47b) as a direct causative. Thus, the former contains an embedded unergative structure which lacks an internal argument and thus receives an atelic reading. On the other hand, the direct causative in (47b) contains an internal argument which makes a telic reading available. The fact that we cannot enforce a telic reading in an indirect causative, unlike in the intransitive base case (46), confirms the claim made earlier that the availability of a telic interpretation in the latter is the result of an unaccusative use of the verb: since indirect causatives cannot embed unaccusatives in Turkish, they must always contain an unergative structure which only allows for an atelic interpretation. To summarize, the direct causative licenses temporal modifiers with *içinde/-te*, the latter only surface with telic events, those in turn are licensed by an internal argument, and thus, we can infer that the direct causative contains an internal argument.

Path arguments and cognate objects The claim that the object is merged as an internal argument makes another prediction, namely, that direct causativization should be blocked for unergatives with true path arguments or cognate objects which occupy the internal argument position. Example (48) shows that this is borne out for a cognate object in Hindi-Urdu:

- (48) a. Rohan do tarah-ke tango naach-egaa.
 Rohan two type-GEN tango dance-FUT.3MSG
 ‘Rohan will dance two types of tango.’
- b. *Shama Rohan-ko do tarah-ke tango nach-aa-egii.
 Shama Rohan-DOM two type-GEN tango dance-CAUS-FUT.3FSG
 Intended: ‘Shama will make Rohan dance two types of tango.’

The ungrammaticality of (48b) cannot be attributed to case but persists regardless of whether the object, the path argument, neither or both receive direct object marking. Moreover, the case marking in (48b) in which one argument is marked with *-ko* and the other one is bare is the same pattern as observed for Hindi-Urdu ditransitives. Hence, there is nothing illicit about this case configuration as such. Note also that the path argument can raise to subject under passivization, confirming that it is a true argument of the verb instead of an adjunct (49):

- (49) kal yahaan do tarah-ke tango naach-e jaa-enge.
 tomorrow here two kind-GEN tango dance-PFV.MPL PASS-FUT.MPL
 ‘Tomorrow, two kinds of tango will be danced here.’

Given that in (48b), the internal argument position is thus already occupied by the path argument, the object cannot be merged in this position, and causativization fails.

Sason Arabic patterns with Hindi-Urdu in equally not allowing unergatives with path arguments or cognate objects to causativize. This is shown below for the verb ‘run’: under the direct reading, the causer takes the object, typically a child, by the hands and helps them run by providing balance, whereas under the indirect reading, the causer makes the object run in some other unspecified way. However, the direct interpretation disappears once a path argument is added, as in (50):

- (50) faqqız-tu-a 10m.
 run.CAUS-1SG-her 10m
 ‘I made her run 10m.’
 #‘I ran her 10m.’

The fact that the path argument construction in (51a) can passivize to form (51b), with the path argument raising to subject, confirms the complement as opposed to adjunct status of the argument:

- (51) a. kemal i-fqez 10m.
 Kemal 3M-run.IPFV 10m
 ‘Kemal runs 10m.’
- b. 10m in-fqez.
 10m PASS.IPFV-run.IPFV
 ‘10m is run.’

Overall, direct causativization is thus blocked whenever the internal argument position which would need to host the object is already occupied by a true path argument or cognate object.

Lastly, in Turkish, the status of path arguments is unclear. Generally, in Turkish, causatives of unergatives – both direct and indirect –, the object receives accusative case, whereas in causatives of transitives, the

object is marked with dative, the embedded direct object with accusative case. Unergatives with path arguments can surface with two different case markings (52):

- (52) a. Çocuğ-a 10m-yi yürü-t-tü-m.
 child-DAT 10m-ACC walk-CAUS-PST-1SG
 %‘I walked the child 10m.’
 ‘I made the child walk the 10m.’
- b. Çocuğ-u 10m yürü-t-tü-m.
 child-ACC 10m walk-CAUS-PST-1SG
 ‘I walked the child 10m.’
 ‘I made the child walk 10m.’

Example (52a) aligns with transitives in that the object receives dative, the path argument accusative case. In (52b), on the other hand, the object is assigned accusative case while the path argument does not receive any case marking. One of my consultants rejects a direct interpretation in (52a), indicating that in their grammar, the accusative-marked path argument is a true complement of the verb blocking direct causativization. Others, however, accept both readings in both cases, which suggests that they allow for accusative case marking on adjuncts. I must leave this question open.

To summarize, I have presented several pieces of evidence supporting the view that the object in direct causatives of unergatives is realized syntactically as a run-of-the-mill internal argument. Next, I turn to the thematic interpretation of this argument.

4.2.2.3 Thematic interpretation

I argue that the object of direct causatives of unergatives is assigned a patient θ -role. In all three languages discussed here, the argument obligatorily receives a deagentivized reading, being depicted as not being in control of the situation or even performing the activity against their will. For instance, in the Hindi-Urdu causative repeated below as (53), Rohan does not actively and volitionally initiate the dancing, but is shoved and twirled around like a puppet on strings. Instead, it is Shama who possesses volition and causal power. This contrast in proto-properties strongly suggests that Shama receives an agent and Rohan a patient θ -role.

- (53) Shama Rohan-ko nach-aa rahii hai.
 Shama.F Rohan-DOM dance-CAUS PROG.F be.PRS.3MSG
 ‘Shama is making Rohan dance/twirling him around (the dance floor).’

(Bhatt and Embick, 2017:124)

In general, we often find that the object of direct causatives of unergatives are children, animals or other borderline-volitional beings (Ramchand, 2008). For instance, the direct causative of Turkish ‘walk’ is typically used for taking a toddler by both hands and thus helping them take a few steps (54):

- (54) Çocuğ-u yürü-t-tü-m.
 child-ACC walk-CAUS-PAST-1SG
 ‘I walked the child.’

Here, the child is physically quite helpless and could not bring about the event on their own. Moreover, there is only a limited sense in which they can be said to act volitionally – toddlers are not able to give informed consent. On the other hand, the speaker possesses the full set of proto-properties. Thus, the speaker is construed as the agent and the child as the patient of the event.

Finally, the contrast in thematic properties associated with the subject of the intransitive and the object of the transitive has become clear in the previous discussion of direct vs. indirect causatives. We have seen that causatives like the Sason Arabic example (55) are ambiguous between a direct and an indirect reading, with the object being an internal argument in the former and an embedded SpecVoiceP causee in the latter. In both readings, the third person participant undergoes a change of state, from being on this to being on the other side of the wall. However, under the direct reading, she is physically lifted over the wall, exerting no energy whatsoever and resembling an inanimate object, whereas under the indirect reading, she might still effectuate the jumping of her own accord. Again, the object of the direct causative resembles a patient rather than an agent, whereas the external causer has the full set of proto-agent properties.

- (55) pattık-tu-a mi haydan.
 jumped.CAUS-1SG-her over wall
 ‘I jumped her over the wall.’
 ‘I made her jump over the wall.’

Overall, analyzing the object argument of direct causatives of unergatives as a patient is not only the null hypothesis given its internal argument status, but is also backed up by systematic interpretative differences to the subject of the corresponding intransitive in terms of thematic properties. In consequence, under a theory that posits individual roles, direct causatives of unergatives would constitute a case of variable linking: a single individual role is mapped onto an agent θ -role in the intransitive but a patient θ -role in the transitive.

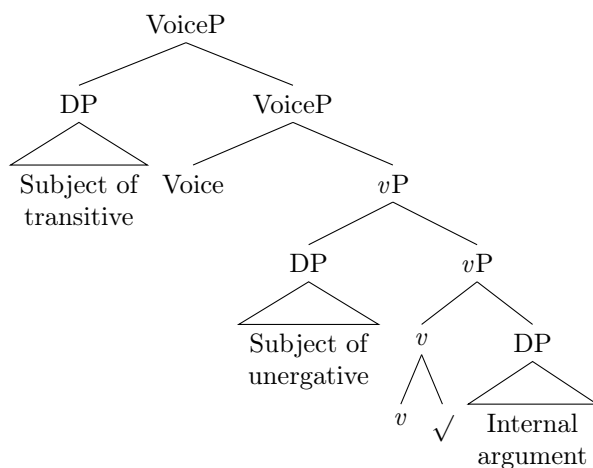
4.2.3 Competing analyses

Before concluding this case study, I review three competing analyses of direct causatives of unergatives, developed in part for languages other than the ones we have looked at above. I begin with Massam (2009) and subsequent work, according to which the subject argument of unergatives is merged in Spec v P rather than SpecVoiceP, and can thus causativize by virtue of adding an argument in the latter position. Next, I discuss Ramchand’s (2008) approach to direct causatives of unergatives in Hindi-Urdu, which analyzes the intransitive unergatives as underlyingly transitives. The last section deals with Biggs’ (2019) proposal for transitives of manner of motion verbs in English.

4.2.3.1 The low subject analysis: Massam (2009) and beyond

I begin with an analysis originally developed by Massam (2009) for Niuean and since adopted by other researchers and expanded to other languages. The core idea behind this approach, which I label the low subject analysis, is that subjects of unergatives differ syntactically from subjects of transitives, contrary to what is commonly assumed. Concretely, it has been argued that while subjects of transitives are generated in SpecVoiceP, subjects of unergatives are generated lower, in Spec*v*P, as seen in (56) (Kouneli, 2021; Kumaran, 2021; Massam, 2009; Myler, 2022; Pineda and Berro, 2020; Tollan, 2018; Tollan and Massam, 2022; Tollan and Oxford, 2018). Internal arguments of unaccusatives and transitives are, as usual, assumed to be merged as complements to the verb:⁸

(56)



In general, what this structure is meant to achieve is to account for systematic differences between unergative and transitive subjects, such as differences in case marking (Massam, 2009), voice morphology (Tollan and Oxford, 2018) and plural marking (Kumaran, 2021), all of which I must sidestep in the following. What matters for our purposes is how the low subject approach is leveraged to account for the fact that in Niuean (Massam, 2009)⁹, Samoan (Tollan, 2018), Algonquian (Tollan and Oxford, 2018), Kipsigis (Kouneli, 2021) and Quechua (Myler, 2022), unergatives but not transitives can causativize. Under the low subject approach, this falls out from the fact that in unergatives, the SpecVoiceP position is vacant and can be filled by the causer, which is not possible with transitives. Unlike under the view argued for in the present paper, the subject of the intransitive unergative and the object of the corresponding causative are assumed to occupy the same position, Spec*v*P.

The low subject analysis of direct causatives of unergatives thus implements the intuition that the subject of the intransitive and the object of the transitive are the ‘same’ argument and thus ought to be merged in

⁸Tollan (2018) and Tollan and Oxford (2018) argue that certain transitive subjects in Samoan and Algonquian, respectively, are also located in Spec*v*P. For Kipsigis, Kouneli (2021) claims that subjects of unaccusatives are equally generated in Spec*v*P, or rather, that the language lacks true unaccusatives altogether.

⁹In Niuean, transitives can causativize if and only if the original direct object remains unexpressed, is incorporated or surfaces with an instrumental applicative marker (Massam, 2009).

the same syntactic position and receive the same θ -role. In this way, the analysis avoids variable linking. The question is what would constitute evidence for or against it. While the *SpecvP* position is novel and its properties thus as of yet ill-defined, we might take the tree in (56) to predict that the subject of the intransitive and the object of the causative ought to have the same semantic and syntactic properties. For the languages investigated here, I have shown that this is not the case. In particular, we have seen that the object receives a deagentivized reading compared to the interpretation of the subject of the intransitive, and that the former but not the latter behaves as an internal argument for the purposes of reduced relatives, resultatives and telicity. In addition, Hindi-Urdu and Sason Arabic block direct causatives of unergatives with path arguments and cognate objects. Under the low subject approach, it is not clear why the presence or absence of a *VoiceP* layer hosting the causer should determine whether or not another DP can be added as a complement of the verb. If the object is located in *SpecvP*, it remains mysterious why its presence co-varies with that of path arguments and cognate objects.

In response, defenders of the low subject proposal might point out that nothing about the tree in (56) strictly necessitates that the subject of the intransitive and the object of the transitive have the same properties. For instance, the θ -role assigned to *SpecvP* might be context-dependent, being lower in agentivity if a *Voice* projection is merged on top.¹⁰ Similar argument could be made about the syntactic properties. However, the question then arises why a shared syntactic position should be posited in the first place if it is not reflected in any shared syntactic or semantic properties. The only sense in which the subject of the intransitive and the object of the transitive are the ‘same’ argument is that, as I have put it here, they are assigned the same individual role.

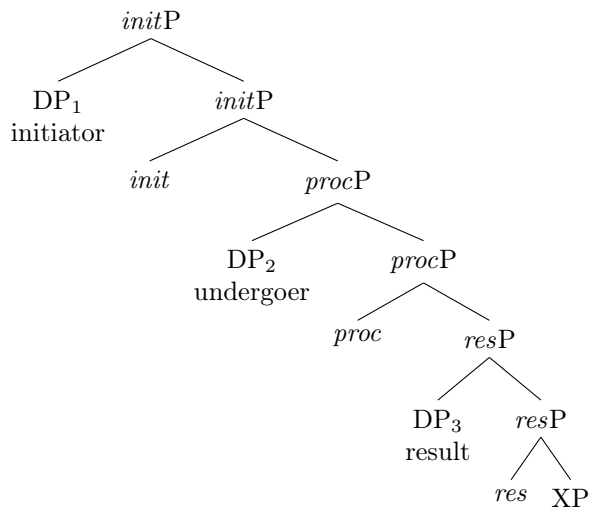
Overall, the low subject approach is not compatible with the data presented above. I do not mean to cast a judgment on its success for the languages for which it was originally designed, and it also ought to be highlighted that the above discussion has set aside all other kinds of data that have been taken as evidence for this analysis, such as case marking. What I do conclude is that the low subject approach, which would avoid the specter of variable linking, cannot account for direct causatives of unergatives in Hindi-Urdu, Turkish and Sason Arabic.

4.2.3.2 Covertly transitive unergatives: Ramchand (2008)

I now turn to Ramchand’s (2008) analysis of direct causatives of unergatives in Hindi-Urdu, developed as part of a novel and highly innovative approach to verbal syntax. Ramchand proposes to split up the verbal domain into three projections, namely *initP* for the initiation, *procP* for the process and *resP* for the result of the event. Each of these heads takes an argument in its specifier: *init* introduces an initiator, *proc* an undergoer and *res* a result (57). The position marked here as XP can host various material; we can ignore it for now.

¹⁰I credit Julie Legate for this perspective.

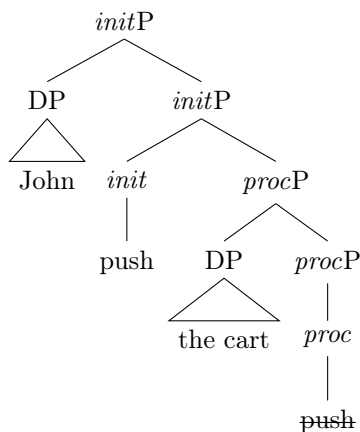
(57)



(Ramchand, 2008:39)

A verbal root can be merged in one or several of these head positions but is only spelled out in the highest position that it occupies. Which heads a verb can associate with is specified in its lexical entry. For instance, the verb *push* is listed with the category features [*init*, *proc*] and thus projects the structure in (58):

(58)

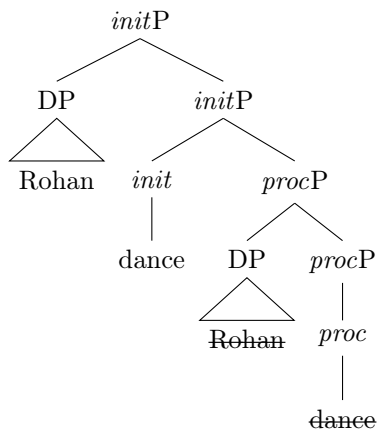


(Ramchand, 2008:65)

While every verbal domain needs to contain a *procP*, neither *initP* nor *resP* are obligatory. For instance, (58) does not have a result state encoded by a *resP*, and unaccusatives lack an *initP* with an initiator.

In the same way as heads, arguments can equally be merged in more than one position simultaneously while again only being spelled out in their highest position. This assumption is crucial to Ramchand's analysis of intransitive unergatives. The latter, she argues, project a transitive structure containing an *initP* and a *procP* – just as seen for *push* above – but their lexical entry requires the two argument positions to be filled by the same argument. Thus, the derivation of 'Rohan is dancing' is as in (59):

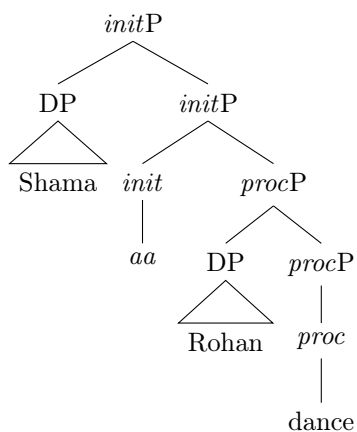
(59)



Hence, the argument *Rohan* is interpreted as both initiating and undergoing the event.

To causativize unergatives in languages such as Hindi-Urdu, Ramchand relies on a mechanism she labels underassociation. While the lexical entry of a verb specifies in which head positions it can be merged, nothing requires it to merge in all of them. A subset of features may remain underassociated as long as the feature is expounded by an independent syntactic element. Ramchand argues that in Hindi-Urdu causatives of unergatives, the *init* feature dissociates from the lexical root and instead associates with the causative morpheme *-aa*. Since *init* and *proc* are now filled by distinct morphemes, the requirement that their two arguments be identical is lifted and a distinct causer can be merged in *SpecinitP* (60):

(60)



Hence, while in the intransitive, *Rohan* both initiates and undergoes the dancing, in the transitive, the initiator is now *Shama*.

For the purposes of variable linking, Ramchand's analysis is an interesting hybrid. Unlike the low subject analysis reviewed earlier, it does not claim that the subject of the intransitive and the object of the transitive are simply assigned the same single θ -role. Nonetheless, it does claim that the arguments have a θ -role in common – that of an undergoer –, which is responsible for the entailments shared by the two. The problem, I argue, is that undergoing, as Ramchand uses the term, is not a θ -role or a property associated with a θ -role.

This is not because ‘undergoer’ happens to be absent from Dowty’s short list of proto-roles, but because it is hard to see what the undergoers of a pushing event, a dancing event or a stabbing event have in common. Dowty’s strategy of diagnosing θ -roles by means of entailments is instructive here: undergoers do not come with any general, verb-independent entailments that would differentiate them from other θ -roles. What the shared undergoer role attributed to *Rohan* in (59) and (60) is meant to encode is a verb-specific entailment, that of moving one’s feet. It is what I have termed here an individual role; it should not be represented as a θ -role.

Moving beyond these theoretical considerations, Ramchand’s proposal also faces a number of more tangible problems, all stemming from the claim that base unergatives contain a silent undergoer argument co-indexed with the initiator. The first question this raises, not addressed by Ramchand, is what determines whether an argument co-indexed with a higher argument is realized as null or spelled out as an overt anaphor. We might argue that an argument must be silent if and only if it is specified on a head as being *obligatorily* co-indexed. However, note that it is possible for the overt and the silent instance of an argument to be introduced by different heads (Ramchand, 2008:102, 118, 179), and it is hard to see what would require the arguments of two distinct heads to be co-indexed and where this information would be stored. What is more, this approach only begs the question: those arguments considered to be obligatorily co-indexed with a higher argument – such as the undergoer of an unergative – are precisely those kinds of arguments whose existence is debatable, and claiming that they can simply forego spell-out would be stipulative. In short, it remains an open question why the undergoer argument of an unergative, if it did exist, would not be realized as an overt anaphor.

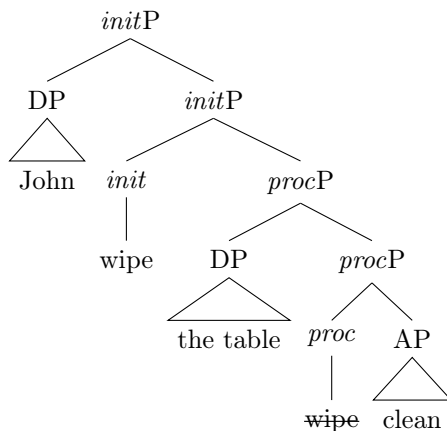
Relatedly, a puzzle raised by having a single DP being merged as an argument of two different heads concerns the interpretation of quantifiers. For instance, in (61), something needs to ensure that the variables in the two argument positions are bound by the same quantifier, rather than having two distinct operators each bind their own variable:

(61) Everyone danced.

In other words, in such cases, it is not sufficient to merely sample the same argument twice from the numeration and thereby guarantee their coindexation. Ramchand is not explicit about such cases, and I cannot preclude that some solution could be constructed, but quantificational DPs are yet another reason to be skeptical about the presence of silent, co-indexed arguments.

Next, if unergatives had an undergoer argument, which corresponds roughly to an internal argument, another puzzle this poses is why they fail to pass standard diagnostics for internal arguments such as resultative secondary predicates. Ramchand proposes two distinct resultative structures. First, certain adjectives can be merged directly as a complement of *proc*, as in *John wiped the table clean* (62):

(62)



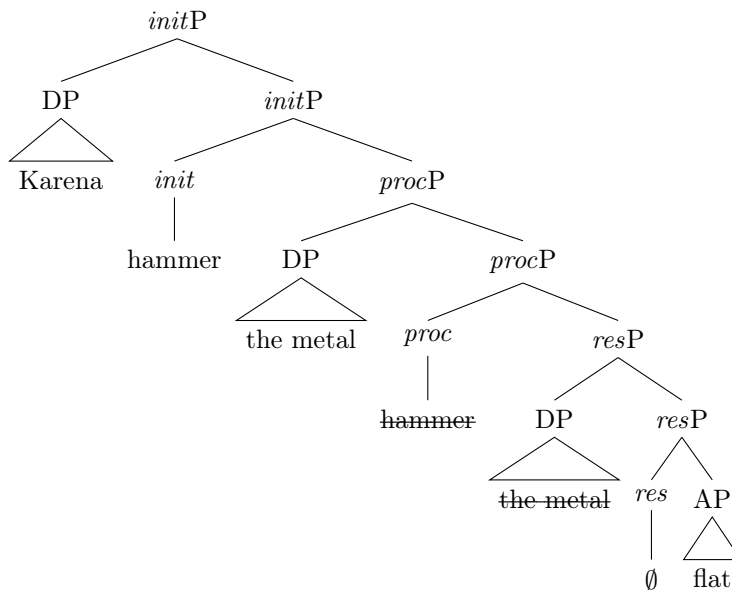
(Ramchand, 2008:122)

Given that under Ramchand's view, unergatives equally have an *init-proc* structure – with the sole difference being that the arguments are co-indexed –, they are expected to equally license a resultative predicate in the complement position of *proc*. This prediction is not borne out, even if the adjective is kept constant. Example (63) can only have a depictive reading:

(63) John showered clean.

The second type of resultative structure Ramchand proposes is built by merging an AP in the complement position of *res*. This can be achieved in two ways. If the verb already has a *res* feature, the AP can be added without any further ado. If, on the other hand, the verb itself lacks *res*, languages such as English have a silent *res* head available that can project a *resP* licensing the resultative AP. The latter is demonstrated in (64) for *Karena hammered the metal flat*:

(64)



(Ramchand, 2008:127)

As before, we would expect unergatives to be able to appear in the same structure. An *init-proc* verb such as *run* should be able to form a resultative via the addition of a *resP* headed by a silent element; in fact, however, such structures are unavailable. Example (65) can only have a depictive reading:

(65) Karena ran tired.

Resultatives with unergatives are of course possible with unselected objects, including anaphoric ones (66):

(66) Karena ran herself tired.

However, this brings us back to the first problem discussed above, namely why one but not the other of the two lower *Karena* arguments in (66) is realized as an anaphor. In sum, the proposal that unergatives have a silent undergoer argument cannot explain why resultatives are either not licensed at all in such structures or require the presence of an anaphor.

The final problem for Ramchand's analysis of unergatives concerns path arguments. Ramchand assumes that path arguments are merged as complements of *proc*, thus in a position distinct from the undergoer argument.¹¹ As a result, she fails to account for the fact that in Hindi-Urdu, as well as in Sason Arabic, unergatives with path arguments are unable to causativize. One would need to argue that path arguments are only licensed if initiator and undergoer are co-indexed, but such a restriction lacks any motivation or basis. More generally, Bruening (2010) has noted that the distinction Ramchand makes between path and undergoer argument positions is not backed up by any kind of syntactic support: while there are clear semantic differences between the two kinds of arguments, there is no evidence that they occupy distinct positions in the structure. If, however, we give up on the idea that path and undergoer arguments are merged in different positions, the idea that unergatives take a silent undergoer argument collapses. In sum, Ramchand's approach to direct causatives of unergatives faces both theoretical and empirical problems.

4.2.3.3 An 'in-motion' θ -role: Biggs (2019)

Before concluding this case study, we will look at a phenomenon that I argue should be included in the group of direct causatives of unergatives, namely, transitives of manner of motion verbs in English, and their analysis in Biggs (2019). As is well known, English manner of motion verbs can sometimes surface with a non-path and non-cognate object (67):

- (67) a. Kemal waltzed Matilda *(around the ballroom).
 b. The general marched the soldiers *(to the battlefield).

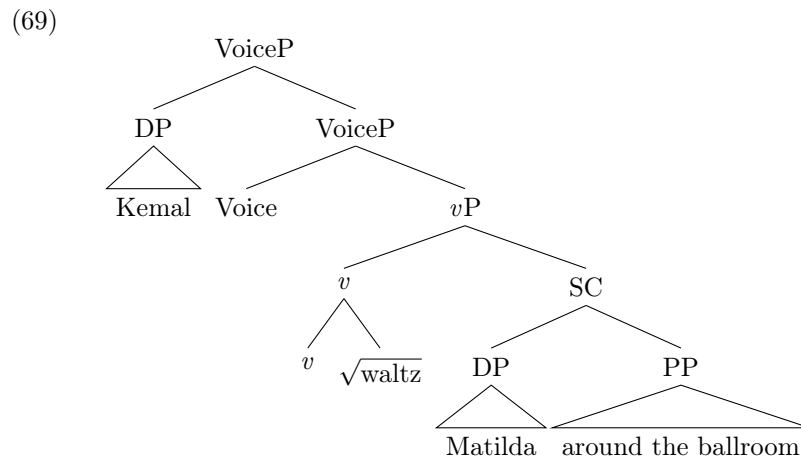
¹¹Under Ramchand's view, the category of path arguments is broader than commonly assumed and also includes arguments that are more commonly regarded as patients or themes. However, there is no reason to believe that for Ramchand, the path arguments discussed in the present paper would not qualify as such; hence, the following argument is pertinent nonetheless.

Examples such as (67) raise two questions: first, why manner of motion verbs, normally analyzed as unergatives, can take a direct object, and second, why they can only do so in a limited set of circumstances. For instance, the transitives in (67) are ungrammatical in the absence of the PPs.

A popular analysis that provides an answer to both of these question has been to analyze transitives of manner of motion verbs as resultatives, analogous to examples such as (68) (Beavers, 2012; Folli and Harley, 2006; Ramchand, 2008):

- (68) a. Kemal danced his feet sore.
b. Ali ran himself tired.

Concretely, both the examples in (67) and those in (68) have been argued to contain a small clause structure. Under this view, the argument *Matilda* in (67a) is not selected by the verb *waltz* but by the resultative secondary predicate *around the ballroom*, as shown in (69):



However, this small clause analysis was dealt an arguably lethal blow by Biggs (2019), who showed that, contrary to what is predicted by the analysis in (69), the object and the PP do not form a syntactic constituent. Rather, it is the verb and the object at the exclusion of the PP which pass constituency diagnostics, namely *do so* anaphora (70a), *though* movement (70b), V-fronting (70c) and intervening adverbials (70d):

- (70) a. Mary waltzed John around the ballroom, and Sarah did so around the garden.
b. Waltz John though Mary did around the ballroom, ...
c. (Mary thought she would waltz John around the ballroom – and) waltz him she did around the ballroom.
d. Mary waltzed John daintily/for hours around the room. (Biggs, 2019:7)

In addition, Biggs (2019) demonstrates that besides PPs, a broad range of other factors equally license causatives of unergatives. These include negation (71a), manner adverbs (71b), modals (71c) and temporal/frequency modifiers (71d):

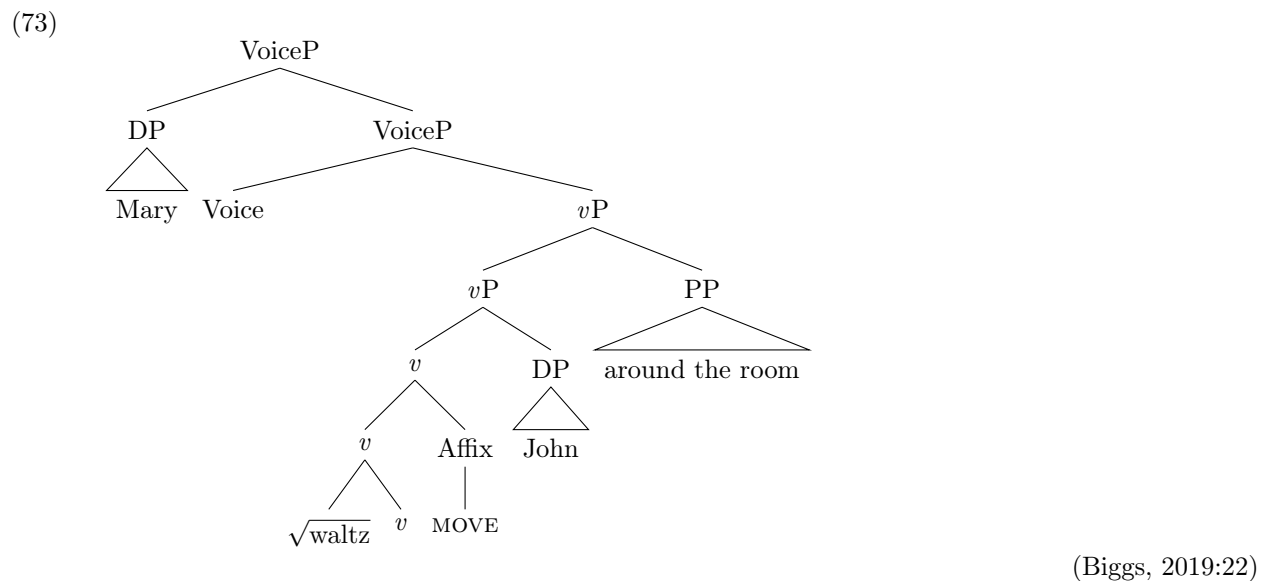
- (71) a. The teacher hasn't somersaulted the second team yet.
 b. The choreographer diligently pirouetted the ballerinas.
 c. Mary can cartwheel John.
 d. Coach swims the first team more often than the second team. (Biggs, 2019:14)

Moreover, object weight also improves acceptability (72):

- (72) a. Coach swam the team that had not performed well at the meet.
 b. The teacher vaulted the gymnasts who had warmed up. (Biggs, 2019:15)

None of this is compatible with a small clause analysis. Biggs concludes that the objects in (67) are complements of the verb and that the PPs have adjunct status.

What distinguishes Biggs' proposal from the view espoused here is that she does not assume the internal argument of transitives of manner of motion verbs to be interpreted as a regular patient. Instead, she argues that the argument receives a novel, 'in-motion' θ -role, understood as a particular kind of patient or theme θ -role assigned to objects which are in motion. This in-motion θ -role is not assigned by the root itself but by a silent affix, termed MOVE, which merges with the verb and introduces the internal argument. The resulting structure is given in (73).



The analysis in (73) leaves the question open why transitives of manner of motion verbs are only licensed in certain contexts, as with PPs. Contrary to what one might assume, it also does not account for the fact that the alternation is limited to motion verbs. As Biggs concedes, nothing prevents the θ -role-bearing affix from combining with any other root as well, citing the following example brought up by a reviewer which cannot be used to express that the passengers slept on the plane while also being in motion (74):

- (74) *The pilot slept the passengers smoothly. (Biggs, 2019:23)

To rule out examples such as (74), Biggs proposes that the affix can only combine with roots with a manner of motion denotation. She does not specify how concretely the root-selecting properties of the head are encoded, arguing that the mechanics of root distribution are generally an open question not limited to her analysis. The latter is certainly correct, but the fact remains that the analysis merely stipulates why only a subset of verbs undergo the alternation. The in-motion θ -role as such does not have any explanatory power in this respect.

Against this background, the question arises what the new θ -role achieves at all and why the object argument of transitives of manner of motion verbs cannot simply be analyzed as a patient. Biggs argues that this is not possible. Regular patient arguments, she maintains, are always acted upon in the manner described by the verb. For instance, in *Mina hits the wall*, the wall is acted upon in the manner of hitting and does not do any hitting itself. This contrasts, Biggs argues, with sentences like *The general marched the soldiers to the battlefield*, in which the soldiers do the marching. Therefore, *the soldiers* cannot be a patient.

I agree with the contrast between these two examples, but I do not follow Biggs' interpretation of it. What is at stake here is the contrast between two properties: 'being acted upon in a certain manner' (according to Biggs, a property of patients) and 'acting in a certain manner' (according to Biggs, not a property of patients). To begin with, this simply does not seem to be true empirically. For instance, in *The girl fell*, the girl – a patient argument – is of course not acted upon in a certain manner, but it does seem felicitous to say that she acts in a certain manner. In general, what it means to act or be acted upon is not explicitly operationalized; the intuitions are taken to be self-evident. It is furthermore not clear what the grammatical status of these properties (acting and being-acted-upon) is – whether these are meant to be thematic properties, or are encoded in projectionist-style lexical entries, or constitute purely descriptive generalizations, etc. Biggs' argument here is just not fully developed.

Taken together, all of this indicates that English transitives of manner of motion verbs do not require a novel θ -role, but should be treated on a par with other direct causatives of unergatives. If this is correct, it would leave two questions open. First, we might ask why motion verbs in particular can productively causativize under a particular set of circumstances. I suspect that this is ultimately not a particularly interesting question, at least not from the perspective of synchronic formal linguistics. Constructions tend to spread to semantically similar verbs, and often for quite arbitrary reasons. The second, more intriguing question is why transitives of manner of motion verbs are only licensed in certain contexts, as with modifiers, modals and negation. There is an interesting parallel, pointed out by Biggs, to English middles, which are licensed under very similar conditions (75) (I. G. Roberts, 1987):

- (75) a. *Bureaucrats bribe.
 b. Bureaucrats bribe easily.
 c. The bureaucrats will bribe.

- d. Bureaucrats don't bribe.

One factor that licenses transitives of manner of motion verbs but at first does not seem to have an equivalent in the realm of middles is object weight, as seen in (72). What is notable, however, is that the heavy objects in Biggs' examples both contain restrictive relative clauses, which might induce a contrastive focus interpretation. Contrastive focus, in turn, is known to make middles more acceptable (76):

- (76) a. ?Bureaucrats BRIBE.
 b. ?This bread DOES cut.
 c. ?CHICKENS kill. (I. G. Roberts, 1987:232)

The parallel between direct causatives of unergatives and middles might be less arbitrary than it seems. From a simple-minded perspective, middles are intransitives of verbs that normally only license a transitive use. Direct causatives of unergatives are transitives of verbs that normally only license an intransitive use. The question remains what modifiers, modals, negation and contrastive focus have in common, but their effect across the board appears to be to consistently allow verbs to surface with a non-standard valency. In sum, English transitives of manner of motion verbs clearly are in need of further research, but I believe that considering them in the context of the broader phenomenon of direct causatives of unergatives is a promising avenue.

4.3 Direct causatives of ingestives

4.3.1 Data and analysis

This case study discusses a construction that is, in a sense, a natural extension of direct causatives of unergatives. We will be concerned with a class of transitives commonly labeled ingestives or ingesto-reflexives, such as 'eat,' 'drink,' 'swallow,' 'taste,' 'see,' 'learn,' 'understand,' etc. – that is, verbs that describe an either literal or metaphorical process of ingestion. In many languages, ingestives are able to form direct causatives, seen below for Hindi-Urdu (77) and Sason Arabic (78). Note the causative morphology and, in (77), the overt dative case marking in the causative alternant.

- (77) a. Mina-ne angrezii siikh-ii
 Mina-ERG English.F learn-PFV.F
 'Mina learned English.'
 b. Tina-ne Mina-ko angrezii sikh-aa-yii.
 Tina-ERG Mina-DAT English.F learn-CAUS-PFV.F
 'Tina taught Mina English.' (Lit. 'Tina learned Mina English.')

(Bhatt and Embick, 2017:128)

- (78) a. ayal-e Leyla badincan.
eat-F Leyla tomatoes
'Leyla ate tomatoes.'
- b. ayyil-tu Leyla badincan.
eat.CAUS-1SG Leyla tomatoes
'I fed Leyla tomatoes.' (Lit. 'I ate Leyla tomatoes.')

Direct causatives of ingestives are a surprisingly robust phenomenon cross-linguistically. In a number of typologically unrelated languages, ingestives have been reported to be the only transitives that can undergo direct causativization, including in Amharic (Amberber, 2002), Acehnese (Legate, 2014), Ahtna (Rice, 2000), Eastern Armenian (Megerdumian, 2002), Berber (Guerssel, 1986), Cora (Vázquez Soto, 2008), Egyptian Arabic (Zaheed, 2011), Hindi-Urdu (Bhatt and Embick, 2017; Masica, 1976), Jarawara (Dixon, 2000), Lubukusu (Jerro, 2019), Malayalam (K. P. Mohanan, 1983), Marathi (Shibatani and Pardeshi, 2002) and Tariana (Aikhenvald, 2000).¹² Here, I focus on Hindi-Urdu and Sason Arabic, two of the languages already discussed in the previous section on direct causatives of unergatives.

Following previous work, I analyze direct causatives of ingestives in Hindi-Urdu and Sason Arabic as ditransitives. Thus, while *Leyla* in (78a) behaves as an external argument, the same argument in (78b) occupies what I label here an applicative position, patterning with the indirect object of standard ditransitives. Moreover, I argue that *Leyla* is assigned an agent θ -role in the transitive, (78a), but a goal θ -role in the ditransitive, (78b). In the latter construction, the agent role is assigned to the first person subject.

Under this analysis, direct causatives of ingestives thus constitute another instance of variable linking. I take it that under the individual role theory, *Leyla* would be associated with the individual role of an eater in both alternants in (78). At the same time, *Leyla* is assigned different θ -roles – agent and goal – in the two alternants. What is more, the agent θ -role is associated with two different individual roles, an eater in the transitive and a feeder in the ditransitive. In short, the pattern is largely the same as what we saw previously for direct causatives of unergatives. The main difference is that direct causatives of ingestives have an additional argument – in (78b), 'tomatoes' – whose syntactic position (internal) and θ -role (patient) remains constant.

In the literature, direct causatives of ingestives have often been documented rather than analyzed, being commonly portrayed as a curious phenomenon in need of further research. To the extent that previous work is explicit about their syntactic structure, direct causatives of ingestives have been analyzed as ditransitives, as established most robustly by Bhatt and Embick (2017) for Hindi-Urdu and Legate (2014) for Acehnese. I am not aware of a fully developed counterproposal. While the syntax of these constructions has thus

¹²Based on a sample of 32 languages, Krejci (2020) reports an implicational hierarchy for the formation of lexical (i.e., morphologically synthetic) causatives (i) (see also Shibatani, 2002):

(i) unaccusatives > ingestives/middles > unergatives > non-ingestive transitives

While the contrast between morphologically synthetic and analytic causatives does not necessarily align with the contrast between semantically direct and indirect causatives, these typological facts further support a special status of ingestives compared to other transitives with respect to causativization. For 'middles,' see Chapter 4.3.3.2 on Krejci (2012).

sparked little controversy, the same does not hold for the θ -roles involved in the alternation. Amberber (2002), Guerssel (1986), Jerro (2019), Krejci (2012), Masica (1976), and Ramchand (2008) have all proposed thematic analyses different from the one sketched out above.

In the following, I first briefly introduce direct causatives of ingestives in Hindi-Urdu and Sason Arabic. I then present syntactic evidence, largely gleaned from Bhatt and Embick (2017) and Akkuş (2021a, 2022), that the causative realizes a ditransitive structure in which the ingester argument occupies an indirect object position, and I further argue that this argument is assigned a goal θ -role. I end by considering two competing analyses: Amberber (2002) and Krejci (2012), with brief remarks on Ramchand (2008).

4.3.2 Evidence for the analysis

4.3.2.1 Background

Hindi-Urdu To briefly recap what was presented earlier in Chapter 4.2.2.1, Hindi has two different causativization strategies: direct causatives are marked with the suffix *-aa* and indirect causatives with *-vaa*. Unlike other transitives, ingestives can form direct causatives with *-aa* (Bhatt and Embick, 2017). The latter obligatorily receive a direct interpretation: for instance, (79a) is not felicitous in a context in which Ram hires a teacher to give Sita English lessons or forces her to learn English on her own, but it can only mean that he personally imparts English skills to her. Accordingly, unlike standard indirect causatives (79b), direct causatives of ingestives are incompatible with intermediate causers bearing instrumental *-se*.

- (79) a. Ram-ne (*Bill-se) Sita-ko angrezii sikh-aa-yii
 Ram-ERG Bill-INST Sita-DAT English learn-CAUS-PERF.F
 ‘Ram taught Sita English.’
 b. Ram-ne Bill-se Sita-ko angrezi sikh-vaa-yii
 Ram-ERG Bill-INST Sita-DAT English learn-CAUS2-PERF.F
 ‘Ram had Sita taught English by Bill.’ (Bhatt and Embick, 2017:127)

Besides ‘learn,’ other ingestives that show the same behavior in Hindi-Urdu are ‘eat,’ ‘drink,’ ‘taste,’ ‘see,’ ‘hear,’ ‘hold,’ ‘read,’ ‘understand’ and ‘wear’ (Bhatt and Embick, 2017). All of these verbs form causatives that both morphologically and semantically clearly qualify as direct.

Sason Arabic Turning to Sason Arabic, recall that the language has two morphological causativization strategies, ablaut and gemination. Geminate causatives obligatorily receive a direct interpretation with an unaccusative base, an indirect interpretation with a (non-ingestive) transitive base, and are ambiguous between the two with an unergative base. For ingestive transitives, the picture is again complex. The geminate causative of ‘eat,’ seen earlier in (78) and repeated below in (80), can only receive a direct reading according to which the speaker literally feeds Leyla, rather than causing her to eat in a more indirect way.

- (80) a. ayal-e Leyla badincan.
eat-F Leyla tomatoes
'Leyla ate tomatoes.'
- b. ayyil-tu Leyla badincan.
eat.CAUS-1SG Leyla tomatoes
'I fed Leyla tomatoes.'

Further support for the fact that (80b) must receive a direct interpretation comes from adverbials. As outlined earlier in Chapter 4.2.2.1, only indirect causatives contain two distinct events that can independently be targeted by adverbial modification. Example (81) features two contradictory adverbs, 'slowly' and 'quickly.' It is not possible for one of them to describe the speaker and the other to describe Leyla. Rather, both describe the same event, resulting in a contradiction.

- (81) #xıfey ayyil-tu Leyla badincan hedi.
quickly eat.CAUS-1SG Leyla tomatoes slowly
'I quickly slowly fed Leyla tomatoes.'

In contrast, the geminate causative of 'drink' is ambiguous between a direct and an indirect causative interpretation. The causative (82b) of the transitive (82a) can either describe a situation in which the speaker directly inserts liquid into Leyla's mouth – perhaps because she is a child or for some other reason unable to hold a glass of water by herself – or a situation in which the speaker orders, persuades, or in some other way causes Leyla to drink.

- (82) a. şarab-e Leyla mayn.
drank-F Leyla water
'Leyla drank water.'
- b. şarıp-tu Leyla mayn.
drank.CAUS-1SG Leyla water
'I gave Leyla water to drink.'
'I made Leyla drink water.'

Modification by two contradictory adverbs, 'quickly' and 'slowly,' supports this ambiguity (83). Under the first reading described above, the effect is odd. Under the second, however, the speaker can be acting quickly and Leyla slowly. For instance, knowing that Leyla is prone to drink water too fast, the speaker might rush to her to ensure she drinks it slowly.

- (83) xıfey şarıp-tu Leyla mayn hedi.
quickly drank.CAUS-1SG Leyla water slowly
'I quickly slowly gave Leyla water to drink.'
'I quickly made Leyla drink water slowly.'

Other ingestives, such as 'read,' do not form direct causatives in Sason Arabic at all. The causative in (84b) only receives a generic indirect interpretation according to which the teacher makes Kemal read the book in some unspecified way. Contradictory modifiers raise no problem.

- (84) a. kemal ku i-qri lala kitab.
kemal be.3M 3M-read.IPFV this.M book
'Kemal is reading this book.' (Akkuş, 2021a:93)
- b. (xıfey) oratman ki tı-qarri kemal lala kitab (hedi).
quickly teacher be.3F 3F-read.CAUS Kemal this.M book slowly
'The teacher is (quickly) making Kemal read this book (slowly).' (Yakut, 2012:14)

In sum, geminate causatives of ‘eat’ must, those of ‘drink’ can, receive a direct interpretation. Other ingestives appear to be unable to form direct causatives in Sason Arabic.

In terms of morphosyntax, the ‘causee’ argument of geminate causatives of ingestives is always assigned dative case, both under a direct and an indirect interpretation. In Sason Arabic, as in other Arabic varieties, case marking is only realized overtly with pronominal arguments, as in (85):

- (85) şarrip-to-lla mayn.
 drank.CAUS-1SG-her.DAT water
 ‘I gave her water to drink.’
 ‘I made her drink water.’

Although the syntax of direct and indirect causatives of ingestives might thus appear identical at first sight, the next section will offer further syntactic evidence that they differ in their underlying structure, aligning with the differences in their semantics that we have already seen.

To summarize, Hindi-Urdu and Sason Arabic both allow some ingestive transitives to form direct causatives. In Hindi-Urdu, the morphology already makes this point evident; for Sason Arabic, where morphology is ambiguous, semantic interpretation and evidence from adverbials allows us to draw the same conclusion. While in Hindi-Urdu, a sizable class of ingestives forms direct causatives, in Sason Arabic the alternation appears to be restricted to the two most prototypical ingestives, ‘eat’ and ‘drink.’¹³ In the following, I turn to the syntax of direct causatives of ingestives.

4.3.2.2 Syntactic diagnostics

Following Bhatt and Embick (2017) and Legate (2014), I argue that direct causatives of ingestives – at least in Hindi-Urdu and Sason Arabic – have a ditransitive structure identical to that of transfer-of-possession verbs such as *give*. In this section, I show that the dative-marked object of direct causative of ingestives behaves like the indirect object of standard ditransitives for the purposes of four diagnostics: case marking, object shift, depictives and null objects.

Case marking For Hindi-Urdu, the first piece of evidence for this claim comes from case marking (Bhatt and Embick, 2017). The morpheme *-ko*, which surfaces on the ‘causee’ argument, both serves as a dative marker in ditransitives and as a direct object marker (DOM) on specific direct objects of transitives. But these two usages can be teased apart. While DOM *-ko* is optional (86a), the dative marker in ditransitives is not (86b). Direct causatives of ingestives pattern with ditransitives in also obligatorily requiring *-ko* (86c).

- (86) a. Tina haar(-ko) uṭh-aa rahii hai.
 Tina necklace-ACC lift-CAUS PROG.F is
 ‘Tina is lifting a/the necklace.’

¹³Interestingly, this matches the findings from the previous chapter, where we saw that unergatives also form direct causatives more easily and more widely in Hindi-Urdu than in Sason Arabic. While this correlation is hardly accidental, I must leave it open what motivates it.

- b. Tina-ne Mina*(-ko) kitaab di-i.
Tina-ERG Mina-DAT book give-PFV.F
'Tina gave a book to Mina.'
- c. Tina-ne Mina*(-ko) angrezii sikh-aa-yii.
Tina-ERG Mina-DAT English learn-CAUS-PFV.F
'Tina taught Mina English.'

(Bhatt and Embick, 2017:127–128)

A confound is presented by the fact that the *-ko*-marked arguments of direct causatives of ingestives are always animate, and animate arguments always require DOM (87):

- (87) Tina-ne Ron*(-ko) dekh-aa.
Tina-ERG Ron-ACC see-PFV
'Tina saw Ron.' (Without *-ko*: 'Tina saw the object/the blob Ron.')

(Bhatt and Embick, 2017:128)

However, conclusive evidence comes from passivization. DOM *-ko* does not surface when the relevant argument is promoted to subject position under passivization (88a), whereas dative *-ko* is retained (88b). Again, direct causatives of ingestives pattern with ditransitives (88c).

- (88) a. haar uṭh-aa-yaa gayaa.
necklace.M raise-CAUS-PFV.MSG PASS.PFV.MSG
'The necklace was lifted.'
- b. Mina*(-ko) kitaab di-i gayii.
Mina-DAT book give-PFV.F PASS.PFV.F
'The book was given to Mina.'
- c. Mina*(-ko) angrezii sikh-aa-yii gayii.
Mina-DAT English.F learn-CAUS-PFV.F PASS.PFV.F
'English was taught to Mina.'

(Bhatt and Embick, 2017:128–129)

Thus, case marking supports a ditransitive analysis.

Object shift In addition to case marking, Bhatt and Embick (2017) draw on evidence from object shift to corroborate a ditransitive analysis of direct causatives of ingestives. Specific direct objects marked with *-ko* have been argued to obligatorily scramble out of the verb phrase (Bhatt and Anagnostopoulou, 1996). While in transitives, the movement is normally string-vacuous, in ditransitives, the direct object lands in a position preceding the indirect object (89):

- (89) a. Tina-ne Mina-ko vo kitaab di-i.
Tina-ERG Mina-DAT that book give-PFV.F
'Tina gave that book to Mina.'
- b. Tina-ne [us kitaab-ko]_i Mina-ko t_i di-yaa.
Tina-ERG that book-ACC Mina-DAT give-PFV
'Tina gave that book to Mina.'

(Bhatt and Embick, 2017:129)

Direct causatives of ingestives show the same pattern (90):

- (90) a. Ram-ne Sita-ko vo bhaashaa sikh-aa-yii.
 Ram-ERG Sita-DAT that language learn-CAUS-PFV.F
 ‘Ram taught Sita that language.’
 b. Ram-ne [us bhaashaa-ko]_i Sita-ko t_i sikh-aa-yaa.
 Ram-ERG that language-ACC Sita-DAT learn-CAUS-PFV
 ‘Ram taught Sita that language.’ (Bhatt and Embick, 2017:129)

In sum, object shift constitutes further evidence for a ditransitive syntax.

Depictives For Sason Arabic, secondary depictive predicates help further clarify the syntax of direct causatives of ingestives. Recall that in this language, direct causatives of ingestives and indirect causatives are morphosyntactically identical, unlike in Hindi-Urdu. However, the dative-marked arguments in the two constructions differ with respect to secondary predication: while the embedded causee in indirect causatives can be modified by a depictive (91a), the indirect object of a ditransitive cannot (91b) (Akkuş, 2021a, see also Pylkkänen, 2008 for English):

- (91) a. nana xassal-na kemal potad sarxoş.
 we wash.CAUS-1PL kemal clothes drunk
 ‘We₁ made Kemal₂ wash the clothes drunk_{1/2}.’
 b. varr-e kemal xabar raxu-e/*raxu.
 showed-3F Kemal.M news sick-F/*sick.M
 ‘She₁ showed Kemal₂ the news sick_{1/*2}.’

In direct causatives of ingestives, the dative-marked argument cannot be modified by a depictive (92):

- (92) şarrip-to-lla mayn raxu-(*e).
 drank.CAUS-1SG-her.DAT water sick-F
 ‘I₁ gave her₂ water to drink sick_{1/*2}.’ (Akkuş, 2021a:184)

Hence, direct causatives of ingestives pattern with ditransitives for the purposes of this diagnostic.

Null arguments Lastly, ditransitives and indirect causatives also differ in Sason Arabic with respect to the interpretation of null arguments (Akkuş, 2022). In indirect causatives, null arguments are interpreted as existential (93a), but in ditransitives, as pronominal (93b):

- (93) a. Leyla qarr-e alu kitabad.
 Leyla read.CAUS-3F these.M books
 ‘Leyla made someone read these books.’
 Not: ‘Leyla made him/her/them read these books.’ (Akkuş, 2022:10)
 b. ams adi-tu potad-i.
 yesterday gave-1SG clothes-my
 ‘Yesterday I gave him/her/them my clothes.’
 Not: ‘Yesterday I gave someone my clothes.’ (Akkuş, 2022:12)

In direct causatives of ingestives, null arguments are interpreted as pronominal (94):

- (94) ayyil-tu badincan.
 feed.CAUS-1SG tomatoes
 ‘I fed her/him/them tomatoes.’
 Not: ‘I fed someone tomatoes.’

To summarize, in both Hindi-Urdu and Sason Arabic, the dative-marked argument of direct causatives of ingestives patterns with the indirect object of ditransitives, not with the causee argument of indirect causatives.

4.3.2.3 Thematic interpretation

Having established that the dative-marked argument in direct causatives of ingestives shows the same syntactic behavior as the indirect object of standard ditransitives, I now argue that it also receives the same θ -role. For the sake of concreteness, let us look again at example (78), repeated below as (95). My claim is that *Leyla* is an agent in the transitive (95a), but a goal in the ditransitive (95b), the same θ -role assigned to the indirect object in transfer-of-possession constructions.

- (95) a. ayal-e Leyla badincan.
 eat-F Leyla tomatoes
 ‘Leyla ate tomatoes.’
 b. ayyil-tu Leyla badincan.
 eat.CAUS-1SG Leyla tomatoes
 ‘I fed Leyla tomatoes.’

In both sentences, the argument is sentient, as are all ‘ingerter’ arguments: that is, eating, drinking, hearing, etc., all presuppose a sentient participant. In the transitive, Leyla is also acting volitionally, another proto-agent property. However, in the ditransitive, the subject is clearly more volitional and more in control of the situation than Leyla. This indicates that the subject is now associated with the agent role.

Crucially, in a ditransitive, one of the arguments must be interpreted as a goal (in the broad sense). For transitivized ingestive verbs, this is naturally a possibility: goal arguments can receive food, drink, visual stimuli, etc. For instance, in the case of eating, the food ends up inside the eater, or in other words, the eater is the destination of the food. On the other hand, transitives other than ingestives such as *kill*, *cut* or *sweep* are not compatible with this interpretation since it is not clear how the causee of, say, a cutting event could be understood as a goal. This line of reasoning predicts that of all transitives, only ingestives can form direct causatives. As reported earlier, this prediction is borne out in several typologically unrelated languages; I am not aware of counterexamples.

To conclude, dative marked arguments of direct causatives of ingestives pattern both syntactically and semantically with indirect objects of standard transfer-of-possession ditransitives such as *give*. I thus conclude that the argument is assigned a goal θ -role, making the alternation a case of variable linking: the same individual role would be mapped onto an agent θ -role in the transitive but onto a goal θ -role in the ditransitive.

4.3.3 Competing analyses

Finally, I review two competing analyses according to which direct causatives of ingestives do not constitute an instance of variable linking, Amberber (2002) and Krejci (2012). While discussing Krejci's work, I also briefly touch on the approach to direct causatives of ingestives developed by Ramchand (2008), whose analysis of direct causatives of unergatives I examined earlier in Chapter 4.2.3.2.

4.3.3.1 Coindexation of agent and goal: Amberber (2002)

Amberber (2002) develops an analysis of direct causatives of ingestives in the Semitic language Amharic. By way of example, the transitive and corresponding ditransitive of 'eat' are given in (96); other causativizing ingestives in the language include 'drink,' 'lick,' 'suck,' 'taste,' 'swallow' and others. No non-ingestive transitives can form causatives with the prefix *a-*, which otherwise only surfaces on intransitives.

- (96) a. *ləmma dabbo bəlla*
 Lemma bread eat.PF.3M
 'Lemma ate some bread.' (Amberber, 2002:10)
- b. *aster ləmma-n dabbo a-bəlla-čč-iw*
 Aster Lemma-ACC bread CAUS-eat.PERF-3F-3M.O
 'Aster fed Lemma some bread.' (Amberber, 2002:3)

In addition to the morpheme *a-*, both intransitives and transitives can also form transitives with the prefix *as-*, which Amberber consistently translates periphrastically with *make*. I take it that *a-* causatives are direct, those formed with *as-* indirect.

For the verb 'eat,' Amberber adopts the denotation in (97) proposed by Jackendoff (1990); INCH stands for 'inchoative':

- (97) *eat*
 V
 $[_{\text{Event}} \text{CAUS } ([\text{Thing}_i], [\text{INCH } ([\text{Thing}], [_{\text{Path}} \text{TO}[\text{IN}[\text{MOUTH-OF } [\text{Thing}_i]]]])])]$
 (Amberber, 2002:13)

According to this denotation, eating involves an agent argument causing a theme argument to enter the mouth of the goal argument. The agent and the goal argument are obligatorily coindexed, meaning that the participant who causes the eating is also the one in whose mouth the food ends up. Amberber simplifies this denotation as in (98):

- (98) *bəlla* 'eat'
 $[_{x_i} \text{CAUS}^H (y) \text{INCH } z_i \text{PATH}]$
 (Amberber, 2002:13)

Amberber argues that when an agent and a goal argument are coindexed, only the former, being higher on the thematic hierarchy, is projected onto the syntax.¹⁴ Therefore, (98) surfaces with a transitive syntax.

The transitive in (97) is causativized, Amberber proposes, via the addition of a second CAUS predicate, spelled out as the causative morpheme *a-*, which takes a second agent argument (99):

- (99) *a-bəlla* ‘feed’
 [w CAUS [x_i CAUS (y) INCH z_i PATH]] (Amberber, 2002:13)

To explain why (99) surfaces with a ditransitive syntax, Amberber argues that the original agent argument (i.e., x_i in (99)) does not project onto the syntax because it is semantically recoverable from the goal argument. The result is a ditransitive structure where the new agent (w) is realized as a subject, the goal argument (z_i) as an indirect object and the theme (y) as a direct object. In sum, according to this analysis, ingestives are causativized simply by adding a causing event with an agent argument, analogous to what is commonly assumed for standard causatives of unaccusatives.

Most of the heavy lifting in this analysis is done by the linking rules that connect the verb’s semantic entry with its argument structure: in the transitive, the goal argument does not project, while in the ditransitive, the original agent argument doesn’t. Why any of this happens is not made explicit. In particular, in the transitive, the agent is projected at the expense of the goal by virtue of being higher on the thematic hierarchy – and why this should matter in the first place is not clear –, but in the ditransitive, the goal is projected after all despite being lower on the thematic hierarchy. No explanation is given for this asymmetry. I do not want to engage here in a sweeping debate about the merits and demerits of positing a semantic denotation that is independent from the syntax, but it is hard to see what would motivate the kind of mapping between semantics and syntax that needs to be assumed for this analysis.

The more straightforward problem with the coindexation analysis, also pointed out by Krejci (2012), is that it makes direct causatives of ingestives semantically identical to indirect causatives. Like all other transitives in Amharic, ingestives can form indirect causatives with the morpheme *-as* (100):

- (100) *aster ləmma-n dabbo as-bəlla-çç-iw.*
 Aster Lemma-ACC bread CAUS2-eat.PF-3F-3MO
 ‘Aster made Lemma eat some bread.’ (Amberber, 2002:3)

Amberber does not provide a denotation for examples like (100), but it is not clear how it would differ from the one in (99) for direct causatives. This does not square with the obvious fact that the two constructions are not semantically identical. The translations that Amberber gives for them are different, and we have also seen for Hindi-Urdu and Sason Arabic that they can be clearly teased apart. Overall, according to the coindexation analysis, direct causatives of ingestives are nothing other than semantically indirect causatives that happen to be squeezed into a ditransitive morphosyntax. Leaving aside the arbitrary nature of the

¹⁴I assume that the superscript H on CAUS stands for Hierarchy.

syntax-semantics mapping assumed here, the proposal does not account for the interpretation assigned to these constructions.

4.3.3.2 Antireflexivization: Krejci (2012)

Krejci's (2012) analysis, later adopted by Jerro (2019), is a purely semantic proposal for direct causatives of ingestives that does not attempt to explicitly account for their syntactic realization. It takes from Amberber the idea that ingestives always underlyingly have three arguments, seen for *eat* in (101). According to this denotation, eating involves a participant manipulating food in such a way as to cause another participant to potentially digest a third participant. By default, the first and the second argument are coindexed; that is, the person who manipulates food to cause the potential digestion event is also the one who potentially digests the food.

$$(101) \quad \llbracket \text{eat} \rrbracket = \llbracket [\text{ACT}_{\langle \text{manipulate food} \rangle} (x)] \text{ CAUSE } [\text{BECOME}_{\langle \text{potentially digest} \rangle} (x,y)] \rrbracket \quad (\text{Krejci, 2012:50})$$

Unlike Amberber, Krejci argues that causativizing an ingestive consists not in the addition of a second agent, but in an antireflexivization operation. In the causative, the two arguments of BECOME are no longer obligatorily coindexed, resulting in the reading that one person manipulates food in such a way as to cause a different person to potentially digest something (102). Note that Krejci develops her analysis primarily for English verbs that lexicalize causativized ingestives, such as *feed* (from *eat*) and *teach* (from *learn*), but her proposal is meant to extend to morphologically transparent causatives in languages like Hindi-Urdu and Amharic as well.

$$(102) \quad \llbracket \text{feed} \rrbracket = \llbracket [\text{ACT}_{\langle \text{manipulate food} \rangle} (x)] \text{ CAUSE } [\text{BECOME}_{\langle \text{potentially digest} \rangle} (y, z)] \rrbracket \quad (\text{Krejci, 2012:51})$$

In proposing this analysis, Krejci pursues a parallel to verbs she describes as middles, such as *wash*, *dress*, *shave*, etc., which receive a (roughly) reflexive interpretation when used intransitively (103):

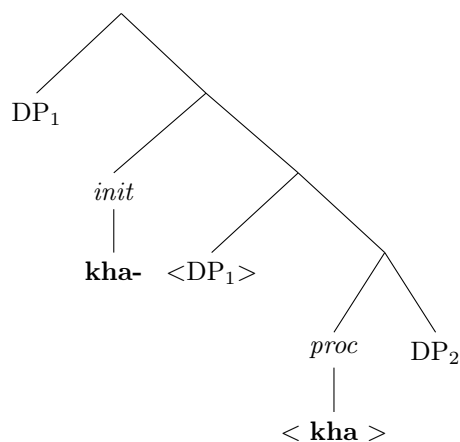
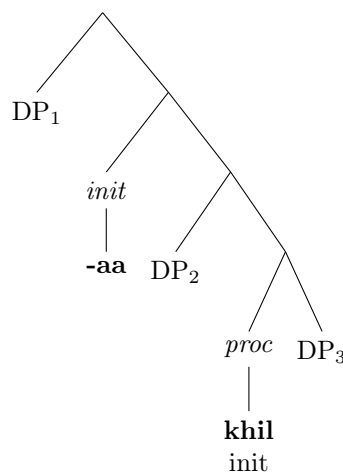
- (103) a. Amir washed. ($\approx$ Amir washed himself)
 b. Leyla washed Amir.

Krejci argues that such verbs are underlyingly transitives with two coindexed arguments, while in the overt transitives, the two roles are assigned to different arguments (104):

$$(104) \quad \begin{array}{ll} \text{a. } \llbracket \text{wash}_{itr} \rrbracket = \llbracket [\text{ACT}_{\langle \text{manipulate water} \rangle} (x)] \text{ CAUSE } [\text{BECOME}_{\langle \text{washed} \rangle} (x)] \rrbracket & (\text{Krejci, 2012:50}) \\ \text{b. } \llbracket \text{wash}_{tr} \rrbracket = \llbracket [\text{ACT}_{\langle \text{manipulate water} \rangle} (x)] \text{ CAUSE } [\text{BECOME}_{\langle \text{washed} \rangle} (y)] \rrbracket & (\text{Krejci, 2012:51}) \end{array}$$

Both ‘middles’ and ingestives thus causativize via an antireflexivization operation. In the remainder of this section, I will not engage with the former in any more detail, but it is generally an argument in Krejci's favor that her analysis of ingestives finds a parallel in seemingly unrelated data.

Krejci's proposal is reminiscent of Ramchand's (2008) treatment of Hindi-Urdu direct causatives of unergatives. Recall that according to Ramchand, the sole argument of an intransitive like *dance* receives both an initiator and an undergoer role, whereas in the causative, this argument only receives the undergoer role, with the initiator role now being assigned to the external causer. And in fact, Ramchand proposes a parallel analysis for Hindi-Urdu direct causatives of ingestives, shown for the root *kha* 'eat' in (105)–(106). In the transitive, the subject argument (the eater, DP₁) is merged in two positions simultaneously, *SpecinitP* and *SpecprocP*, being thus interpreted as both an initiator and an undergoer argument. The object (the food, DP₂) occupies the complement position of *proc* and is interpreted as a path. In the ditransitive, on the other hand, the subject (the feeder, DP₁) is the initiator, the direct object (the food, DP₂) the undergoer, and the indirect object (the eater, DP₃) the path.¹⁵

(105) *Transitive*(106) *Ditransitive*

Under both Krejci's and Ramchand's analysis, then, transitive ingestives are underlyingly ditransitives in that the subject argument is assigned two θ -roles. In the causativized variant, these two θ -roles come apart. Having discussed Ramchand's analysis of direct causatives of unergatives extensively in the previous chapter, I will not examine her approach to ingestives in any more detail in the following, but only use it as a point of comparison for Krejci's work.

While the general intuition underlying the two proposals is the same, the implementation varies significantly. Specifically, Ramchand assumes that in the transitive, the argument is indeed merged twice over the course of the derivation and is syntactically present, albeit covertly, in both the initiator and the undergoer position, *SpecinitP* and *SpecprocP*. In my earlier discussion of Ramchand's analysis of direct causatives of unergatives, which relies on the same idea, I argued that such a claim is not supported by the syntactic evidence. For instance, the covert lower argument cannot be detected by any standard diagnostics, and it remains unclear why it is not spelled out as an overt anaphor. But Krejci, working in a projectionist framework, makes no such commitments. For her, non-causativized ingestives are syntactically pure transitives;

¹⁵Note that according to this analysis, direct causatives of ingestives would still constitute an instance of variable linking in that the food is variably merged as a path and as an undergoer. Ramchand does not comment on this change of roles.

the ditransitivity merely exists in the semantics. Accordingly, the absence of a third argument in the syntax is not an argument against her account.

On the flip side, this raises the question of how and why concretely a ditransitive semantics would translate into a transitive syntax. As highlighted earlier, Krejci focuses exclusively on the semantics of these constructions and does not address their syntactic realization. I cannot rule out that within a projectionist architecture, it would be possible to formulate a linking rule to this effect, although as with Amberber's coindexation analysis, I am not sure what would motivate it. The fact remains that if one pursues the Ramchand-Krejci strategy of letting a single argument be assigned several θ -roles by the same verb to avoid variable linking, one either needs to show that the argument is indeed multiply present in the syntax or provide a principled account of why an argument that is multiply present in the semantics nonetheless only surfaces as one in the syntax. So far, neither of these two conditions has been met.

Another challenge for any analysis that posits multiple θ -roles for a single argument, already discussed for Ramchand, concerns quantificational arguments as in (107):

(107) Everyone ate nattoo.

Here, the two arguments in the underlying semantic entry that are spelled out as a universal quantifier – for Krejci, the argument of ACT and the first argument of BECOME – need to be realized as variables bound by the same operator. This is not trivially achieved by just inserting the same argument at two different points in the semantic derivation. Again, the projectionist lexicon is powerful, and it might be possible to adjust the lexical entries to produce the desired denotation, but this is an outstanding task. I currently do not see what kind of machinery could achieve it.

A final and major problem I raised for Ramchand's analysis was that the shared undergoer role she ascribed to both the subject of the intransitive and the object of the transitive did not come with any abstract thematic properties. Rather, in the context of *dance*, what the undergoer role seemed to encode were only non-thematic, verb-specific entailments such as moving one's feet. As a result, there is nothing that the undergoer arguments of dancing, breaking or pushing events would have in common which would warrant positing a shared θ -role associated with abstract entailments. In the remainder of this section, I consider how this problem presents itself in the context of Krejci's antireflexivization analysis.

Krejci provides an in-depth discussion of the different entailments associated with the different roles. While this discussion focuses on English verbs such as *eat* and *feed*, Krejci also confirms her findings with data from Marathi, which has morphologically transparent direct causatives of ingestives. To begin with *eat*, the verb normally entails that the subject argument puts food into their mouth (108):

(108) #John ate the food, but he didn't put any of it into his mouth. (Krejci, 2012:44)

However, Krejci also notes that if we were to imagine an alien species that ingested food via their palms,

nothing would force this entailment. Rather, one entailment of *eat* is that ‘the subject is actively involved in a canonical intake process’ (Krejci, 2012:47), where for humans, their canonical intake process is via the oral route. This entailment, Krejci argues, is associated with the argument of ACT; what I would refer to as an agent role.

Second, *eat* further entails that the food somehow enters the subject’s digestive system. It is not enough for the food to be in the subject’s mouth: if they spit it out immediately, their activity does not qualify as eating. On the other hand, vomiting does not preclude that there was an eating event (109):

(109) John ate the food, but vomited it up. (Krejci, 2012:46)

However, Krejci also maintains that if the subject merely swallows something with the intent of vomiting it up later – as in the case of drug smugglers –, *eat* is infelicitous (110):

(110) John swallowed/#ate the drugs in order to store them in a small plastic bag in his stomach. (Krejci, 2012:47)

To take all these possibilities into account, Krejci proposes that another entailment of *eat* is that the subject potentially digests a substance. This entailment is associated with the first argument of BECOME.

Turning to the verb *feed*, Krejci argues that the entailment of being actively involved in a canonical intake process, associated with the argument of ACT, now applies to the feeder, not the feedee. For instance, *feed* is commonly used for a parent feeding their child, where it is the parent, not the child, who is in charge of transferring the food from the plate into the child’s mouth. Accordingly, Krejci notes that for very young children, it can be infelicitous to say that the child ate because in this construction, the child is associated with the ACT entailment of actively manipulating food. At the same time, the baby still potentially digests the food (although, Krejci admits, babies who spit out their food are still being fed, whereas adult eaters need to keep food down to qualify as such – double standards apply). All of this, Krejci concludes, supports the antireflexivization analysis: in the transitive, the subject argument is associated with both the ACT and the BECOME entailments – actively manipulating food and potentially digesting it –, while in the ditransitive, the causer subject is associated with the ACT entailment, the dative argument with the BECOME entailment.

To summarize, Krejci’s claim is that for a given verb, each abstract role is associated with a particular verb-specific entailment, or in our terminology, that the θ -roles and individual roles of a verb stand in a one-to-one mapping relationship: the argument of ACT is an active food-manipulator, and the first argument of BECOME a potential digester. However, one key point to emerge from her discussion is how difficult it is to define what concretely the relevant verb-specific entailments are. The entailment proposed for the argument of ACT – being actively involved in a canonical intake process – is simply a description of the event (a canonical intake process) and what is essentially Dowty’s agent proto-role (being actively involved). Which activity concretely the argument performs varies widely, as Krejci recognizes, and does not reduce

to placing food in someone's mouth. Setting aside aliens who ingest food manually, animals are usually fed by pouring food into their bowls or troughs, and hospital patients can be fed intravenously. By the same token, what exactly qualifies as potential digestion is highly unclear: recall that babies can spit out food, but adults cannot, and that vomiting does not preclude potential digestion, unless it was intended from the start. As discussed earlier in Chapter 3.2 in the context of verbs like *kill* and *take*, it is very difficult in practice to make individual roles such as eaters, killers and takers concrete and clarify what activity specifically a participant would need to perform to qualify as such.

What about thematic entailments? As pointed out just now, Krejci recognizes that the argument of ACT comes with abstract agentive entailments. However, she does not try to isolate a set of general thematic properties that are associated with the first argument of BECOME in both alternants and that would generalize to other verbs. In fact, the few times she refers to this role using θ -vocabulary, she describes it as a patient, not as a goal. I doubt that this terminological choice reflects any deeper theoretical commitment; rather, whether this role comes with any abstract, verb-independent content is just not at issue here.

But unlike Ramchand in her analysis of direct causatives of unergatives, I believe that Krejci could make the argument that there are abstract entailments associated with the shared role. It might be argued that the subject argument of the transitive does not necessarily seem to have fewer goal entailments than the indirect object of the ditransitive. Regardless of whether someone eats or is being fed, they are always the (at least interim) endpoint of the path that the food takes. Thus, in both cases, we might say that such an argument comes with the same abstract thematic properties, those of a goal. To reiterate, this is not the claim that Krejci makes because she does not consider the argument a goal, but it is a claim that could be made by adapting her analysis. The charge leveled against Ramchand that the alleged shared θ -role does not come with any shared thematic properties would then not apply; rather, the abstract goal role would apply to both.

The general question this raises is whether shared thematic properties are always evidence for a shared θ -role. They are clearly a necessary condition, but we need to ask if they are sufficient. If θ -roles are conceptualized as prototypes following Dowty (1991), the answer is no. Under the prototype view, an argument that is assigned a given θ -role does not necessarily have all the properties associated with it, but might possess only a proper subset of them. At the same time, an argument can have properties associated with a θ -role it does not receive. For instance, while sentience is an agent property, plenty of patients are sentient, and no one would argue that all sentient arguments must share a θ -role. By the same token, the fact that the subject argument of *eat* has properties associated with the goal θ -role does not entail that this is the θ -role it is in fact assigned; rather, its overall profile matches that of an agent. On the other hand, the indirect object of *feed* would make for a poor agent but is much better described by the goal θ -role. Overall, shared thematic properties do not necessarily indicate a shared θ -role.

To conclude, leaving aside the implementational details of Krejci's analysis, any proponent of the idea that the subject of non-causative ingestives is assigned two θ -roles – including Ramchand (2008) – faces clear

challenges, such as how and why an argument that is realized twice in the semantics comes to surface as one in the syntax, how to deal with quantifiers, and how an argument with two θ -roles is different from an argument with a reflexive. I cannot rule out that these challenges might be solvable. However, the question remains what exactly such an analysis would aim to achieve. If it is to capture shared thematic properties, there is no need for a shared θ -role. In a proto-role world, thematic properties inevitably overlap all the time. But if it is to capture a shared *individual* role – and this is arguably the key motivation behind this analysis –, the discussion of Krejci’s work above has shown that it is not clear what content precisely would be associated with roles such as eaters and feeders, besides the theoretical problems with the concept of individual roles in the first place.

4.4 The agent-affectee alternation

4.4.1 Data and analysis

The fourth case study concerns a moderately well-studied phenomenon that I will refer to as the agent-affectee alternation, exemplified in (111) and (112) for Spanish and German, respectively. The (a) examples contain standard direct causatives, while the (b) examples feature an inchoative verb with an additional dative-marked argument.

- (111) a. El tintorero quemó los pantalones de Carolina.
 the dry-cleaner burnt.SG the trousers of Carolina
 ‘The dry-cleaner burnt Carolina’s trousers.’
 b. Al tintorero se le quemaron los pantalones de Carolina.
 to.the dry-cleaner REFL CL.DAT burnt.PL the trousers of Carolina
 ‘Carolina’s trousers got burnt on the dry-cleaner.’
 ‘The dry-cleaner (accidentally) burnt Carolina’s trousers.’ (Cuervo, 2003:186)
- (112) a. Ich habe die Vase zerbrochen.
 I have the vase broken
 ‘I broke the vase.’
 b. Mir ist die Vase zerbrochen.
 me.DAT is the vase broken
 ‘The vase broke on me.’
 ‘I (accidentally) broke the vase.’

Morphologically, the Spanish example (111b) bears standard reflexive anticausative marking, unlike the causative in (111a). The verb agrees with *los pantalones* ‘the trousers.’ In German, the unaccusative construal of the verb in (112b) is evidenced by the auxiliary ‘be,’ whereas the corresponding causative in (112a), as all transitives, takes ‘have.’ The auxiliary in the unaccusative (112b) agrees with *die Vase* ‘the vase.’ Semantically, the (a) examples receive a regular direct causative interpretation, compatible with both intentional and unintentional causation. The (b) examples have been argued to allow for two readings: the dative-marked argument can either be interpreted as being affected by the verbal event or as unintentionally causing it, not, however, as causing it intentionally.

Dative-marked unintentional causers are common in Indo-European languages. Besides Spanish (Cuervo, 2003; Suárez-Palma, 2020; Suárez-Palma and Santamaría-Gallego, 2024) and German (Bosse, 2017; Schäfer, 2008), they have been attested in Balkan languages such as Albanian, Bulgarian, Greek, Romanian, Slovenian and Serbo-Croatian (Kallulli, 2008; Rivero, 2004) as well as Italian (Schäfer, 2008) and Polish (Frąckowiak and Rivero, 2008). Similar facts have also been documented for Basque (Suárez-Palma and Santamaría-Gallego, 2024), Hungarian (Rákosi, 2014) and the Caucasian language Agul (Ganenkova et al., 2008; Schäfer, 2008). In the following, I focus on Spanish and German.

Let us consider the agent-affectee alternation from the perspective of variable linking, focusing on the Spanish example (111). Both the direct causative and the inchoative-cum-dative construction – under the unintentional causer reading – contain an argument that we might label a burner, someone who burns something. However, the syntactic realization of this argument clearly differs. In (111a), the burner occupies the external argument position, but in (111b) this cannot be the case, as clearly indicated by the dative case marking on the argument, its failure to control agreement, and the anticausative verbal morphology. Here, I describe the dative-marked argument as a high applicative argument, following Pykkänen (2008).

Moreover, I argue that the burner argument receives two different θ -roles in the two alternants. In the causative, it is realized as an agent. For the inchoative, I argue that the dative argument is always assigned what I label an affectee θ -role, and that one of the ways in which a participant can be affected by the event is by unintentionally causing it. Some situations can thus be described by either of the two variants, with some slight shifts in focus: under the agent construal, the emphasis is on the fact that the participant's actions caused the event, whereas the affectee construal highlights the involuntary nature of the causation and the fact that the participant was affected by the event. Under a theory that has individual roles, the agent-affectee alternation would thus constitute a case of variable linking, with the role of a burner being variably mapped onto an agent and an affectee θ -role in the two alternants.

The most comprehensive analysis of datives in general and the agent-affectee alternation in particular remains Cuervo (2003), whose work I heavily draw on in the following. However, unlike under the analysis that I develop here, Cuervo argues that the difference between the affectedness reading and the unintentional causer reading of the inchoative-cum-dative alternant corresponds to a structural difference in the position of the applicative projection introducing the dative argument in the clausal spine. I discuss why I diverge from Cuervo's account in this respect in Chapter 4.4.3. There, I also review previous work on the agent-affectee alternation by Rivero (2004), Kallulli (2008) and Schäfer (2008). Before doing so, I introduce the data more in detail and then, as always, turn to syntactic and semantic evidence for the analysis proposed here.

4.4.2 Evidence for the analysis

4.4.2.1 Background

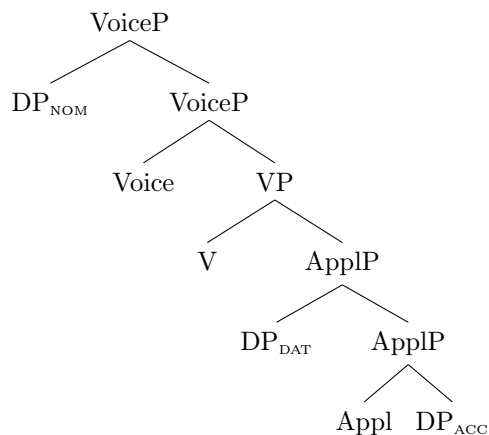
Low and high applicatives I begin with an overview over dative arguments with a focus on Spanish and German. Following Pykkänen (2008), datives are widely assumed to fall into two main groups, low and

high applicative arguments (see also Marantz, 1993; McGinnis, 2001, a.m.o.). The first group is canonically exemplified by transfer-of-possession constructions such as (113) and (114). Note that in Spanish, dative arguments are marked with the preposition *a* and are doubled by a dative clitic, the latter of which Cuervo (2003) analyzes as the spell-out of the applicative head.

- (113) Pablo le mandó un diccionario a Gabi.
 Pablo CL.DAT sent a dictionary to Gabi.DAT
 ‘Pablo sent Gabi a dictionary.’ (Cuervo, 2003:29)
- (114) Ersin gibt dem Jungen ein Buch.
 Ersin gives the.DAT boy.DAT a book
 ‘Ersin gives the boy a book.’

A low applicative structure for examples such as (113) and (114) is sketched out in (115). Here, the verb takes as its complement a functional projection headed by Appl, which takes the accusative and the dative DPs as its arguments.

- (115) *Low applicative*



(Pylkkänen, 2008:14)

Low applicatives thus relate two individuals – the dative and the accusative argument – to each other, besides also relating the accusative argument to the verb. Pylkkänen posits two different low Appl heads depending on whether the dative argument is interpreted as a source or as a recipient of the accusative argument; Cuervo (2003) adds to the low Appl inventory a head that predicates a (stative) possessive relation between the two.

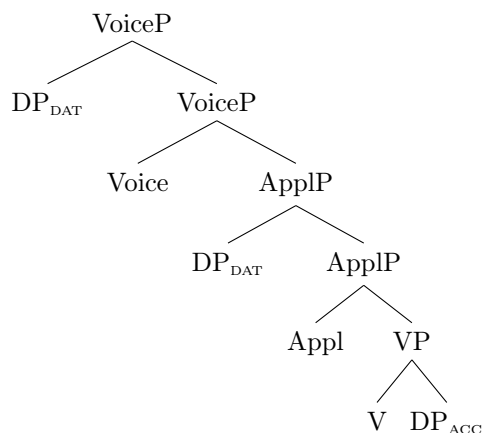
The second group of datives, high applicatives, is exemplified in (116) and (117):

- (116) Emilio le rompió la radio a Carolina.
 Emilio CL.DAT broke the radio to Carolina.DAT
 ‘Emilio broke the radio on Carolina.’ (Cuervo, 2003:30)

- (117) Ersin hat mir das Boot versenkt.
 Ersin has me.DAT the boat sank
 ‘Ersin sank the boat on me.’

High applicative arguments are broadly interpreted as being affected by the verbal event. In both (116) and (117), the most salient interpretation is a malefactive reading, under which the event happens to the detriment of the dative-marked argument. A possessive relation between the accusative and the dative argument is possible but not entailed. For instance, in (116), Carolina might be affected by the event because the radio that Emilio broke was hers or because she was responsible for safekeeping someone else’s radio (see Cuervo, 2003). Pykkänen’s syntax for high applicatives is shown in (118).

- (118) *High applicative*



(Pykkänen, 2008:14)

Unlike low Appl, high Appl thus does not relate two individuals, but an individual and an event. For instance, in (116), a relation is predicated to hold between Carolina and the event of breaking the radio. Both low and high applicative heads have thus been argued to denote a ‘have’ relation: in a low applicative such as (113), Gabi has a dictionary, whereas in a high applicative such as (116), Carolina ‘has’ a broken radio (Cuervo, 2003; Schäfer, 2008). Overall, the difference in interpretation between low and high applicatives is assumed to be due to a difference in where the Appl head is merged in the syntax.

Datives with inchoatives Having introduced standard low and high applicatives, let us now return to the constructions that are the focus of this chapter: inchoatives with an additional dative argument, such as (119) and (120).¹⁶

- (119) Al tintorero se le quemaron los pantalones de Carolina.
 to.the dry-cleaner REFL CL.DAT burnt.PL the trousers of Carolina
 ‘Carolina’s trousers got burnt on the dry-cleaner.’
 ‘The dry-cleaner (accidentally) burnt Carolina’s trousers.’

(Cuervo, 2003:186)

¹⁶In the following, I will only be concerned with unaccusatives that undergo the causative alternation. In German, not all inchoatives allow for the construction in (120) to the same extent (Schäfer, 2008); I discuss this restriction in Chapter 4.4.3.4.

- (120) Mir ist die Vase zerbrochen.
 me.DAT is the vase broken
 ‘The vase broke on me.’
 ‘I (accidentally) broke the vase.’

As noted earlier, the Spanish verb surfaces with reflexive anticausative marking; in German, the unaccusative status of the verb is indicated by the auxiliary *be*. The simple unaccusatives are given in (121) and (122):

- (121) Los pantalones de Carolina se quemaron.
 the trousers of Carolina REFL burnt
 ‘Carolina’s trousers burnt.’
- (122) Die Vase ist zerbrochen.
 the vase is broken
 ‘The vase broke.’

Inchoative-cum-dative examples such as (119) and (120) have been reported to have two possible readings, but to anticipate, I will argue that these two readings are merely variations of a single underlying interpretation that is subject to vagueness. First, like high applicative datives, they allow for an affectedness interpretation such as a malefactive or benefactive reading, which again can but does not have to come with a possessive relation between the dative and the accusative argument. Here, I do not gloss the possessive reading separately. Second, the dative arguments can also be interpreted as unintentionally causing the event. For instance, in (120), the speaker might have dropped the vase out of carelessness. The construction is strictly impossible if the action was carried out intentionally, as also evidenced by the impossibility of adjectives such as *on purpose* (123)–(124) (Cuervo, 2003):¹⁷

- (123) *Al tintorero se le quemaron los pantalones de Carolina a propósito.
 to.the dry-cleaner REFL CL.DAT burnt.PL the trousers of Carolina on purpose
 Intended: ‘The dry-cleaner burnt Carolina’s trousers on purpose.’
- (124) *Mir ist absichtlich die Vase zerbrochen.
 me.DAT is on.purpose the vase broken
 Intended: ‘I broke the vase on purpose.’

¹⁷Ana Arregui (p.c.) points out that examples such as (123) are acceptable under the reading that the dry-cleaner *made it so that* he accidentally burnt Carolina’s trousers; that is, if he intentionally created a situation in which he would be more likely to accidentally burn the trousers. For instance, he might turn the iron a few degrees higher, knowing that this increases the risk of burning the clothes, because he secretly dislikes Carolina (or her trousers) but does not dare to take more direct action. In this scenario, it is technically true that the burning was an accident, but the dry-cleaner would nonetheless be responsible for it. Another construction that yields the same meaning is (i):

- (i) El tintorero hizo que se le quemaron los pantalones de Carolina.
 the dry-cleaner made that REFL CL.DAT burnt the trousers of Carolina
 ‘The dry-cleaner made it so that Carolina’s trousers burnt on him.’

I concur with these judgments for German. I assume that the reading in question is a coercion effect that arises due to conflicting information – the dative argument is presented as both intentional and unintentional –, but I cannot give a more detailed explanation here.

This contrasts with standard direct causatives, which are compatible with both intentional and unintentional causation (125)–(126):

- (125) El tintorero quemó los pantalones de Carolina a propósito/sin querer.
 the dry-cleaner burnt the trousers of Carolina on purpose/without wanting
 ‘The dry-cleaner burnt Carolina’s trousers on purpose/by accident.’
- (126) Ich habe absichtlich/aus Versehen deine Vase zerbrochen.
 I have on.purpose/by accident your vase broken
 ‘I broke your vase on purpose/by accident.’

By the same token, unlike direct causatives, inchoatives with datives are incompatible with instrument phrases (127)–(128) and purpose clauses (129)–(130):

- (127) a. *A Juan se le quebró el florero con un martillo.
 to Juan REFL DAT.CL broke the vase with a hammer
 Intended: ‘Juan broke the vase with a hammer.’
 b. Juan quebró el florero con un martillo.
 Juan broke the vase with a hammer
 ‘Juan broke the vase with a hammer.’
- (128) a. *Mir ist deine Vase mit einem Hammer zerbrochen.
 me.DAT is your vase with a hammer broken
 Intended: ‘I broke your vase with a hammer.’
 b. Ich habe deine Vase mit einem Hammer zerbrochen.
 I have your vase with a hammer broken
 ‘I broke your vase with a hammer.’
- (129) a. *Al tintorero se le quemaron los pantalones de Carolina para vengarse.
 to.the dry-cleaner REFL CL.DAT burnt the trousers of Carolina to take.revenge
 Intended: ‘The dry-cleaner burnt Carolina’s trousers to take revenge.’
 b. El tintorero quemó los pantalones de Carolina para vengarse.
 the dry-cleaner burnt the trousers of Carolina to take.revenge
 ‘The dry-cleaner burnt Carolina’s trousers to take revenge.’ (Cuervo, 2003:187)
- (130) a. *Mir ist die Vase zerbrochen, um dich zu ärgern.
 me.DAT is the vase broken in.order you to annoy
 Intended: ‘I broke the vase in order to annoy you.’
 b. Ich habe die Vase zerbrochen, um dich zu ärgern.
 I have the vase broken in.order you to annoy
 ‘I broke the vase in order to annoy you.’

To recap, datives in ditransitives have been broadly categorized into two groups: low applicatives that establish a relation (source, recipient or possession) between two individuals, and high applicatives that establish an affectedness relation between an individual and an event. Against this background, we have then looked at dative arguments with inchoatives, which can receive either an affectedness reading, parallel to the high applicatives introduced above, or an unintentional causer reading, which is also available in

standard direct causatives. Next, we turn to evidence for the syntactic position and thematic interpretation of unintentional causer datives.

4.4.2.2 Syntactic diagnostics

I argue that dative unintentional causers are syntactically realized as what I will refer to as high applicative arguments. I do not specifically endorse Pylkkänen's exact analysis of these structures as shown in (118), but my claim is simply that the syntactic status of dative arguments with inchoatives is the same under the affectedness reading and the unintentional causer reading. In the next section, I will then make the case that they are also assigned the same θ -role.

Demonstrating that the syntax of the affectedness reading and the unintentional causer reading is identical is a somewhat odd task: the two constructions are morphosyntactically perfectly identical and cannot be teased apart by any syntactic diagnostic I am aware of. It has never been claimed that they can. In the following, I merely report one diagnostic that might be expected to yield evidence for a structural difference between the two readings if a difference did indeed obtain, namely, *again* modification (see also Bosse, 2017). To anticipate the finding of this section, the diagnostic detects no difference between the two readings.

***Again* modification** As is well known, *again* can give rise to two different interpretations, repetitive and restitutive (e.g., Beck and Johnson, 2004; von Stechow, 1995, 1996, a.m.o.). Under the repetitive reading of (131), what happens again is the event of Sally opening the door, whereas under the restitutive reading, it is the state of the door being open that holds again.

(131) Sally opened the door again.

- a. Sally opened the door, and that had happened before. (*repetitive*)
- b. Sally opened the door, and the door had been open before. (*restitutive*)

(Beck and Johnson, 2004:107)

The two readings can be derived by assuming that *again* can attach at two different points in the syntax. Under the high attachment, *again* modifies the eventuality described by the entire verbal domain, giving rise to a repetitive reading. In contrast, under the low attachment, it only modifies the lower subeventuality, the result state, giving rise to a restitutive reading.

In German, the readings available depend on the position of *again* in the sentence (132):

- (132) a. ... dass Ali Baba Sesam wieder öffnete.
 that Ali Baba Sesam again opened
 'that Ali Baba opened Sesam again.'
(*restitutive/repetitive*)

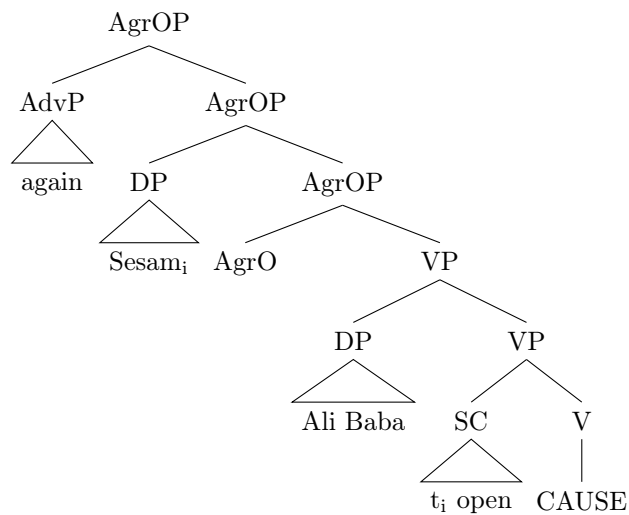
- b. ... dass Ali Baba wieder Sesam öffnete
 that Ali Baba again Sesam opened
 ‘that Ali Baba opened Sesam again.’

(*only repetitive*)

(von Stechow, 1996:87–88)

To account for the effect of word order, von Stechow (1995, 1996) argues that objects undergo movement to a high SpecAgrOP position outside of the VP (133); note that movement of the subject to SpecTP is not shown. The fact that *again* precedes the object indicates that it is higher than AgrOP, thus outside of the VP. In consequence, only the repetitive reading is available.

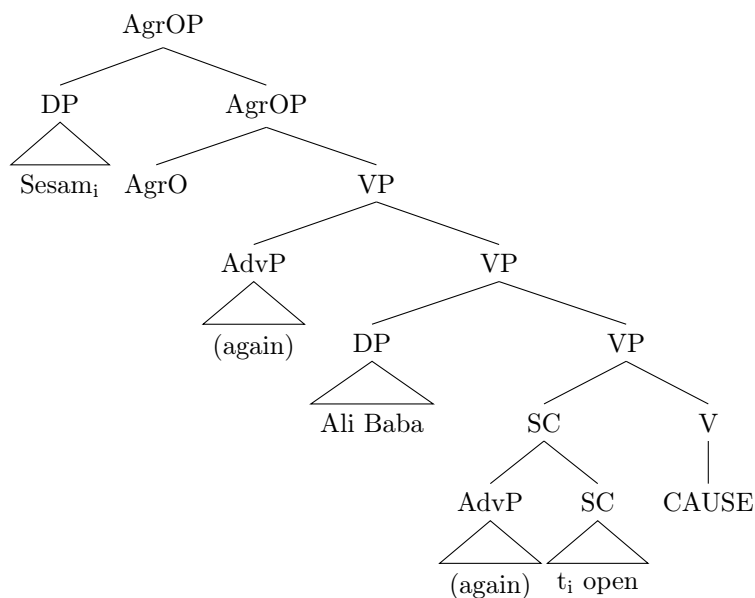
(133)



(von Stechow, 1996:88)

On the other hand, if the adverb follows the object, it could either have merged with the VP or with the stative small clause (134):

(134)



(von Stechow, 1996:88)

Thus, both the repetitive and the restitutive reading are possible in this case.

Since *again* modification can help diagnose the structural height of an argument in the syntax, we can ask whether it yields different results for datives with inchoatives under an affectedness and an unintentional causer interpretation, which would be evidence for a structural difference between the two. For German, I will use the *again* diagnostic for the sentence in (135):

- (135) Mir ist das Segel zerrissen.
 me.DAT is the sail torn
 ‘The sail tore on me.’
 ‘I accidentally tore the sail.’

To keep the affectedness reading and the unintentional causer reading distinct – which, as the next section will argue, is not at all straightforward – I set up two scenarios that exemplify as clearly as possible the two interpretations (136):

- (136) a. Affectedness scenario:
 While I am out at sea on my sailing ship, a heavy wind tears the sail, putting me in danger.
 b. Unintentional causer scenario:
 While trying to take down the sail to store it away for the night, I accidentally tear it.

Combining these two scenarios with the two possible readings of *again* – repetitive and restitutive –, we arrive at four potential interpretations:

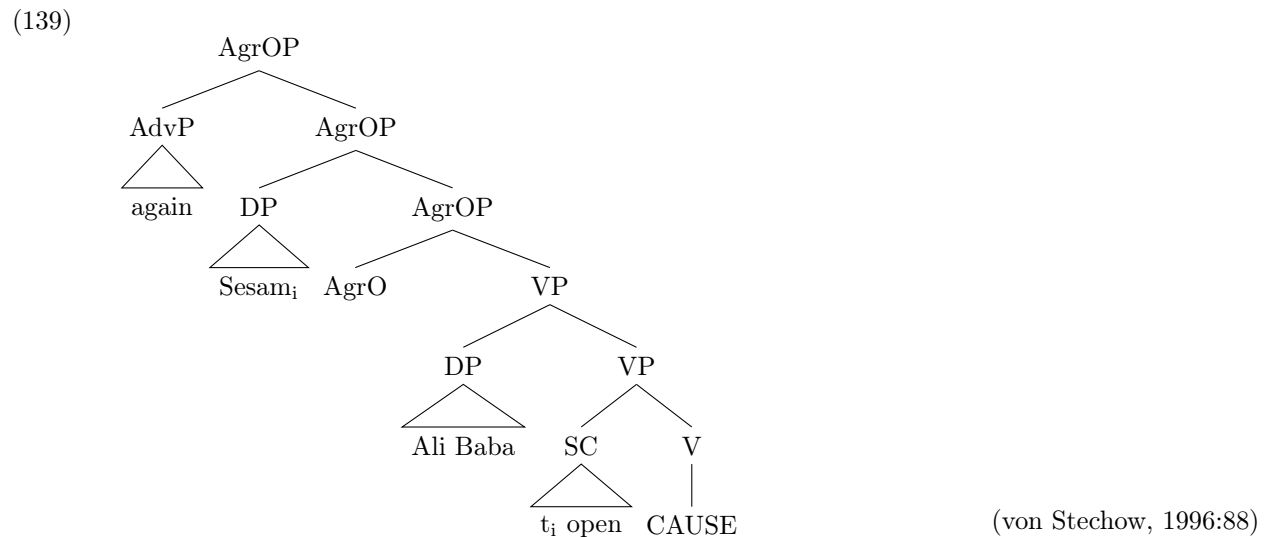
- (137) a. Affectedness scenario, repetitive reading:
 While I am out at sea on my sailing ship, a heavy wind tears my sail, putting me in danger. I manage to return to land and repair it. This happens again the next day.
 b. Unintentional causer scenario, repetitive reading:
 While trying to take down the sail to store it away for the night, I accidentally tear it. I repair it afterwards. This happens again the next day.
 c. Affectedness scenario, restitutive reading:
 While my friend is out at sea on his sailing ship, a heavy wind tears his sail, putting him in danger. He manages to return to land and repair it. The next day, I borrow his sailing ship. The sail tears again in the wind, which now puts me in danger.
 d. Unintentional causer scenario, restitutive reading:
 While trying to take down the sail to store it away for the night, my friend accidentally tears it. He repairs it afterwards. The next day, I borrow his sailing ship. While trying to take down the sail, I now accidentally tear it.

The two relevant sentences for the diagnostic are given in (138), with *again* realized in two different positions in the sentence:

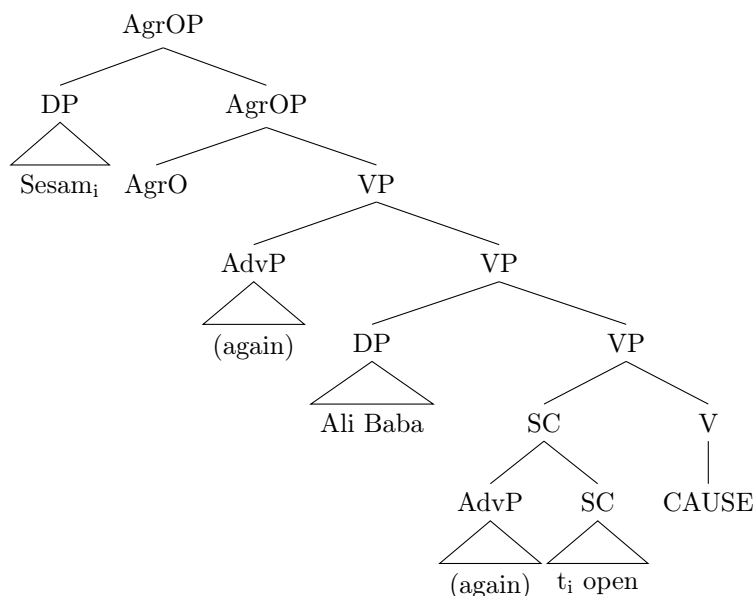
- (138) a. ... dass mir das Segel wieder zerrissen ist.
 that me.DAT the sail again torn is
 ‘that the sail tore me (dative) again.’ (*restitutive/repetitive*)
- b. ... dass mir wieder das Segel zerrissen ist.
 that me.DAT again the sail torn is
 ‘that the sail tore me (dative) again.’ (*only repetitive*)

Example (138a) is compatible with all four readings in (137); example (138b) only with the two repetitive readings. This matches what was observed by von Stechow (1995, 1996), and importantly does not distinguish between the affectedness reading and the unintentional causer reading.

In consequence, *again* does not offer any evidence for a structural difference between dative arguments with inchoatives under the affectedness reading and the unintentional causer reading, respectively, even if they are kept apart as clearly as possible. Recall Stechow’s trees in (139) and (140):



(140)



(von Stechow, 1996:88)

Like the external argument, the dative argument must occupy a position somewhere between what Stechow represents here as a small clause and the surface position of the object argument. There is no evidence that this position depends on the interpretation of the dative.

As for Spanish, there are two candidates for expressing the meaning of *again* in the language, the adverb *otra vez* ‘another time’ and the verbal periphrastic construction *volver a* ‘return to,’ both shown in (141):

- (141) a. Juan rompió el florero otra vez.
 Juan broke the vase another time
 ‘Juan broke the vase again.’
 b. Juan volvió a romper el florero.
 Juan returned to breaking the vase
 ‘Juan broke the vase again.’

For the direct causative baseline in (141), my informants found both the repetitive and the restitutive reading easily accessible with the periphrastic construction *volver a* in (141b), whereas for the adverb *otra vez* in (141a), the restitutive reading was difficult to obtain. Hence, I use *volver a* in the following. While for *again*-type adverbs, the ambiguity between the repetitive and the restitutive reading has been analyzed as being due to attachment height of the adjunct, as outlined above, I propose that for the *volver a* construction, the ambiguity is due to complement size. That is, *volver a* can embed either a smaller constituent describing only the result state, which lacks the external argument (resulting in a restitutive reading), or a full verbal domain that describes an event resulting in a state (resulting in a repetitive reading). In Stechow’s representation above, the complement would be either a small clause or (at least) a VP.

I apply the *again* diagnostic to the sentence in (142):

- (142) A mí se me volvió a parar el auto.
 to me REFL CL.DAT returned to breaking.down the car
 ‘The car broke down on me again.’

The affectedness scenario and the unintentional causer scenario are given in (143):

- (143) a. Affectedness scenario:
 On my way to work, my car breaks down due to an unforeseeable technical problem.
 b. Unintentional causer scenario:
 I have been told by my car mechanic that my car is sensitive and that I need to shift gears carefully. On my way to work, I forget about their advice and shift gears quickly. The car breaks down.

Combining those two scenarios with the repetitive and restitutive reading of *again*, we arrive at the Latin square design in (144):

- (144) a. Affectedness scenario, repetitive reading:
 On my way to work, my car breaks down due to unforeseeable technical problems. Later that day, I have it fixed. The next day, it breaks again.
 b. Unintentional causer scenario, repetitive reading:
 I have been told by my car mechanic that my car is sensitive and that I need to shift gears carefully. On my way to work, I forget about the mechanic’s advice and shift gears quickly. The car breaks down. Later that day, I have it fixed. The next day, it breaks down again.
 c. Affectedness scenario, restitutive reading:
 While my friend is on her way to work, her car breaks down due to unforeseeable technical problems. Later that day, she has it repaired. The next day, I borrow her car, and it breaks down again.
 d. Unintentional causer scenario, restitutive reading:
 My friend has been told by her car mechanic that her car is sensitive and that she needs to shift gears carefully. On her way to work, she forgets about the mechanic’s advice and shifts gears quickly. The car breaks down. The next day, I borrow her car. She reminds me to shift gears carefully, but I forget and shift gears quickly. The car breaks down again.

All four readings are available. As in German, the *again* diagnostic thus fails to distinguish between the affectedness reading and the unintentional causer reading of the dative argument. While absence of evidence is not evidence of absence, there are no syntactic reasons for positing a structural difference between the two interpretations.

4.4.2.3 Thematic interpretation

We now turn to the θ -role assigned to the dative argument under the unintentional causer reading. I aim to defend two claims in the following: first, the θ -role assigned to unintentional dative causers is not an agent, and second, it is the same affectee θ -role that is assigned to affected datives with inchoatives, as well as to other high applicative arguments.

To begin with the first claim, we have seen that unintentional dative causers never possess the full set of agent entailments but lack volition or purpose: they cannot be interpreted as intentionally causing the event and are incompatible with agent-oriented adverbs, instrument phrases and modifiers. In all these respects, they differ from the external arguments of direct causatives, which I assume to be true agents. While it is true that situations of unintentional causation can often be described both by a direct causative and by an inchoative with a dative argument, a difference in interpretation remains. For instance, both questions in (145) would be felicitous in a situation in which the parents come home to find a broken vase (Cuervo, 2003). However, the two variants highlight different aspects of the situation: using the dative variant signals that the vase is assumed to have been broken unintentionally, that the action that ultimately gave rise to the breaking might have been quite minor – e.g., being slightly clumsy in handling the vase – and that the effect could not easily have been foreseen. The same effect obtains in German, seen in (146).

- (145) a. ¿Quién rompió el florero?
 who broke the vase
 ‘Who broke the vase?’
 b. ¿A quién se le rompió el florero?
 to who REFL CL.DAT broke the vase
 ‘Who broke the vase accidentally?’ (Cuervo, 2003:189)
- (146) a. Wer hat die Vase zerbrochen?
 who has the vase broken
 ‘Who broke the vase?’
 b. Wem ist die Vase zerbrochen?
 who.DAT is the vase broken
 ‘Who broke the vase accidentally?’

In short, the meaning of the two sentences in (145), as well as in (146), is simply not the same, and I argue that the difference between the two stems from the θ -role assigned to the breaker argument in the two variants.

One piece of evidence that might be taken to undermine this claim is the fact that the unintentional causer reading is not available for dative arguments with causatives (147a) or passives (147b) (Kallulli, 2008), shown here for German:

- (147) a. Du hast mir die Vase zerbrochen.
 you have me.DAT the vase broken
 ‘You broke the vase, and I was affected by this.’
 Not: ‘I accidentally broke the vase.’
- b. Die Vase ist mir zerbrochen worden.
 the vase is me.DAT broken been
 ‘The vase was broken by someone, and I was affected by this.’
 Not: ‘I accidentally broke the vase.’

In sentences in which an agent argument is either explicitly (147a) or implicitly (147b) represented in the denotation, the dative argument cannot be interpreted as causally responsible. If unintentional causer datives were equally assigned an agent θ -role, this effect would straightforwardly fall out from thematic uniqueness (Carlson, 1984, 1998; Williams, 2015): under the readings blocked in (147a) and (147b), the agent θ -role would need to be assigned to more than one argument in a single event, causing the derivation to crash. If, as I have argued, the θ -roles are distinct, then an additional constraint needs to be posited that blocks two arguments that are part of the same event and are both interpreted with causal power. Assuming that thematic uniqueness is an independently available constraint, such an analysis might seem disadvantageous from the point of view of economy.

However, the same blocking effect obtains for possessor readings. Recall that dative arguments with inchoatives can also be interpreted as possessing the object undergoing a change of state. There is a consensus in the literature that this possessor reading is not truth-conditionally distinct from the affectedness reading but rather a contextually salient inference: one way in which someone can plausibly be affected by, say, the breaking of a vase is if they own the vase. Therefore, I have not glossed this reading separately so far, again in keeping with the literature. But it is well-known that a possessor reading becomes unavailable if an overt possessor is present (148) (Kallulli, 2008, a.o.):

- (148) a. Mir ist die Vase zerbrochen.
 me.DAT is the vase broken
 ‘The vase broke on me.’
 ‘I unintentionally broke the vase.’
 ‘My vase broke.’
- b. Mir ist deine Vase zerbrochen.
 me.DAT is your vase broken
 ‘Your vase broke on me.’
 ‘I unintentionally broke your vase.’
 Not: ‘My vase broke.’

In the same way that the unintentional causer reading is thus blocked by explicit or implicit external arguments, the possessor reading is blocked by overt possessors. But if there is no distinct possessor θ -role for the dative argument in (148a), then thematic uniqueness cannot explain this effect. As a result, some sort of mechanism is needed anyways to block the possessor reading for (148) based on broader contextual inference. This mechanism can then also rule out the unintentional causer reading with causatives and passives in (147) without the need for a single θ -role assigned to dative unintentional causers and external arguments.

To give an interim summary, unintentional causer datives have a very different semantic profile from external arguments, a fact that is widely acknowledged in the literature (for one exception, Rivero (2004), see Chapter 4.4.3.1). While it is true that the two cannot co-occur, I have argued that this is not evidence

that they are assigned the same θ -role. I conclude that the θ -role of unintentional causer datives is not an agent.

The second claim I make here – that they are assigned the same θ -role as affected datives – is more controversial. In previous work, Cuervo (2003), Kallulli (2008) and Schäfer (2008) (all discussed below in Chapter 4.4.3) have argued that the difference between the unintentional causer reading and the affectedness reading is one of true ambiguity; or in other words, that the two readings are represented differently in the denotation of the sentence. Under this view, unintentional causer datives and affected datives would be assigned different θ -roles. Ultimately, this is difficult to falsify. What is more, the agent-affectee alternation would remain a case of variable linking either way. However, I will make the case that there are clear continuities between the unintentional causer reading and the affectedness reading, that they prove impossible to tease apart, and that their relation is better described in terms of vagueness, with the choice between them being governed by contextual inference.

To begin with, there is no hard and fast line that could be drawn between the two readings. Example (150) lists a number of scenarios under which the sentence in (149) would be true. I challenge the reader to identify the scenarios in which I (unintentionally) cause the vase to break.

- (149) Mir ist die Vase zerbrochen.
 me.DAT is the vase broken
 ‘The vase broke on me.’
 ‘I (accidentally) broke the vase.’

- (150) a. For fun, I juggle with the vase because I am confident that I can catch it. It turns out that I can’t.
 b. I put the vase on the shelf, but because I am in a hurry I handle it carelessly and the vase drops.
 c. I put the vase on the shelf. I am not in a hurry and I do not handle the vase particularly carelessly, but it drops nonetheless.
 d. My friend hands me the vase. I take it carefully but unbeknownst to me, the vase is wet and heavy and slips through my fingers.
 e. I am walking through my apartment holding the vase when suddenly, my puppy runs into my path. I stumble and drop the vase.
 f. I put the vase on the windowsill, ignorant of the weather forecast. A strong wind comes up and knocks the vase down.

At the same time, in all scenarios it must be true that I am affected by the breaking of the vase: be it that the vase is mine, that I am responsible for its safekeeping, or even simply that its breaking puts me in an awkward position. Thus, the unintentional causer reading does not seem to be dissociable from an affectedness component. This is evidenced most clearly by the fact that inchoatives with datives are usually impossible for inanimate arguments (151) (Cuervo, 2003; Kallulli, 2008; Schäfer, 2008):

- (151) a. *Der Baum ist der Sonne/Erde verrottet.
 the.NOM tree is the.DAT sun/soil rotten
 Intended: ‘The sun/soil caused the tree to rot.’ (Bosse, 2017:41)
- b. #Der Waschmaschine ist meine Hose eingelaufen.
 the.DAT washing.machine is my trousers shrunk
 Intended: ‘The washing machine caused my trousers to shrink.’

In (151), the sun and the soil are not affected by the harm they cause to the tree and are thus not licensed as dative arguments even under an unintentional causer reading. Similarly, (151b) is only marginally available under an anthropomorphic reading under which the washing machine, perhaps a long-time companion in the household, feels guilty for inadvertently shrinking its owner’s trousers. All of this indicates that the unintentional causer reading cannot come without an affectedness component.

Interestingly, counter-examples to the animacy requirement are attested (152):

- (152) a. Dem Stuhl brachen zwei Beine ab.
 the.DAT chair broke two legs off
 ‘Two legs of the chair broke.’
 Not: ‘The chair accidentally broke two legs.’ (Schäfer, 2008:87)
- b. A la mesa se le rompieron dos patas.
 to the table REFL CL.DAT broke two legs
 ‘Two legs of the table broke.’
 Not: ‘The table accidentally broke two legs.’ (Cuervo, 2003:188)

Cuervo (2003) and Schäfer (2008) argue that in (152), the inanimate arguments (the chair and the table) are interpreted as affected and thus can be used in the dative construction. Nonetheless – so the argument goes –, an unintentional causer reading is unavailable, which is thus evidence that there must be an animacy requirement holding on unintentional causers. Examples such as (151) would then not be evidence that the datives always must receive an affectedness reading, as I have argued just now, but could be attributed to this animacy requirement.

I do not see that this line of argument goes through. It is true that in (152), the chair and the table are affected by the breaking in a literal sense (not, arguably, emotionally). However, there is no reason to be surprised that an unintentional causer reading is not available in these examples since it is not clear how chairs and tables, not being prone to playing a very active role in the world, could causally break their own legs. One might argue that, perhaps, the weight of the chair or the table might cause their legs to collapse. This is true – causality is a broad concept, and with sufficient imagination, causal relations can be drawn between almost anything. However, not all of those causal relations can be expressed in language in the same way. Importantly, the causal meaning that is absent in (152) cannot be expressed with a direct causative either (153). Of course, no animacy requirement obtains for direct causatives.

- (153) a. #La mesa rompió dos patas.
 the table broke two legs
 Intended: ‘The table broke two legs.’
- b. #Der Stuhl brach zwei Beine ab.
 the chair broke two legs
 Intended: ‘The chair broke two legs.’

I propose that the absence of an unintentional causer reading in (152) is due to the same constraint that is also behind the oddness of (153). Specifically, I suspect that the causal relation that would need to obtain is too indirect to be expressed in either of the two constructions.

To recap, then, datives with inchoatives must always be interpreted as affected even under an unintentional causer reading, which rules out distant inanimate causers such as the sun, the soil and emotionally unavailable washing machines in this position. The fact that some inanimate arguments like chairs and tables, which can surface as datives in inchoatives under an affectedness reading, nonetheless do not allow an unintentional causer reading is not evidence for an animacy requirement on the latter interpretation, but is due to the fact that a causal reading is contextually difficult to construe and cannot be expressed with a direct causative either. All of this is difficult to reconcile with the idea that the relation between the unintentional causer reading and the affectedness reading is one of true ambiguity, rather than mere vagueness.

Again, it is hard to disprove the idea that the unintentional causer reading and the affectedness reading involve distinct θ -roles. I also wish to reiterate that θ -roles can come at different levels of granularity. Nothing necessarily rules out that there is an unintentional causer role and an affectee role that are semantically similar but not identical, and that some but not other aspects of the grammar distinguish between them. But I argue that there is no single, discrete θ -role that picks out the same individual role, such as a breaker, in both the causative and the inchoative-cum-dative alternant, and further, that the unintentional causer interpretation of the dative argument is not random, but is in line with the semantic properties of other dative arguments with the same syntactic profile. I conclude that the agent-affectee alternation constitutes a case of variable linking, where the same individual role is mapped onto an agent θ -role in the transitive but on an affectee θ -role in the inchoative-cum-dative.

4.4.3 Competing analyses

In the final part of this case study, I review four competing approaches to the agent-affectee alternation: Kallulli (2008), Rivero (2004), Cuervo (2003) and Schäfer (2008). Of all of them, only Rivero (2004) explicitly argues that dative unintentional causers are agents. In contrast, Kallulli (2008) and Cuervo (2003) are more in line with the approaches by Ramchand (2008) and Krejci (2012) discussed in the previous two case studies. Both Kallulli and Cuervo pursue the idea that dative unintentional causers and external arguments of direct causatives are not semantically identical but overlap, sharing a semantic property – causal force – that is explicitly represented in the denotation of the sentence. Unlike Ramchand (2008) and Krejci (2012), neither of them represents this shared semantic property as a role, but instead as a feature (Kallulli, 2008) and in terms of event structure (Cuervo, 2003), respectively.

The final piece of previous work I discuss here, Schäfer (2008), largely adopts Cuervo's analysis of dative unintentional causers but contributes an additional data point from German to the debate. This data point is unexpected under any analysis I am aware of. While I will argue against Schäfer's account of it, I will not have a better answer myself. However, I will make the case that the general approach to the agent-affectee

alternation defended here is just as well-equipped to solve this puzzle as any other approach.

4.4.3.1 Pragmatic inference: Rivero (2004)

I first review an analysis of datives with inchoatives developed by Rivero (2004), which is in a sense very close to the approach I advocate here, but with some important differences. Rivero compares two uses of the dative in Balkan languages: with inchoatives, as discussed in this case study so far, and with passives. An example of the latter construction from Albanian is given in (154). The sentence can receive the reading that the dative-marked argument is inclined or would like to perform the activity described by the verb:

- (154) Anës i-u punua (këtu).
 Ann.DAT 3SG.DAT-NACT.AOR work.3SG here
 Preferred interpretation: ‘Ann felt like working (here).’ (Rivero, 2004:3)

The parallel between datives with inchoatives and datives with passives that Rivero is interested in is as follows: in the same way that in inchoatives, datives can be interpreted as unintentional causers, in passives, they can be interpreted as agents, such as the agent of working in (154). These are both interpretations that are more commonly associated with external arguments. Rivero argues that in passives and inchoatives, dative arguments are non-selected and thus not assigned an interpretation by the verb itself but through other means. However, the mechanism through which the dative argument receives an interpretation differs between the two constructions. Rivero labels the two operations D-Disclosure (Dative Existential Disclosure, for passives) and E-strategy (Ethical Strategy, for inchoatives), respectively.

Rivero assumes that passives contain an implicit, existentially closed agent argument. D-disclosure operates on this argument, eliminates the existential quantifier and binds the now free variable to the dative argument, resulting in obligatory control. For instance, in (154), passive ‘work’ contains an implicit agent argument that is bound to ‘Anës,’ who thus comes to be interpreted as the agent of the working event. Since our focus here is on datives with inchoatives, I leave the technical details of D-Disclosure aside, and I will not attempt to evaluate this part of the analysis in the following.

Unlike passives, Rivero argues that inchoatives do not contain an implicit external argument. Thus, the interpretative mechanism available for the dative must be a different one: the E-strategy. This mechanism allows dative arguments to be associated with whichever semantic role is contextually felicitous. Whereas D-Disclosure is a formal process taking place in compositional semantics, the E-strategy is an extragrammatical, purely pragmatic inferential mechanism that adjusts the interpretations available to a dative argument. Whereas D-Disclosure results in a clear-cut, fixed interpretation, the E-strategy gives rise to multiple vague readings that can be hard to dissociate – unintentional causation, affectedness, possession, etc. Whereas D-Disclosure has to operate on arguments that are formally (although implicitly) represented in the derivation, the E-strategy cannot: as discussed earlier, datives cannot be interpreted as unintentional causers if the denotation already contains an explicit or implicit external argument, nor as possessors in the presence of

an overt possessor.

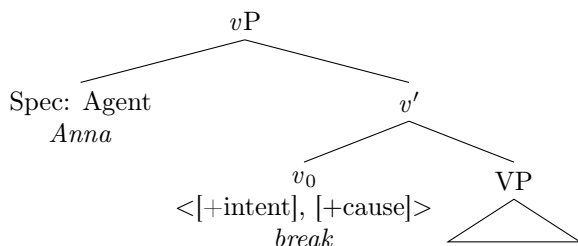
I believe that Rivero's overall intuition here is correct. What determines the interpretation of dative arguments with inchoatives is not a formal transfer of θ -roles, as we seem to see it for datives with passives or in other cases of obligatory control, but a mechanism that is complex, inferential in nature, contextually determined and results in vagueness. I disagree with Rivero on two points. First, under her analysis, there is no θ -role, however broad, or other kind of semantic property that would be shared by all dative arguments *qua* dative arguments, or even a particular kind of dative arguments such as high applicatives. As a result, nothing constrains the readings that can be associated with them. While the range of possible interpretations is indeed wide, it is not unrestricted: for instance, we never find narrow, canonical agents – sentient and volitional participants with causal power – realized as datives with inchoatives. In other words, the analysis fails to account for the shared set of proto-properties of the dative arguments in question. The second and related point is, of course, that even under an unintentional causer reading, the dative argument is not an agent. If it were, it should receive exactly the same interpretation as the external argument in the standard direct causative variant, but in fact, the two are not semantically identical. To summarize, it is true that the interpretation of datives is resolved through an inferential process, but I have argued here that this process does not operate on θ -roles.

4.4.3.2 A feature-based analysis: Kallulli (2008)

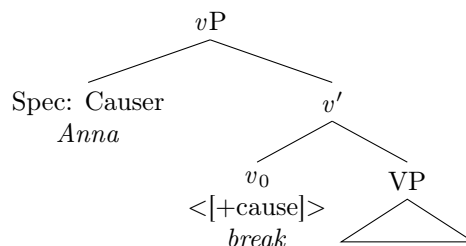
Next, I move on to Kallulli (2008), who proposes a feature-based analysis according to which unintentional causer datives are represented with a [+cause] and a [+affect] but no [+intent] feature. Like Rivero, Kallulli contrasts the inchoative-cum-dative construction with another phenomenon widely attested in Balkan languages, involuntary states; in the following, I leave this comparison aside.

Kallulli's analysis relies on two basic ideas. First, causatives fall into two classes depending on the features on the head that introduces the external argument (for Kallulli, *v*). Agentive causatives come with both a [+intent] and a [+cause] feature, whereas non-agentive causatives only have [+cause].¹⁸ The two structures are given in (155) and (156), respectively.

(155) *Agentive causative*



(156) *Non-agentive causative*

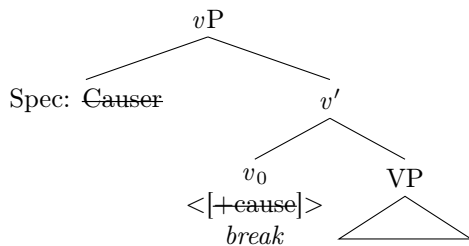


(Kallulli, 2008:286f.)

¹⁸As a minor technical point, Kallulli represents [+intent] and [+cause] as binary features. However, the fact that non-intentionality and non-causation are represented not by [−intent] and [−cause], respectively, but by the absence of these features suggests that they are actually to be understood as privative features.

The second assumption is that the function of non-active morphology is to suppress the first feature in a predicate structure. For instance, anticausatives are derived from non-agentive causatives by deleting the only feature on v , [+cause] (157):

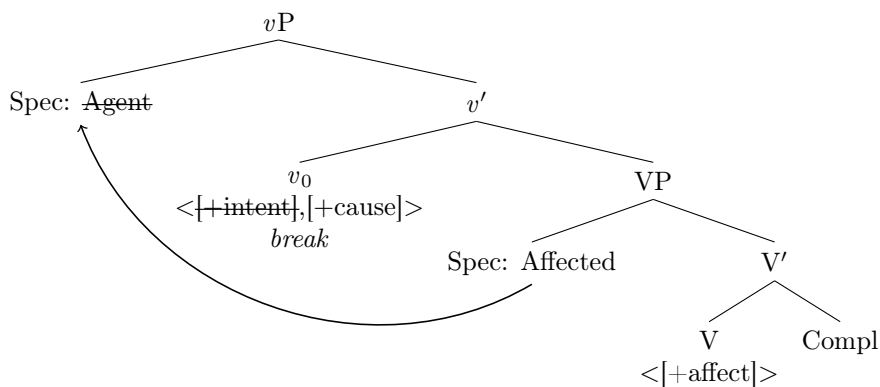
(157) *Anticausative*



(Kallulli, 2008:292)

With these two assumptions in place, dative unintentional causers are then derived as in (158). The head introducing an external argument is merged with both [+intent] and [+cause] features. Unaccusative morphology deletes the first member of this set, viz., [+intent]. Kallulli argues that since the original tuple of features on v is now destroyed, the original Spec v P argument is no longer licensed and is deleted from the derivation. However, the [+cause] feature on v must still be licensed. Therefore, the dative argument – which Kallulli represents here as being merged in SpecVP, where it receives a [+affect] feature – raises to the Spec v P position and there is associated with the [+cause] feature.

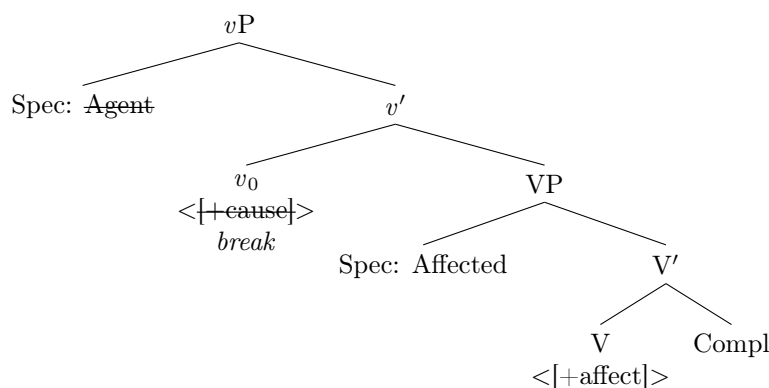
(158) *Unintentional causer dative*



(Kallulli, 2008:289)

Accordingly, the argument is interpreted as [+affect] and [+cause], whereas standard agentive external arguments are [+cause] and [+intent]. Similar to Ramchand (2008) and Krejci (2012), Kallulli thus argues that the two arguments overlap in their semantic interpretation, but are not identical.

The affectedness reading of the dative, on the other hand, is derived by letting v be merged with [+cause] only. As a result, unaccusative morphology now deletes this [+cause] feature, which is the first – and only – member of the set. As under the unintentional causer reading, the argument in Spec v P disappears, but nothing triggers movement of the dative argument to this position since there are no additional features to license. As a result, the argument is only interpreted as [+affect].

(159) *Affectedness dative*

There are two main problems with Kallulli's analysis, both discussed more extensively by Schäfer (2008). The first problem concerns some broader architectural assumptions that the analysis rests on. Kallulli argues that non-active morphology itself *changes* the denotation of the sentence from active to non-active, in contrast to the more common view that non-active morphology *spells out* non-active semantics. This assumption is crucial since under her analysis, non-active morphology does not correspond to a consistent feature set that could be targeted by a spell-out rule, but to a particular kind of reduction in the feature set. It is not clear how concretely morphology and semantics could interact in this non-standard way. On a related note, contemporary syntactic theory does not usually assume that arguments, once they have been merged, can then vanish from the derivation, with other arguments moving to take their place. In short, the analysis relies on quite unorthodox theoretical assumptions that do not seem warranted on other grounds.

The second problem is the somewhat arbitrary nature of the analysis. For it to deliver the right results, it must be the case that the $[+intent]$ feature is ordered before the $[+cause]$ feature, which ensures that when both features are present, it is $[+intent]$ that is deleted. However, nothing motivates this specific ordering. Moreover, there is no particular reason why after the deletion of the first feature on *v* – e.g., $[+intent] -$, the argument in its specifier can no longer appear. Both of these assumptions have to be stipulated solely on the basis of the data to be explained.

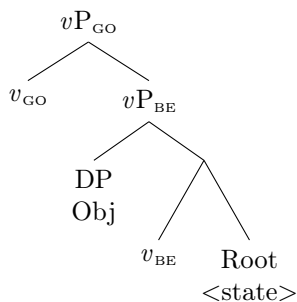
In brief, Kallulli's analysis attempts to derive the agent-affectee alternation using three features, $[+cause]$, $[+intent]$ and $[+affect]$. The rules according to which these features can be shuffled around the derivation are architecturally problematic and rest on rather ad-hoc assumptions. More in general, I have argued above against the idea that the distinction between the affectedness reading and the unintentional causer reading is a genuine ambiguity, which is what the feature system is designed to derive in the first place.

4.4.3.3 Affected vs. high applicatives: Cuervo (2003)

In this section, I discuss previous work by Cuervo (2003), who posits that the difference between the affectedness reading and the unintentional causer reading of datives with inchoatives corresponds to two different underlying structures. Focusing on Spanish, Cuervo takes from Pytkänen (2008) the idea that all dative

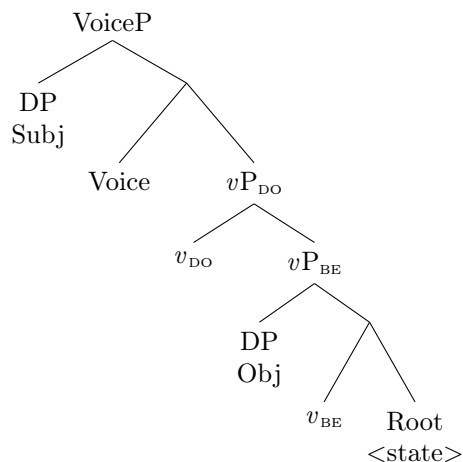
arguments are introduced by an applicative head but differ in where this head is merged in the tree. Besides Pylkkänen's low and high applicatives, reviewed above in Chapter 4.4.2.1, she introduces a third type of applicative arguments which she labels affected applicatives. Her analysis relies on a more fine-grained structure of the verbal domain than assumed in Pylkkänen's work, in that she argues that both inchoatives and direct causatives are formed by combining an eventive and a stative sub-eventuality, with the event leading to or resulting in the state. Eventuality types are encoded as flavors of v : both inchoatives and causatives have a stative v_{BE} , which in inchoatives is embedded by a v_{GO} and in causatives by a v_{DO} which is then topped by a VoiceP (160)–(161):¹⁹

(160) *Inchoative*



(Cuervo, 2003:27)

(161) *Causative*



(Cuervo, 2003:26)

The new type of applicative, affected applicatives, is derived by merging ApplP between the eventive and the stative vP layer in both inchoatives and causatives (162)–(163):

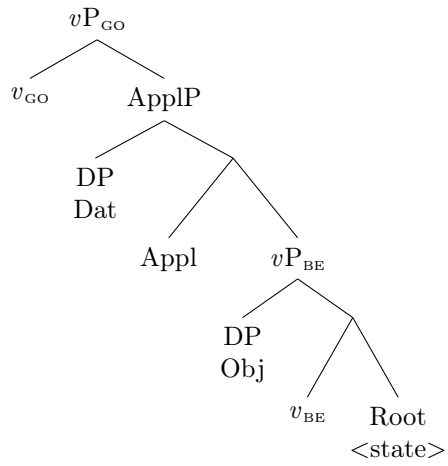
¹⁹Having different eventive v heads for inchoatives and causatives is meant to account for contrasts such as the one between (i) and (ii) (Cuervo, 2003:134).

- (i) a. I broke the glass
b. I broke my promise.

- (ii) a. The glass broke.
b. *My promise broke.

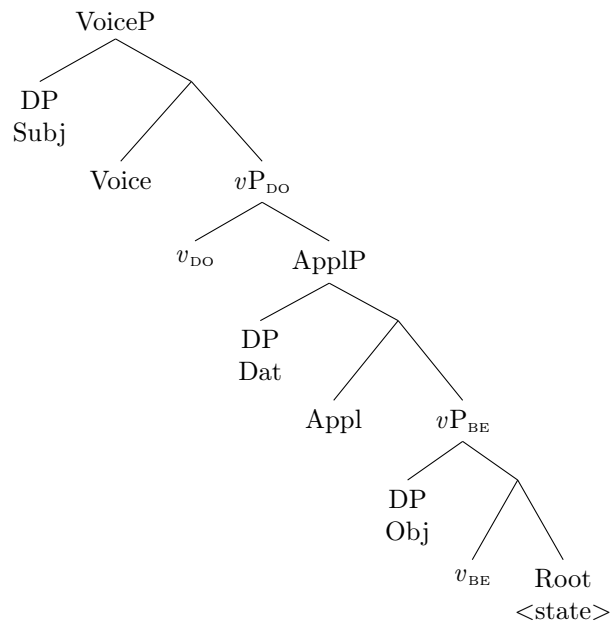
The figurative reading of *break* in the context of *promise* is only available in the causative. Cuervo argues that if the causative and the inchoative contained the same v heads, the absence of the figurative reading in the inchoative would be unexpected. I leave this matter aside.

(162) *Inchoative with affected applicative*



(Cuervo, 2003:141)

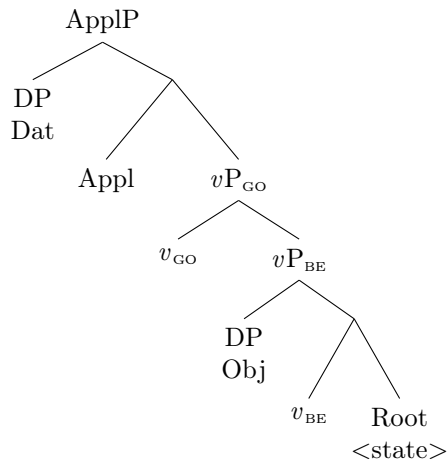
(163) *Causative with affected applicative*



(Cuervo, 2003:130)

This contrasts with high applicatives, in which $ApplP$ is merged above the higher vP , shown for inchoatives in (164):

(164) *Inchoative with high applicative*



(Cuervo, 2003:188)

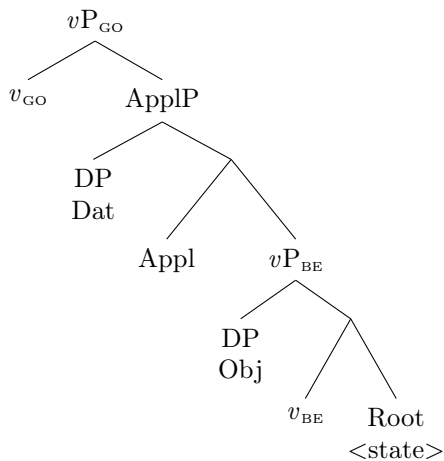
The structural difference between affected and high applicatives crucially affects how they are integrated in the event structure. Affected applicatives are part of the lower, stative sub-eventuality, whereas high applicatives are part of the higher, eventive sub-eventuality. Cuervo leverages this difference to account for what she argues is a true ambiguity in inchoatives with dative arguments. The affectedness reading corresponds to the affected applicative structure where the argument is integrated into the state. The unintentional causer reading, on the other hand, is derived via the high applicative structure where the argument is part of the higher, causing event and thus can be interpreted with causal power. Hence, what unintentional dative causers and external arguments of direct causatives have in common, and what separates them from affected datives, is that they are both part of the higher, eventive sub-eventuality.

Before discussing what I believe are shortcomings of this analysis, it ought to be noted that the addition of a third type of applicative to Pykkänen's original typology is motivated in part by considerations that are unrelated to the data that are relevant for our purposes. In the following, I only focus on the role played by the high/affected distinction in the analysis of inchoatives with datives. In brief, I argue that the high/affected distinction does not do any work in deriving the different interpretations of the dative argument and lacks independent support.

To begin with, there is no direct syntactic evidence for a structural ambiguity in inchoatives with datives. The purpose of positing a syntactic distinction is, in the context of the data we are concerned with, exclusively to account for the semantic contrast between the affectedness reading and the unintentional causer reading. As one piece of evidence that might be expected to weigh in on this question, I have discussed earlier in Chapter 4.4.2.2 that *again* modification does not detect any difference between the affectedness reading and the unintentional causer reading. Under both interpretations, it is possible for the repeated event to either include or not include the dative argument, indicating that *again* can attach either above or below the argument-introducing head. Unfortunately, *again* does not appear to provide clear evidence for or against a

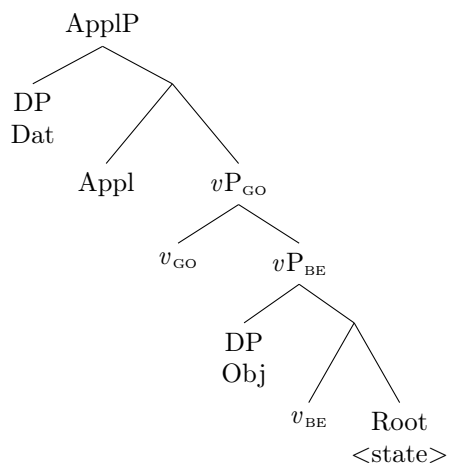
distinction between affected and high applicatives. For convenience, the two structures are repeated below in (165) and (166).

(165) *Inchoative with affected applicative*



(Cuervo, 2003:141)

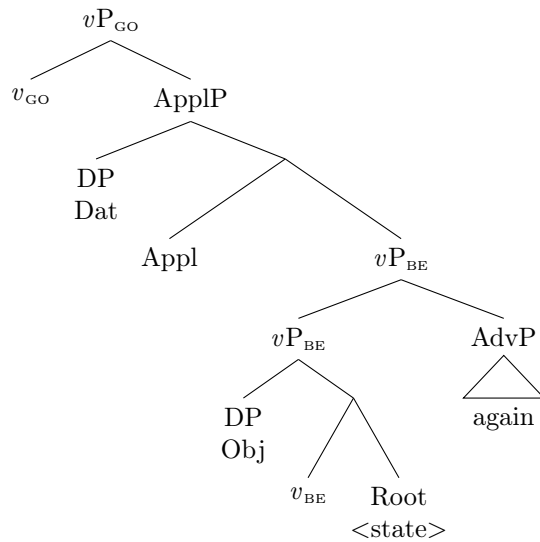
(166) *Inchoative with high applicative*



(Cuervo, 2003:188)

In the high applicative structure in (164), *again* should be able to modify either the eventuality encoded by the entire verbal domain, containing the dative argument, or the state only, excluding the dative argument. This correctly predicts the availability of two different readings with *again* under the unintentional causer reading. In the affected applicative structure in (162), a reading where the repeated eventuality includes the dative argument is trivially available. Since the argument is part of the lower, stative sub-eventuality, the question is whether low attachment of *again* could nonetheless generate the alternative reading under which the argument is not part of the repeated eventuality. I believe that the relevant reading could still be derived by merging *again* with vP_{BE} before the applicative argument is added to the state.

(167)



If this is correct, *again* modification does not provide evidence for the absence of a structural distinction between high and affected applicatives. Nonetheless, the absence of evidence for this distinction is concerning.

Since the purpose of the high/affected distinction is thus only to account for the contrast between the unintentional causer reading and the affectedness reading, we need to ask whether it at least successfully derives this interpretative difference. It is not obvious that it does. As for the affectedness reading, Cuervo argues that this interpretation arises by virtue of the argument being introduced between two eventualities, the causing event and the result state; or in other words, that this specific event-structural configuration naturally enforces an affectedness reading of the argument. The intuition that Cuervo appears to articulate is that the dative argument is both affected by the higher causing event and, as with Pylkkänen's original high applicatives surveyed above, stands in a 'have' relationship to the result state. For instance, in our go-to Spanish example (168) under the affectedness reading, the dry-cleaner 'has' Carolina's burnt trousers.

- (168) Al tintorero se le quemaron los pantalones de Carolina.
 to.the dry-cleaner REFL CL.DAT burnt.PL the trousers of Carolina
 'Carolina's trousers got burnt on the dry-cleaner.' (Cuervo, 2003:186)

I admit that I find myself unable to grasp this intuition. It is unclear to me why the position of the dative argument in the affected applicative structure, where it is sandwiched between two sub-eventualities, should by itself generate an affectedness interpretation.

What is more, one of the interpretations that Cuervo argues are available with high applicatives – besides the unintentional causer reading – is a benefactive, malefactive or ethical reading, as in (169) below. Here, the dative argument of interest is the first person singular, who is negatively affected by her son's life being ruined.

- (169) Me le arruinaron la vida a mi hijo.
 CL.1SG.DAT CL.3SG.DAT ruined the life to my son.DAT
 ‘They ruined my son’s life on me.’ (Cuervo, 2003:198)

Nonetheless, Cuervo analyzes (169) as an instance of a high applicative, not an affected applicative. It is hard to see what distinguishes affected datives from benefactive, malefactive and ethical datives, and why they should require a different event structure.

Second, turning to the unintentional causer reading, Cuervo argues that it arises because the argument is part of the higher, causing event. Like Kallulli (2008), Cuervo thus assumes that unintentional dative causers and external arguments of causatives share a semantic property – i.e., causal power – explicitly represented in the denotation of the sentence. Unlike under Kallulli’s feature-based analysis, this is encoded by virtue of where the argument is integrated in the event structure. However, note that for Cuervo, unintentional causation is only one interpretation available for high applicatives. Benefactive/malefactive/ethical datives – such as the first person dative in (169) – are also part of the higher causing event but are not, of course, interpreted with causal force. Being part of the higher event does not ‘mean’ having causal power. As with the affectedness reading, the event structure for the unintentional causer reading does not itself give rise to the observed interpretation. But if some other mechanism needs to be posited anyway to derive the interpretations in question – regardless of where in the grammar this mechanism is located –, then it is not clear what is still achieved by the syntactic difference that Cuervo posits.

In sum, there is no independent evidence for a structural difference between the affectedness reading and the unintentional causer reading. Moreover, while the attempt to represent the shared causal component of external arguments and unintentional dative causers in terms of event structure is undeniably more elegant than Kallulli’s somewhat heavy-handed feature-based approach, it is hard to see how the difference in event structure between affected and high applicatives would actually derive the interpretative difference in question.

4.4.3.4 A remaining puzzle from German: Schäfer (2008)

Lastly, I address a remaining puzzle from German, and the solution offered by Schäfer (2008). German has two types of anticausatives, marked and unmarked; the former surface with a reflexive pronoun. The two types are exemplified in (170a) and (170b) for ‘break’ and ‘open,’ respectively. Note also that marked anticausatives, like all reflexives, take the auxiliary ‘have,’ as opposed to the standard unaccusative ‘be.’

- | | | | | |
|-------|----|---|----|--|
| (170) | a. | Die Vase ist zerbrochen.
the vase is broken
‘The vase broke.’ | b. | Die Tür hat sich geöffnet.
the door has REFL opened
‘The door opened.’ |
|-------|----|---|----|--|

Crucially, while unmarked anticausatives allow for both an affectedness reading and an unintentional causer reading as discussed so far (171a), marked anticausatives can only receive an affectedness reading (171b) (Schäfer, 2009).

- (171) a. Mir ist die Vase zerbrochen.
 me.DAT is the vase broken
 ‘The vase broke on me.’
 ‘I (accidentally) broke the vase.’
- b. Mir hat sich die Tür geöffnet.
 me.DAT has REFL the door opened
 ‘The door opened, and I was affected by this.’
 Not: ‘I (accidentally) opened the door.’

A natural hypothesis would be that marked anticausatives have an implicit agent argument that is interpreted as causing the event, such that this interpretation is no longer available for the dative. As seen earlier, this effect holds for passives, which cannot receive an unintentional causer reading (172):

- (172) Die Vase ist mir zerbrochen worden.
 the vase is me.DAT broken been
 ‘The vase was broken by someone, and I was affected by this.’
 Not: ‘I accidentally broke the vase.’

However, Schäfer shows conclusively that marked anticausatives do not have an implicit agent. For instance, they are not compatible with *by*-phrases, agent-oriented adverbs and purpose clauses. Moreover, note that the presence of an agent is also not entailed with marked anticausatives (173):

- (173) Die Tür öffnete sich, aber niemand hat sie geöffnet.
 the door opened REFL but nobody has it opened
 ‘The door opened, but nobody opened it.’

Hence, the unavailability of the unintentional causer reading remains a puzzle, especially since in other languages with reflexively marked anticausatives, the same effect does not obtain. Schäfer takes these data as further evidence for the claim that unintentional causer datives differ structurally from affected datives, and he follows Cuervo (2003) in her analysis of both. Furthermore, he proposes an analysis of marked German anticausatives that accounts for their incompatibility with the unintentional causer reading, which I sketch out in the following.

In brief, Schäfer analyzes marked German anticausatives as containing an expletive Voice head. Rather than assuming that expletive Voice merges directly, he proposes that the Voice head is expletivized over the course of the derivation. Specifically, in marked anticausatives, Voice is originally merged with a categorial D-feature and unvalued ϕ -features. This Voice head thus requires a specifier in the same way as in regular transitives and unergatives. But what is special about marked anticausatives under Schäfer’s analysis is that this specifier position is occupied by a variable with unvalued ϕ -features.

Schäfer argues that the Voice head, having unvalued ϕ -features, probes for a DP to agree with in its m-command domain. It first finds the variable in its specifier position. Since the latter has itself unvalued ϕ -features, no valuation can take place, but a connection between Voice and SpecVoiceP is established. Next, Voice probes again and finds the internal argument. This DP values the features on Voice as well as on the

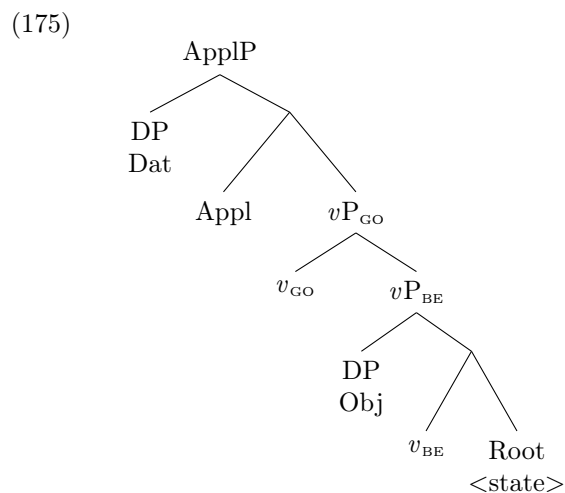
variable in SpecVoiceP, due to the connection that resulted from the unsuccessful attempt of Voice to agree with this variable. This is how Schäfer derives the fact that both the verb and the variable – realized as a reflexive – agree with the internal argument; in (174), for third person singular:

- (174) Die Tür hat sich geöffnet.
 the door has REFL opened
 ‘The door opened.’

So far for the morphosyntax. As for the semantics, Schäfer assumes that θ -role assignment takes place at LF. That is, it is not the case that a Voice head is merged already bearing an agent θ -role; rather, an agent θ -role is what arises at LF from a particular syntactic configuration involving a Voice head provided certain constraints are met. In particular, Schäfer takes θ -role assignment to be blocked for DPs that have no reference. Since in marked anticausatives, the variable in SpecVoiceP has no c-commanding antecedent, it is uninterpretable and thus cannot be assigned a θ -role.

In sum, under Schäfer’s analysis of marked anticausatives, the SpecVoiceP position is occupied by a variable which does not receive a θ -role. This results in an anticausative reading despite the fact that technically, there is a DP argument in SpecVoiceP. Schäfer derives this constellation not by immediately merging an expletive Voice head, but via a somewhat complex interplay of agreement operations and conditions on θ -role assignment at LF, the details of which are not particularly crucial for our purposes.

Schäfer then aims to account for the unavailability of the unintentional causer reading as follows. Recall that according to Cuervo (2003), unintentional causer datives are high applicatives, where ApplP merges above vP_{GO} (175):

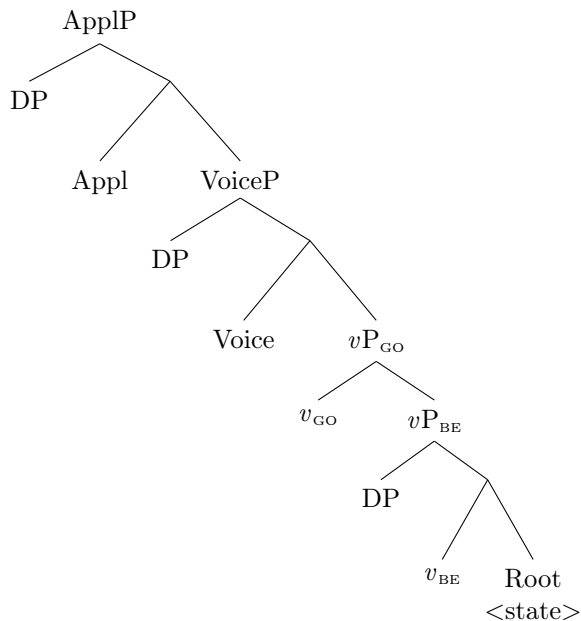


(Cuervo, 2003:188)

In marked anticausatives, Schäfer argues, the structure would thus be as in (176), with the variable that bears unvalued ϕ -features again being merged in SpecVoiceP:²⁰

²⁰Note that there are some differences in how Schäfer represents the vP compared to Cuervo, which are not relevant for our purposes. In (176), I retain Cuervo’s vP for simplicity’s sake.

(176)



The reason why this tree would be malformed, according to Schäfer, is that the DP in SpecApplP would bind the DP in SpecVoiceP. Since the latter is thus contentful, it would receive a θ -role and be interpreted as an agent, while the dative DP would be interpreted as an unintentional causer. As a result, the derivation would contain two arguments – albeit coindexed – that both receive an agent-/causer-type θ -role. This results in a conceptual clash: it is not possible for two entities to simultaneously be represented as causally responsible for an event.

The question raised by (176) is why a high ApplP cannot be merged between vP_{GO} and VoiceP instead, as also canonically assumed since Pykkänen (2008).²¹ In this case, no problems would arise for marked anticausatives. In response, Schäfer points out that VoiceP cannot combine with a stative event description but must always directly select an eventive verbal head. Since ApplP expresses a stative relation, it thus cannot interfere between VoiceP and vP_{GO} . Schäfer's discussion of this point is very brief and confined to a footnote (Schäfer, 2008:272). His claim is not, I believe, that Appl genuinely acts as a stativizer that existentially closes the causing event and introduces a new stative eventuality, which is evidently not the case. But if this is not what is intended, then it is thoroughly unclear in which sense the applicative relation is 'stative' and how or why it should interrupt the connection between the causing event and the external argument. Hence, the question remains why ApplP should have to dominate VoiceP, but if it does not, then nothing rules out the unintentional causer reading.

Thus, leaving aside the problems with Cuervo's original analysis, it does not provide a solution to the puzzle of marked anticausatives in German either. I have no solution to offer myself. But I take issue with Schäfer's claim that the unavailability of the unintentional causer reading is evidence for a syntactic distinction between affected and unintentional causer datives. There is wide agreement that the interpretation of datives with

²¹Cuervo (2003), to my knowledge, is not explicit about where VoiceP would be merged in the context of high applicatives.

inchoatives is subject to some degree of vagueness – for instance, whether or not the dative is interpreted as a possessor –, and we generally do not understand how and where in the grammar this vagueness is resolved. Whatever the difference between marked and unmarked German anticausatives is, there is no reason why it should be able to affect the choice between two truth-conditionally distinct denotation associated with different structures, but not between two variants of a single underlying denotation that is vague.

None of this bears on the central claim of this case study, which is that under a theory that posits individual roles, the agent-affectee alternation is an instance of variable linking. Regardless of which and how many different kinds of θ -roles can be assigned to the dative argument in the inchoative, and whether they correspond to different syntactic structures, the fact remains that none of them is an agent role. But in order to better understand how abstract θ -roles are given a specific interpretation in the context of an event description, Schäfer’s data remain an interesting case to consider that I hope future research will tackle.

4.5 Pseudo-symmetric predicates

4.5.1 Data and analysis

This final case study deals with a class of verbs that I will refer to as pseudo-symmetric predicates, which allow for both an intransitive use with a plural subject and a transitive use with a (potentially) singular subject and object. Examples from English and Dutch are given in (177) and (178), respectively. Other pseudo-symmetric predicates include verbs like *kiss*, *break up (with)*, *speak (to)* and *collide*.

- (177) a. Joel and Clementine hugged.
 b. Clementine hugged Joel.
 c. Joel hugged Clementine.

- (178) a. Fleur en Mark hebben geknuffeld.
 Fleur and Mark have hugged
 ‘Fleur and Mark hugged.’
 b. Fleur heeft Mark geknuffeld.
 Fleur has Mark hugged
 ‘Fleur hugged Mark.’
 c. Mark heeft Fleur geknuffeld.
 Mark has Fleur hugged
 ‘Mark hugged Fleur.’

(Kruitwagen, 2024:129–130)

The vast majority of the previous literature on pseudo-symmetric predicates has focused on English (e.g., Carlson, 1998; Dowty, 1991; Gleitman, 1965; van Craenenbroeck and Johnson, 2026; Wehbe, 2025; Winter, 2016, 2018); for broader but more cursory cross-linguistic surveys see Siloni (2008) and Dimitriadis (2008). The phenomenon itself appears to be highly pervasive, although the morphological realization of the alternants differs across languages. While in English, the intransitive verb is morphologically identical to the transitive, other languages employ a variety of marking strategies for the intransitive, such as suffixes (e.g.,

Russian, Hungarian), SE clitics (e.g., Romance, German) or morphological templates (e.g., Hebrew) (Siloni, 2008). In the following, I focus on English and Dutch; for the latter, I draw on recent work by Kruitwagen (2024).

A widely held view is that pseudo-symmetric predicates are always underlyingly transitive with an agent and a patient θ -role (e.g., Dimitriadis, 2008; Gleitman, 1965; Siloni, 2008; Wehbe, 2025). Intransitive pseudo-symmetric predicates would then be similar – perhaps semantically identical – to overt reciprocals, with both arguments being interpreted as both agent and patient (179):

(179) Joel and Clementine hugged ~~each other~~ $\leftrightarrow$ Joel hugged Clementine and Clementine hugged Joel.

The covert reciprocal analysis would give us a very simple picture of the individual roles and θ -roles involved in these alternations. Pseudo-symmetric predicates would come with two individual roles – e.g., a hugger and a huggee –, which are always mapped onto an agent θ -role and a patient θ -role, respectively. In the transitive, one argument is a hugger and an agent, the other a huggee and a patient. In the intransitive, each of the singular arguments making up the plural subject is both hugger and huggee, both agent and patient. No variable linking would obtain.

In the following, I argue against this picture. Instead, I follow the alternative view that has been proposed in the literature, which is that intransitive pseudo-symmetric predicates are true intransitives with a single θ -role assigned to the plural subject (Carlson, 1998; Dowty, 1991; Kruitwagen, 2024; Winter, 2016, 2018). Transitives, on the other hand, involve two different θ -roles. If verbs like *hug* indeed came with two individual roles as part of their lexical entry, a hugger and a huggee, these would be mapped onto different θ -roles in the transitive, but the same θ -role in the intransitive. As a result, pseudo-symmetric predicates would constitute a case of variable linking.

To anticipate the upshot of this analysis, pseudo-symmetric predicates allow us to see particularly clearly that which θ -role a participant in a given event receives depends on this participant's degree of proto-properties (Dowty, 1991), rather than some sort of discrete individual role. If the two participants involved in the event are sufficiently similar in terms of proto-properties, speakers can construe either of them as the agent or the patient, respectively, or also assign them the same θ -role. The more the two participants differ in their degree of proto-properties, the more speakers are biased towards one particular asymmetric construal. If the participants differ radically – for instance, if Clementine hugs Joel's dead body –, a construal in which Clementine is the agent and Joel the patient of the event is virtually enforced. All of this is at odds with the idea that verb-specific individual roles such as huggers and hugges are consistently mapped onto the same θ -role. Rather, speakers evaluate the felicity of a given thematic construal by judging how closely the participants involved resemble the agent and patient prototypes, in a way that is gradient and prone to inter-speaker variation.

In the following, I first introduce pseudo-symmetric predicates more in detail by contrasting them with strictly symmetric predicates. Next, I provide syntactic and semantic evidence for the analysis outlined

above. Finally, I discuss competing (and pseudo-competing) analyses by Kruitwagen (2024) and Wehbe (2025), as well as related work by Winter (2016, 2018) and van Craenenbroeck and Johnson (2026) on strictly symmetric predicates which has implications for pseudo-symmetric predicates as well.

4.5.2 Evidence for the analysis

4.5.2.1 Background

I begin by introducing symmetric predicates in the strict sense, following Winter (2016, 2018). I then turn to pseudo-symmetric predicates, which are the focus of this case study.²²

Strictly symmetric predicates Strictly symmetric predicates are a class of verbs, nouns and adjectives that yield the same truth-conditional output regardless of the order of their arguments (180):

- (180) $R(x, y) \leftrightarrow R(y, x)$, where R is a binary predicate
- a. Joel married Clementine. $\leftrightarrow$ Clementine married Joel.
 - b. Sebastian is Tabea's cousin. $\leftrightarrow$ Tabea is Sebastian's cousin.
 - c. A is identical to B. $\leftrightarrow$ B is identical to A.

Most strictly symmetric predicates have an intransitive variant with a plural subject that entails the conjunction of the corresponding transitives (181).²³ Winter (2016, 2018) labels intransitives that show this entailment pattern plain reciprocals.

- (181) $P(x \oplus y) \leftrightarrow R(x, y) \wedge R(y, x)$, where P is a unary-collective predicate and R a binary predicate
- a. Joel and Clementine married. $\leftrightarrow$ Joel married Clementine and Clementine married Joel.
 - b. Sebastian and Tabea are cousins. $\leftrightarrow$ Sebastian is Tabea's cousin and Tabea is Sebastian's cousin.
 - c. A and B are identical. $\leftrightarrow$ A is identical to B and B is identical to A.

Winter (2016, 2018) hypothesizes that only strictly symmetric predicates can form plain reciprocals, or in other words, that all predicates that participate in plain reciprocity are symmetric (182):

²²A complication that I will not address in the following overview concerns the behavior of both strictly and pseudo-symmetric predicates under VP ellipsis. Unlike fully asymmetric verbs, they permit what appears to be a violation of parallelism, shown for strictly symmetric predicates in (i) (Stockwell, 2020):

- (i) a. Joel can marry Clementine, but Clementine SHOULDN'T ~~marry Joel~~. (She should marry PATRICK instead.)
 b. *Joel can invite Clementine, but Clementine SHOULDN'T ~~invite Joel~~. (She should invite PATRICK instead.)

In (ia), *marry Joel* can be elided although the overt antecedent is *marry Clementine*. I have no explanation for this fact.

²³Exceptions include verbs like *resemble* (i):

- (i) This resembles that. $\leftrightarrow$ That resembles this. $\neq$ *They resemble. (Winter, 2018)

(182) Reciprocity-Symmetry Generalization

A reciprocal alternation between a unary-collective predicate P and a binary predicate R is plain if and only if R is truth-conditionally symmetric. (Winter, 2018:13)

In short, the hallmark of strictly symmetric predicates is that each of the three variants – the intransitive and the two transitives – always entail every other member of this set.

For concreteness, a list of verbal strictly symmetric predicates in English is given in (183). The prepositions only surface in the transitive variant.²⁴

(183) Strictly symmetric predicates: *date, marry, get married (to), meet (with), fight (with), talk (with), make love (with), share NP (with), cross, match, rhyme, differ (from)* (Winter, 2018:7)

In addition, some ditransitives also fall under the Reciprocity-Symmetry Generalization in that the two object arguments can be expressed in any order or alternatively as a single plural argument, all without any truth-conditional differences (184):

- (184) a. She mixed camphor with menthol. ↔ She mixed menthol with camphor. ↔ She mixed camphor and menthol.
 b. She introduced you to me. ↔ She introduced me to you. ↔ She introduced us.
 c. She combined the beans with the rice. ↔ She combined the rice with the beans. ↔ She combined the beans and the rice. (van Craenenbroeck and Johnson, 2025)

In the following, I limit myself to transitive verbs and leave ditransitives, nouns and adjectives aside.

Strictly symmetric predicates are not the focus of this case study for the following reason. For pseudo-symmetric predicates, I will argue that the intransitive contains a single θ -role assigned to the plural subject, but the transitive variant two different θ -roles assigned to subject and object, respectively. Crucially, if a verb like *hug* indeed came with two individual roles such as a hugger and a huggee, this would result in a case of variable linking. But for strictly symmetric predicates, it is not obvious that in the transitive, subject and object are indeed assigned two different θ -roles. There are no clear thematic differences between the two arguments of the transitive, or between either of them and the plural argument of the intransitive. For

²⁴Anecdotally, it appears that *marry* does not behave as a strictly symmetric predicate in languages that are spoken in societies where marriage is not considered a union between (approximately) equals. Latin has two distinct lexical items, *nubeo* for female subjects and (dative) male objects, and *marito* for male subjects and (accusative) female objects. Ancient Greek has a single verb, *gameō*, which takes active voice marking with male subjects and middle voice marking with female subjects. Note that the gender of the subject is to be understood in terms of social roles, not in terms of grammatical features. Lucian of Samosata (2nd c. CE), writing about a fictional society where men only marry men, describes the following custom (*A True Story*, Sect. 22) (i):

- (i) mechri men oun pente kai eikosi etōn gameitai hekastos, apo de toutōn gamei autos.
 until PTCL PTCL five and twenty years marry.3SG.MID each.M, after PTCL these marry.3SG.ACT INTENS
 ‘Until (the age of) twenty-five years, every man marries (middle); after that, he himself marries (active).’

instance, it is doubtful that in (185a), Joel is the more agentive participant of the marrying event, whereas in (185b), Clementine is:

- (185) a. Joel married Clementine.
 b. Clementine married Joel.

Rather, which of the two sentences a speaker chooses to produce appears to be governed solely by information-structural constraints such as givenness (but see Carlson, 1998). In general, the fact that with strictly symmetric predicates, the intransitive and both transitives mutually entail each other is difficult to reconcile with the idea that their arguments are assigned different θ -roles, which would make the alternants semantically distinct.

Accordingly, previous work has argued that strictly symmetric predicates are always underlyingly intransitive regardless of how their arguments are realized in the surface syntax of the sentence (van Craenenbroeck and Johnson, 2026; Winter, 2016, 2018). Under this view, there is only a single θ -role assigned to both participants in each of the predicate's syntactic guises. If this is true, then strictly symmetric predicates are not an interesting case to consider in the context of variable linking: they could simply come with two identical individual roles, each of which would be mapped onto the same θ -role regardless of the syntactic construal. I briefly discuss the proposals by Winter and by van Craenenbroeck and Johnson below in Chapter 4.5.3.3, but for the main purposes of this case study, I sidestep strictly symmetric predicates. However, I argue that pseudo-symmetric predicates cannot be analyzed as involving a single θ -role. I turn to these verbs next.

Pseudo-symmetric predicates On the surface, pseudo-symmetric predicates appear to participate in the same alternation as strictly symmetric predicates (186):

- (186) a. Joel and Clementine kissed.
 b. Joel kissed Clementine.
 c. Clementine kissed Joel.

Some further examples from English are given in (187):

- (187) Pseudo-symmetric predicates: *hug, embrace, kiss, fuck, fight, divorce, break up (with), collide (with), fall/be in love (with), talk (to), speak (to), chat (to), make love (to)* (Winter, 2018:22)

Pseudo-symmetric predicates used to be analyzed on a par with strictly symmetric predicates (e.g., Gleitman, 1965), but on closer inspection, they form a distinct class. In contrast to strictly symmetric predicates, the three variants in (186) do not entail each other. For instance, Joel might kiss Clementine against her will, in which case (186b) would be true but (186c) plainly false. Moreover, transitive pseudo-symmetric predicates do not entail their intransitive variant, a point made famously for the verb *embrace* by the example

in (188), attributed to Chomsky:

- (188) a. The drunk embraced the lamppost.
 b. #The drunk and the lamppost embraced.

Overall, pseudo-symmetric predicates differ semantically from strictly symmetric predicates and require a different account.

It is against this background that the covert reciprocal analysis of pseudo-symmetric predicates has emerged, which posits that pseudo-symmetric predicates are always underlyingly transitive with an agent and a patient role. The intransitive variants are analyzed as being similar or semantically identical to overt reciprocal constructions, in which both arguments are interpreted as both agent and patient (189):

- (189) Joel and Clementine kissed ~~each other~~.

Crucially for our purposes, this would allow the agent and patient θ -role to consistently map onto a single individual role each, such as a kisser and a kissee. This is the approach I argue against.

The covert reciprocal analysis does not require each transitive to entail the corresponding other transitive and the intransitive, as with strictly symmetric predicates. Instead, it predicts intransitive pseudo-symmetric predicates to behave just like reciprocals, namely, to stand in a bidirectional entailment relation to the conjunction of transitives (190):

- (190) a. Joel and Clementine kissed each other. $\leftrightarrow$ Joel kissed Clementine and Clementine kissed Joel.
 b. Joel and Clementine kissed. $\leftrightarrow$ Joel kissed Clementine and Clementine kissed Joel.

However, the entailment in (190b) has been shown to not be valid in either direction, in contrast to (190a). In the following, I first briefly discuss the non-entailment from the conjunction of the transitives to the intransitive. In the remainder of this case study, I then focus on the less well-known fact that the intransitive does not entail the conjunction of transitives. I argue that this non-entailment crucially undermines the covert reciprocal analysis of pseudo-symmetric predicates.

One difference between overt reciprocals and intransitive pseudo-symmetric predicates is that the latter cannot describe two spatio-temporally distinct actions. Consider a situation in which Joel kisses Clementine while she is asleep. Later that night, she kisses him while he is asleep. In this context, the overt reciprocal is felicitous while the intransitive is not (191):

- (191) a. Joel and Clementine kissed each other.
 b. #Joel and Clementine kissed.

There are a number of similar scenarios in which both transitives as well as the overt reciprocal are true, but the intransitive is not. Thus, while the conjunction of the transitives entails the overt reciprocal, the same

does not hold for the intransitive, (192), because the latter is subject to the additional requirement that the kissings must take place at the same time, besides other similar constraints.

- (192) a. Joel kissed Clementine and Clementine kissed Joel. $\rightarrow$ Joel and Clementine kissed each other.
 b. Joel kissed Clementine and Clementine kissed Joel. $\nrightarrow$ Joel and Clementine kissed.

In recent work, Wehbe (2025) has proposed a way of accounting for the contrast in (192) while still maintaining the idea that intransitive pseudo-symmetric predicates are fundamentally covert reciprocals. I review Wehbe's analysis in detail in Chapter 4.5.3.2, where I highlight several challenges for his account.

In addition, however, the reverse entailment does not hold either. Intransitive pseudo-symmetric predicates do not entail the conjunction of the corresponding transitives, again contrasting in this respect with overt reciprocals (193):

- (193) a. Joel and Clementine kissed each other. $\rightarrow$ Joel kissed Clementine and Clementine kissed Joel.
 b. Joel and Clementine kissed. $\nrightarrow$ Joel kissed Clementine and Clementine kissed Joel.

I discuss the non-entailment in (193b), which has been confirmed experimentally by Kruitwagen (2024), in detail in Chapter 4.5.2.3. Following Kruitwagen, I will argue that the contrast in (193) is due to a difference in the θ -roles involved in the two variants. While in the overt reciprocal, each argument is assigned both an agent and a patient θ -role, the intransitive involves only a single θ -role assigned to the plural subject argument, giving rise to the variable linking profile described earlier.

To summarize, pseudo-symmetric predicates differ from strictly symmetric predicates in that the three variants – the intransitive and the two transitives – do not entail each other. Instead, it is often assumed that intransitive pseudo-symmetric predicates are covert reciprocals. This analysis predicts that the intransitive should stand in a bidirectional entailment relationship with the conjunction of the transitives, but this entailment has been shown to not be valid in either direction. In the following, I focus on the fact that unlike overt reciprocals, intransitive pseudo-symmetric predicates do not entail the conjunction of the corresponding transitives, which I will take as evidence that they involve only a single θ -role. To substantiate this claim, I first look at the syntax and semantics of intransitive pseudo-symmetric predicates more in detail.

4.5.2.2 Syntactic diagnostics

There are two issues that this section needs to tackle: the syntax of transitive pseudo-symmetric predicates and the syntax of their intransitive variants; specifically, whether the latter are unergatives or unaccusatives. As for the first question, I assume without further ado and in line with the overt morphosyntax that the transitives are simply that, transitives, with an external argument position occupied by the agent and an internal argument position occupied by the patient. Depending on the verb, the latter position is either realized as a DP or as a PP – for verbs like *talk to* or *break up with* –, and I have nothing interesting to say

about this variation.

As for the syntax of intransitives, two points are worth highlighting from the start. First, whether or not pseudo-symmetric predicates constitute an instance of variable linking does not intrinsically hinge on the unergative or unaccusative status of the intransitive variant and what θ -role is assigned to the argument. What is crucial is merely that unlike the transitive, the intransitive involves only a single θ -role, whatever it may be. Second, there is no particular reason to expect all strictly and pseudo-symmetric predicates to behave as a coherent class with respect to the syntax of their intransitives. Some might behave as unergatives, others as unaccusatives, the facts might differ cross-linguistically, and nothing precludes variable unaccusativity with these verbs.

By way of example, for English alone, the relevant evidence is scarce but conflicting. On the one hand, van Craenenbroeck and Johnson (2026) have pointed out that the strictly symmetric predicates *marry* and *meet* allow for *re*-prefixation (194), assumed to only be available in the presence of an internal argument (195) (Horn, 1980):

- (194) a. Raj and Joan remarried.
 b. Raj and Joan remet every spring.
- (195) a. They repainted the house.
 b. They reappeared.
 c. *They resneezed. (van Craenenbroeck and Johnson, 2026)

On the other hand, resultative secondary predicates, an unaccusativity diagnostic, point in the opposite direction for the pseudo-symmetric predicate in (196). While the transitive allows for the resultative construction, the reading is unavailable in the intransitive, suggesting that the plural subject of the latter occupies the external argument position:²⁵

- (196) a. He fucked her senseless.
 b. #They fucked senseless.

By the same token, Siloni (2008) has shown that in Hebrew, French, Italian and Russian, the semantically similar verbs *kiss* and *hug* pass a range of unergativity diagnostics, but whether this finding extends to other pseudo-symmetric predicates, let alone strictly symmetric predicates, is an open question. Overall, the syntactic status of the intransitives needs to be established on a case-by-case basis.

Given this preamble, the remainder of this section focuses on the set of Dutch verbs in (197), for which we will see experimental evidence from Kruitwagen (2024) on their semantic properties in the next section:

²⁵Similarly, I once overheard (ia) at a poetry slam. In contrast, (ib) can only receive a depictive reading.

- (i) a. Just fuck me happy!
 b. #They fucked happy.

- (197) *vechten* (*tegen* ‘fight (against),’ *fluisteren* (*tegen*) ‘whisper (to),’ *knuffelen* ‘hug,’ *botsen* (*tegen*) ‘hit’/‘collide (with)’

I will argue that the syntactic and semantic evidence in conjunction with each other support the analysis I advance here, namely, that intransitive pseudo-symmetric predicates involve only a single θ -role, whereas the transitives involve two, an agent and a patient role.

I now survey three diagnostics for the unergative/unaccusative status of these verbs in their intransitive forms. Note that as all such diagnostics, the tests I survey below are known to not necessarily always return the same results for a given verb but to be sensitive to the broader context of the sentence, as discussed earlier under the label of variable unaccusativity in Chapter 4.1 (e.g., see Beliën, 2016; Keller and Sorace, 2003; Primus, 2011 for variation with impersonal passives). While I cannot exclude variable behavior, I apply the syntactic diagnostics below to the verbs in (197) in the same minimal contexts, which are similar to the contexts in which they are used in Kruitwagen (2024). Thus, I assume that the syntactic results below and Kruitwagen’s experimental findings can be meaningfully related to each other.

Auxiliary selection The first diagnostic comes from auxiliary selection. Dutch belongs to a group of languages that have been argued to use different auxiliaries for unergative and unaccusative verbs (see Sorace, 2000 for an overview). In Dutch, unergatives surface with the auxiliary *hebben* ‘have’ (198a), while unaccusatives take *zijn* ‘be’ (198b):

- (198) a. De trompettist heeft met bolle vangen geblazen.
 the trumpeter has with puffed-out cheeks blown
 ‘The trumpeter blew with puffed-out cheeks.’ (Sorace, 2000:874)
- b. De leraar is plotseling gestorven.
 the teacher is suddenly died
 ‘The teacher suddenly died.’ (Sorace, 2000:866)

Most verbs in our sample take *hebben*, shown for the verb *knuffelen* ‘hug’ in (199):

- (199) Fleur en Mark hebben geknuffeld.
 Fleur and Mark have hugged
 ‘Fleur and Mark hugged.’ (Kruitwagen, 2024:129)

The sole exception is the verb *botsen* ‘collide,’ which takes *zijn*

- (200) Fleur en Mark zijn gebotst.
 Fleur and Mark are collided
 ‘Fleur and Mark collided.’ (Kruitwagen, 2024:132)

Thus, this diagnostic indicates an unaccusative syntax for *botsen* ‘collide’ and an unergative syntax for the remaining verbs.

Impersonal passives Next, since Perlmutter (1978), impersonal passives in Dutch have been argued to be only licensed with unergatives (201a) but not with unaccusatives (201b) (see also Beliën, 2016 for a more recent corpus study):

- (201) a. Er wordt hier door de jonge lui veel gedanst.
 there becomes here by the young people a.lot danced
 ‘It is danced here a lot by the young people.’ (Perlmutter, 1978:168)
- b. *In dit weeshuis wordt er door de kinderen erg snel gegroid.
 in this orphanage became there by the children very fast grown
 Intended: ‘In this orphanage, it is grown very fast by the children.’ (Perlmutter, 1978:169)

Three out of four verbs can easily form impersonal passives, shown for *fluisteren* ‘whisper’ in (202):

- (202) Er wordt gefluisterd.
 there becomes whisper
 ‘There is whispering, people are whispering.’

However, impersonal passives are marginal for the verb *botsen* ‘collide’:

- (203) ??Er wordt gebotst.
 there becomes collided
 Intended: ‘There is colliding, people are colliding.’

Like the previous diagnostic, impersonal passives thus indicate an unergative syntax for most verbs in the sample, but suggest an unaccusative syntax for *botsen*.

Agent nominalizations Lastly, only unergative verbs have been argued to allow for agent nominalizations (Levin and Rappaport, 1988; Rappaport Hovav and Levin, 1992), seen for Dutch in (204):

- (204) a. Jan en Pieter zijn dansers/hardlopers/kopers.
 Jan and Pieter are dancers/runners/buyers
 ‘Jan and Pieter are dancers/runners/buyers.’
- b. ??Jan en Pieter zijn stervers/vallers/verdwijners.
 Jan and Pieter are diers/fallers/disappearers
 Intended: ‘Jan and Pieter are people who die/fall/disappear.’

Most of our pseudo-symmetric predicates pass this diagnostic; (205) gives the relevant example for *vechten* ‘fight’:

- (205) Jan en Pieter zijn vechters.
 Jan and Pieter are fighters
 ‘Jan en Pieter are fighters.’

Again, the outlier is *botsen* (206):

- (206) ??Jan en Pieter zijn botsers.
 Jan and Pieter are colliders
 ‘Jan en Pieter are colliders.’

Overall, according to all three diagnostics, all verbs in our sample of pseudo-symmetric predicates except *botsen* behave as unergatives. The next section argues that this is in line with the thematic interpretation of the verbs under the analysis adopted here.

4.5.2.3 Thematic interpretation

In the following, I summarize and discuss experimental work by Kruitwagen (2024) on the thematic interpretation of pseudo-symmetric predicates in Dutch.²⁶ Chapter 4.5.3.1 will review in detail Kruitwagen’s theoretical account of these data, which is very similar to what I argue for here.

In brief, Kruitwagen’s experimental evidence supports the view that intransitive pseudo-symmetric predicates involve only a single θ -role assigned to the subject argument. Recall that under the competing covert reciprocal analysis, each individual participant making up the plural subject of the intransitive is assigned both an agent and a patient θ -role. In consequence, the intransitive should entail both transitives. Kruitwagen’s data show that this entailment does not go through: there are scenarios in which speakers judge the intransitive to be felicitous but not the two transitives (207).

- (207) Joel and Clementine hugged. $\nrightarrow$ Joel hugged Clementine and Clementine hugged Joel.

Moreover, the experiment demonstrates that the felicity of each of the three variants depends on the thematic properties of the participants involved. If a participant is depicted as more volitional, speakers are more likely to construe them as agentive even if the specific activity performed remains the same.

In the experiment, Dutch speakers were shown video clips of events that could be described by the pseudo-symmetric predicates in (208):

- (208) *vechten (tegen)* ‘fight (against),’ *fluisteren (tegen)* ‘whisper (to),’ *knuffelen* ‘hug,’ *botsen (tegen)* ‘hit’/‘collide (with)’

Each situation depicted two participants who were asymmetrically involved in the event. For instance, the scene for ‘hug’ showed a woman wrapping her arms around a man, while the man does not do so in return. Speakers were then shown either an intransitive sentence such as (209a) or a transitive sentence such as (209b) – where crucially the *passive* participant is realized as the subject – and were asked whether they judged the sentence true or false. If (209a) is found true significantly more often than (209b), then this is evidence against the alleged entailment from the intransitive to the transitives.

²⁶Kruitwagen (2024) also reports results from a second experiment that targeted more specifically the difference between intransitive pseudo-symmetric predicates and overt reciprocals. While the findings equally support Kruitwagen’s – and my – claim that the former involve only a single θ -role, the data are not as clear, and I believe that they are subject to noise stemming from problems with the experimental design. Thus, I do not include these results here.

- (209) a. Fleur en Mark hebben geknuffeld.
 Fleur and Mark have hugged
 ‘Fleur and Mark hugged.’
- b. Mark heeft Fleur geknuffeld.
 Mark has Fleur hugged
 ‘Mark hugged Fleur.’

The reverse transitive in which the more *active* participant serves as the subject – ‘Fleur hugged Mark’ – was not tested since it can trivially be expected to be judged true in the scenario described.

The experiment involved one additional parameter which Kruitwagen labels collective intentionality. For each verb, two scenes were constructed: one in which both participants displayed the appropriate emotions associated with the verb and one in which the passive participant – here, Mark – remained emotionally neutral or hostile. For instance, in the scenes depicting a hugging event, Mark either smiled and seemed affectionate or showed disgust. In general, passive participants depicted with collective intentionality were shown with facial expressions indicating that they participated in the event of their own accord, rather than against their will.

449 native speakers of Dutch participated in the experiment. Each participant saw only three videos: one filler item and two test items. For one of the test videos, participants were shown the intransitive sentence, for the other, the transitive. The verbs for all three videos were different. Speakers either only saw videos with or only videos without collective intentionality to prevent them from consciously noticing an effect of this parameter, which might affect their response.

The full set of results is summarized in Figure 1; collective intentionality is coded as +CI and –CI:

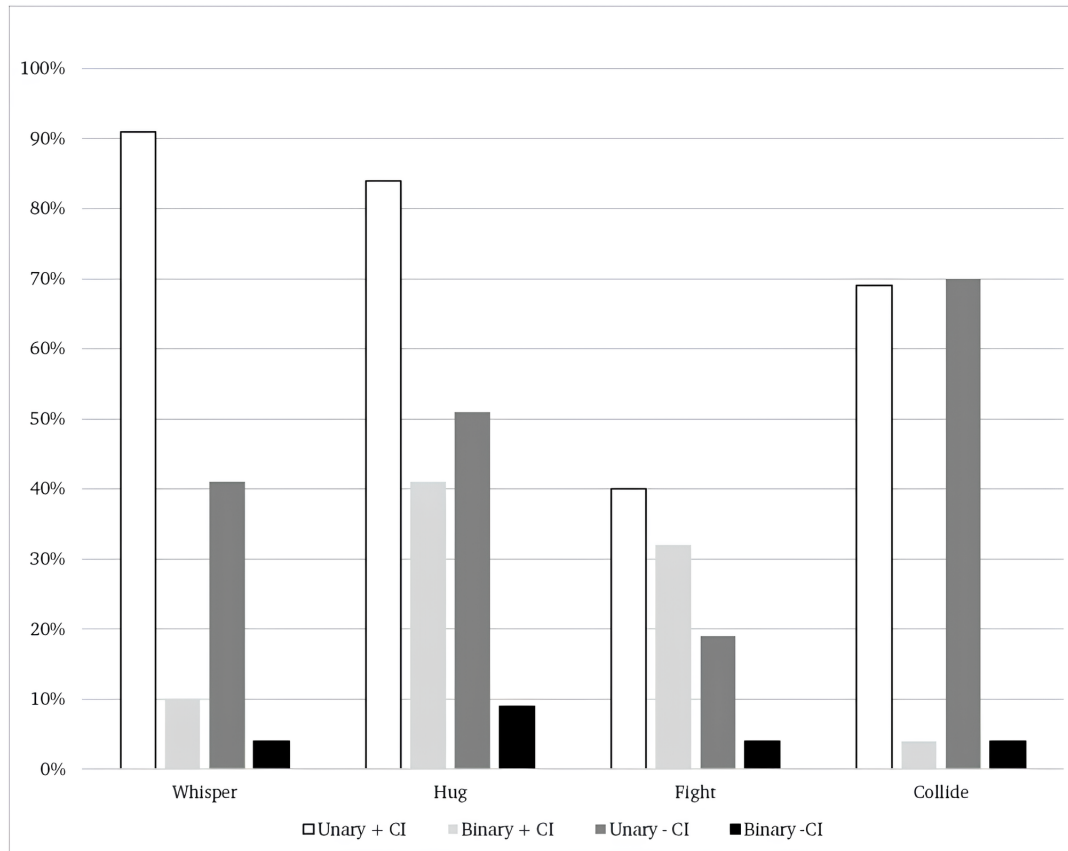


Figure 1: Acceptance rates for intransitives and transitives in asymmetric situations (Kruitwagen, 2024:43)

For all verbs, the intransitive form was judged true significantly more often than the transitive, although the contrast between the two varies across items. It is least pronounced for ‘fight,’ which notably also has low acceptance rates for the intransitive sentence even in the +CI condition. I take this to suggest that the speakers considered the situations depicted in the videos to be too asymmetrical for either variant to be felicitous. For ‘whisper,’ ‘hug’ and ‘collide,’ the data clearly indicate that the intransitive can be judged true in situations in which the transitive is not. This confirms that unlike overt reciprocals, intransitive pseudo-symmetric predicates do not entail the corresponding transitive variants.

In addition, Kruitwagen found a significant effect of collective intentionality on the acceptance rates for both intransitives and transitives, with the exception of *botsen* ‘collide,’ which I discuss further below. I focus first on the intransitive (unary) forms. The data show that the intransitives become more felicitous if the passive participant is depicted with collective intentionality, i.e., as volitional. For instance, if Mark showed affection during the hug rather than disgust, speakers are more likely to judge it true that Mark and Fleur hugged, despite the fact that Mark never wraps his arms around Fleur. Crucially, volition is a proto-agent property. In the scenario with collective intentionality, the difference in agent properties between the two participants is thus diminished, or in other words, they occupy a similar position on the spectrum between the agent and the patient prototypes. As a result, it is felicitous to assign them the same agent θ -role.

Overall, whether or not Mark is conceptualized as an agent – or, more precisely, as one of the participants making up the plural agent – does not depend in a binary fashion on whether or not he is a hugger, but depends in a gradient fashion on whether or not he displays proto-agent properties.

Turning to the transitive (binary) alternant, we see that in situations in which Fleur is the active and Mark the passive hugging participant, speakers are unwilling to judge *Mark hugged Fleur* true in the scenario without collective intentionality; that is, if Mark is depicted as showing disgust and being hugged against his will. However, acceptance rates go up in the scenarios with collective intentionality, in which Mark is smiling and appears to participate in the hug of his own accord. As seen before for the intransitives, if Mark is associated with more proto-agent properties, speakers are more likely to accept him as the agent and Fleur as the patient of the event even if Mark does not wrap his arms around Fleur.

However, even in scenarios with collective intentionality, the active participant – here, Fleur – still resembles a prototypical agent more than the passive participant, Mark, does. The question this raises is why it should nonetheless be possible for speakers to conceptualize Mark as the agent and Fleur as the patient of the event. I argue that if the difference in proto-properties between participants is sufficiently negligible, speakers allow for both transitive variants. In the collective intentionality scenarios, and *a fortiori* in perfectly symmetric hugging events, who makes for a better agent and who for a better patient becomes a moot point, and as a result, both construals are available. The choice between the different alternants might depend more heavily on discourse-related constraints on subjecthood. For instance, speakers might prefer the sentence *Mark hugged Fleur* over its alternatives even if Mark does not actively wrap his arms around Fleur simply because he is the topic of the broader context.

Interestingly, no effect of collective intentionality obtains for *botsen* ‘collide,’ neither in its transitive nor in its intransitive variant. In the relevant scenario with collective intentionality, the passive participant showed anger and initiated an argument when the active participant collided with him. Against the background of the syntactic data discussed above, this non-effect is as expected. *Botsen* was notably the only verb that passed unaccusativity diagnostics, indicating that in the intransitive, Mark and Fleur are both construed as patients. If this is correct, then depicting Mark as actively responding to the collision with anger – i.e., as more agent-like – naturally does not make the intransitive variant more felicitous. Rather, we would expect acceptance rates for the intransitive to be sensitive to the proto-patient property of how causally affected the two participants are. For instance, speakers might judge the intransitive to be more felicitous if both participants stumble and fall as a result of the collision.

To reiterate the two take-home messages from this experiment, first, intransitive pseudo-symmetric predicates can be judged true in a situation where one of the corresponding transitives is not. This demonstrates that the former does not entail the latter, which runs counter to the covert reciprocal analysis. Second, speakers have gradient preferences as to which alternant is felicitous in a given context. Whether or not a participant is associated with a given θ -role does not simply depend on whether or not they perform a particular activity – e.g., whether or not they wrap their arms around another participant –, but on their degree

of proto-properties, with more volitional participants being more likely to be construed as agents. Both of these observations indicate that intransitive pseudo-symmetric predicates are not reciprocals in disguise, but involve a single θ -role.

Overall, what makes pseudo-symmetric predicates special is that the participants in these events tend to be very similar to each other in terms of proto-properties. If they are sufficiently similar, speakers can construe either of them as the agent or the patient, respectively, or alternatively assign both of them the same θ -role. But the more asymmetric the event is, the more speakers tend to identify one participant as the agent, the other as the patient. Again, this is not compatible with the view that a discrete set of individual roles such as hugger and huggee is consistently mapped onto corresponding θ -roles.

4.5.3 Competing analyses

Lastly, I review three competing analysis to pseudo-symmetric predicates. I begin by discussing the theoretical proposal by Kruitwagen (2024), whose experimental work I have reported above. While I identify some differences between our accounts, I also highlight the considerable overlap. The next section then deals with Wehbe (2025), who argues that a covert reciprocal analysis of intransitive pseudo-symmetric predicates can be maintained after all by imposing a restriction on variable binding. Lastly, the main thrust of this case study so far has been to argue against the possibility that pseudo-symmetric predicates are always underlyingly transitive with an agent and a patient θ -role, and that the intransitive variants are merely covert reciprocals. An alternative possibility, however, that would equally run counter to a variable linking analysis is that pseudo-symmetric predicates are always – or at least more often than it might seem – semantically intransitive. I address this idea in the last section of this case study, drawing on previous work by Winter (2016, 2018) and van Craenenbroeck and Johnson (2026) on strictly symmetric predicates.

4.5.3.1 A threshold model: Kruitwagen (2024)

Kruitwagen is concerned with three variants of pseudo-symmetric predicates: intransitives, transitives taking bare DP objects, and transitives taking PP objects, marked with the preposition *met* ‘with’ (210):

- (210)
- a. Fleur en Mark hebben geknuffeld.
Fleur and Mark have hugged
‘Fleur and Mark hugged.’
 - b. Fleur heeft Mark geknuffeld.
Fleur has Mark hugged
‘Fleur hugged Mark.’
 - c. Fleur heeft met Mark geknuffeld.
Fleur has with Mark hugged
‘Fleur hugged (with) Mark.’

In the following, I leave aside the *with* variant in (210c) and focus on the variants in (210a) and (210b).

To account for the experimental results that we saw earlier, Kruitwagen proposes that the verbal root is represented in the lexicon with what she labels a *conceptual core* (CC), which contains (at least) two primitive features or parameters: the agent's *activity* (A_r) and *intentionality* (I_r) towards the patient. These features are specified differently for each verb. For instance, for *fight*, the activity is specified as 'physically oppose' and the intention as 'being hostile to,' whereas for *hug*, the activity is 'wrapping one's arm around' and the intention 'being affectionate towards.'

Moreover, Kruitwagen assumes a typicality approach to verb meaning. Whether or not a given event can be described by a given verb – say, *fight* – depends on how typical a fighting the event is, or in other words, how closely it resembles a prototypical fighting event. Typicality is defined in terms of activity and intentionality. For instance, an event in which one participant merely pokes at the other participant in a slightly antagonistic way would arguably not pass the typicality threshold for fighting, whereas one participant firing a machine gun at the other with the intention to kill certainly would. Crucially, Kruitwagen argues that transitive (211b) and intransitive (211a) construals of pseudo-symmetric predicates come with differently defined thresholds:

- (211) a. $\mathbf{B}_r(x, y, e) \leftrightarrow [A_r(x, y, e) + I_r(x, y, e)]/2 > thr$ *binary/trans.*
 b. $\mathbf{C}_r(x + y, e) \leftrightarrow [A_r(x, y, e) + I_r(x, y, e) + A_r(y, x, e) + I_r(y, x, e)]/4 > thr$ *collective/intrans.*

For the binary/transitive variant, their felicity depends on the average of the activity and intentionality values of only one participant – the one who is realized as the agent –, whereas for the collective/intransitive variant, activity and intentionality of both participants matter.

In general, I fully agree with Kruitwagen's typicality approach to verb meaning in general and pseudo-symmetric predicates in particular. Whether Mark and Fleur hug, Mark hugs Fleur or the two hug intransitively is not a matter of neatly delineated truth conditions but of gradient similarity to a prototypical (transitive or intransitive) hugging event. Kruitwagen also correctly highlights that different speakers can have different threshold values. If two speakers disagree on whether or not Mark in fact hugged Fleur, this does not necessarily entail that one of them is factually wrong; it is a matter of degree, of perspective and of idiosyncratic differences.

One point where I diverge from Kruitwagen is that she implicitly assumes all intransitive pseudo-symmetric predicates to have an agent argument. I have argued above that this does not hold true, for instance, for the verb *botsen* 'collide.' In the experiment reported above, depicting the passive participant in the colliding event as more intentional did not increase acceptance rates for the intransitive. In response, Kruitwagen proposes that for this verb, the intentionality feature (I_r) is not defined and does not factor into the computation of the typicality value, or in other words, that what makes for a proper collision event does not depend on the intentionality of its participants. But a more straightforward conclusion to draw from the experimental finding, in line with the syntactic diagnostics, is that the sole argument of intransitive *botsen* is simply not an agent.

This brings us to some more general limitations of Kruitwagen's theory. According to the calculus described

above, the typicality value of a given event depends solely on the properties of the agent argument, but this seems too narrow. For instance, what makes for a melting event arguably depends on the properties of the patient argument as well. Moreover, the calculus does not allow typicality to be computed for unaccusative event descriptions with only a patient argument. Finally, a lot of importance is attributed to the difference between activity and intentionality, but for many agents – such as inanimates –, intentionality does not matter, whereas for others – such as the arguments of psych verbs –, activity doesn't. It is not clear why the distinction between activity and intentionality should have to be a theoretical primitive.

The discussion in the previous section has tried to integrate Kruitwagen's core intuition with Dowty's theory of proto-roles. What makes for a good hugging, colliding or fighting event depends on many things, but one of them is how closely the participants involved match the proto-roles assigned to the corresponding arguments in the sentence. Whether Mark hugged Fleur depends on how closely Mark matches the agent role and Fleur the patient role. Reasonable people can disagree. But crucially, θ -role assignment is sensitive to proto-properties, not simply to whether or not a participant performs a particular activity associated with an individual role.

4.5.3.2 Covert reciprocals: Wehbe (2025)

This case study has argued extensively that intransitive pseudo-symmetric predicates cannot be analyzed as reciprocals in disguise. This is not a novel insight. The covert reciprocal analysis, going back to Gleitman (1965), has long been recognized to be undermined by the fact that intransitive pseudo-symmetric predicates and overt reciprocals are not truth-conditionally identical. Wehbe's (2025) work is an attempt to resuscitate the covert reciprocal analysis in a way that amends the shortcomings of previous versions of this approach. Specifically, he argues that intransitive pseudo-symmetric predicates and overt reciprocals share the same basic structure, but that the intransitives are subject to an additional restriction on variable binding. For our purposes, reviewing his analysis is thus key.

In brief, I believe that Wehbe's strategy can deal with some but not all problems for the covert reciprocal approach. As we saw earlier, overt reciprocals and intransitive pseudo-symmetric predicates differ in that the former but not the latter stand in a bidirectional entailment relationship to the conjunction of the corresponding transitives (212):

- (212) a. Joel and Clementine kissed each other. $\leftrightarrow$ Joel kissed Clementine and Clementine kissed Joel.
 b. Joel and Clementine kissed. $\nleftrightarrow$ Joel kissed Clementine and Clementine kissed Joel.

Wehbe's proposal aims to derive the non-entailment from the conjunction of the transitives to the intransitive (213):

- (213) Joel kissed Clementine and Clementine kissed Joel. $\nrightarrow$ Joel and Clementine kissed.

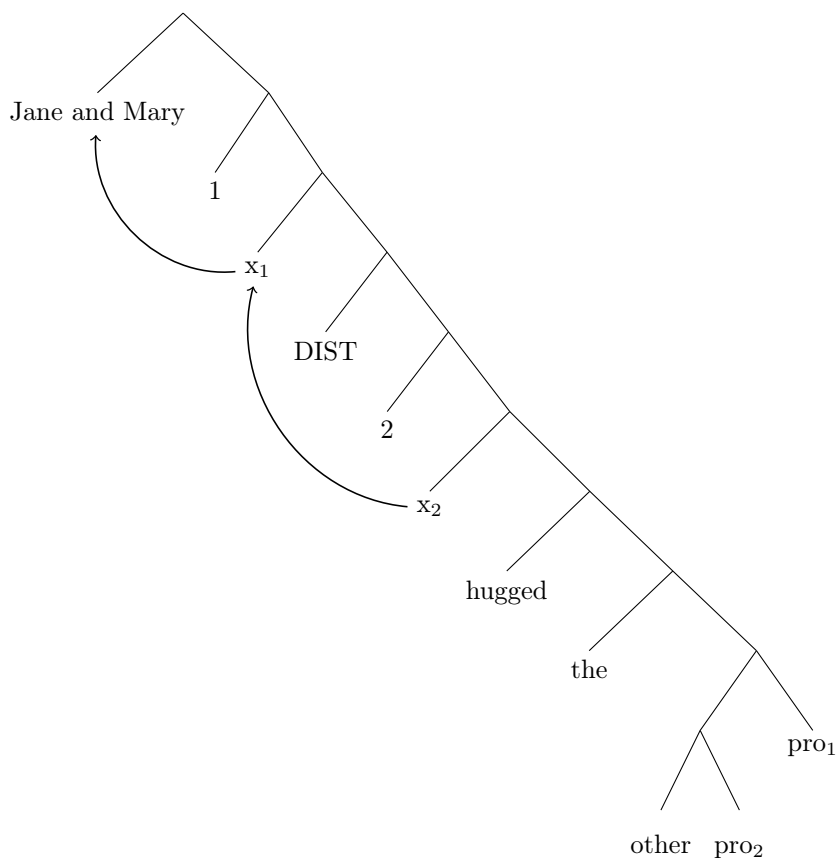
However, Kruitwagen (2024) has shown that furthermore, the intransitive does not entail the conjunction of the transitives (214):

(214) Joel and Clementine kissed. $\nrightarrow$ Joel kissed Clementine and Clementine kissed Joel.

I argue that Wehbe cannot account for the non-entailment in (214). In addition, I discuss some more general challenges for his analysis. In the following, I limit myself to pseudo-symmetric predicates only, but note that Wehbe deals with strictly symmetric predicates as well.

For overt reciprocals, Wehbe assumes the structure in (215), building on Sauerland (1998) (see also Heim et al., 1991; C. Roberts, 1991):

(215)



(Wehbe, 2025:582)

The reciprocal element *each other* is syntactically decomposed into a definite determiner, two silent pronouns, and the predicate *other* with the denotation given in (216). The latter takes two entity arguments as input and returns a property describing an entity which is, informally speaking, the second argument (in (215), somewhat confusingly, pro_1) minus the first argument (pro_2). The definite determiner then returns the corresponding entity if the existence and uniqueness presupposition is met. For instance, if pro_1 is *Mary and Jane* and pro_2 *Mary*, *the other* will denote *Jane*.

$$(216) \quad \llbracket \text{other} \rrbracket = \lambda x. \lambda y. \lambda z. z \leq y \wedge \neg \exists x' : x' \leq x \wedge x' \leq z \quad (\text{Wehbe, 2025:582})$$

The subject argument, *Mary and Jane*, moves twice over the course of the derivation: first, across a distributivity operator to bind the first pronominal argument of *other*, and then a second time to bind the second argument. The distributivity operator combines with a property holding of an entity and returns the same property holding of every atomic part of this entity (217):

$$(217) \quad \llbracket \text{DIST} \rrbracket = \lambda f_{\langle e, t \rangle}. \lambda x. \forall y [y \leq_{AT} x \rightarrow f(y) = 1] \quad (\text{Wehbe, 2025:582})$$

Thus, *Mary and Jane hugged each other* is true if for every atomic part of *Mary and Jane*, said atomic part hugged the respective other atomic part; that is, if Mary hugged Jane and Jane hugged Mary.

Intransitive pseudo-symmetric predicates, Wehbe proposes, differ from overt reciprocals in two respects. First, and trivially, the reciprocal is not spelled out. Second, unlike overt reciprocals, intransitive pseudo-symmetric predicates require the two arguments of *other* to be bound at the lowest possible positions that avoid a contradiction. As we will see now, this postulate has several desirable effects.

First, consider how overt reciprocals and intransitives differ when embedded under attitude verbs such as *say* (218):

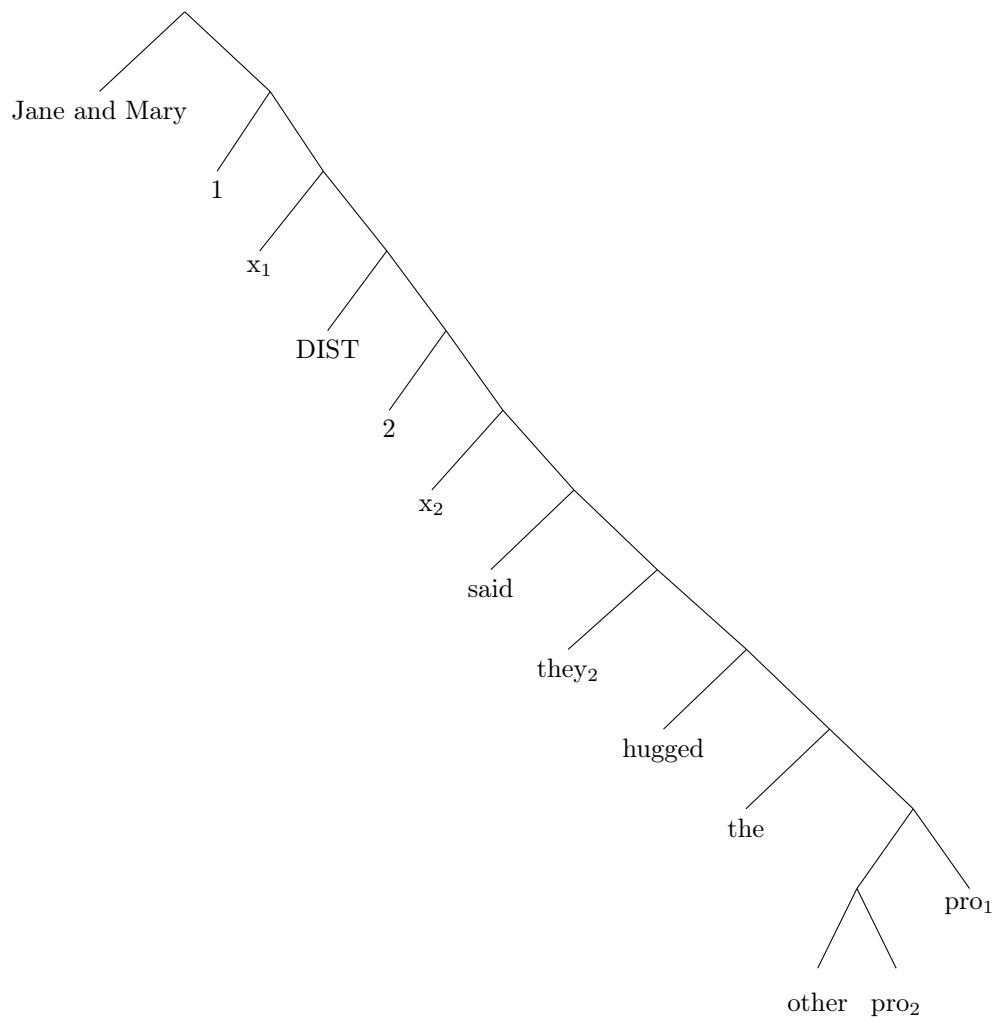
- (218) Context: Jane said that she hugged Mary but that Mary didn't hug her back. Mary said that she hugged Jane but that Jane didn't hug her back.
- a. Jane and Mary said that they hugged each other.
 - b. #Jane and Mary said that they hugged. (Wehbe, 2025:24)

Wehbe follows Heim et al. (1991) in assuming that the overt reciprocal is ambiguous between two readings. Under the long-distance reciprocal reading, the distributivity operator applies to the matrix verb *say* (219a), whereas under the low-scope reciprocal reading, it applies to the embedded verb *hug* (219b):

- (219) Jane and Mary said that they hugged each other.
- a. Long-distance reciprocal: Jane said that she hugged Mary and Mary said that she hugged Jane.
 - b. Low-scope reciprocal: Jane and Mary said that Jane hugged Mary and Mary hugged Jane.

The tree for the long-distance reciprocal reading is given in (220). Here, DIST is merged high, and the two arguments of *other* are bound above *said*.

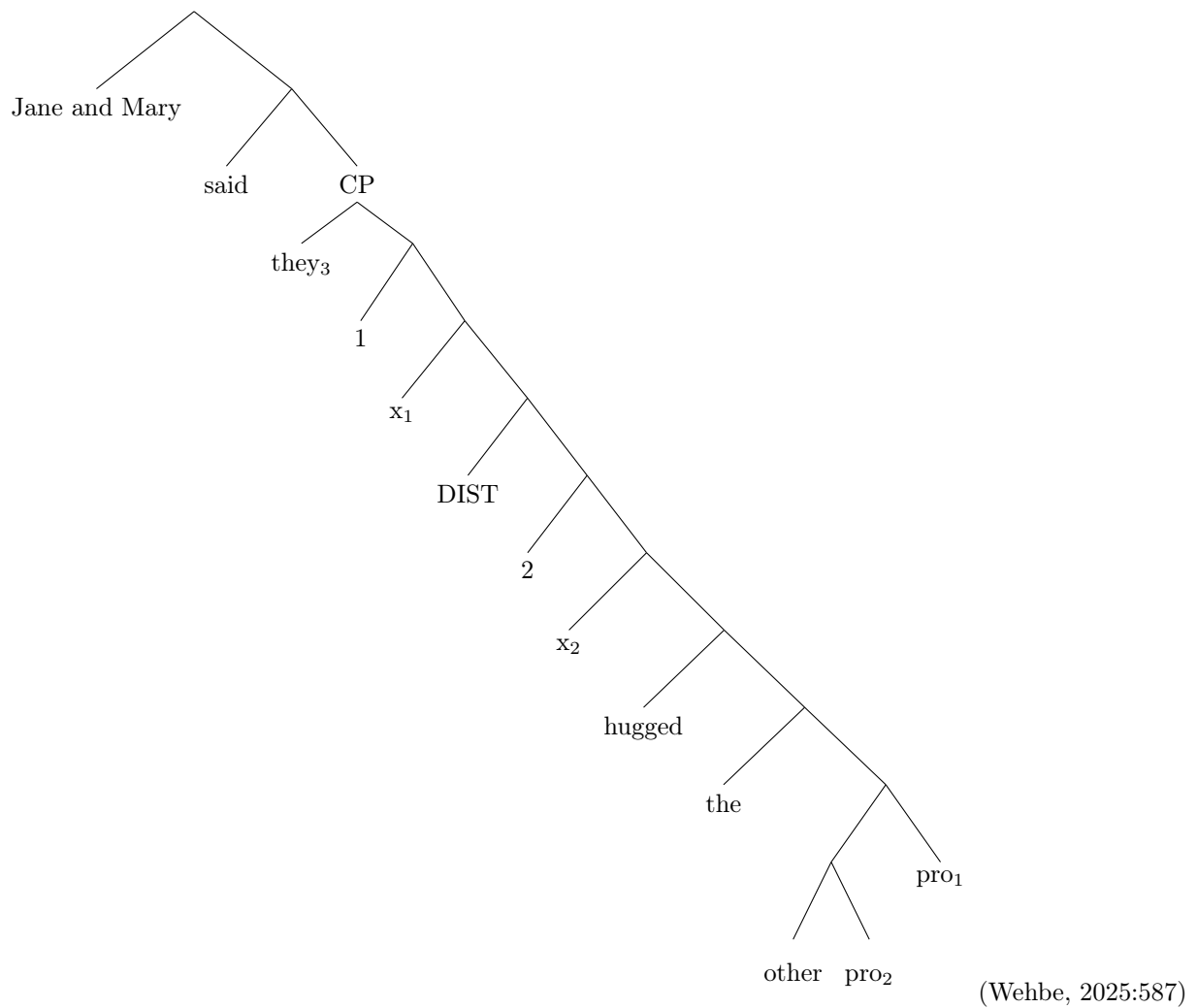
(220) *Long-distance reciprocal reading*



(Wehbe, 2025:588)

In contrast, to derive the low-scope reciprocal reading, DIST is merged low and the arguments of *other* are bound below *said* (221).

(221) *Low-scope reciprocal reading*



Recall that according to Wehbe’s postulate, intransitive pseudo-symmetric predicates (‘covert reciprocals’) require the two arguments of *other* to be bound as low as possible. This requirement is met in the low-scope tree (221) by moving *they* and letting it bind the two *pro* arguments. On the other hand, in the long-distance tree (220), the more distant DP *Mary and Jane* moves, binding the arguments of *other* from a higher position. Accordingly, the intransitive is only compatible with the low-scope structure, correctly predicting the contrast in (218), repeated below as (222):

- (222) Context: Jane said that she hugged Mary but that Mary didn’t hug her back. Mary said that she hugged Jane but that Jane didn’t hug her back.
- a. Jane and Mary said that they hugged each other.
 - b. #Jane and Mary said that they hugged.

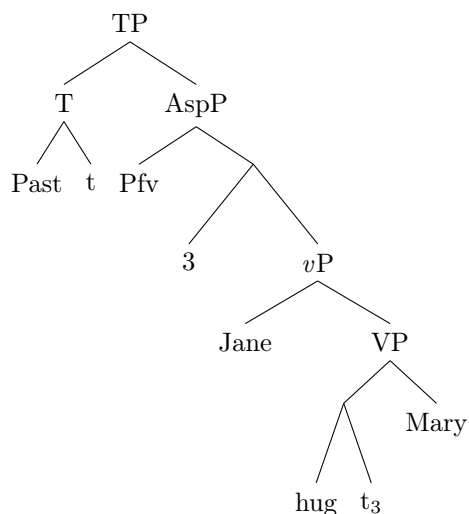
A second semantic difference between overt reciprocals and intransitive symmetric predicates is that the latter cannot denote two actions taking place at different times, seen in (223):

- (223) Context: Jane hugged Mary while Mary was sleeping and then Jane fell asleep and Mary woke up and hugged Jane.
- a. #Jane and Mary hugged.
- b. Jane and Mary hugged each other. (Wehbe, 2025:591)

Wehbe makes the following basic assumptions regarding tense. Verbs take as their first argument a time variable, shown for *hug* in (224a). This variable moves and adjoins to T, with a λ -abstractor being inserted below AspP (225). The perfective aspectual head takes as input a predicate of times and returns another predicate of times, mapping a time variable onto truth iff there exists a time within the time specified by the variable at which the verbal action unfolds (224b); or in other words, if the action is completed within the time span denoted by the variable. Lastly, past tense T denotes the time variable that started out as the first argument of the verb, with the presupposition added that this time is in the past (224c). Overall, then, *Jane hugged Mary* is true iff within a time span presupposed to be in the past, Jane hugged Mary; QED.

- (224) a. $\llbracket \text{hug} \rrbracket = \lambda t. \lambda x. \lambda y. y \text{ hugged } x \text{ at } t$
 b. $\llbracket \text{Pfv} \rrbracket = \lambda P. \lambda t. \exists t' \in t : P(t') = 1$
 c. $\llbracket \text{Past} \rrbracket^c = \lambda t : t < t_c. t$ (Wehbe, 2025:592)

- (225) a. Jane hugged Mary.
 b.



(Wehbe, 2025:592)

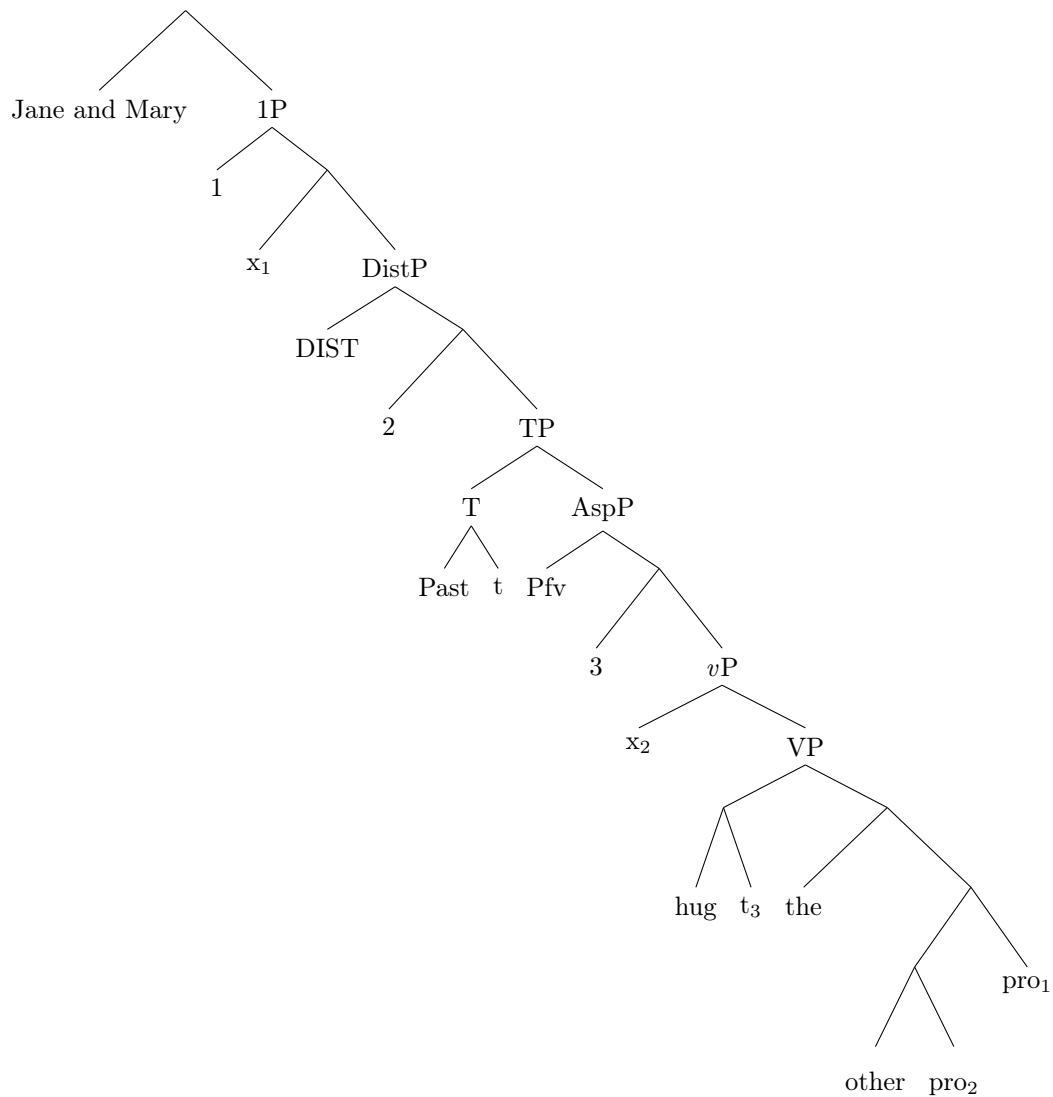
Let us return to the contrast in (223), repeated below as (226):

- (226) Context: Jane hugged Mary while Mary was sleeping and then Jane fell asleep and Mary woke up and hugged Jane.
- a. #Jane and Mary hugged.
 - b. Jane and Mary hugged each other. (Wehbe, 2025:591)

As before, Wehbe derives this contrast from the requirement that in the intransitives, the two arguments of *other* must be bound from the lowest position possible. Consider first the structure in (227).²⁷ Here, the subject *Jane and Mary* moves past AspP – where the time variable is existentially closed – already in the first step, and DIST is thus equally merged above this level. Accordingly, the operator can distribute over times, resulting in the denotation that for every atomic part of *Mary and Jane*, there was a time at which said atomic part hugged the respective other atomic part. Nothing requires the two times to coincide. The denotation derived from this tree is thus compatible with the non-simultaneous reading in (223).

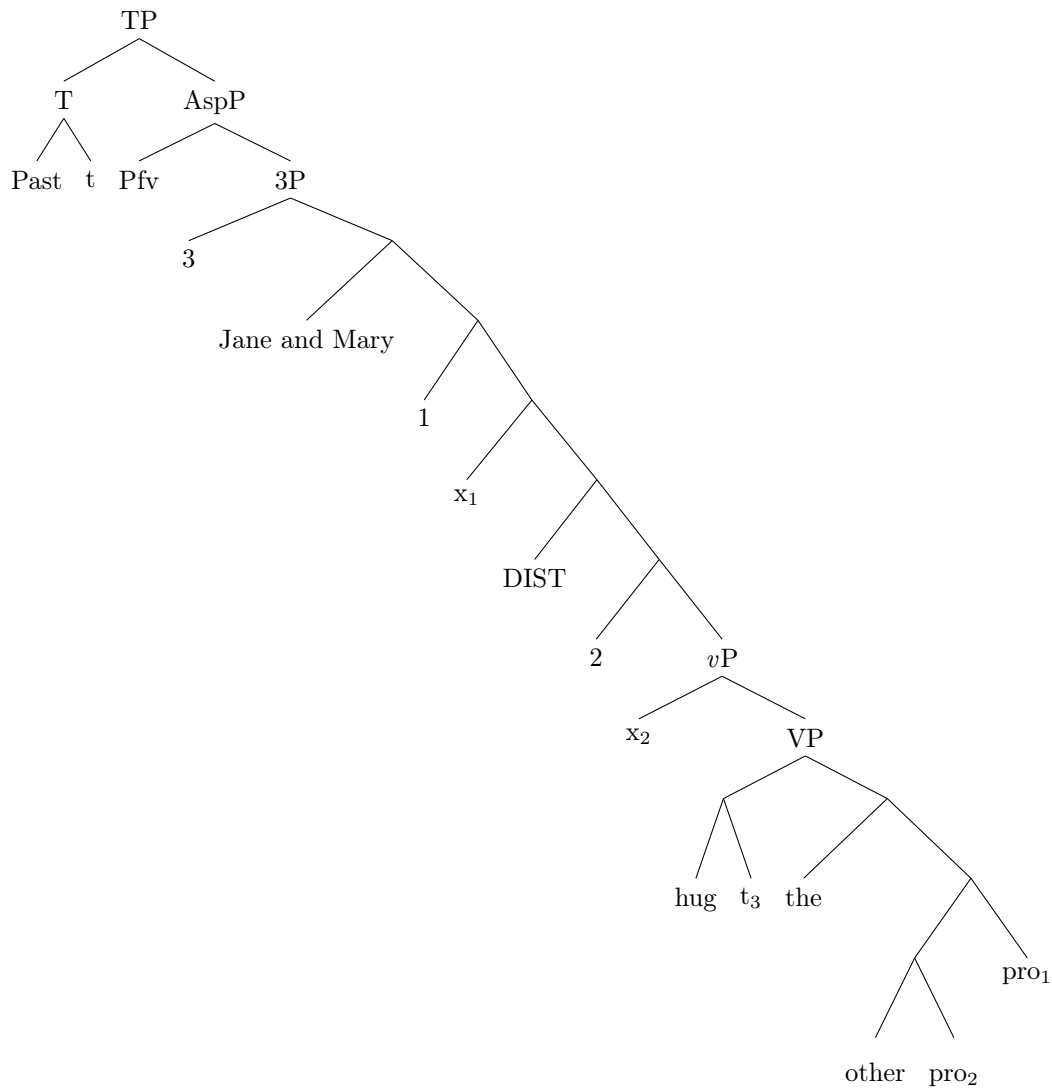
²⁷For unclear reasons, in this tree and the following, Wehbe represents the arguments of *other* as x_1 and x_2 . I use pro_1 and pro_2 in line with previous trees.

(227)



In contrast, in (228), *Jane and Mary* moves twice below *AspP*, and *DIST* is inserted low. Accordingly, the denotation derived is that there was a time at which for every atomic part of *Mary and Jane*, said atomic part hugged the respective other atomic part. The huggings must coincide temporally.

(228)



Only this second structure in (228) is compatible with the constraint that the arguments of *other* must be bound from the lowest position possible; accordingly, only the simultaneous reading is available with intransitive pseudo-symmetric predicates.

A final difference between intransitive pseudo-symmetric predicates and overt reciprocals discussed by Wehbe is that in the former, the two actions must occur not only at the same time but also in the same place, as evidenced by (229):

(229) Context: Mary kissed Jane on the arm and Jane kissed Mary on the arm (at the same time).

a. #Jane and Mary kissed.

b. Jane and Mary kissed each other.

(Wehbe, 2025:596)

Wehbe derives this effect in an analogous fashion to the simultaneity requirement by using a place variable

instead of a time variable; here, I omit the details.

In sum, Wehbe argues that intransitive pseudo-symmetric predicates should be analyzed as covert reciprocals after all, and that their semantic differences to overt reciprocals can be accounted for by imposing a restriction on variable binding. I will not take issue here in technical terms with his solution to the attitude verb embedding effect, the simultaneity requirement and the spatial overlap requirement that I have presented above. However, I see several problems.

First, a very general challenge for Wehbe's analysis is the question of what would motivate the variable binding restriction in the first place and how it would be encoded architecturally. Under his account, two structures can in principle be freely generated, one in which the arguments of *other* are bound low and one in which they are bound high. If the reciprocal is covert – a fact only established at PF –, one of the two possible derivations is ruled out. It is not clear why this would be the case, or where in the grammar such a constraint would be located.

Second, Wehbe focuses on deriving only a subset of the semantic differences between overt reciprocals and intransitive pseudo-symmetric predicates, acknowledging that it remains to be seen to what extent the remaining differences can equally be explained by the variable binding restriction. One contrast that his system fails to account for is the fact that intransitive pseudo-symmetric predicates but not overt reciprocals can take group nouns as arguments (230):

- (230) a. Team A met.
 b. *Team A met each other. (Wehbe, 2025:614–615)

As Wehbe rightly points out himself, this falls out naturally if the overt reciprocal involves distributivity, whereas the intransitive is predicated directly of the group nouns, as argued for here.

More importantly, recall that overt reciprocals but not intransitive pseudo-symmetric predicates stand in a bidirectional entailment relationship with the conjunction of the corresponding transitives (231):

- (231) a. Joel and Clementine kissed each other. $\leftrightarrow$ Joel kissed Clementine and Clementine kissed Joel.
 b. Joel and Clementine kissed. $\leftrightarrow$ Joel kissed Clementine and Clementine kissed Joel.

In deriving the differences between overt reciprocals and intransitive pseudo-symmetric predicates, Wehbe focuses on the fact that the conjunction of transitives does not entail the intransitive (232):

- (232) Joel kissed Clementine and Clementine kissed Joel. $\nrightarrow$ Joel and Clementine kissed.

If Joel kissed Clementine and Clementine kissed Joel, then the two kissings can occur at different times, in different places, and so forth. According to Wehbe's variable binding restriction, this is not possible for the intransitive. Therefore, he correctly predicts that the entailment in (232) does not go through.

In addition, however, the reverse entailment – from the intransitive to the transitive – is not valid for

pseudo-symmetric predicates either (233):

(233) Joel and Clementine kissed. $\nrightarrow$ Joel kissed Clementine and Clementine kissed Joel.

As confirmed experimentally by Kruitwagen (2024), it is possible for speakers to consider the intransitive felicitous while rejecting one of the transitives, namely, the one in which the less agentive participant is realized as the subject. Wehbe's analysis predicts this to be impossible. If Joel and Clementine kissed, then there is a kissing of Clementine by Joel and a kissing of Joel by Clementine; it is just that these two kissings must happen at the same time, in the same place, etc. There could not be a world in which Joel and Clementine kissed, and Joel kissed Clementine, but Clementine did not kiss Joel. In other words, a difference in degrees of agentivity could not have any effect in Wehbe's world because both participants are always both agent and patient. This fails to account for Kruitwagen's experimental results.

Lastly, the covert reciprocal analysis in all of its guises leaves crucially unexplained why unlike their overt counterparts, 'covert reciprocals' should only be licensed for a very specific subset of verbs: strictly and pseudo-symmetric predicates. If all it takes to turn an overt reciprocal into an intransitive was a silent *each other* and a restriction on variable binding, this route should also be open for any other verbs. Leaving aside strictly symmetric predicates, the analysis adopted here naturally accounts for the fact that pseudo-symmetric predicates can form 'covert reciprocals': using two different θ -roles for the two participants in the transitive but only one θ -role for both of them in the intransitive is possible precisely when the two participants are similar to each other in terms of proto-properties, as hugging, kissing and colliding participants commonly are. Thus, the special grammatical profile of pseudo-symmetric predicates can be linked to their lexical semantics. The covert reciprocal analysis has no handle on this, and Wehbe's restriction on variable binding does not help.

4.5.3.3 Covert intransitives: Winter (2016, 2018) and van Craenenbroeck and Johnson (2026)

This final section addresses not a competing analysis per se, but a concern that arises for the analysis I have adopted here from previous work by Winter (2016, 2018) and van Craenenbroeck and Johnson (2026) on strictly symmetric predicates. I will argue against this concern, but part of the purpose of this section is to acknowledge that open questions persist.

To recap, in this case study I have made the case that pseudo-symmetric predicates constitute an instance of variable linking. I have done so by arguing against the covert reciprocal analysis, according to which pseudo-symmetric predicates are always underlyingly transitive with an agent and a patient θ -role. Under this view, a verb like *hug* could denote a relation between a hugger and a huggee, with the individual role of the hugger always being mapped onto an agent θ -role, and the individual role of a huggee always onto a patient θ -role. In intransitives, both participants would be hugger and agent, and huggee and patient. I believe that we have seen good reasons to reject this analysis. However, an alternative possibility that equally runs counter to a variable linking analysis is that pseudo-symmetric predicates are always – or at

least more often than it might seem – underlyingly intransitives, featuring two identical individual roles that map onto the same θ -role.

This idea finds its predecessors in work by Winter (2016, 2018) and van Craenenbroeck and Johnson (2026) on strictly symmetric predicates such as *meet*, *rhyme* and *marry*. With these verbs, the intransitive and the two corresponding transitives mutually entail each other (234), as discussed earlier:

(234) Joel and Clementine married $\leftrightarrow$ Clementine married Joel $\leftrightarrow$ Joel married Clementine

This fact suggests that all three variants have the same LF. Concretely, Winter has proposed that strictly symmetric predicates are always underlyingly unary-collective and that the transitive/binary variant is derived from it via the lexical rule in (235):

(235) Symmetric image of a collective predicate

Let P be a unary-collective predicate meaning. The symmetric image of P is the binary predicate R that is defined as follows:

$R =_{\text{def}} \lambda x.\lambda y.P(x + y)$ (Winter, 2018:16)

According to Winter, transitive strictly symmetric predicates are thus nothing but collective predicates in disguise. Similarly, van Craenenbroeck and Johnson (2026) have argued that the two arguments of strictly symmetric predicates are always related to the verb by the same θ -role, both in the intransitive and the transitives. Unlike Winter, they derive this denotation in the syntax by means of a silent comitative operator. Since strictly symmetric predicates are not the topic of this case study, I omit the details of their analysis.²⁸

Of course, neither of the two proposals straightforwardly carries over to pseudo-symmetric predicates, which are clearly semantically distinct in their transitive and intransitive variants. However, a possibility to consider is whether seemingly transitive pseudo-symmetric predicates are structurally ambiguous between a truly transitive, asymmetric reading with an agent and a patient θ -role on the one hand, and an underlyingly intransitive reading with only a single θ -role on the other. Whenever speakers judge the transitives to be true at the same time as the intransitive, this would be evidence that the symmetric reading of the transitive obtains, which is semantically identical to the intransitive. In asymmetric situations, in contrast, the truly transitive construal would apply.

This is not an obvious possibility, but there is some genuine evidence for it. Strictly symmetric predicates are known to resist passivization (236) (van Craenenbroeck and Johnson, 2026). This is unexpected for transitives but would fall out if strictly symmetric predicates are always, even if seemingly transitive,

²⁸As a sidenote, one challenge for both Winter's and van Craenenbroeck and Johnson's analysis is to account for the fact that (ia) can only describe mixed-gender marriages, while (ib) is also compatible with same-gender (or a mix of same-gender and mixed-gender) marriages. The latter reading can be made more accessible by imagining a society in which same-gender marriage is the norm.

- (i) a. The men married the women.
- b. The men and the women married.

underlyingly intransitive.

- (236) a. *Joan was met by Raj.
 b. *John was dated by George in high school.
 c. *George was made love with by Bill. (van Craenenbroeck and Johnson, 2026)

Similarly, pseudo-symmetric predicates are typically odd when passivized (237):

- (237) a. *Joel was chatted to by Clementine.
 b. *Joel was fallen in love with by Clementine.
 c. ?Joel was hugged by Clementine.
 d. ?Joel was kissed by Clementine.
 e. *Joel was broken up with by Clementine.

Crucially, passivization is licensed if the transitive describes a markedly asymmetric situation (238):

- (238) a. I hate being hugged by people I hardly know.
 b. Sleeping Beauty was kissed by a prince.

These data can be accounted for under the structural ambiguity hypothesis outlined above. If pseudo-symmetric predicates have a clearly asymmetric denotation, they are true transitives and can passivize. On the other hand, if they are underlyingly intransitive with a symmetric meaning, they disallow passivization.

I will argue against the ambiguity hypothesis, but I do not have a solution to the passivization data. The only counterargument I can muster here is that how unacceptable the passive is appears to depend on the individual lexical item. *Hug* and *kiss* are marginal in default contexts and perfectly acceptable in asymmetric situations, as seen in (238); *embrace* and *fuck* are similar. Other verbs are more strongly degraded in default contexts and more difficult to improve (239):

- (239) a. ?Joel looks like a guy who's been broken up with too many times.
 b. *You just want to be fallen in love with.

This variation is not accounted for by the structural ambiguity hypothesis. Nonetheless, the general puzzle persists.

To give an interim summary, the possibility we are considering here is that transitive pseudo-symmetric predicates can be underlyingly intransitive with a single θ -role, and that this is the construal that obtains whenever they can be used interchangeably with the surface intransitive form. I leave it open how and why an intransitive semantics would result from a transitive syntax, but it is not unlikely that some such mechanism needs to be assumed for strictly symmetric predicates anyway. The evidence from passivization does point in this direction for both sets of verbs.

The problem for such an analysis arises, again, from Kruitwagen (2024). Recall that in slightly asymmetric situations, it is possible that speakers judge the intransitive felicitous, but only one of the transitives, and specifically the one in which the more active participant is realized as the subject. If the transitives could be covertly intransitive, both of them should always be licensed when the intransitive is, but as Kruitwagen has shown, this is not the case. Even in situations where the intransitive can apply, the participants in the transitives still remain associated with slightly different agent and patient proto-properties. The transitives do not behave like covert intransitives.

Hence, the structural ambiguity hypothesis does not make the right predictions. However, open questions remain. First, the passivization data still require an explanation. Second, this entire case study has carefully circumvented strictly symmetric predicates, but much of what I have said about pseudo-symmetric predicates would apply to strictly symmetric predicates as well. If we were to give the same account of strictly symmetric predicates that I have adopted here for their pseudo counterparts – that the intransitive involves a single θ -role, but the transitive two –, then this would mean that θ -roles can be semantically bleached to the point where there is no difference in meaning between them anymore. I will later suggest in Chapter 5.1 that this is in principle possible, but whether this is an adequate analysis of strictly symmetric predicates is a question I leave to future work.

4.6 Statives of ingestives

4.6.1 Data and analysis

This final case study deals with statives, a construction that describes a state resulting from a verbal event. For instance, in (240a), the door is in the state resulting from an opening event. Statives are distinct from both eventive passives (240b), which describe events, and plain adjectives (240c), which describe simple states.

- | | | |
|-------|------------------------------------|-------------------------|
| (240) | a. The door is opened. | <i>stative</i> |
| | b. The door was opened (by Robin). | <i>eventive passive</i> |
| | c. The door is open. | <i>adjective</i> |

The argument that surfaces in statives normally corresponds to the internal argument of the transitive, not the external argument (241):

- | | |
|-------|------------------------------|
| (241) | a. The girl opened the door. |
| | b. The door is opened. |
| | c. *The girl is opened. |

The exception to this pattern are certain statives formed from ingestive verbs such as *eat*, *drink* and *learn*, which we already encountered in Chapter 4.3. The Greek example in (242) allows for two different interpre-

tations. Besides the pragmatically odd reading that Mary has been devoured – the interpretation we would expect –, the sentence can also mean that Mary has finished eating.

- (242) I Maria ine fayō-men-i.
 the.NOM Mary.NOM be.3SG eat-PTCP-F.NOM
 ‘Mary is eaten.’
 ‘Mary has eaten.’ (Paparounas, 2025:23)

Under this second reading, the argument of the stative corresponds to the external argument of the corresponding transitive (‘Mary ate something’) – what we might expect to be an agent argument. I will refer to the unexpected interpretation of the stative as the pseudo-agentive reading, and to its argument (here, ‘Mary’) as the pseudo-agentive argument. Other statives in Greek that allow for a pseudo-agentive reading are *pçomenos* ‘drunk,’ *ðiavazmenos* ‘studied’ and *maθimenos* ‘learnèd’ (Paparounas, 2025).

The same phenomenon is seen for Hindi-Urdu in (243) for the statives *parḥaa huaa* ‘read’ and *likhaa huaa* ‘written.’ Here, the argument ‘that man’ is interpreted not as having been read and written – an interpretation that is ruled out on obvious grounds –, but as having read and written something.²⁹

- (243) vo aadmii kaafii parḥ-aa likh-aa hu-aa hai.
 that man a.lot read-PFV write-PFV be-PFV.MSG be.PRS.MSG
 ‘That man is very educated.’ (Lit. ‘That man is very read-written.’)

Again, the argument corresponds to the external argument of the transitive assigned an agent θ -role (‘That man read and wrote’).

As a further complication, in Hindi-Urdu, pseudo-agentive statives of ingestives can take a second argument. Example (244) with the stative *pahnaa huaa* ‘worn’ contains not only a pseudo-agentive argument, ‘that woman,’ but also the additional argument ‘a sari.’

- (244) vo aurat saarīi pehn-e hu-e thii.
 that woman sari wear-PFV.OBL be-PFV.OBL be.PST.F
 ‘That woman was wearing a sari.’ (Lit. ‘That woman was sari-worn.’)

Other statives that allow for a pseudo-agentive interpretation in Hindi-Urdu, including with an additional argument, are *piyaa huaa* ‘drunk,’ *lagaayaa huaa* ‘worn,’ and *siikhaa huaa* ‘learnèd’ (Subbārāo, 2012, see also Zakharyin, 2020).

Pseudo-agentive statives of ingestives have been documented in a number of other languages, including Latin (Haspelmath, 1994), Italian, Hebrew and English (Arad, 1998). Some examples from English are shown in (245); for all of them, *Kim* would be the external argument of the corresponding transitive.³⁰

²⁹Note that in (243), the two verbal forms *parḥaa huaa* ‘read’ and *likhaa huaa* ‘written’ obligatorily occur together; the latter does not license a pseudo-agentive reading on its own.

³⁰*Travel* is not an ingestive verb in the strict sense. However, since the Romantics, Western societies have regarded travelling an experiential process associated with learning and personal growth, which makes *travel* a verb of non-literal ingestion akin to *experience* or *learn*. A similar argument can be made for Hindi-Urdu *likhaa huaa* ‘written,’ which is also not strictly an ingestive.

(245) Kim is drunk/experienced/learnèd/well-read/widely traveled.

In the following, I largely limit myself to Greek and Hindi-Urdu, with some scattered remarks on English.

I argue that the argument of pseudo-agentive statives of ingestives occupies the same position as the argument of regular statives, and that furthermore, it receives the same θ -role (see also Paparounas, 2025 for a similar proposal). I will describe this θ -role as the holder of a result state. For instance, in the Greek example in (242), Mary is in the result state of an eating event – typically, the state of being full. More generally, pseudo-agentive statives of ingestives are licensed because the argument that is realized as the subject of the transitive can be understood as ending up in a non-trivial state that results from the verbal event, unlike with non-ingestive transitives: an event participant who has eaten is in a markedly different state to before, whereas an event participant who has, say, opened a door is not.

To account for cases in which statives take an additional argument besides the pseudo-agentive argument – such as ‘a sari’ in (244) –, I make the novel claim that this additional argument is pseudo-incorporated into the verb: the woman is not in a wearing state, but in a sari-wearing state. What is more, I propose that in both Hindi-Urdu and Greek, pseudo-agentive statives of ingestives without such an overt additional argument still involve incorporation of a null argument interpreted with existential force, as is also commonly assumed for seemingly intransitive variants of ingestive verbs such as *Mary ate*. For instance, in the Greek example discussed earlier, Mary is in the state of having eaten *something*.

If this analysis is correct, pseudo-agentive statives constitute a case of variable linking. Suppose that the Greek verb ‘eat’ came with the individual role of an eater. I take it that in the transitive, the eater would be realized as an agent in the external argument position. But in the stative, I argue that it is realized as a result state holder in the same argument position as the arguments of regular statives. Thus, the same individual role would be mapped onto different θ -roles in the two alternants.

I must warn the reader that this final case study will indulge in even more hedging and agnosticism than the previous ones, and I will not attempt to deliver a fully developed analysis of pseudo-agentive statives of ingestives, or statives in general, for that matter. My goal in the following is merely to offer a tentative discussion of these examples from the perspective of variable linking in the hope that it will prove fruitful in future work. Pseudo-agentive statives are a genuinely intriguing data set, and if none of the case studies so far has made it clear yet that the research question of this study matters for the analysis of specific argument-structural alternations, this one certainly should.

I will proceed as follows. I begin by introducing statives in general by contrasting them with eventive passives on the one hand and adjectives on the other. I then argue that in pseudo-agentive statives of ingestives, the pseudo-agentive argument does not occupy the canonical agent position, SpecVoiceP, and that it is not interpreted as an agent but as a holder of a result state. Lastly, I discuss the only previous analysis of pseudo-agentive statives that I am aware of, Paparounas (2025).

4.6.2 Evidence for the analysis

4.6.2.1 Background

This section introduces statives by comparing them to eventive passives on the one hand and simple adjectives on the other, drawing mainly on Anagnostopoulou (2003), Embick (2004a), and Kratzer (2000). I begin by describing the surface properties of the three constructions in English, Greek and Hindi-Urdu, then discuss their event semantics in more detail, and lastly address the thematic interpretation of their arguments.

Introducing statives Example (246) shows an English stative (246a) together with the corresponding eventive passive (246b) and adjective (246c). The stative describes a state resulting from an event, the eventive passive an event, and the adjective a pure state.

- | | | |
|-------|------------------------------------|-------------------------|
| (246) | a. The door is opened. | <i>stative</i> |
| | b. The door was opened (by Robin). | <i>eventive passive</i> |
| | c. The door is open. | <i>adjective</i> |

Morphologically, English statives are always syncretic with the eventive passive. For a few verbs, they are also syncretic with the corresponding adjective (e.g., *closed*), but for others, as in (246), the stative and the adjective are distinct.

Distributionally, statives pattern with adjectives and differ from eventive passives in that they are able to surface in attributive position (e.g., *the opened door*). In the following, I only consider statives in predicative position. It has been shown that attributive statives in at least some languages can receive, oddly enough, an eventive passive interpretation (see Biggs and Embick, 2025b for English), which presents a potential confound. For simplicity's sake, I thus leave attributive statives aside.³¹

A note on terminology. In the literature, statives have often been described as ‘stative *passives*’ (e.g., Biggs and Embick, 2025b; Embick, 2023; Paparounas, 2025) or ‘adjectival *passives*’ (e.g., Alexiadou et al., 2015; Meltzer-Asscher, 2011).³² The term ‘passive’ describes an argument-structural configuration characterized by the presence of an existentially closed external argument. Eventive passives are thus defined both by their eventive semantics and by the presence of an implicit external argument. For instance, (246b) easily licenses a *by*-phrase. However, not all statives are passives in this sense, as we will also see below. Hence, I use ‘stative’ for any verbal form that describes a state resulting from an event, remaining neutral with respect to its argument structure. Note that my usage of this term should not be taken to include verbs like *know*, which denote pure states.

³¹Paparounas (2025) reports that in Greek, a pseudo-agentive reading is only available for predicative, not attributive statives. I have no explanation for this asymmetry.

³²As a further terminological conundrum, Embick (2004a) uses ‘stative’ to describe what I call a simple adjective, and ‘resultative’ for what I call a stative. The infelicity of the term ‘resultative,’ which is more commonly used for resultative secondary predicates, is already noted by Embick himself.

Turning to Greek, the language uses distinct morphological markers for statives, eventive passives and adjectives. Statives are formed analytically with an auxiliary and a specialized participial affix *-men* (247a). In contrast, eventive passives are synthetic, bearing the same non-active morphology that surfaces on reflexives and some unaccusatives (247b). Pure adjectival states are formed with yet another suffix, *-t* (247c).

- (247) a. I porta ine aniy-men-i.
the door be.PRS.3SG open-PTCP-F.NOM
'The door is opened.' *stative*
- b. I porta anix-θ-ik-e (apo tin Maria).
the door open-PFV.NACT-PST-3SG by the Mary
'The door was opened (by Mary).' *eventive passive*
- c. I porta ine anix-t-i.
the door be.PRS.3SG open-ADJ-F.NOM
'The door is open.' *adjective*

The situation in Hindi-Urdu is more complex. Statives are formed with the perfective form of the verb together with the auxiliary *huua*. In the following, I gloss the latter as perfective ‘be,’ but note that in the context of statives, there is no actual perfective reading. The reader might recall from previous case studies that Hindi-Urdu systematically distinguishes between intransitive and causative verbs morphologically; in (248), the contrast is based on vowel length (*kaṭe* vs. *kaaṭe*). Statives can be formed from both unaccusatives and causatives (248), although statives of causatives can be perceived as degraded for reasons I touch on later:

- (248) a. yeh phal (*Mina-dwaaraa) kaṭ-e hu-e haĩ.
 this fruit by-Mina cut.UNACC-PFV.MPL be-PFV.MPL be.PRS.MPL
 ‘These fruits are cut (*by Mina).’ *stative of unaccusative*
- b. ?yeh phal Mina-dwaaraa kaaṭ-e hu-e haĩ.
 this fruit Mina-by cut.CAUS-PFV.MPL be-PFV.MPL be.PRS.PL
 ‘These fruits are cut by Mina.’ *stative of causative*

As indicated both by the verbal morphology – unaccusative in (248a), causative in (248b) – and the availability of *by*-phrases, only (248b) is a stative *passive* in the strict sense, while (248a) is a stative unaccusative.³³

Without the auxiliary *huaa*, a stative reading of the perfective form of the verb is in principle still available, but the more easily accessible interpretation is that of an unaccusative (249a) or causative (249b) active perfect. In the causative (249b), the subject can be missing due to pro-drop.

³³With causative verbs, the stative auxiliary *huaa* can also co-occur with an overt subject (i):

- (i) yeh phal kisi-ne kaat-e hu-e haĩ.
 this fruit someone-ERG cut.CAUS-PFV.MPL be-PFV.MPL be.PRS.PL
 ‘Someone has cut these fruits, and they are now in a cut state.’

Thus, (i) appears to be a stative active transitive. Note that this is a different construction from pseudo-agentive statives of ingestives. Besides the fact that it is not restricted to ingestive verbs, the subject here surfaces with ergative case, and the verb and auxiliary regularly agree with the object (for agreement with pseudo-agentive statives of ingestives, see Chapter 4.6.2.2). I leave (i) to future research.

Statives result in a contradiction after verbs of creation such as *build*, *create* or *make*, which assert that the state held from the beginning (252a). Simple adjectival states are acceptable in these contexts (252b).

- (252) a. *This door was built opened. *stative*
 b. This door was built open. *adjective*

The same obtains in Greek. The stative of ‘boil,’ formed with *-men*, cannot surface after a verb of creation (253a). However, Greek also has a corresponding adjective, formed with *-t*, which is acceptable in this context (253b).

- (253) a. *Eftiak-sa to kotopoulo vras-men-o.
 made-1SG the chicken boil-PTCP-N.NOM
 Intended: ‘I made the chicken boiled.’ *stative*
 b. Eftiak-sa to kotopoulo vras-t-o.
 made-1SG the chicken boil-ADJ-N.NOM
 ‘I made the chicken boiled.’ *adjective*

(Anagnostopoulou, 2003:14–15)

For Hindi-Urdu, where adjectives are syncretic with unaccusative statives, I exemplify this contrast with an unaccusative stative that does not have a corresponding adjective (‘washed’) and an adjective that does not have a corresponding verbal form (‘clean’) (254). Only the latter can surface with the verb ‘produce’:

- (254) a. *yahãã kapre dhul-e ban-aa-ye jaa-te haĩ.
 here clothes wash.UNACC-PFV.MPL make-CAUS-PFV.MPL PASS-IMPV.MPL be.PRS.3PL
 Intended: ‘Here, clothes are produced washed.’ *stative of unaccusative*
 b. yahãã kapre saaf ban-aa-ye jaa-te haĩ.
 here clothes clean make-CAUS-PFV.MPL PASS-IMPV.MPL be.PRS.3PL
 ‘Here, clothes are produced clean.’ *adjective*

Moreover, unlike adjectives, statives allow for adverbial modification usually reserved for verbs (255)–(257):

- (255) a. The package remained carefully opened. *stative*
 b. *The package remained carefully open. *adjective*
- (256) a. Ta keftedakia ine prosektika tiganis-men-a.
 the meatballs are carefully fry-PTCP-N.PL.NOM
 ‘The meatballs are fried carefully.’ *stative*
 b. *Ta keftedakia ine prosektika tigani-t-a.
 the meatballs are carefully fry-ADJ-N.PL.NOM
 Intended: ‘The meatballs are fried carefully.’ *adjective*

(Anagnostopoulou, 2003:15)

- (257) a. yeh picture baarikii-se ban-i hu-i hai.
 this drawing.FSG meticulousness-INST make-PFV.FSG be-PFV.FSG be.PRS.3SG
 ‘This drawing is made meticulously.’ *stative of unaccusative*
- b. *yeh picture baarikii-se sundar hai.
 this drawing.FSG meticulousness-INST beautiful be.PRS.3SG
 Intended: ‘This drawing is beautiful in a meticulous way.’ *adjective*

Kratzer (2000) has taken this to indicate that unlike simple adjectives, statives contain an event variable that can be accessed by the modifier: they denote states that result from events, not simply states as such.

However, this conclusion has been called into question. Embick (2023) has pointed out that many adverbial modifiers that are acceptable with actives and eventive passives are severely degraded with statives (258):

- (258) a. The Romans destroyed the city rapidly/2035 years ago/with fire arrows.
 b. The city was destroyed rapidly/2035 years ago/with fire arrows/by the Romans.
 c. #The city is destroyed rapidly/2035 years ago/with fire arrows/by the Romans.

(Embick, 2023:289)

Embick notes further that such modifiers become available with statives under a ‘state relevance’ reading; that is, if they can be interpreted as describing the state resulting from the event (see also Alexiadou et al., 2015; Gehrke, 2015; McIntyre, 2015; Meltzer-Asscher, 2011; Rapp, 1996). For instance, *The city is destroyed with fire arrows* is felicitous if the fire arrows have left visible scorch marks on the city walls. The same point is made by the contrast in (259):

- (259) a. The note is hastily typed.
 b. ??The note is slowly typed.

If a note is typed hastily, this might result in typos or other signs of carelessness, whereas a note does not typically betray whether or not it was slowly typed. That is, the felicity of the adverb depends on the availability of a state relevance reading. Embick concludes that, *pace* Kratzer, adverbial modification with statives does not actually target the event variable, but the state variable.

At the same time, of course the fact remains that the adverb must describe the event, even if the manner in which the event was executed is reflected in the state. For instance, in (259b), it is the typing event that is slow; states cannot be slow. What is more, we still need to account for Kratzer’s observation that such adverbial modifiers are unacceptable with adjectives altogether. In other words, the data from adverbial modification do support the idea we started with that adjectives denote simple states whereas statives denote states resulting from events. However, this eventive component is not directly accessible to modification in the way that it is in eventive passives and actives.³⁵

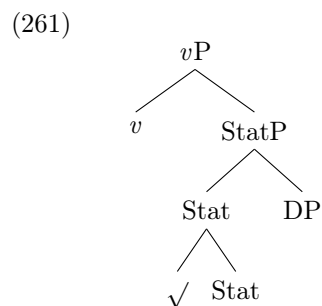
³⁵To account for this finding, Embick (2023) proposes a complex head analysis of statives, according to which the eventive *v* head that has merged with the root immediately combines with a stative Stat head without projecting a *v*P layer. Embick assumes that a head that merges with a phrase would necessarily project a phrase. Accordingly, *v* cannot combine with (phrasal)

Turning to the contrast between eventive passives and statives, an eventive passive such as (260a) describes a past event which, in the case of the change-of-state verb *open*, resulted in a state, but nothing requires the state to still hold. On the other hand, a stative like (260b) asserts the existence of a state.

- (260) a. The door was opened (by Robin).
 b. The door is opened.

But this neat dichotomy between events on the one hand and states resulting from events on the other is complicated by the fact that change-of-state verbs like *open* and *break* are usually assumed to denote complex events consisting of several subeventualities, including a result state. At the same time, as discussed just now, statives also appear to contain an eventive component. In a sense, both eventive and statives thus denote a state resulting from an event, raising the question how to represent the difference between the two.

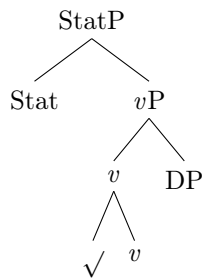
A solution commonly adopted in the literature is to reverse the order of the eventive and stative subeventualities in the two constructions. A simplified version of this analysis is shown in (261) and (262). In eventive actives and, by extension, eventive passives of change-of-state verbs, the root first combines with a head depicted here as *Stat*, which introduces a state. This head then merges further with a *v*, which existentially closes the state and introduces an event that causes, or ends in, the state (e.g., Alexiadou et al., 2006; von Stechow, 1996). The resulting construction thus denotes an event that results in a state.



In contrast, in statives, the root would first merge with *v*, introducing an event, and then combine with *Stat*. The latter existentially closes the event and introduces a state which is caused by or results from this event (262) (e.g., Anagnostopoulou, 2003; Biggs and Embick, 2025b; Embick, 2004a, 2023; Paparounas, 2025). Thus, the stative denotes a state that results from an event. Note that analyses differ with respect to whether *v* projects and whether the DP is merged as an argument of *v* or *Stat* (see also footnote 35); the tree shown in (262) is based on Embick (2004a).

modifiers because this would project a *vP*, running counter to the complex head structure that statives require. This complex head analysis is taken up by Paparounas (2025), discussed in Chapter 4.6.3.1.

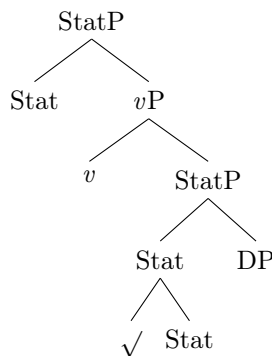
(262)



However, this analysis also faces some difficulties. Suppose that the root merged in (262) is *open*. Thus, the *v* that has merged with the root would denote an opening event that does not result in a state. It is not obvious that there could be such a thing – an opening might seem to be intrinsically defined by the state it results in.

An alternative possibility would be to combine the representations in (261) and (262) and argue that statives have the structure in (263). Here, we begin with the root denoting a state, then add a causing event, and top the resulting structure with a second StatP layer.

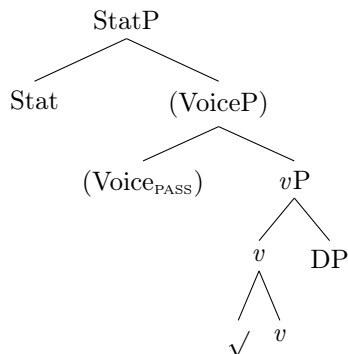
(263)



But now, the question arises why exactly the lower and the higher state – the state of being open and the state of being opened – match each other. Unlike in (262), the Stat head merging with *vP* here does not simply close the event variable and introduce a result state, but this result state needs to somehow correspond to the state already introduced earlier by the lower Stat head. Nothing in (263) seems to intrinsically enforce this.

To summarize, statives contain both an eventive and a stative component, but unlike in eventive passives, it is the stative component that is asserted while the eventive component is backgrounded and not directly accessible to modification. As flagged earlier, I will not endorse any particular analysis of the event structure of statives here. However, for the sake of concreteness, the reader is welcome to tentatively assume the tree in (264) as a rough structure for statives, despite the problem for this analysis raised earlier. I bracket the VoiceP layer for reasons that will become apparent in Chapter 4.6.2.2.

(264)



Against this background, we can now turn to θ -roles.

θ -roles This section discusses the θ -role assigned to the (overt) argument of adjectives, eventive passives and statives, such as *the door* in (265):

- | | | | |
|-------|----|---------------------------------|-------------------------|
| (265) | a. | The door is opened. | <i>stative</i> |
| | b. | The door was opened (by Robin). | <i>eventive passive</i> |
| | c. | The door is open. | <i>adjective</i> |

I begin by showing in purely descriptive terms that the arguments of statives, eventive passives and adjectives have overlapping but not wholly identical thematic properties. At the end of this section, I then discuss how this empirical picture can be captured formally.

One property that Dowty (1991) identified as being associated with proto-patients is undergoing a change of state. I take it that ‘undergoing a change of state’ means that the argument ends up in a result state. For instance, in *Mei broke the glass*, the glass ends up in the state of being broken. Of course, not all patient arguments in Dowty’s sense qualify for this property. For instance, the object argument in *Toma kicked the tires* does not since there is nothing that systematically distinguishes kicked from unkicked tires, other than the fact that the former happened to have undergone a kicking at some point in the past. Being kicked is not in and of itself a result state. In short, the argument that surfaces overtly in eventive passives needs to be a patient, but this does not necessarily mean that it undergoes a change of state.

This contrasts with the argument of statives. As is well known, statives are infelicitous if the argument does not undergo a change of state.³⁶ Accordingly, (266) is odd:

- (266) #The tires are kicked.

³⁶Parsons (1990) has proposed a distinction between ‘target states’ and ‘resultant states,’ where a target state corresponds roughly to what I call here a result state – e.g., the state of the glass being broken – while a resultant state is the state that an argument trivially finds itself in just by virtue of having participated in the event. E.g., in *Toma kicked the tires*, the tires are in the resultant state of having been kicked. Kratzer (2000) has argued that statives can denote both target states and resultant states, and that target statives and resultant statives differ in their syntax and semantics (see also Alexiadou et al., 2015; Anagnostopoulou, 2003). I take this claim to have been disproven by Paparounas (2025); see also Biggs and Embick (2025a). The argument of statives needs to be in a genuine result (i.e., ‘target’) state; being in the trivial state of having undergone an event is not sufficient to license this construction.

At the same time, it is also well-known that examples such as (266) become felicitous if a salient result state can be induced from the context. For instance, (266) could be uttered in a tire factory, where all tires need to be kicked for reasons of quality control before leaving the factory (e.g., Embick, 2004a; Kratzer, 2000, a.m.o.). Biggs and Embick (2025a) have shown that under this reading, (266) is indeed a true stative that denotes a state resulting from an event. Overall, for a stative to be felicitous, the argument must undergo a change of state.

Lastly, turning to adjectives, their arguments are holders of states, but not all states are result states. For instance, not all things that are open were opened, and not all things that are black used to be something other than black at some point in the past.

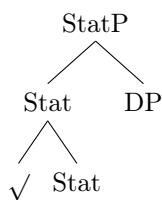
To recap, the overt argument of an eventive passive, as well as of an eventive active, can be associated with the wide range of thematic properties that Dowty (1991) described with the concept of a proto-patient, which might or might not include undergoing a change of state. In contrast, the argument of a stative must always undergo a change of state, while the argument of an adjective must be a holder of a state, which might or might not result from a change of state.

Against this background, consider the fact that it is the ‘same’ argument that surfaces with eventive passives, statives and adjectives, and that it corresponds to the internal argument of the transitive (302):

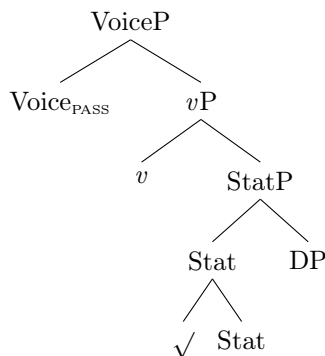
- (267) The girl opened the door.
- | | | |
|----|----------------------------|-------------------------|
| a. | The door/*girl is opened. | <i>stative</i> |
| b. | The door/*girl was opened. | <i>eventive passive</i> |
| c. | The door/*girl is open. | <i>adjective</i> |

This parallel can in principle be captured in two different ways. One is to argue that in all three constructions, the argument is related to the same eventuality by the same θ -role. A very simple implementation of this idea is shown in (268)–(270). Here, the DP argument is always merged with a pure state – being perhaps assigned the θ -role of a holder –, and the differences between the three constructions arise from the material that is built on top. Note that the structure for the eventive passive in (269) is specifically for a change-of-state verb like *open*; the eventive passive of a verb like *kick* that does not contain a result state would of course be different.

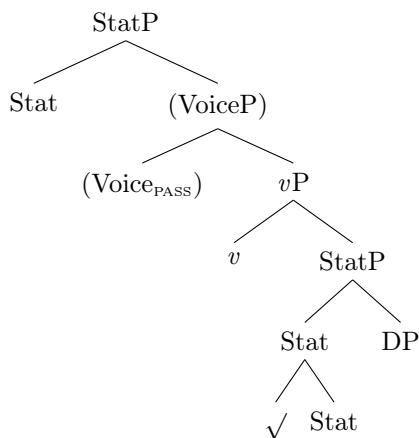
- (268) *Adjective*



(269) *Eventive passive (for change-of-state verbs)*



(270) *Stative*



The correspondence between the arguments of statives, eventive passives and adjectives could thus be directly read off the syntax. While it might seem appealing, this view is not without challenges; for instance, I have noted problems earlier with (270). In general, the approach makes certain predictions about what the structure of statives, eventive passives and adjectives can be, and these predictions might not be borne out.

The alternative approach, more commonly adopted in the literature, is by means of correspondence rules (e.g., Biggs and Embick, 2026; Paparounas, 2025; Parsons, 1990; Williams, 2015). Under this view, the arguments of statives, eventive passives and adjectives are arguments of different eventualities and/or assigned different θ -roles, but an additionally imposed rule requires these arguments to correspond to each other. For instance, Parsons posits that the theme of a BECOME event must be identical to the theme of the resulting state. Given the discussion in this section so far, such a correspondence rule could be motivated by overlapping proto-properties, but what exactly the status of these rules in the grammar would be is an open question.

In short, matters are complicated. In the following, I assume that the arguments of statives are assigned a θ -role, and that this role can be meaningfully described as the holder of a result state. I leave it open whether this role is a primitive or can be reduced to that of a patient or that of a holder, and how it relates

in formal terms to the θ -role of eventive passives and of adjectives. What is crucial for our purposes is that as I show later in Chapter 4.6.2.3, being the holder of a result state is exactly the interpretation assigned to the pseudo-agentive argument of statives. First, however, we turn to syntactic evidence for the proposal.

4.6.2.2 Syntactic diagnostics

The main claim of this case study is that under the pseudo-agentive reading of examples like (271) and (272), the argument does not in fact receive an agent θ -role. Since I take it that in the corresponding transitive, the argument is assigned an agent θ -role, the alternation constitutes a case of variable linking.

- (271) I Maria ine fayō-men-i.
 the.NOM Mary.NOM be.3SG eat-PTCP-F.NOM
 ‘Mary is eaten.’
 ‘Mary has eaten.’
- (272) vo aadmī kaafī parh-aa likh-aa (hu-aa) hai.
 that man a.lot read-PFV write-PFV be-PFV.MSG be.PRS.MSG
 ‘That man is very educated.’

In support of this claim, this section argues that pseudo-agentive arguments – ‘Mary’ in (271), ‘that man’ in (272) – do not occupy the syntactic position canonically associated with an agent interpretation, SpecVoiceP. On some level, this argument is difficult to make since standard unaccusativity diagnostics do not apply in this environment. For instance, the three unaccusativity diagnostics I used earlier for Hindi-Urdu in Chapter 4.2 on direct causatives of unergatives – impersonal passives, reduced relative clauses, and active vs. passive syntax with inabilitatives – cannot be used in the context of statives.

Instead, my argument will be as follows. It is well-known that arguments which occupy SpecVoiceP in the transitive normally cannot surface in statives (273), a pattern only defied by pseudo-agentive statives.³⁷ If statives were compatible with overt external arguments, the ungrammaticality of (273b) would be unexpected.

- (273) a. The girl opened the door.
 b. *The girl is opened.

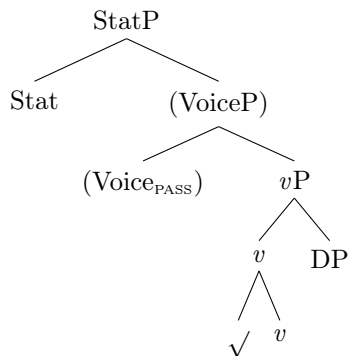
I have no explanation for why this generalization holds, but it is robust. Thus, unless there are any compelling syntactic data indicating that pseudo-agentive arguments defy this generalization, it would be stipulative to simply posit that they occupy SpecVoiceP.

In this section, I look at data which might be taken as evidence that pseudo-agentive arguments are merged in SpecVoiceP after all. Specifically, I address two issues: the question of passive Voice in statives, and the status of additional arguments that can surface in pseudo-agentive statives in Hindi. In both cases, I conclude that the data do not contradict my analysis.

³⁷Another exception are the statives of active transitives seen for Hindi-Urdu in footnote 33, but these constructions are transitives with an ergative-marked subject; as noted, this is clearly a different construction.

No obligatory passive Voice I begin by examining the question of whether pseudo-agentive statives of ingestives contain passive Voice. Recall that in the tentative structure for statives seen earlier, I bracketed the VoiceP layer (274):

(274)



The previous literature has been divided on the question of whether statives contain a passive Voice head introducing an existentially closed external argument. For Greek, Paparounas (2025) has argued that all statives are unaccusatives without Voice, whereas Alexiadou et al. (2015) and Anagnostopoulou (2003) have argued that some statives in the language are true passives and contain Voice.

We need to consider what predictions a passive analysis would make for pseudo-agentive statives of ingestives. The relevant examples from Greek and Hindi-Urdu are shown again in (275) and (276):

- (275) I Maria ine fayō-men-i.
 the.NOM Mary.NOM be.3SG eat-PTCP-F.NOM
 ‘Mary is eaten.’
 ‘Mary has eaten.’

- (276) vo aadmī kaafī paṛh-aa likh-aa (hu-aa) hai.
 that man a.lot read-PFV write-PFV be-PFV.MSG be.PRS.MSG
 ‘That man is very educated.’

I have argued here that pseudo-agentive arguments – ‘Mary,’ ‘that man’ – occupy the same position as the overt arguments of regular statives. In (274), this would be the complement position of *v*. However, under the passive analysis, the question would then arise how the implicit external argument is interpreted, or in other words, what it would mean to be the agent of an event that results in Mary having finished eating, or of the man being educated. It might seem that such an argument could not be assigned an interpretation, which would pose a problem for the analysis proposed here. At least, the interpretation of this argument would need to be clarified. In the following, I show that while a passive analysis of statives might be correct for some statives in Greek, and certainly is for some statives in Hindi-Urdu, neither language requires statives to contain passive Voice. Stative unaccusatives are clearly attested. Accordingly, I propose that pseudo-agentive statives also do not contain Voice. As a result, the challenge outlined above does not arise.

To begin with Hindi-Urdu, we have already seen that statives can surface either with unaccusative or with causative morphology (277). In the latter case, they can also take a *by*-phrase, indicating the presence of an implicit external argument.

- (277) a. yeh phal (*Mina-dwaaraa) kaṭ-e hu-e haĩ.
 this fruit by-Mina cut.UNACC-PFV.MPL be-MPL be.PRS.MPL
 ‘These fruits are cut (*by Mina).’ *stative of unaccusative*
- b. ?yeh phal Mina-dwaaraa kaat-e hu-e haĩ.
 this fruit Mina-by cut.CAUS-PFV.MPL be-PFV.MPL be.PRS.PL
 ‘These fruits are cut by Mina.’ *stative of causative*

Interestingly, the *by*-phrase in (277b) is subject to state relevance effects. That is, the sentence is only felicitous if the fact that it was specifically Mina who cut the fruits affects the state that the fruits are in. For instance, Mina might always cut fruits very creatively. Looking at fruits cut in a creative way, a speaker can utter (277b).

This ties in with my earlier observation that statives of causatives – true stative *passives* – are sometimes perceived as degraded. If speakers simply want to assert that the fruits are in a cut state, the unaccusative stative is more felicitous. On the other hand, if what is to be communicated is that the fruits were cut by someone – e.g., Mina –, speakers would opt for the eventive passive. The stative of the causative is only appropriate if the fruits are specifically in the state of having been cut by a particular event participant, in that the fact that it was one person and not another who cut the fruits affects the state that the latter are in. In contrast, *by*-phrases are acceptable with eventive passives regardless of state relevance effects, and they are strictly unacceptable with unaccusative statives such as (277a), again regardless of state relevance effects.

In sum, Hindi-Urdu statives can but do not have to contain Voice, but even with true stative passives, the availability of *by*-phrases is subject to state relevance.³⁸ As seen earlier for adverbial modification with statives in (258), *by*-phrases technically modify the eventive component of statives, but they are only licensed if they have a bearing on the state as well.

Turning to Greek, Anagnostopoulou (2003) has argued on the basis of evidence from agent-oriented modifiers, instruments and *by*-phrases that some statives in the language do contain a passive Voice head introducing an implicit external argument (278) (see also Alexiadou et al., 2015):

- (278) I porta ine aniy-men-i viea / me losto / apo ton ðiarikti.
 the.NOM door.NOM be.3SG open-PTCP-F.NOM violently / with crowbar.ACC / by the burglar.ACC
 ‘The door is opened violently/with a crowbar/by the burglar.’ (Paparounas, 2025:44)

This evidence has been questioned by Paparounas (2025), who has pointed out that these agent-oriented

³⁸Note that if it is assumed that *by*-phrases always attach at the level of Voice, this would entail that the complex head analysis proposed by Biggs and Embick (2025b), Embick (2023), and Paparounas (2025) (see footnote 35) could not apply for true stative passives: having combined with a *by*-phrase, Voice would need to project, which is incompatible with the claim that Stat (above Voice) must be a complex head.

diagnostics are subject to state relevance, shown in (279) for *by*-phrases. The sentence in (279a) is only felicitous if the fact that it was Roman soldiers who destroyed the city affects the state the city is in now. In contrast, no state relevance effect obtains for these diagnostics in eventive passives such as (279b).

- (279) a. I poli ine katestra-men-i (#apo Romeus stratiotes).
 the.NOM city.NOM be.3SG destroy-PTCP-F.NOM from Roman soldiers
 ‘The city is destroyed (by Roman soldiers).’ *stative*
- b. I poli eçi katastrafi (apo Romeus stratiotes).
 the.NOM city.NOM have.3SG destroy.NACT.PFV from Roman soldiers
 ‘The city has been destroyed (by Roman soldiers).’ *eventive passive*

(Paparounas, 2025:48)

Paparounas takes these data to indicate that Greek statives do not contain passive Voice.

This conclusion does not seem necessarily warranted. We have seen that in Hindi-Urdu, state relevance effects on *by*-phrases obtain even with statives that are unambiguously formed from morphologically transitive verbs. Just like in Greek, Hindi-Urdu statives differ in this respect from eventive passives. Thus, the mere fact that diagnostics indicating the presence of an implicit external argument are only licensed with statives if this external argument affects the state described does not allow us to conclude that there is no passive Voice. By the same token, we have seen earlier that eventive modifiers with statives are subject to state relevance effects, but this does not entail that statives have no eventive component at all.

However, just as in Hindi-Urdu, this does not mean that all Greek statives are passives, as is well-established in the literature (e.g., Alexiadou et al., 2015; Anagnostopoulou, 2003; Lascaratou and Philippaki-Warburton, 1984). Greek allows statives of non-alternating unaccusatives such as ‘break’ that do not form an eventive passive:

- (280) a. To bukali ine spas-men-o.
 the bottle is break-PTCP-N.NOM
 ‘The bottle is broken.’ *stative passive*
- b. To bukali espase (*apo enan pelati).
 the bottle break.ACT.3SG (from one customer)
 ‘The bottle broke (*by a customer).’ *eventive active*
- c. *To bukali spastike (apo enan pelati).
 the bottle break.NACT.3SG (from one customer)
 Intended: ‘The bottle was broken (by a customer).’ *eventive passive*

It might be true that some Greek statives are passives, but as in Hindi-Urdu, not all of them are.

However, the reader might still insist that if Greek and Hindi-Urdu have both passive and non-passive statives, those formed from ingestives certainly must be in the former category since they are formed from obligatorily transitive verbs. But this does not entail that they obligatorily take a SpecVoiceP argument. By way of analogy, a variable behavior intransitive obligatorily takes at least one argument, but this argument does not always correspond to the same syntactic position or the same θ -role. There is a constraint on the

minimal *number* of arguments, but this constraint itself says nothing about where an argument is realized syntactically. By the same token, the fact that ingestive verbs like *eat*, *drink*, *read*, etc., always take at least two arguments does not entail that one of them always needs to be a SpecVoiceP argument, or, for that matter, an agent argument.

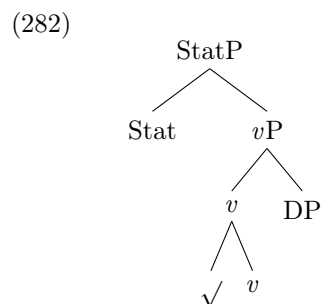
To summarize, I have shown that in general, both Greek and Hindi-Urdu allow for stative unaccusatives, and I have proposed that pseudo-agentive statives in these languages do not contain passive Voice. Accordingly, the question of how the implicit external argument would be interpreted in these constructions does not apply. The next section addresses another challenge for the analysis proposed here.

Pseudo-noun incorporation To give an interim summary, statives normally surface with the argument that corresponds to the internal argument of the active (281):

- (281) a. The girl opened the door.
 b. The door is opened.
 c. *The girl is opened.

Pseudo-agentive statives of ingestives at first sight defy this pattern. However, I have argued that the pseudo-agentive argument in fact occupies the same position and receives the same θ -role as regular arguments of statives. Pseudo-agentive arguments have no syntactic properties that would set them apart from other stative arguments, and I later also show that in terms of thematic properties, the argument is not interpreted as an agent, but as a result state holder.

But in addition, pseudo-agentive statives of ingestives upset our expectations about how statives should behave in yet another way. Recall our tentative tree for statives in (282); in line with the above discussion, I omit the VoiceP layer for pseudo-agentive statives:



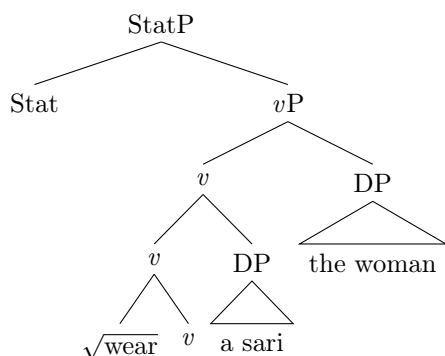
This tree contains a single overt argument position. But as seen earlier, Hindi-Urdu statives can surface with two arguments; in (283), ‘that woman’ and ‘a sari’:

- (283) vo aurat saari pehn-e (hu-e) thii.
 that woman sari wear-PFV.OBL be-PFV.OBL be.PST.F
 ‘That woman was wearing a sari.’ (Lit. ‘That woman was sari-worn.’)

In such examples, the position occupied by arguments of ordinary statives – in (282), the DP in the complement position of v – might seem to host ‘a sari.’ If this were the case, then the question would arise what position the argument ‘that woman’ is in, and a natural candidate would be SpecVoiceP. This is the claim I argue against.

Instead, I propose that arguments such as ‘a sari’ are pseudo-incorporated into the verb. A sketch of this analysis is shown in (284), still assuming the structure in (283). Here, ‘that woman’ occupies the regular position of stative arguments, the complement position of v , but ‘a sari’ has incorporated into the v :

(284)



To reiterate, this is a tentative structure, but the key idea is that the woman is not in a wearing state, but in a sari-wearing state: just as in the transitive, she is an argument of wearing a sari. What is more, I will argue that both in Hindi-Urdu and in Greek, pseudo-agentive statives that do not contain an overt second argument still involve incorporation, but of a null implicit argument. These claims are novel, and when we revisit statives of ingestives later in Chapter 5.2.5, (pseudo-)incorporation will turn out to play a key role.

To begin with overt pseudo-incorporation, Hindi-Urdu is what Dayal (2011) calls a ‘liberal pseudo-incorporation language’ (see also Dayal, 1999; T. Mohanan, 1995). Pseudo-incorporation does not lead to a change in valency, it can target phrasal arguments, including those modified by adjectives, and the incorporated argument does not need to be adjacent to the verb. One characteristic property of pseudo-incorporated arguments that does obtain in the language is that they obligatorily refer to non-definites rather than to specific entities. For instance, in (285), there is no particular sari that the speaker has in mind; rather, the woman is wearing some unspecified sari.

- (285) vo aurat saarii pehn-e (hu-e) thii.
 that woman sari wear-PFV.OBL be-PFV.OBL be-PST.FSG
 ‘That woman was wearing a sari.’ (Lit. ‘That woman was sari-worn.’)

Besides interpretation itself, the obligatorily non-definite interpretation of the argument can be corroborated further using two diagnostics: case marking and discourse anaphora. To begin with case marking, inanimate objects in Hindi-Urdu optionally bear the direct object marker *-ko*. Objects with *-ko* are interpreted as definites, those without as narrow scope indefinites (286) (Butt, 1993):

- (286) Anu kitaab/kitaab-ko parh-egii.
 Anu book/book-DOM read-FUT
 ‘Anu will read a book/the book.’ (Dayal, 2011:127)

In pseudo-agentive statives of ingestives, the argument that I argue to be pseudo-incorporated cannot take direct object marking (287):

- (287) vo aurat saari-(*ko) pehn-e (hu-e) thii.
 that woman sari-DOM wear-PFV.OBL be-PFV.OBL be.PST.FSG
 Intended: ‘That woman was wearing the sari.’

This indicates that the object is obligatorily interpreted as a narrow scope indefinite, not as a definite object.

Second, since incorporated arguments cannot refer to specific objects, they are normally not felicitous as antecedents for discourse anaphora. This is shown in (288) for the expression ‘to girl-look,’ meaning ‘to look for a prospective bride.’ Here, the pseudo-incorporated argument ‘girl’ cannot be referred back to by the genitive pronoun *us-kaa* ‘her.’

- (288) Anu apne bete keliye lar̥kii dekh rahii hai. #vo us-kaa swabhaav jaannaa caahthii
 Anu self’s son for girl look PROG be.PRS.3SG she she-GEN nature to.know want.IMP
 hai.
 be.PRS.3SG
 Intended: ‘Anu is girl-looking (looking for prospective brides) for her son. She wants to know her (i.e. the girl’s) temperament.’ (Dayal, 1999:46)

The same holds for the additional argument in pseudo-agentive statives of ingestives. Example (289) contrasts a regular active transitive (289a) with a stative containing what I argue is a pseudo-incorporated argument (289b):³⁹

- (289) a. Mina-ne laal juute pehn-e the. ve bahut mehenge the.
 Mina-ERG red shoe.MPL wear-PFV.MPL be.PST.MPL they very expensive.MPL be.PST.MPL
 ‘Mina wore red shoes. They were very expensive.’
 b. Mina laal juute pehn-e hu-e thii. #/?ve bahut mehenge
 Mina red shoe.MPL wear-PFV.OBL be-PFV.OBL be.PST.FSG they very expensive.MPL
 the.
 be.PST.MPL
 Intended: ‘Mina was red-shoes-wearing. They were very expensive.’

In the active transitive in (289a), ‘red shoes’ is a regular direct object which can be referred back to by the demonstrative *ve* ‘they’ in the following sentence. In contrast, in the stative (289b), the demonstrative cannot refer back to ‘red shoes.’ This is as expected if the argument is pseudo-incorporated and thus cannot denote a specific entity.

³⁹Dayal (2011) notes that discourse anaphora is not always a reliable test for pseudo-incorporation with inanimates and that furthermore, it can be sensitive to aspect. However, since (289) is a minimal pair, I take it that the diagnostic is probative in our case.

In addition, pseudo-agentive statives with an incorporated argument have another curious property related to their morphological realization. Normally, in predicative statives, the perfective-marked verbal form as well as the auxiliary *huua* agree with the main argument, here the feminine singular *kitaab* ‘book’ (290):

- (290) *kitaab* *Shaunak-dwaaraa* *likh-ii* *hu-ii* *hai*.
 book.F Shaunak-by write-PFV.FSG be-PFV.FSG be.PRS.3SG
 ‘The book is written by Shaunak.’

In contrast, pseudo-agentive statives of ingestives with an additional argument allow for a variety of different case marking patterns, summarized in (291):

- (291) a. *Mina* *laal juute* *pehn-e/??pehn-ii* *hue/huii* *thii*.
 Mina.F red shoe.MPL wear-PFV.OBL/wear-PFV.FSG be-PFV.OBL/be-PFV.FSG be.PST.F
 ‘Mina was red-shoes-wearing.’
 b. *Mina* *laal juutaa* *pehn-e/??pehn-ii/*pehn-aa* *hue/huii*
 Mina.F red shoe.MSG wear-PFV.OBL/wear-PFV.FSG/wear-PFV.MSG be-PFV.OBL/be-PFV.FSG
 thii.
 be.PST.F
 ‘Mina was red-shoe-wearing.’

One option, which is what was shown in (289b), is for the verbal form and the auxiliary to take oblique marking (here, *pehn-ee hu-ee*). Oblique marking normally surfaces on nouns ending on *-a* when these occur in the presence of case markers such as ergative *-ne*. It is not clear what licenses the oblique in the examples in (291). A confound is presented by the fact that oblique marking is syncretic with the masculine plural. Accordingly, in (291a), *pehn-ee hu-ee* might rather signal agreement with the masculine plural pseudo-incorporated argument, *juute* ‘shoes.’ To control for this, (291b) uses a masculine singular incorporated argument instead (*juutaa* ‘shoe’). But this argument cannot control agreement (**pehn-aa*), showing that *pehn-ee hu-ee* is in fact oblique rather than spelling out agreement with the incorporated argument. Alternatively, the auxiliary can also agree with the non-incorporated argument; here, for feminine singular (*hu-ii*). Feminine singular on the verb itself (*??pehn-ii*) is degraded.

To summarize, neither the verb nor the auxiliary can agree with the incorporated argument. The auxiliary but not the verb can show the expected pattern of agreeing with the non-incorporated argument, and both of them can show oblique marking for reasons that are unclear. I have no analysis of these data, but they further support the claim that arguments such as ‘shoes’ in the examples above are not regular arguments of statives, but pseudo-incorporated into the verb, thus triggering different agreement patterns.

Against this background, I further argue that even pseudo-agentive statives without an overt additional argument still involve incorporation, but of a null implicit argument. As a baseline, consider example (292):

- (292) The man read, #but there was nothing that he read.

Although the verb *read* does not take an overt object argument in this example, the existence of something

that was read is nonetheless entailed and cannot be denied without contradiction. Accordingly, (292) is commonly analyzed as containing a null argument in object position interpreted with existential force, as in *The man read something*.

The same holds for pseudo-agentive statives of ingestives. Example (293) cannot be followed by the continuation ‘... but there was nothing that he read and wrote’:

- (293) vo aadmii kaafii parh-aa likh-aa hu-aa hai.
 that man a.lot read-PFV write-PFV be-PFV.MSG be.PRS.MSG
 ‘That man is very educated.’ (Lit. ‘That man is very read-written.’)

By the same token, Greek examples such as (294) under the pseudo-agentive reading cannot be followed by a continuation such as ‘... but there was nothing that she ate’:

- (294) I Maria ine fayomen-i.
 the.NOM Mary.NOM be.3SG eat-PTCP-F.NOM
 ‘Mary is eaten.’
 ‘Mary has eaten.’ (Paparounas, 2025:23)

The entailment test thus indicates that there is an additional argument just like in the pseudo-incorporation examples from Hindi, but that this argument is implicit. The pseudo-agentive argument of statives of ingestives is always in the result state of eating, drinking and learning *something*. In Hindi-Urdu, a language that allows for overt pseudo-incorporation, this something can but does not have to be overtly realized; in Greek, it must be implicit.

To briefly recap, the original concern of this section was that if arguments like ‘a sari’ occupied the standard argument position of statives corresponding to *the door* in *The door is opened*, it might appear that there must be an additional position for the second argument – e.g., ‘that woman’ –, and a plausible candidate for this position would be SpecVoiceP. This is the claim I have argued against. I have countered it by making a novel proposal, namely, that arguments like ‘a sari’ are pseudo-incorporated, and I have further argued that pseudo-agentive statives without an overt second argument in fact involve incorporation of a null implicit argument.

Developing this idea further in future work will require determining what exactly the verbal element is that the incorporated argument combines with; specifically, whether it is a state or an event. This of course hinges on the broader analysis of statives adopted. Moreover, the proposal raises the puzzle why pseudo-agentive statives do not violate thematic uniqueness (e.g., Carlson, 1984, 1998; Williams, 2015). For instance, in the sari-wearing example, it appears that the two arguments – the sari and the woman – receive the same θ -role, in that they are both interpreted as holders of a result state: the sari of the state resulting from the wearing (or putting-on) event, and the woman of the state resulting from the sari-wearing event. I need to leave this question open here.

Lastly, a general fact about the pseudo-agentive interpretation that I have not made explicit so far is that

it is strictly unavailable in eventive passives (295)–(296). Similarly, I noted earlier that in Hindi-Urdu, the absence of *huaa* usually makes an active perfect reading available, but in (296a), *huaa* can be absent without resulting in an eventive interpretation.

- (295) a. I Maria ine fayō-men-i.
 the.NOM Mary.NOM be.3SG eat-PTCP-F.NOM
 ‘Mary is eaten.’
 ‘Mary has eaten.’ *stative*
- b. I Maria fayō-θ-ik-e.
 the Mary eat-PFV.NACT-PST-3SG
 ‘Mary was eaten.’ (Not: ‘Mary has eaten.’) *eventive passive*
- (296) a. vo aadmīi kaafīi paṛh-aa likh-aa (hu-aa) hai.
 that man a.lot read-PFV write-PFV be-PFV.MSG be.PRS.MSG
 ‘That man is very educated.’ *stative*
- b. *vo aadmīi kaafīi paṛh-aa likh-aa gay-a hai.
 that man a.lot read-PFV write-PFV PASS-MSG be.PRS.MSG
 Intended: ‘That man is very educated.’ *eventive passive*

Against the background of this section, a plausible reason why this might hold is that eventive passives do not allow for their first argument to be incorporated. This might again relate to thematic uniqueness and how it interacts with the different event-structural properties of statives and eventive passives.

To conclude, I have argued that pseudo-agentive arguments of statives do not occupy the canonical agent position, SpecVoiceP. This has given rise to the novel claim that additional arguments that can surface with pseudo-agentive statives in Hindi-Urdu are pseudo-incorporated, although I have left the details of this analysis open for now. In the following, I show that pseudo-agentive arguments are also semantically not interpreted as agents, but as holders of a result state.

4.6.2.3 Thematic interpretation

Some examples of pseudo-agentive statives of ingestives are shown again in (297) and (298) for Greek and Hindi-Urdu, respectively:

- (297) a. I Maria ine fayō-men-i.
 the.NOM Mary.NOM be.3SG eat-PTCP-F.NOM
 ‘Mary is eaten.’
 ‘Mary has eaten.’ (Paparounas, 2025:23)
- b. I Maria ine maθi-men-i.
 the.NOM Mary.NOM be.3SG learn-PTCP-F.NOM
 ‘Mary is learned.’
- (298) a. vo aadmīi kaafīi paṛh-aa likh-aa (hu-aa) hai.
 that man a.lot read-PFV write-PFV be-PFV.MSG be.PRS.MSG
 ‘That man is very educated.’ (Lit. ‘That man is very read-written.’)

- b. vo aurat saari pehn-e (hu-e) thii.
 that woman sari wear-PFV.OBL be-PFV.OBL be.PST.F
 ‘That woman was wearing a sari.’ (Lit. ‘That woman was sari-worn.’)

I argue that the non-incorporated argument in these examples is interpreted not as an agent but as the holder of a result state, just like any other argument of statives (see also Paparounas, 2025 for a similar proposal discussed later in Chapter 4.6.3.1). What is predicated of these arguments is not that they performed a certain activity – eating, drinking, learning, etc. –, but that they are in a state that results from these activities.

Crucially, we find pseudo-agentive readings specifically with statives of ingestive verbs, not with standard causatives such as *break* or *open*. With the latter, the internal argument ends up in a salient result state – a glass is shattered, a door stands ajar –, but the external argument does not. Typically, nothing distinguishes people who have broken a glass or opened a door from those who have not. But with ingestive verbs, people who have eaten, drunk or learned something are in a markedly different state from those who have not. Accordingly, such arguments can surface with statives.

In support of this claim, Paparounas (2025) notes that (297) is only felicitous if the eating is completed. Having eaten some food in the recent past does not in and of itself make for a salient result state, but having finished a meal is a specific state both physiologically and socially. For instance, if Mary has finished eating, this means that her blood sugar is elevated, she might be tired, and we would expect her to turn down a dinner invitation.

Similarly, statives such as *mathimenos* ‘learned’ in Greek and *parhaa likhaa huua* ‘read-written’ in Hindi-Urdu – similar to English *learned* and *well-read* – do not simply mean that the participant learned, read or wrote something, but that they pursued their learning, reading and writing to a point where they became, in some sense, a different person. Someone who has read a newspaper article is not markedly different from someone who has not, but someone who has undergone an extensive education will differ in their conversation style, their vocabulary, their outlook on current events, etc.

Moreover, Zakharyin (2020) reports that in Hindi-Urdu, the stative of ‘drink’ is licensed with the incorporated argument ‘wine’ but not with ‘milkshake’ (299):

- (299) a. vo aadmi saraab piy-aa hu-aa hai.
 that man wine drink-PFV.MSG be-PFV.MSG be.PRS.MSG
 ‘That man is drunk on wine.’ (Lit. ‘That man is wine-drunk.’)
 b. %vo aadmi lassii piy-aa hu-aa hai.
 that man milkshake drink-PFV.MSG be-PFV.MSG be.PRS.MSG
 ‘That man has drunk milkshake.’ (Lit. ‘That man is milkshake-drunk.’)

This is as expected if it is assumed that drinking milkshake does not result in a salient result state: there does not appear to be anything that systematically distinguishes people who have drunk milkshake from those who have not, other than their recent milkshake consumption. On the other hand, drinking wine results in

noticeable changes in the participant with respect to behavior, cognitive function, legal liability, etc. (In a similar vein, English *drunk* means specifically ‘inebriated.’) However, one of my informants does in fact find (299b) acceptable if the person described has white marks around the lips which visibly indicate the recent consumption of milkshake. Thus, like any other argument of statives, the non-incorporated argument of ingestive statives must undergo a change of state in order to be licensed in this position, and speakers vary in the extent to which they are able to construe such a state contextually. Of course, none of this holds for the subjects of the corresponding transitives.

Overall, the fact that pseudo-agentive arguments pattern semantically with the arguments of regular statives in that they are interpreted as holders of result states is very clear. The more general question remains how the content of a result state is determined by the content of the verbal root, together with – in the case of pseudo-agentive statives – (pseudo-)incorporated nominal material. For instance, for the pseudo-agentive stative *The scholar is well-read*, we can ask why this sentence has the interpretation that the scholar is educated rather than that they have poor eyesight, which might equally be considered a state resulting from extensive reading. This is a genuine question, but it is not limited to pseudo-agentive statives. For instance, in *The door is opened*, we can ask why the sentence cannot mean that the door is broken as the result of a particularly violent opening. To my knowledge, we do not have a fully satisfactory account of the lexical semantics of result states, and developing it is beyond the scope of this work. What I argue here is merely that pseudo-agentive arguments are not agents but holders of result states – making them a case of variable linking –, and that unlike with regular statives, these result states contain (pseudo-)incorporated material.

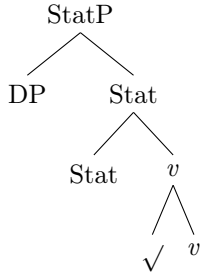
4.6.3 Competing analyses

As always, I end by discussing competing analyses, or rather, in this case only one, Paparounas (2025). In doing so, I focus strictly on Paparounas’ analysis of the pseudo-agentive interpretation; I will not attempt to evaluate his proposal more broadly.

4.6.3.1 Agentivity via a meaning postulate: Paparounas (2025)

In his work on statives in Greek, Paparounas (2025) adopts a complex head analysis first developed by Embick (2023) for English. Under this approach, statives are built by merging the root with a *v* head which then combines directly with a stativizing head Stat, without projecting a *v*P. The DP argument is merged in the specifier of StatP (300).

(300)



Paparounas proposes the following denotation for Stat (301):

$$(301) \quad \llbracket \text{Stat} \rrbracket = \lambda P. \lambda x. \lambda s. \exists e'. P(e') \wedge \text{CAUSE}(e', s) \wedge \text{STATE}(s) \wedge \text{HOLDER}(s) = x$$

(Paparounas, 2025:27)

The stativizer existentially closes the event argument introduced by v , introduces a state that results from the verbal event, and further introduces an entity argument that is the holder of this state. In short, the DP is thematically integrated in the state, not the event; and the θ -role it receives is that of a holder, not a patient/theme.

However, this alone leaves unexplained the oddity of (302):

(302) *The girl is opened.

In other words, something needs to account for the fact that the holder argument of the state typically corresponds to the patient of the event rather than the agent. Paparounas takes statives of ingestives to be crucial evidence for solving this puzzle. Assuming that in ingestive statives, the argument can truly be interpreted as an agent, he argues that the correspondence between the holder of the state and the patient/theme of the event is merely a defeasible inference, with ingestive statives constituting the exception. He proposes that while the status of the argument as a holder is syntactically encoded, its status as a patient/theme – or, in the case of ingestives, agent – comes about by means of a meaning postulate (303):

$$(303) \quad \begin{array}{ll} \text{a.} & \forall x. \forall e. \forall s. [\text{event}(e) \wedge \text{CAUSE}(e, s) \wedge \text{STATE}(s) \wedge \text{HOLDER}(s) = x] \rightarrow \text{THEME}(e) = x \\ \text{b.} & \text{OR } \forall x. \forall e. \forall s. [\text{event}(e) \wedge \text{CAUSE}(e, s) \wedge \text{STATE}(s) \wedge \text{HOLDER}(s) = x] \rightarrow \text{AGENT}(e) = x \\ & \text{in the context of } \sqrt{\text{eat}}, \sqrt{\text{learn}} \dots \end{array}$$

(Paparounas, 2025:28f.)

To put matters differently, letting the DP merge as an argument of Stat – and thus of the state rather than the event – *prima facie* raises the problem of how it nevertheless comes to correspond to one argument of the event rather than another. Paparounas interprets ingestive statives as evidence that this thematic underspecification with respect to the event is in fact a welcome state of affairs.

Above, I have followed Paparounas in arguing that the pseudo-agentive argument is interpreted as the

holder of a result state. Where we diverge is that he proposes that the argument is additionally interpreted as the agent of the event, ending up with two θ -roles. This is thus yet another variant of the double- θ -role analysis previously proposed by Ramchand (2008), Krejci (2020) and, to some extent, Kallulli (2008) and Cuervo (2003) for other cases of variable linking: the intuition that both in ‘Mary is eaten’ (under the pseudo-agentive reading) and ‘Mary eats,’ ‘Mary’ is interpreted with the individual role of an eater is captured by the fact that the argument receives an agent θ -role in both alternants.

Unlike Ramchand, Paparounas does not argue that the argument occupies two positions in the syntax, and unlike Krejci, he does not propose that it is assigned two θ -roles as a result of a systematic lexical rule. Neither does he try to make the case that pseudo-agentive arguments of statives have abstract semantic properties associated with the agent θ -role, or syntactic properties associated with SpecVoiceP. Instead, the argument is simply declared an agent in an admittedly somewhat ad-hoc post-hoc fashion via a meaning postulate. There is not much of a point in arguing against this analysis on empirical grounds.

At the same time, there is a very real motivation behind this proposal, which is to account for the fact that the Greek version of ‘Mary is eaten’ can have two different readings, only one of which involves cannibalism. In other words, we need to solve the puzzle that being the result state holder of an eating event can apparently mean two different things: having eaten and being eaten. The proposal that a meaning postulate requires the argument of statives to be interpreted either as the agent or the patient of the event, in addition to it being a result state holder, aims to explain this ambiguity.

In Chapter 5.2.5, I will later propose a different solution to this puzzle, which I argue avoids the ad-hoc-ness of Paparounas’ analysis. As everything in this case study, this solution will remain a sketch. But overall, what the above discussion of pseudo-agentive statives should have made very clear is that the research question pursued in this study – what it means to be interpreted with a given θ -role in the context of a given event description, such as what it means to be the result state holder of an eating event – is not just an abstract theoretical concern but matters for the analysis of concrete argument-structural alternations. The next chapter finally turns to the analysis I propose.

Chapter 5: Analysis

Against the background of the discussion of previous work in Chapter 3 and the empirical case studies in Chapter 4, this chapter now presents the main proposal of this work. Recall the distinction drawn earlier between the meaning of θ -roles in isolation – what it means to be an agent or a patient – and the meaning of θ -roles in the context of a specific event description – what it means to be the agent or patient *of*, say, a stabbing event. The question we have set out to solve is how knowledge of the latter kind is represented in speakers’ grammars. Earlier, I have sketched out a potential answer to this question: that θ -roles interpreted in the context of a given event description are *individual roles*, which are stored as part of the meaning of the verb and systematically map onto more abstract θ -roles. I hope to have dissuaded the reader from this view by now. In the following, I lay out my counterproposal.

This chapter comes in two parts. Chapter 5.1 presents the proposal in general terms, making reference to variable unaccusativity throughout. In Chapter 5.2, I then briefly revisit each of the remaining five case studies from Chapter 4 and outline how they can be analyzed against the background of my proposal.

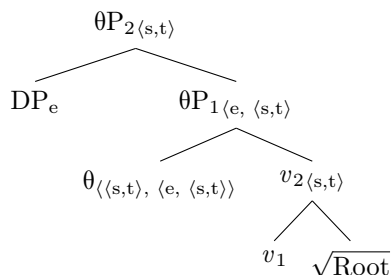
5.1 Rethinking argument interpretation

5.1.1 Argument interpretations can be located in the tree

I begin with an architectural assumption that is commonplace in recent generative work on argument structure, which is that arguments are introduced not by the verb itself, but by functional heads. Since Marantz (1984) and Kratzer (1996), this idea is standardly adopted for external arguments, and Pykkänen (2008) has popularized it for applicative arguments as well. The last holdout are internal arguments, which are more commonly – albeit not always – assumed to be introduced by the verb itself. Here, I follow the most radical view that all arguments are external to the verb. The verb itself does not introduce any arguments.

The syntax we will work with in the following is shown in (1), for now limited to a single argument. The θ -head projecting a θP is a schematic representation of an argument-introducing head, abstracting over both the specific θ -role with which it is associated (agent, patient, etc.) and the syntactic category of this head.

(1)



I make no commitments regarding spell-out, leaving it open which node(s) in the verbal domain are realized as the verbal stem, which as affixes or other functional morphemes, and which are phonologically null. The morphological realization of this structure will vary across languages and it does not concern us here. Moreover, I do not rule out that some of the heads in (1) might be bundled in some languages (see, e.g., Harley, 2017).

I further assume the denotations in (2); here, $\theta(e)(x)$ stands for $\text{Agent}(e)(x)$, $\text{Patient}(e)(x)$, etc., depending on the θ -role assigned by the relevant head. Unlike in the semantics for argument-introducing heads originally proposed by Kratzer (1996), θ -heads combine with their complement via Function Application.

- (2)
- a. $\llbracket v_2 \rrbracket = \lambda e. \text{Root}(e)$
 - b. $\llbracket \theta \rrbracket = \lambda P. \lambda x. \lambda e. P(e) \wedge \theta(e)(x)$
 - c. $\llbracket \theta P_1 \rrbracket = \lambda x. \lambda e. \text{Root}(e) \wedge \theta(e)(x)$
 - d. $\llbracket \theta P_2 \rrbracket = \lambda e. \text{Root}(e) \wedge \theta(e)(DP)$

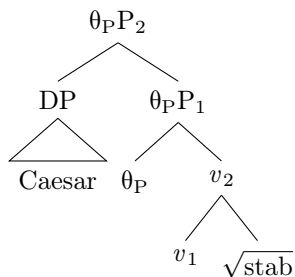
Lexical meaning takes the form of acategorical roots which then merge with a category-forming head; here, the verbalizer v (Halle and Marantz, 1993, 1994). The resulting node, which I will refer to as the *prethematic verb*, is equally of category v and semantically of type $\langle s, t \rangle$ (2a). Next, we merge an argument-introducing θ -head of type $\langle \langle s, t \rangle, \langle e, \langle s, t \rangle \rangle$ (2b): this head combines first with an event description and then with a DP, and it returns – at the level of θP_2 , (2d) – another event description which now contains this DP as an argument interpreted with the corresponding θ -role.¹ As a first stab, let us say that a θ -head relates an argument to an event description by providing a prototype that serves as a blueprint for the way in which the argument is integrated into the event. For instance, the agent θ -head adds an argument that is interpreted in the context of the event as resembling a prototypical agent.

The crucial observation about this architecture is as follows. The level of θP_1 , (2c), corresponds to a θ -head that has combined with an event description and still awaits an argument. Accordingly, it denotes a θ -role interpreted in the context of a given event description. The entity argument with which it combines next will not only be, say, a patient, but a patient *of* a specific event. Thus, the meaning that the research question is after – what it means for an argument to be interpreted with a given θ -role in the context of a given event description – can be located at a particular node in the tree, θP_1 .

For the sake of concreteness, let us assume that the root in (1) is *stab* and that the θ -head is a patient θ -head, θ_P (3):

¹I leave clausal arguments to future work.

(3)



The denotation of $\theta_P P_1$ is then as in (4):

$$(4) \quad \llbracket \theta_P P_1 \rrbracket = \lambda x. \lambda e. \text{stab}(e) \wedge \text{Patient}(e)(x)$$

According to this denotation, what holds true of x is that x is the patient of an event which also happens to be a stabbing event, or in other words, that x is the patient of a stabbing event. In the sentence *Brutus stabbed Caesar*, in which Caesar is the patient of the stabbing event, we know what is predicated of Caesar: that he is stabbed, or penetrated by a sharp weapon. This is the meaning of $\theta_P P_1$, which is merged with *Caesar*.

Following the denotations given, this meaning is generated by merging the prethematic verb *stab* with a patient θ -head. Accordingly, the meaning of *stab* must be something that generates the meaning of being penetrated by a sharp weapon when serving as input to a patient θ -head. The challenge, which we will tackle in the next section, will be to understand what exactly this something is. For the time being, I define the meaning of the verb as something that generates the interpretation assigned to its first argument when serving as input to its first θ -head. Note that this θ -head does not necessarily have to be a patient θ -head; in the case of unergatives, it would be an agent θ -head.

We can now answer the question of how speakers know what it means to be interpreted with a given θ -role in the context of a given event. We know what it means to be the patient of a stabbing event because we know the meaning of the verb *stab*, and this meaning, as defined just now, is something that gives rise to the meaning of being the patient of a stabbing event when serving as input to a patient θ -head. Unlike under the individual role theory of θ -roles, what it means to be the patient of a stabbing event is thus not as such already stored on the verb. Instead, this meaning arises from combining the verb *stab* with a patient θ -head, and the verb means whatever is needed to make this happen.

In sum, the crucial switch is from thinking about the verb as something that *prespecifies* the interpretation of its arguments in the form of stored individual roles to something that *generates* the interpretation of its arguments when combining with θ -heads. Next, we look more in detail at the meaning of prethematic verbs by returning to the data from variable unaccusativity. This discussion will still be concerned solely with the verb's first argument. The section after then extends the proposal to any additional arguments.

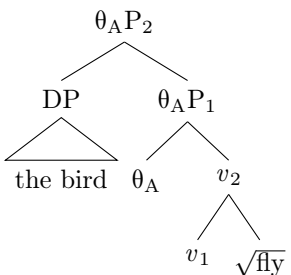
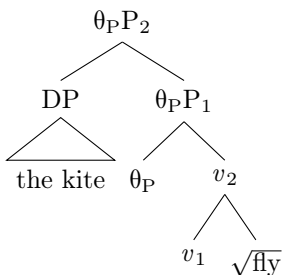
5.1.2 Variable linking and the nature of prethematic verb meaning

In this section, I will only discuss the first and most basic case of variable linking, variable unaccusativity; the remaining case studies are dealt with in Chapter 5.2. An example of variable unaccusativity that we saw earlier is repeated in (5). Both sentences feature the same verb, ‘fly,’ which variably comes with an agent or a patient θ -role: the bird flies as an agent (5a), the kite as a patient (5b).

- (5) a. *ur-ii (huu-ii) ciryaa
 fly-PFV.FSG be-PFV.FSG bird.FSG
 Intended: ‘the flown bird’
- b. ur-ii (huu-ii) patang
 fly-PFV.FSG be-PFV.FSG kite.FSG
 ‘the flown kite’ (Ahmed, 2010:8f.)

In the case study on variable unaccusativity in Chapter 4.1, I have emphasized that such examples are highly common and productive; they are not isolated exceptions. It is thus not sufficient to simply assume that speakers store the unergative and unaccusative variants of ‘fly’ as two separate lexical items. Moreover, if ‘fly’ with an agent θ -head and ‘fly’ with a patient θ -head were simply two independently stored idioms, nothing would explain why they clearly share a semantic core.

Rather, speakers must regularly and productively form a representation of a verb without θ -roles that can then be used in the context of different θ -roles. Hindi-Urdu must have a prethematic verb ‘fly’ that generates different but related meanings depending on which θ -head it merges with (6):

- (6) a. 
 b. 
- Diagram (6a) is a syntax tree for the agent role. The root is $\theta_A P_2$, which branches into DP and $\theta_A P_1$. DP branches to the triangle 'the bird'. $\theta_A P_1$ branches into θ_A and v_2 . θ_A branches to the triangle 'the bird'. v_2 branches into v_1 and $\sqrt{\text{fly}}$. v_1 branches to the triangle 'the bird'.
- Diagram (6b) is a syntax tree for the patient role. The root is $\theta_P P_2$, which branches into DP and $\theta_P P_1$. DP branches to the triangle 'the kite'. $\theta_P P_1$ branches into θ_P and v_2 . θ_P branches to the triangle 'the kite'. v_2 branches into v_1 and $\sqrt{\text{fly}}$. v_1 branches to the triangle 'the kite'.

In (6a), the prethematic verb ‘fly’ combines with an agent θ -head and derives what it means to be the agent of flying (7a) – to fly actively, holding true of the bird. By contrast, in (6b), it combines with a patient θ -head, deriving what it means to be the patient of flying (7b) – to float passively, holding true of the kite.

- (7) a. $\llbracket \theta_A P_1 \rrbracket = \lambda x. \lambda e. \text{fly}(e) \wedge \text{Agent}(e)(x)$
 b. $\llbracket \theta_P P_1 \rrbracket = \lambda x. \lambda e. \text{fly}(e) \wedge \text{Patient}(e)(x)$

The question is how the meaning of the prethematic verb needs to be defined in order for this to be possible.

As a first stab, it might appear that we can pinpoint this meaning very concretely: the verb contributes the information that there is some entity that flies, meaning that this entity moves through the air. Whether this movement happens in a more agent- or more patient-like manner is information contributed by the θ -heads.

(9) $\lambda e. \text{fly}(e)$

Prima facie, (9) denotes a set of events without arguments. However, it is hard to see what a flying event would be that does not contain something or someone that flies. In general, we seem to be unable to conceptualize events that do not involve any participants, with the exception of the event descriptions denoted by weather verbs, which never take any arguments. If a prethematic verb indeed denoted a set of events without arguments, then each of the events in that set would be a cognitively impenetrable mystery.

But I believe that this conclusion is wrong. Assume that the prethematic verb *fly* merges with a patient θ -head, yielding (10):

(10) $\lambda x. \lambda e. \text{fly}(e) \wedge \text{Patient}(e)(x)$

In order to form a non-empty intersection of flying events and events that have x as a patient, a subset of flying events must have a patient argument. This holds true even though the representation in (9) does not contain any arguments.

The issue can be sharpened by looking at the verb *cut*. In English, *cut* is an obligatorily transitive verb. There is no such thing as a cutting event that does not have an agent and a patient argument. In consequence, the three event descriptions in (11) denote the same set:

- (11) a. $\lambda e. \text{cut}(e)$
 b. $\lambda e. \exists x. \text{cut}(e) \wedge \text{Patient}(e)(x)$
 c. $\lambda e. \exists x. \exists y. \text{cut}(e) \wedge \text{Patient}(e)(x) \wedge \text{Agent}(e)(y)$

This means that in a sense, all events in the set denoted by (11a) have an agent and a patient argument despite the fact that there is no argument in the actual denotation.

Unlike English, the verb corresponding to *cut* in Hindi-Urdu has an unaccusative variant. The equivalent of (12) is grammatical in the language:

(12) The bread cuts.

However, speakers report that an event described by the Hindi-Urdu equivalent of (12) still must include someone who cuts the bread; that is, an agent. Even in this language, there is no such thing as a cutting that happens by itself. Both (11b) and (11c) are well-formed in the language, but they denote the same set of events. The question, then, is what the semantic difference between them is, and in what sense the events described by (11b) can be said to include agent arguments.

I argue that the two denotations in (11b) and (11c) differ with respect to which properties of the events in this set they make accessible to the grammar. In the unaccusative denotation in (11b), the fact that each event in the set has an agent is a fact about the lexical meaning of *cut*, whereas in (11c), it is a fact about

the compositional denotation of the sentence. By way of (imperfect) analogy, the noun phrases *bachelor* and *unmarried bachelor* denote the same set of entities, but in one, the property of being unmarried that holds of all bachelors is encoded as part of the lexical meaning of the noun, whereas in the other, it is explicitly expressed with a modifier.

It is thus in some sense misleading to speak of functional heads as ‘introducing arguments,’ as if the events denoted by the verb itself did not already contain arguments to begin with. What it means to introduce an argument is to select from the original set of events a subset – which does not have to be a proper subset – that contains such an argument, which makes the property of having such an argument part of the compositional denotation of the sentence and thereby accessible to the grammar. This makes it possible to specify the identity of the argument – say, Caesar –, describe their actions further using modifiers, and so forth.

To summarize, this section has examined what exactly the meaning of a verb is before it combines with a θ -head. This meaning should not be equated with intuitions about a prototypical instance of an event described by this verb: for instance, the prethematic verb *fly* does not mean ‘move through the air.’ Instead, I have proposed that prethematic verbs refer to *kinds* of events that are defined by all tokens that instantiate them. Some events in the set denoted by *fly* contain an agent argument, others a patient argument, others both, or other kinds of arguments. Some can be adequately described as events of something moving through the air, others cannot. All of them are flying events, in the sense of being instantiations of this particular kind of event. Complete event descriptions such as *The bird flies*, *Gennaro flies a kite*, *The news flew across town*, etc., are derived by combining the prethematic verb with θ -heads and arguments, thereby selecting a subset of the set of flying events.

5.1.3 Beyond intransitives

So far, we have only considered the verb’s first argument. This section expands the proposal to event descriptions with more than one argument. I argue that by adding arguments to the event description, θ -heads create new event descriptions, which in turn affects how higher arguments are interpreted. This idea will be crucial for analyzing several of the variable linking alternations that we will revisit in Chapter 5.2, and it adds further support to the claim that what it means to be interpreted with a given θ -role in the context of a given event is not prespecified on the verb, but generated in the syntax.

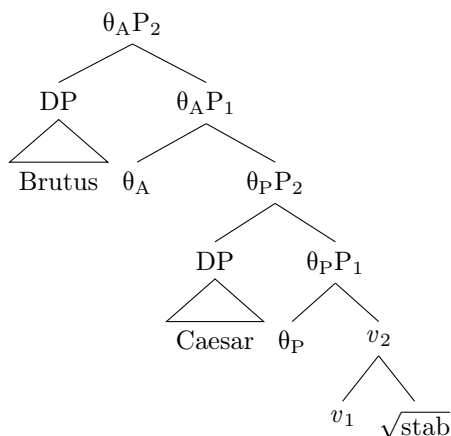
Let us return to a baseline example, (13):

- (13) Brutus stabbed Caesar.

So far, I have argued that what holds true of Caesar is generated by merging a patient θ -head with the event description it combines with, namely, the prethematic verb *stab*. I now propose that the same holds, *mutatis mutandis*, for the second argument: the interpretation that is assigned to Brutus is generated by merging

the agent θ -head with the event description in its complement position. Crucially, this constituent denotes not a stabbing event, but a stabbing-Caesar event (14):

(14)



Simply as a consequence of constituency, the patient argument is contained in the event description that the agent θ -head combines with ($\theta_P P_2$).

We will see this effect of constituency play out repeatedly when revisiting the case studies in Chapter 5.2; here, I demonstrate it using Marantz's examples in (15) and (16):

- (15)
- a. take a book from the shelf
 - b. take a bus to New York
 - c. take a nap
 - d. take an aspirin for a cold
 - e. take a letter in shorthand

- (16)
- a. kill a cockroach
 - b. kill a conversation
 - c. kill an evening watching TV
 - d. kill a bottle (i.e., empty it)
 - e. kill an audience (i.e., wow it)

(Marantz, 1984:25)

In these examples, the interpretation of the agent (external) argument depends on the patient (internal) argument even if the verb is kept constant. That is, the subjects of these phrases perform very different activities: what it means to kill a cockroach is markedly different from what it means to kill a conversation, and so forth. This would be unexpected if the interpretation assigned to the agent argument was prespecified on the verb in the form of an individual role such as a killer or a taker.

On the other hand, under the assumption that the interpretation of the agent arguments is generated by combining an agent θ -head with an event description, these data fall out naturally. The denotations of some of the examples in (16) are given in (17); for simplicity, I use definites as objects:

- (17)
- a. $\llbracket \text{kill the cockroach} \rrbracket = \lambda e. \text{kill}(e) \wedge \text{Patient}(e)(\text{the cockroach})$
 - b. $\llbracket \text{kill the conversation} \rrbracket = \lambda e. \text{kill}(e) \wedge \text{Patient}(e)(\text{the conversation})$
 - c. $\llbracket \text{kill the evening} \rrbracket = \lambda e. \text{kill}(e) \wedge \text{Patient}(e)(\text{the evening})$

The set of killing events that have the cockroach as a patient, those that have the conversation as a patient and those that have the evening as a patient are different sets. When merging an agent θ -head, the agent argument will thus be related to different event descriptions (18). As a result, the interpretation of the argument differs.

- (18) a. $\lambda x. \lambda e. \text{kill}(e) \wedge \text{Patient}(e)(\text{the cockroach}) \wedge \text{Agent}(e)(x)$
 b. $\lambda x. \lambda e. \text{kill}(e) \wedge \text{Patient}(e)(\text{the conversation}) \wedge \text{Agent}(e)(x)$
 c. $\lambda x. \lambda e. \text{kill}(e) \wedge \text{Patient}(e)(\text{the evening}) \wedge \text{Agent}(e)(x)$

In his account of the examples in (15) and (16), Marantz (1984) proposed that the external argument is an argument of the entire verb phrase rather than of the verb itself. This is precisely the intuition I argue for here: the argument is an agent of a killing-a-cockroach event, or a killing-a-conversation event, not of a killing event in isolation. What holds true of the argument depends both on the θ -head and the entire event description in its complement position.

Crucially, Marantz also shows that the agent argument does not affect the interpretation of the patient argument, as seen in (19):

- (19) a. Harry killed NP.
 b. Everyone is always killing NP.
 c. The drunk refused to kill NP.
 d. Silence can certainly kill NP.
 e. Cars kill NP. (Marantz, 1984:26)

While the choice of external argument might suggest a certain interpretation of the verb phrase – e.g., in (19d), the killing is most likely not a literal one, but perhaps a killing of a conversation –, the agent argument does not determine the interpretation of the patient argument in the same way that the patient argument determines the interpretation of the agent argument. This is as expected: due to constituency, the patient argument is part of the event description that the agent θ -head combines with, but not vice versa.

Under the analysis proposed here, the data from Marantz (1984) thus fall out as a simple effect of constituency, given the assumption that an argument is always an argument of the event description that its θ -head combines with in the syntax. In previous work, Kratzer (1996) has argued against this interpretation of these examples. Instead, she has taken the latter as evidence for the idea that there is a fundamental difference between external and internal arguments, in that only the latter are actually arguments of the verb. In the remainder of this section, I briefly review and reject Kratzer's objection.

Kratzer sketches out two scenarios that would explain Marantz's data. Either a verb like *kill* has a range of homophonous variants associated with different senses, and each variant selects a different kind of complement: for instance, the variant of *kill* with the meaning 'waste' selects a time interval as an argument.

Or such verbs are functions that return different outputs for different input arguments: for instance, when the verb *kill* takes a time interval *x* as an argument, it would return true when someone wastes *x*, etc. In both scenarios, Kratzer argues, any argument of the verb – including the external argument – should be able to trigger a particular reading of the verb. Since this is not the case, the external argument cannot be an argument of the verb.

Note that I am of course not advocating the view that the external argument is equally an argument of the verb, which according to Kratzer would wrongly allow it to trigger a polysemous reading. The point I am making is that neither argument is an argument of the verb in Kratzer's sense. Thus, under the reasoning above, neither of them should be able to trigger a polysemous reading, contrary to fact.

But Kratzer's argument rests on a very narrow theory of polysemy that faces independent problems. To begin with, it is a sense enumeration approach according to which the various senses associated with polysemous items are simply listed in the lexicon, with one of them being selected in a given context (see, e.g., Katz, 1972; Lakoff, 1987). While sense enumeration is still commonly adopted in the syntactic literature, in lexical semantics it is widely recognized as a deeply deficient theory. Setting aside the vast amount of lexical storage it requires, sense enumeration has nothing to say about the semantic relationships between different senses of polysemous items. At least in terms of mental representation, it conflates the difference between polysemous items and homonymous items such as *bank*. More importantly, it does not account for the fact that speakers extend the meaning of items productively, and that similar items show similar patterns of polysemy.

But even if we did adopt a sense enumeration approach to polysemy – which, to reiterate, I do not think we should –, it is not clear why only arguments of the verb should be able to trigger a polysemous reading. For instance, one implementation of sense enumeration that has become popular recently is contextual allosemy (e.g., Wood, 2023). Under this view, lexical items are listed with different senses – allosemes –, one of which is selected depending on the context. While this context needs to be in some sense local, nothing requires it to be an argument of the item in question. What is more, Kratzer's reasoning could not apply for items other than verbs in the first place. For instance, in (20), the noun *book* is associated with different senses, but this is not due to the fact that *book* takes different arguments in the two sentences.

- (20) a. This book is riveting.
 b. This book has 587 pages.

The context that determines the sense of a polysemous item is quite obviously not limited to this item's arguments, and it is not clear why we should expect it to be.

By contrast, my analysis of Marantz's examples assumes that the prethematic verbs *kill* and *take* denote kinds of events, which are defined by the totality of tokens that instantiate them rather than by shared properties. Accordingly, nothing requires the event of Gennaro killing a cockroach and the event of Gennaro killing a conversation to have a clearly defined shared semantic core. I do not mean to pretend that this

is a satisfactory account of polysemy as a whole. Specifically, it has no handle on what makes different instantiations of a kind of event semantically similar to each other, in the same way that analyzing *dogs* as a kind does not immediately explain why dogs are similar to each other and different from, say, hippos. I also leave it open whether an analysis based on kinds can account for other kinds of polysemy such as the examples in (21) discussed just now:

- (21) a. This book is riveting.
 b. This book has 587 pages.

What I do argue is that Marantz's data can be analyzed as an effect of constituency, as long as an argument is assumed to be an argument of the event description that its θ -head combines with in the syntax rather than of the verb as such.

To summarize, I have proposed that analogously to what we saw earlier for the verb's first argument, the interpretation of the second argument (and further arguments) is equally not prespecified on the verb, but generated by combining the relevant θ -head with the event description in its complement position. The interpretation assigned to an argument should thus depend not only on the θ -role and the root – as under the individual role theory –, but also on any arguments that are already contained in the event description merged with the θ -head, and we have seen data indicating that this prediction is correct.

5.1.4 Proto-properties and exemplars

A remaining question that we need to discuss is how different θ -heads differ from each other. Above, I have simply posited that they generate a different output: when fed the same event description, the agent θ -head returns what holds true of an argument if it is the agent of this event, the patient θ -head what holds true of an argument if it is the patient of this event, etc. However, nothing so far encodes that different θ -heads are associated with different thematic properties, and that arguments interpreted with the same θ -role are semantically similar.

Let us focus on the agent θ -head for simplicity's sake, but the same line of reasoning applies to any other θ -role. An interesting property of the framework proposed above is that there is no node in the entire tree that encodes what it means to be an agent as such. There is merely a θ -head that generates different kinds of agents *of* specific events from an input event description: for instance, when taking as input the set of dancing events, it returns what it means to be the agent of a dancing event. If a θ -role is defined as a relation between arguments and event descriptions, thus being of type $\langle e, \langle s, t \rangle \rangle$, then it is not the θ -head itself that is a θ -role, but the projection immediately dominating it, and this node denotes a θ -role already interpreted in the context of a given event description.

By way of example, take the argument interpretations generated by the agent θ -head (22):

- (22) a. $\lambda x. \lambda e. \text{fly}(e) \wedge \text{Agent}(e)(x)$

- b. $\lambda x. \lambda e. \text{dance}(e) \wedge \text{Agent}(e)(x)$
- c. $\lambda x. \lambda e. \text{stab}(e) \wedge \text{Patient}(e)(\text{Caesar}) \wedge \text{Agent}(e)(x)$
- d. ...

Imagine these various denotations, each of which is of type $\langle e, \langle s, t \rangle \rangle$, being represented as points in a semantic space. This semantic space would encode semantic similarity such that argument interpretations that sit closer to each other in the space are more semantically similar to each other. While it is not crucial for our purposes, we can assume that as in standard computational models, the degree of semantic similarity between two argument interpretations is a function of how similar the contexts are in which they occur. Note, however, that this space is meant to not merely be a computational artifact, but to provide on some level of abstraction an appropriate representation of this aspect of speakers' mental grammars.

My expectation is that the various argument interpretations generated by a single θ -head would not be scattered randomly throughout this semantic space but to cluster together. What it means to be the agent of flying, the agent of dancing, the agent of stabbing Caesar, and so forth, is not identical, but these interpretations have overlapping semantic properties. We can identify these properties with Dowty's proto-properties, and the center of the cluster formed by the interpretations generated by a single θ -head with the corresponding prototype. Argument interpretations generated by other θ -heads are associated with different properties and form different prototypes.

As a result, thematic prototypes are not explicitly encoded as such in the grammar, but they are simply the center of a distribution generated by a θ -head. This is, at its essence, an exemplar-theoretic view on θ -roles: concepts like agent or patient are generalizations formed over the output of the function that is a θ -head. What it means to be an agent as such is an abstraction over what it means to be the agent of flying, the agent of dancing, the agent of stabbing Caesar, etc. In the following, I will continue to describe arguments as being, say, the agent of an event, or being assigned an agent θ -role. But this only means that such an argument is the argument of a predicate derived via an agent θ -head.

A consequence of this view is that there are no hard and fast boundaries on how semantically coherent arguments with the same θ -role must be, or, on the flip side, to what extent different θ -roles result in differences in interpretation. A θ -head simply generates a certain cluster of argument interpretations, but nothing prevents this cluster from having outliers. In principle, it is possible for an agent θ -head to generate an argument interpretation that is far removed from the agent-prototype, or for an agent and a patient θ -head to generate the exact same argument interpretation. We would expect these cases to be rare – including for reasons related to learning –, but nothing rules them out altogether.

If two arguments could receive the same interpretation despite the fact that they are associated with different θ -roles, then the question arises whether it necessarily matters how we categorize them in terms of θ -roles. That is, if an argument interpretation generated by an agent θ -head could also have been generated by a patient θ -head, the question is whether we might just as well call it a patient. I believe that this would

not be correct. By way of analogy, phonemes can be defined by features, but from the point of view of phonetics, phonemes are simply generalizations formed over clusters of exemplars, phones that are produced by a given speaker in a given context. These clusters overlap, but there is a fact of the matter as to whether a particular phone is a high or a mid vowel, even if some exemplars of high vowels happen to be lower than some exemplars of mid vowels.

Lastly, while in this study I have dealt with θ -roles as purely linguistic categories, at least core roles such as agent and patient have also been argued to play a role in other areas of cognition (see Rissman and Majid, 2019 for an overview), including for prelinguistic infants (Papeo et al., 2024). This might seem to contradict the idea that θ -roles are abstractions formed over linguistic exemplars. I argue that the way in which θ -roles as broader cognitive concepts enter the picture is by shaping the possible distribution of argument interpretations, forcing them to cluster in some areas of the semantic space rather than in others. Conversely, if θ -roles are hard-wired or acquired based on non-linguistic evidence, this would also facilitate the acquisition of argument interpretations and, by extension, of verb meaning.

To conclude, this section has sketched out an exemplar-theoretic way of thinking about θ -roles according to which abstract roles like agent or patient are generalizations formed over all the different agents or patients *of* specific events. This is not at all a radical thought, and it is very similar to Dowty's idea of proto-roles. What I have argued for here is only novel in two ways. First, I have made a very concrete proposal as to what concepts like agent and patient are abstractions of: they are abstractions formed over the interpretations assigned to arguments in the context of the event that their θ -head combines with in the syntax, denotations of type $\langle e, \langle s, t \rangle \rangle$. Second, I have argued that an exemplar-theoretic view on θ -roles can be grounded in the architecture of the verbal domain adopted above, in that different θ -heads generate different clusters of argument interpretations.

5.2 Revisiting the case studies

So far, I have developed a proposal for argument interpretation that accounts for variable unaccusativity. In the following, I briefly revisit each of the five other case studies of variable linking – direct causatives of unergatives, direct causatives of ingestives, the agent-affectee alternation, pseudo-symmetric predicates and statives of ingestives – and show how they can be analyzed against the background of the previous sections. Having given a proof of concept of the formal machinery in the previous section, I will keep the discussion below informal.

5.2.1 Direct causatives of unergatives

I begin with the second case study, direct causatives of unergatives, covered in Chapter 4.2. An example of an unergative verb with its corresponding causative variant is repeated in (23):

- (23) a. Rohan naach rahaa hai.
 Rohan.M dance PROG.MSG be.PRS.3MSG
 ‘Rohan is dancing.’
- b. Shama Rohan-ko nach-aa rahii hai.
 Shama.F Rohan-DOM dance-CAUS PROG.F be.PRS.3MSG
 ‘Shama is making Rohan dance/twirling him around (the dance floor).’

(Bhatt and Embick, 2017:124)

The individual role theory of θ -roles would declare the verb ‘dance’ to come with the individual role of a dancer, which would be variably mapped onto an agent or a patient θ -role. In addition, the agent θ -role would correspond to two different individual roles in the two alternants.²

By contrast, under the analysis proposed here, direct causatives of unergatives can be made sense of as follows. First, these data are yet another instance of variable unaccusativity as analyzed earlier. Analogous to the prethematic verb ‘fly,’ the prethematic verb ‘dance’ generates the meaning of dancing agentively when combining with an agent θ -head, and the meaning of dancing in a patient-like manner – being pushed and twirled around – when combining with a patient θ -head. Again, there is a temptation to label the shared component of meaning as dancing or being a dancer – perhaps defined as moving one’s feet to the music –, but again, this wrongly narrows down the range of interpretations that can be associated with these arguments by evoking a prototypical concept of dancing.

In addition, and going beyond simple variable unaccusativity, the agent of the intransitive (*Rohan*) and the agent of the transitive (*Shama*) perform different activities. Thus, a single θ -role is associated with two different interpretations although the verb remains the same. I argue that analogous to Marantz’s *take* and *kill* examples, this is because the two agent arguments are arguments of different events. If argument interpretation is stored on the verb in the form of an individual role, this is unexpected, but if this interpretation is generated in the syntax by merging a θ -head with an event description, it falls into place.

At first sight, *Rohan* in the intransitive and *Shama* in the transitive are agents of different events for the very same reason that the subject arguments in Marantz’s examples are agents of different events, which is that the event descriptions that the agent θ -head combines with in the two alternants do not contain the same internal argument. In the causative in (23b), the internal argument is *Rohan*; in the intransitive in (23a), there is no internal argument. Accordingly, the subject arguments are agent arguments of different events, and their interpretation differs.

However, matters are more complicated. Direct causatives are assumed to contain at least two subeventualities, a causing and a caused eventuality, which are summed up to form the eventuality that constitutes the domain of thematic uniqueness (24):

²Direct causatives of unergatives also raise a puzzle for verb distribution: not all unergatives that form causatives also have an unaccusative variant. The fact that the verb can thus only combine with a patient θ -head if an agent θ -head is latter added seems to create a lookahead problem. A similar concern applies for direct causatives of ingestives. In Chapter 6.2, I show that this pattern is not problematic.

$$(24) \quad \llbracket \text{Shama danced Rohan} \rrbracket = \lambda e_2. \exists e_1. \text{dance}(e_1) \wedge \text{Patient}(e_1)(\text{Rohan}) \wedge \text{Cause}(e_2, e_1) \\ \wedge \text{Agent}(e_2)(\text{Shama})$$

On the other hand, the intransitive does not contain two such subeventualities that are linked by a causal relation. Thus, the event descriptions that the agent θ -head combines with in the two alternants differ also in terms of their internal event structure: Rohan is the agent of a dancing event, but Shama the agent of a causing event. One might argue that nothing more needs to be said to explain why what holds true of Shama and what holds true of Rohan in the two alternants in (23) differs.

But by itself, this is not yet a satisfactory account. Compare (24), repeated below as (25a), with (25b):

$$(25) \quad \begin{array}{ll} \text{a. } \llbracket \text{Shama danced Rohan} \rrbracket = \lambda e_2. \exists e_1. \text{dance}(e_1) \wedge \text{Patient}(e_1)(\text{Rohan}) \wedge \text{Cause}(e_2, e_1) \wedge \\ \quad \text{Agent}(e_2)(\text{Shama}) \\ \text{b. } \llbracket \text{Shama bounced the ball} \rrbracket = \lambda e_2. \exists e_1. \text{bounce}(e_1) \wedge \text{Patient}(e_1)(\text{the ball}) \wedge \text{Cause}(e_2, e_1) \wedge \\ \quad \text{Agent}(e_2)(\text{Shama}) \end{array}$$

In both sentences, Shama is represented as the agent of a causing event, but the interpretation assigned to the argument – what we know to be true about Shama – is very different. Since in (25), the root describes only the lower eventuality, knowing that Shama is the agent of a causing event is vacuous since we do not know what the latter is a causing event *of*. Thus, it is not sufficient to attribute the semantic difference between the two agent arguments in the causative-unergative alternation to the fact that one of them is the agent of a causing event, because nothing would semantically distinguish this argument from the agent of any other causing event.

What this demonstrates, I argue, is that analogous to θ -roles, causing events must always be causing events *of* specific eventualities. An event that causes a dancing of Rohan is different from an event that causes a bouncing of the ball. In (25a), Shama is the agent of a causing event of a dancing-Rohan event. This is different both from being the agent of a dancing event and from being the agent of a causing event of a bouncing-the-ball event. If a causing event was not lexically specified as being the causing event *of* a particular eventuality, then nothing would account for the fact that what holds true of the agent arguments of different (direct) causatives differs.

Overall, the fact that the agent argument of direct causatives of unergatives is interpreted differently to that of the corresponding intransitive does not fall out for free from a complex event structure for causatives alone, but also requires causing events to always be causing events *of* a particular eventuality. This suggests that much of the proposal developed in this paper for argument interpretation might transfer to event structure: in the same way that argument interpretations are always lexically specified, event-structural components such as causing events are too. I leave it to future work to pursue this idea further, including whether it is correctly captured by the semantic framework adopted here.

5.2.2 Direct causatives of ingestives

Chapter 4.3 discussed a case study that has much in common with direct causatives of unergatives: direct causatives of ingestives. An example is shown in (26b):

- (26) a. ayal-e Leyla badincan.
eat-F Leyla tomatoes
'Leyla ate tomatoes.'
- b. ayyil-tu Leyla badincan.
eat.CAUS-1SG Leyla tomatoes
'I fed Leyla tomatoes.' (lit. 'I ate Leyla tomatoes.')

The individual role theory of θ -roles would posit that the verb 'eat' comes with two individual roles, an eater and an eatee. In (26a), the eater is mapped onto an agent θ -role, and in (26b) onto a goal θ -role. Furthermore, in (26a), the agent corresponds to an eater, but in (26b) to what we might call a feeder.

As in the previous section on direct causatives of unergatives, there are two factors at play. First, a θ -head – that of an agent – combines with two different event descriptions, thereby giving rise to two different meanings: being the agent of eating tomatoes is different from being the agent of feeding Leyla tomatoes, in the same way that it is different from being the agent of dancing, dancing Rohan, petting porcupines, etc. What it means to be the agent of these events is not prespecified on the verb, but generated by merging the agent θ -head with an event description. If the event description differs – both in terms of arguments and in terms of event semantics –, then the properties ascribed to the argument differ as well.

The second factor at play in direct causatives of ingestives can again be understood as a variant of variable unaccusativity, although here, it involves not an agent and a patient θ -role, but an agent and a goal θ -role (in (26), the roles that are assigned to *Leyla*). Both of these θ -heads combine not with the verb itself, but with an event description that already contains a patient argument: the set of eating-tomatoes events. Earlier, I have argued that the meaning of a prethematic verb is something that gives rise to the interpretation assigned to its first argument when combining with its first θ -head. Following the same logic, we can now argue that the meaning of an event description already containing a first argument is something that gives rise to the interpretation assigned to the second argument when combining with a second θ -head. Specifically, what it means to eat tomatoes is something that gives rise to the meaning of eating tomatoes agentively when combining with an agent θ -head (26a), and to the meaning of being fed tomatoes when combining with a goal θ -head (26b). In sum, direct causatives of ingestives simply rehash the lessons learned from direct causatives of unergatives, only at a different level.

5.2.3 The agent-affectee alternation

Turning to our fourth case study from Chapter 4.4, which I have labeled the agent-affectee alternation, the example in (27) shows a regular direct causative in (27a) and an unaccusative with a dative argument in

(27b). This dative argument can be interpreted as unintentionally causing the event and/or as being affected by it:

- (27) a. El tintorero quemó los pantalones de Carolina.
 the dry-cleaner burnt.SG the trousers of Carolina
 ‘The dry-cleaner burnt Carolina’s trousers.’
 b. Al tintorero se le quemaron los pantalones de Carolina.
 to.the dry-cleaner REFL CL.DAT burnt.PL the trousers of Carolina
 ‘Carolina’s trousers got burnt on the dry-cleaner.’
 ‘The dry-cleaner (accidentally) burnt Carolina’s trousers.’ (Cuervo, 2003:186)

Under the individual role theory of θ -roles, the verb ‘burn’ would come with two individual roles, a burner and a burnee. In the transitive in (27a), the burner role would be mapped onto an agent θ -role, and under the unintentional causer reading of (27b), onto whatever θ -role is assigned to the dative argument. As I discussed, it is a nontrivial question what this role is. I have labeled it an affectee, highlighting its similarities to other affectee-style θ -roles like benefactives and malefactives equally associated with these dative positions. Whichever specific term is chosen, the thematic interpretation of the argument is different from that of the agent in (27a).

We can analyze the agent-affectee alternation as simply another instance of variable unaccusativity in the broad sense. The meaning of *burn Carolina’s trousers* is something that generates the meaning of burning Carolina’s trousers agentively when combining with an agent θ -head, and burning them unintentionally or being affected by their burning when combining with an affectee θ -head. There is no single burner role already stored on the verb, but similar argument interpretations are generated when the same event description combines with different θ -heads.

Comparing these data to variable unaccusativity proper, the following stands out. For variable unaccusativity, I have highlighted that the intuitions we have about the meaning underlying both the agentive and the patient-like construal of a verb such as *fly* are only poor approximations that are distorted by our tendency to focus on a prototypical construal of the verbal event. *Fly* does not mean ‘move through the air.’ But for the agent-affectee alternation, this does not appear pertinent. We seem to know very clearly what it means for Carolina’s trousers to burn, not least because this can be an independent unaccusative event description (including in Spanish), and this is largely unaffected by which θ -head is merged next.

I suggest that this difference is not categorical, but one of degree. The higher we climb up the tree, the more we are able to form an explicit, context-independent representation of the meaning associated with a constituent. This does not change the picture proposed here, but it is an open question why these differences should obtain – that is, why the interpretation of lower layers of the tree appears to be so much less intuitable and so much more prone to polysemy than higher layers. A reasonable answer might be that every compositional operation narrows down the range of possible senses associated with an item. Under this view, *fly* can mean so many things because we do not know yet what it will combine with.

5.2.4 Pseudo-symmetric predicates

Next, an alternation with a pseudo-symmetric predicate, discussed in Chapter 4.5, is shown in (28):

- (28) a. Joel and Clementine hugged.
 b. Joel hugged Clementine.
 c. Clementine hugged Joel.

Under the individual role theory of θ -roles, *hug* would come with two individual roles, a hugger and a huggee. To maintain the claim that the hugger role is consistently mapped onto an agent θ -role and the huggee onto a patient θ -role, the intransitive in (28a) would have to be analyzed as a covert reciprocal, in which both Joel and Clementine are assigned two θ -roles: a hugger agent role and a huggee patient role (e.g., Dimitriadis, 2008; Gleitman, 1965; Siloni, 2008; Wehbe, 2025). In Chapter 4.5, I have argued against the covert reciprocal analysis and instead endorsed the view that the intransitive involves only a single θ -role assigned to the subject (see also Carlson, 1998; Dowty, 1991; Kruitwagen, 2024; Winter, 2016, 2018).

From the perspective I have proposed here, the *spiel* should by now be familiar. The prethematic verb *hug* generates the meaning of hugging agentively when combining with an agent θ -head and hugging in a patient-like manner when combining with a patient θ -head. As with variable unaccusativity and all other cases of variable linking, the distinction between what it means to hug as an agent and what it means to hug as a patient is gradient, and either can be applicable in a given situation. Someone who vigorously wraps their arms around another person is clearly participating in the hug as an agent, and someone who is forced into someone's arms clearly as a patient, but someone who shows a mix of agent and patient properties could be conceptualized as either, and the speaker's perspective and intention matters.

However, open questions remain, and I will not solve them here. For one, we must ask how pseudo-symmetric predicates differ from verbs that undergo the causative alternation. Under my analysis, both classes of verbs allow for an intransitive variant with only an agent or a patient θ -role and for a transitive variant with both, with the sole difference being that causatives have a causative event structure. But pseudo-symmetric predicates are special. Most notably, as intransitives, they require a plural argument and receive an obligatorily collective reading. *Joel and Clementine hugged* cannot mean that Joel and Clementine pursued their individual hugging, in the way that *Joel and Clementine ran* means that they pursued their individual running. I have argued extensively against a covert reciprocal analysis of pseudo-symmetric predicates, but their peculiar interpretation still requires some sort of explanation.

Pseudo-symmetric predicates have a range of other properties that remain unaccounted for. In Chapter 4.5, I have mentioned that they tend to resist passivization and show unexpected behavior under VP ellipsis. Moreover, the interpretation of intransitive pseudo-symmetric predicates sometimes differs from the interpretation of the transitive in specific ways. For instance, in (29a), Joel and Clementine must agree with each other on the same thing, whereas in (29b), Joel might agree with Clementine's view on abortion and

Clementine might agree with Joel's view on recreational drugs:

- (29) a. Joel and Clementine agreed.
 b. Joel and Clementine agreed with each other.

Lastly, Chapter 6.1 on entailments will later raise another concern for the analysis of pseudo-symmetric predicates adopted here.

For future work, there are two avenues for tackling these problems. The first is to enrich the functional structure of pseudo-symmetric predicates beyond what I have assumed here. The second, which I am admittedly partial to, is to try to better understand the prethematic verb meaning of pseudo-symmetric predicates. That is, the challenge is to figure out what prethematic *hug* must mean for it to generate the interpretations it does when combining with different θ -heads and arguments.

5.2.5 Statives of ingestives

The final case study in Chapter 4.6 dealt with pseudo-agentive statives of ingestives such as (30) from Greek and (31) from Hindi-Urdu:

- (30) I Maria ine faɣo-men-i.
 the.NOM Mary.NOM be.3SG eat-PTCP-F.NOM
 'Mary is eaten.'
 'Mary has eaten.'
- (31) vo aurat saari pehn-e (hu-e) thii.
 that woman sari wear-PFV.OBL be-PFV.OBL be.PST.F
 'That woman was wearing a sari.' (Lit. 'That woman was sari-worn.')

I have argued that the arguments 'Mary' and 'that woman,' which are the agent arguments of the corresponding active eventive forms of the verbs, are interpreted as holders of result states. Under the individual role theory of θ -roles, eaters (30) and wearers (31) would thus be variably linked to an agent θ -role or a result state holder θ -role.

Comparing the two interpretations available for the stative in (30), it might seem that being the result state holder of an eating event can mean two very different things: being eaten and having eaten. Thus, just as in previous case studies, a single θ -role – that of a result state holder – is associated with different interpretations despite the fact that the verb remains constant. This is the puzzle that Paparounas (2025) tried to solve by postulating that in addition to being a result state holder, the argument of statives is also interpreted as either the agent of the event (in pseudo-agentive statives) or as the patient of the event (in regular statives).

But again, just as in the previous case studies, these result state holders are arguments of different events. Crucially, I have argued that pseudo-agentive statives always involve (pseudo-)incorporation of either a null (30) or overt argument (31). Under the pragmatically odd reading of (30) that Mary is eaten, there is no

incorporation, and Mary is the holder of the state resulting from an eating event in the same way that an apple or a slice of pizza might be. On other hand, under the pseudo-agentive reading, Mary is in the state resulting from eating *something*, which is the state of having eaten. The θ -head merges with two different event descriptions – one that does not and one that does contain an incorporated argument –, resulting in two different interpretations.

This final case study has remained a sketch, largely because even garden-variety statives remain elusive. But I hope to have shown that the approach proposed here is a promising avenue for future work on these data. Pseudo-agentive statives exemplify particularly clearly that the interpretation of arguments is not stored on the verb but generated bottom-up in the syntax, being sensitive to event structure and the presence of other arguments in a way that reflects constituency.

Chapter 6: Ramifications

Before concluding, I address some further questions and complications that arise for the proposal. Chapter 6.1 looks at various challenges that data from entailments between different alternants of a verb appear to pose for variable linking. Chapter 6.2 deals with the question of how knowledge about the different syntactic usages of a verb is encoded in speakers' grammars.

6.1 Entailments

One of the functions that θ -roles serve in formal semantics is to capture entailments between different alternants of a verb. In the following, I first demonstrate this general point. I then consider three challenges that evidence from cross-alternant entailments appears to present for the variable linking alternations analyzed earlier. I conclude that only the last challenge – concerning pseudo-symmetric predicates – is genuine, suggesting either that the analysis of pseudo-symmetric predicates adopted so far needs to be amended or, perhaps, that we need a more fundamental overhaul of neo-Davidsonian event semantics.

As a baseline case, consider (1):

- (1) a. Brutus stabbed Caesar yesterday.
- b. Brutus stabbed Caesar.

Example (1a) entails (1b). This fact can be accounted for in very simple terms by positing the denotations in (2), according to which the meaning of (1a) is a proper subset of (1b) (Parsons, 1990):

- (2) a. $\llbracket \text{Brutus stabbed Caesar yesterday} \rrbracket = \exists e. \text{stab}(e) \wedge \text{Subject}(e)(\text{Brutus}) \wedge \text{Object}(e)(\text{Caesar}) \wedge \text{yesterday}(e)$
- b. $\llbracket \text{Brutus stabbed Caesar} \rrbracket = \exists e. \text{stab}(e) \wedge \text{Subject}(e)(\text{Brutus}) \wedge \text{Object}(e)(\text{Caesar})$

By including Brutus and Caesar in the denotation as the subject and the object of the stabbing, we can track the identity of the arguments across the two sentences and ensure that (1a) does not entail that Caesar stabbed Brutus. So far, θ -roles are not needed.

However, surface-level descriptions such as subject and object are insufficient to predict entailments across valency-changing argument-structural alternations. In the causative-inchoative alternation, the transitive entails the intransitive. This entailment does not go through if the arguments are individuated by referring to them as subject and object (3):

- (3) a. $\llbracket \text{Ahmad broke the glass} \rrbracket = \exists e. \text{break}(e) \wedge \text{Subject}(e)(\text{Ahmad}) \wedge \text{Object}(e)(\text{the glass})$
- b. $\llbracket \text{The glass broke} \rrbracket = \exists e. \text{break}(e) \wedge \text{Subject}(e)(\text{the glass})$

Since *the glass* is the object of the transitive but the subject of the intransitive, no entailment should obtain. But if the arguments are individuated by their θ -roles, the causative correctly entails the inchoative because the events denoted by (4a) are a subset of the events denoted by (4b). Note that in the following, I also adopt a more fine-grained representation of causative event structure.

- (4) a. $\llbracket \text{Ahmad broke the glass} \rrbracket = \exists e. \exists s. \text{broken}(s) \wedge \text{Patient}(s)(\text{the glass}) \wedge \text{Cause}(e, s) \wedge \text{Agent}(e)(\text{Ahmad})$
 b. $\llbracket \text{The glass broke} \rrbracket = \exists e. \exists s. \text{broken}(s) \wedge \text{Patient}(s)(\text{the glass}) \wedge \text{Cause}(e, s)$

Thus, θ -roles are needed to capture entailments such as those from a causative to an inchoative.

A first apparent problem posed by entailments for the present proposal comes from the fact that in several variable linking alternations as analyzed here, there are pairs of alternants between which no entailment is predicted, but nonetheless, some speakers judge that an entailment relation holds. For instance, as discussed extensively in Chapter 4.5, intransitive pseudo-symmetric predicates have often been assumed to entail the corresponding transitives (5), although I have argued, with many others, that the arguments are interpreted with different θ -roles in the different alternants.

- (5) a. Joel and Clementine kissed.
 b. Joel kissed Clementine and Clementine kissed Joel.

Similarly, it might seem that an entailment relation holds between a direct causative of an unergative and the corresponding intransitive in which the patient argument of the causative is realized as the agent, shown for pseudo-Hindi-Urdu in (6).

- (6) a. Shama danced Rohan.
 b. Rohan danced.

To give yet another example, unaccusatives with affected dative arguments might be taken to entail the corresponding causative, as in the pseudo-German example (7). Recall that I have analyzed the argument *Ana* as an affectee in (7a), but as an agent in (7b).

- (7) a. The vase broke on Ana.
 b. Ana broke the vase.

If such entailments were indeed valid, despite the fact that an argument is assigned different θ -roles in the two alternants, this could be taken as evidence against the present proposal.

I have two points to make in response. First, I do not believe that these entailments are valid. For pseudo-symmetric predicates, I have discussed earlier in Chapter 4.5 experimental evidence from Kruitwagen (2024) that the intransitive can be felicitous in a situation in which one of the transitives is not. By the same token,

direct causatives of unergatives can be licensed while the intransitive is not, namely, if the participant in question is much more easily construed as a patient rather than as an agent. For instance, if Rohan is being twirled around against his will, the intransitive of ‘dance’ is odd. Lastly, an unaccusative-cum-dative can be used if the participant is affected without being clearly causally responsible, whereas the corresponding causative cannot. In all of these cases, the relevant judgments are subtle. The alternants clearly overlap in meaning, and in some situations, both can be felicitous. However, as shown in Kruitwagen’s experiment, setting up specific scenarios in which one alternant is appropriate while the other is not demonstrates that the thematic properties ascribed to the arguments are not the same in the two alternants.

But moreover, note that such entailments are not strictly ruled out. Nothing in the semantic denotations I have proposed *enforces* an entailment from a transitive pseudo-symmetric predicate to the corresponding intransitive, or from a causative of an unergative to the intransitive, or from an unaccusative-cum-dative to the causative, but nothing *prevents* them from arising spuriously on the level of lexical semantics either. Entailments between propositions can arise for all kinds of reasons: for instance, *I have the flu* entails *I am sick*, but this does not result from the formal semantic structure of the two predicates. By the same token, it is in principle possible that what it means to be an affectee of a breaking-the-vase event happens to be a proper subset of what it means to be an agent of this event, and this might even depend on the specific event. Both argument interpretations are ultimately pieces of lexical meaning, and they can stand in all kinds of semantic relations to each other. It is true, and unfortunate, that this conclusion weakens the predictive power that cross-alternant entailments have as a diagnostic for θ -roles, but I do not see that it can be avoided.

We can now turn to the second apparent problem arising from entailments. According to the analysis of direct causatives of unergatives adopted here, in the pseudo-Hindi-Urdu examples in (8), *Shama* is interpreted with an agent θ -role in both alternants. Nonetheless, the transitive does not entail the intransitive.

- (8) a. Shama danced Rohan.
b. Shama danced.

Similarly, in direct causatives of ingestives – shown here for pseudo-Saxon Arabic –, I have analyzed the subject argument as an agent argument in both (9a) and (9b), but again, the former alternant does not entail the latter.

- (9) a. I ate Leyla (dative) tomatoes. (i.e., I fed Leyla tomatoes)
b. I ate tomatoes.

This worry is easily dispersed. As can be seen in (10) for direct causatives of unergatives, *Shama* is an agent of two different events in the two alternants: a causing event of a dancing-Rohan event in the transitive, and a dancing event in the intransitive. Thus, the denotation in (10a) is not a subset of the denotation in (10b),

and the entailment is correctly predicted to not be valid.

- (10) a. $\llbracket \text{Shama danced Rohan} \rrbracket = \exists e_2. \exists e_1. \text{dance}(e_1) \wedge \text{Patient}(e_1)(\text{Rohan}) \wedge \text{Cause}(e_2, e_1) \wedge \text{Agent}(e_2)(\text{Shama})$
 b. $\llbracket \text{Shama danced} \rrbracket = \exists e. \text{dance}(e) \wedge \text{Agent}(e)(\text{Shama})$

The same reasoning applies, *mutatis mutandis*, for direct causatives of ingestives.

But a similar predicament applies for pseudo-symmetric predicates, and this is the last, and genuine problem posed by entailments. For simplicity's sake, let us assume that the English verb *hug* behaves as an unergative in its intransitive guise; if this assumption turned out to be incorrect, the following argument could also be made with any of the Dutch verbs I have shown to be unergatives in Chapter 4.5. Consider the fact that (11a) does not entail (11b) although *my parents* is interpreted with an agent θ -role in both alternants:

- (11) a. My parents hugged me.
 b. My parents hugged.

On the one hand, the failure of this entailment is as expected under the present account. I have argued that the interpretation of the external argument is determined by merging an agent θ -head with an event description. Since the two event descriptions in (11) that the agent θ -head combines with are distinct – the set of hugging-the-speaker events and the set of hugging events –, the interpretation of *my parents* is too. This exactly mirrors what we saw for Marantz's *take* and *kill* examples.

However, according to the denotations in (12), the set of events denoted by the transitive should be a proper subset of those denoted by the intransitive, wrongly predicting a valid entailment:

- (12) a. $\llbracket \text{My parents hugged me} \rrbracket = \exists e. \text{hug}(e) \wedge \text{Agent}(e)(\text{the speaker's parents}) \wedge \text{Patient}(e)(\text{speaker})$
 b. $\llbracket \text{My parents hugged} \rrbracket = \exists e. \text{hug}(e) \wedge \text{Agent}(e)(\text{the speaker's parents})$

The problem is that unlike direct causatives of unergatives or ingestives, pseudo-symmetric predicates do not have a complex event structure containing different subeventualities. For instance, recall that in *Shama danced Rohan/Shama danced*, the fact that *Shama* is an agent of different events is transparently represented in the denotation of the sentences in that the argument is connected to different eventuality variables (13):

- (13) a. $\llbracket \text{Shama danced Rohan} \rrbracket = \exists e_2. \exists e_1. \text{dance}(e_1) \wedge \text{Patient}(e_1)(\text{Rohan}) \wedge \text{Cause}(e_2, e_1) \wedge \text{Agent}(e_2)(\text{Shama})$
 b. $\llbracket \text{Shama danced} \rrbracket = \exists e. \text{dance}(e) \wedge \text{Agent}(e)(\text{Shama})$

By contrast, for the case of pseudo-symmetric predicates, the same eventuality variable is used for the set of hugging events and the set of hugging-the-speaker events. As a result, the two cannot be distinguished from

each other for the purposes of entailments, giving rise to the wrong prediction.

A possible conclusion to draw from this finding is that the analysis of pseudo-symmetric predicates adopted here must be mistaken. For reasons discussed extensively in Chapter 4.5, I still maintain that the covert reciprocal analysis is not a viable alternative, but one might argue that the subject argument of the intransitive is interpreted with a θ -role that is distinct from the one assigned to the subject of the transitive. For instance, we could call it a participant θ -role (14):

- (14) a. $\llbracket \text{My parents hugged me} \rrbracket = \exists e. \text{hug}(e) \wedge \text{Agent}(e)(\text{the speaker's parents}) \wedge \text{Patient}(e)(\text{speaker})$
 b. $\llbracket \text{My parents hugged} \rrbracket = \exists e. \text{hug}(e) \wedge \text{Participant}(e)(\text{the speaker's parents})$

This successfully blocks the entailment. Of course, the alternation remains a case of variable linking either way.

I have no decisive arguments to level against (14), but I have two qualms. First, without a clear account of what semantic and associated syntactic properties set the ‘participant’ θ -role apart from the agent θ -role, the label remains stipulative and ill-founded. Second, as argued earlier, the overall proposal for argument interpretation developed in this study gives us a much more systematic way of explaining the data: the interpretation of an argument depends on the event description its θ -head combines with, and if the event description is different, then so is the interpretation of the argument. The lack of a valid entailment from transitive to intransitive pseudo-symmetric predicates crucially contrasts with the valid entailment from a causative to its inchoative, repeated in (15):

- (15) a. $\llbracket \text{Ahmad broke the glass} \rrbracket = \exists e. \exists s. \text{broken}(s) \wedge \text{Patient}(s)(\text{the glass}) \wedge \text{Cause}(e, s) \wedge \text{Agent}(e)(\text{Ahmad})$
 b. $\llbracket \text{The glass broke} \rrbracket = \exists e. \exists s. \text{broken}(s) \wedge \text{Patient}(s)(\text{the glass}) \wedge \text{Cause}(e, s)$

Here, the argument that is missing in one of the alternants (*Ahmad*) is higher than the one that remains (*the glass*). Thus, the former is not contained in the event description that the latter is an argument of: whether or not the external argument is present does not affect the interpretation of the internal argument, and the entailment from the transitive to the intransitive is valid. But with pseudo-symmetric predicates, the argument that is missing in one of the alternants (*me*) is *lower* than the one that remains (*my parents*), and whether or not it is present affects what holds true of *my parents* because it is contained in the event description that the latter is an argument of. Therefore, the entailment does not go through. This appears to be a systematic pattern, and simply inventing a new θ -role for the subject of intransitive pseudo-symmetric predicates gives us no handle on it.

Thus, the other possible response to the challenge from entailments with pseudo-symmetric predicates is to maintain the general analysis argued for earlier – the subject of the intransitive is an agent –, but to place the blame on the formal framework in which this analysis has been couched so far. In this study, I have

relied on a very standard neo-Davidsonian event semantics in which both the predicate of the eventuality itself and its arguments are represented as separate conjuncts. Instead, we might need a formal framework in which the entire event description that a θ -head combines with is fed as an argument to the θ -role itself, as represented very schematically in (16):

- (16) a. $\lambda e. \text{Agent}(\lambda e. \text{hugging-the-speaker}(e))(\text{the speaker's parents})(e)$
 b. $\lambda e. \text{Agent}(\lambda e. \text{hugging}(e))(\text{the speaker's parents})(e)$

Such denotations would transparently reflect the fact that being the agent of a hugging event and being the agent of a hugging-the-speaker event is semantically distinct, and no entailment would be predicted from one to the other.

Clearly, these data from pseudo-symmetric predicates alone hardly warrant such a radical overhaul of event semantics, especially given the other challenges raised by this alternation discussed earlier in Chapter 5.2.4. But in Chapter 5.2.1, I have also pointed out that the proposal developed here for argument interpretation might need to be adopted for event semantics as well: in the same way that an agent is always an agent *of* a particular eventuality, a causing event is always a causing event *of* a particular eventuality. I have left it open whether this idea is compatible with standard neo-Davidsonian semantics. If it is not, this would be another motivation for developing a different formal framework.

To summarize, in this section I have discussed three problems that data from cross-alternant entailments might appear to raise for the proposal developed in this study. For the first two, I have concluded that we can safely set them aside. However, the data from pseudo-symmetric predicates have suggested that their analysis needs amending, although I have argued that the alternation would remain a case of variable linking either way. Alternatively, we might place the blame on the formal semantic framework, but tossing it out would require stronger arguments than I have presented here.

6.2 Verb distribution

A general problem in argument structure that is distinct from, but related to, the one I have tackled here is how knowledge about the licit syntactic uses of verbs is encoded. Not every verb can be used in any syntax or with any set of θ -roles: for instance, *give* cannot be an intransitive and *fall* cannot be a transitive. The question this raises is how this knowledge is encoded in speakers' grammars.

The simplest approach to this problem would be to list each verb (or root) in the lexicon with the syntactic usages which it allows. This faces a number of well-known challenges. First, the syntactic structures in which a verb can appear are not rigidly determined but subject to coercion (e.g., Borer, 2005b). Second, not all possible usages of a verb are equally likely. For instance, a verb might usually act as an unergative but occasionally surface as an unaccusative, while another has the opposite profile, being predominantly unaccusative and occasionally unergative (Sorace, 2000, 2004, 2011). Under a listing approach, such verbs

cannot be distinguished from each other. Lastly, simply storing a verb in the lexicon with the syntactic frames in which it can appear fails to capture any systematic regularities in the mapping between meaning and syntax. For instance, nothing would truly explain why *run*, *dance*, *walk* and *swim* would all happen to be listed as predominantly unergative verbs.

In this section, I sketch out an answer to the problem of verb distribution that leverages the approach to verb meaning developed in the previous chapter, with a focus on variable linking. In short, I argue that the answer lies in turning the problem on its head: it is not that verbs have a certain syntactic distribution because they have a certain semantics, but they have a certain semantics because they have a certain syntactic distribution. The following is a very preliminary account; I hope that it can be developed fruitfully in future work.

Let us assume a distributional approach to semantics according to which the meaning of a linguistic item is defined by the contexts in which it can occur. This view is encapsulated in the Distributional Hypothesis, which holds that the degree of semantic similarity between two items is a function of the similarity of the contexts in which they occur. For instance, *dog* is more similar to *cat* than to *cockroach* because they occur in more similar contexts – for instance, those relating to pet ownership. The Distributional Hypothesis comes in two flavors (Lenci, 2008). Under the Weak Distributional Hypothesis, distribution *correlates* with meaning and can thus be used as a tool in semantic research to operationalize meaning. Under the Strong Distributional Hypothesis, an item's distribution *determines*, or is constitutive of, its meaning. Let us assume the latter view.

So far, I have been treating θ -roles as being associated with independent functional heads entering the derivation. Thus, under a distributional approach, θ -heads can be included in the set of items to which verbs (or roots) stand in a co-occurrence relation. Accordingly, the meaning of a verb is determined, among other things, by how often it co-occurs with which θ -head. Verbs that occur at similar rates with different θ -heads are expected to be semantically similar.

For variable unaccusativity in particular, Sorace (2000, 2004, 2011) has argued that intransitives with a more strongly agentive meaning are more likely to behave as unergatives, while those with a more telic meaning – a proto-patient property – tend to behave as unaccusatives. I suggest that what creates the sense that strongly unergative-leaning verbs have a strongly agentive meaning is that they strictly co-occur in the context of an agent θ -head, and *mutatis mutandis* for unaccusative verbs and a patient θ -head. And by the same token, variable behavior verbs evoke the meanings they do because they can co-occur with both an agent and a patient θ -head. Verbs that show a similar syntactic distribution – e.g., the class of strictly unergative verbs – have a similar semantics *because* they have a similar distribution.

Note that this has a consequence that is not necessarily intuitive. It has been observed that cross-linguistically, some verbs behave as unaccusatives in one language and as unergatives in another; *blush* is one example (Sorace, 2004). Under the view proposed here, this entails that these verbs have similar but ultimately different meanings in different languages. This might not square with our intuitions that these

verbs are intertranslatable, but as should have become clear by now, I am not confident that our intuitions about root meaning are in any way reliable in the first place. The cross-linguistic differences between meanings can be subtle, but if meaning is defined in terms of co-occurrence, and if functional elements like θ -heads are part of the set of items to which the verb stands in a co-occurrence relation, then verbs with different syntactic distributions cross-linguistically have different meanings.

Two other cases of variable linking that were discussed above, direct causatives of unergatives and ingestives, seem to pose a special puzzle for how knowledge about verb distribution is encoded. Not all unergative verbs that can causativize also have an unaccusative variant. That is, they can only merge with a patient θ -head if they then further combine with an agent θ -head. By the same token, ingestive verbs can combine with a goal θ -head only if an agent θ -head is added later. At first sight, this creates a lookahead problem.

However, the same problem would hold for obligatorily transitive verbs like English *kill* or *cut*. Here as well, there is a node in the tree that denotes an event description with only a patient argument, which is only felicitous if an agent θ -head is merged after. The same holds for pseudo-symmetric predicates that are unergatives in their intransitive form. More generally, for every verb, the lowest event description derived contains only the root and no arguments at all. But such an event description is felicitous in isolation only if the verb is a weather verb; for all other verbs, θ -heads need to be added. Overall, thematic well-formedness is never evaluated bottom-up in a step-by-step fashion: it is not the case that every event description that is generated over the course of the derivation must be licensed in isolation. Thematic well-formedness can be evaluated only once the entire verbal domain is closed off.

From a distributional perspective, there is no particular problem. For instance, the meaning of causativizing unergatives is simply determined, among other things, by the fact that they can occur in the context of an agent θ -head, or of both an agent and a patient θ -head, but not in the context of a patient θ -head only. One of the questions this leaves open is how to account for cross-linguistic variation in this domain. Direct causatives of unergatives are not attested to the same degree across languages: Hindi-Urdu licenses them very productively, Turkish and Sason Arabic less so, and in English, they appear to be even more restricted. It is highly likely that speakers have abstract and productive knowledge of these generalizations.¹ Against the background of what I argue for here, these cross-linguistic differences would reflect constraints on possible distributions, patterns in how verbs can and cannot co-occur with different θ -heads. There would be certain kinds of distributions that are permitted in some languages but not in others. Of course, this still leaves open the question why certain languages pattern one way or another, or in other words, which other properties of a language are predictive of the degree to which it licenses direct causatives of unergatives or other argument-structural alternations.

I reiterate that this proposal is a sketch, and it comes with a number of limitations. First, I have here reduced the syntactic distribution of a verb to its distribution with respect to θ -heads. One reason why this is

¹See, e.g., Wonnacott (2011) and Wonnacott et al. (2008) for evidence from artificial language learning experiments that speakers can learn higher-level generalizations such as how likely verbs in their language are to vary between two different syntactic frames.

too simplistic concerns clausal arguments, which are not usually analyzed as receiving a θ -role. Nonetheless, the same general reasoning could still apply. There would still be some functional element that introduces a clausal argument – say, a head selecting a CP –, and the extent to which a verb co-occurs with this element would be constitutive of its meaning.

More importantly, there remains a noticeable tension between the distributional, usage-based-learning approach espoused in this section where meaning is defined in terms of global co-occurrence patterns, and the strictly derivational, compositional, bottom-up framework in which the analysis in the previous chapter was couched. For one, the notion of context that is assumed in distributional approaches is commonly that of a string rather than a hierarchical structure (but see, e.g., Padó and Lapata, 2007 for a model in which context is calculated based on syntactic relations in a dependency grammars). Moreover, standard computational implementations of distributional models operate on overt word or subword forms, which are further subject to the vagaries of orthography: they encode the relation of *dog* to *cat*, but not to a null nominalizer, a T head or a question morpheme. There is no representation of, say, an agent θ -head that could be directly extracted from such models. However, it might be possible to build a distributional model that quantifies and formalizes the co-occurrence relations holding between what we believe are the atoms of syntactic and semantic composition, regardless of whether they receive an (independent) spell-out.

Lastly, one might argue that it is not clear how distributional information associated with verbs is operative in the derivation: that is, how concretely the distributional properties of a verb affect which functional structure can and cannot be built on top of it. I believe that this argument rests on a false premise. I have argued earlier in the discussion of direct causatives of unergatives and ingestives as well as of obligatory transitives that thematic well-formedness is not evaluated step-by-step in the derivation in the first place, but by constraints holding on the verbal domain as a whole. Such output constraints are perfectly standard in parallelist architectures. It is true that they are not popular with derivational purists, but I do believe that they are unavoidable for encoding speakers' knowledge about verb distribution regardless of whether or not the specific approach proposed here is adopted.

Overall, under a distributional approach, the meaning of a verb is a function of how it is used in context. Here, I have proposed that θ -heads can be understood as elements to which the verb can stand in a co-occurrence relation, and that for instance, a verb has a more agentive meaning because it occurs more frequently in the context of an agent θ -head. Verbs that occur in similar thematic contexts have a similar meaning *because* they occur in similar thematic contexts. Distributional knowledge about the syntactic usages of verbs is not encoded as an appendix to their lexical meaning that the latter must somehow map onto, but as constitutive of their lexical meaning. It is important to highlight that all of this poses a challenge for acquisition. Learning the meaning of roots – or prethematic verbs – appears to require already knowing θ -roles, but learning θ -roles appears to require already knowing the lexical meaning of roots. As discussed at the end of Chapter 5.1, this task might be made easier if the search for θ -roles is constrained by hard-wired principles of general cognition.

Chapter 7: Conclusion

The research question I set out to answer is how speakers know what it means for an argument to be interpreted with a given θ -role in the context of a specific event description; for instance, what it means to be the patient of a stabbing event. Concretely, I identified three goals for this study. First, it should convince the reader that the research question is a genuine, nontrivial question, with repercussions for argument structure and the syntax-semantics interface more broadly. My second goal was to refute a previous approach to this question – to my knowledge, the only one –, which I have labelled the individual role theory of θ -roles. Third, I aimed to make progress with developing an alternative account.

To begin with the first goal, the case studies in Chapter 4 should have made it clear that whether or not variable linking is expected to be possible affects how we analyze a wide range of specific argument-structural alternations. Many if not most of the competing analyses reviewed at the end of each case study were motivated, albeit often implicitly, by the individual role theory of θ -roles. In the final case study on statives of ingestives in Chapter 4.6, I have presented only a sketch of an analysis, and I hope to have shown that having a theory of how θ -roles are interpreted in context is indispensable for gaining a better understanding of these data. Furthermore, in Chapters 5 and 6, we have seen that the research question also has repercussions for larger architectural issues such as the nature of lexical meaning and the status of θ -roles in the grammar.

Turning to the second goal, refuting the individual role theory of θ -roles, I have noted a number of significant shortcomings. Besides the fact that it is crucially inconsistent with variable linking, it also presupposes a lexicalist view on word meaning that fails to account for the distributional flexibility and productivity that verbs allow for in the wild. Relatedly, the theory predicts that the interpretation of an argument is determined exclusively by the verb and the θ -role, but I have adduced various pieces of evidence indicating that it depends on the entire event description that the argument's θ -role combines with. Lastly, I have generally questioned the idea that the image a verb conjures up for us when we contemplate its meaning in isolation is an adequate representation of its meaning, as assumed, for instance, in Goldberg's implementation of the individual role theory. Rather, I have suggested that such an image represents a prototypical usage of the verb in context.

It bears reiterating that the individual role theory is not a proposal that is explicitly adopted and defended by the majority – or even a sizeable minority – of researchers in the field. There is a reason why I invented my own term for it. Even in the context of Goldberg's work, its status is arguably rather marginal. To the extent that I have failed to completely dissuade some readers from this view, I believe to have at the very least demonstrated that it needs sharpening if it is to be at all defensible. It would be interesting to see and engage with a revised and strengthened version of the individual role theory.

My final goal was to make progress with developing an alternative approach to the research question. This goal was formulated cautiously for a reason. I hope to have identified a fruitful perspective on the problem that is coherent, compelling and superior to the individual role theory, but I also see open questions for future work.

The key idea I have relied on is that the interpretation of an argument with a given θ -role in the context of a given event is generated in the syntax by merging a θ -head with an event description. The resulting node describes what holds true of an argument interpreted with this θ -role in the context of this event. Unlike under the individual role theory, the interpretation of an argument is thus not already stored on the verb and then, as it were, looked up by the θ -role, but only arises from the combination of the two. This seemingly small change makes crucially different predictions.

I have first outlined how this approach plays out at the very bottom of the tree. The meaning of the prethematic verb – a root that has combined with a verbalizer – is something that generates the interpretation of its first argument when merging with a first θ -head. I have argued that some representation of prethematic verbs that can combine with different θ -heads must be posited to account for variable unaccusativity as a highly common and productive alternation, and to explain why the unaccusative and unergative variant of a verb clearly share a semantic core. At the same time, due to polysemy, the meaning of a prethematic verb is hard to pin down. In response, I have suggested that prethematic verbs can be usefully described as referring to kinds of events, being defined not by a set of semantic properties but by all the tokens that instantiate them.

Against this background, I have proposed that the research question can be answered as follows. Speakers know what it means to be the patient of a stabbing event because they know the meaning of the verb *stab*, and the latter is something that gives rise to the meaning of being the patient of a stabbing event – roughly, being penetrated by a sharp weapon – when combining with a patient θ -head. Being the patient of a stabbing event is not information that is already prespecified on the verb, but it is generated by merging the verb with the relevant θ -head.

So far, this only deals with the verb's first argument. By the very same logic, the interpretation assigned to a verb's second (or third) argument is equally generated by merging the relevant θ -head with an event description, but crucially, this event description now already contains the verb's first (or first and second) argument. Thus, what it means to be the agent of a given event depends not only on the verb, but also on any arguments contained in the event description that the agent θ -head combines with as well as its event-structural make-up. This proved crucial for the analysis of Marantz's *kill* and *take* examples and several cases of variable linking.

Lastly, under the analysis proposed, θ -heads are functions that take an event description and return what holds true of an argument interpreted with a certain θ -role in the context of the input event; for instance, what it means to be the agent of a dancing event, a flying event or a stabbing-Caesar event. An agent θ -head thus does not directly denote the agent prototype, but merely generates various exemplars of agents

of specific events. This naturally captures the prototype structure and fuzziness of θ -roles, and it aligns with the view that they are represented – and acquired – as abstractions over exemplars.

Overall, I consider the main selling points of this analysis to be as follows. First, variable linking falls out naturally. Second, unlike the individual role theory of θ -roles, my proposal does not rely on rigid structured lexical representations, which have long been argued to face challenges. This also means that although nothing in the present study requires adopting the current DM orthodoxy in its entirety, I believe that it is fully compatible with it. Third, the analysis correctly predicts constituency-based asymmetries in argument interpretation: lower arguments can affect the interpretation of higher arguments but not vice versa. Fourth, the proposal naturally aligns with an exemplar-theoretic perspective on θ -roles, according to which the role of, say, an agent is an abstraction formed over agents *of* specific events. I believe that this is a compelling view, and that the fact that it can be grounded in the formal denotation of θ -heads is a strength.

I now turn to open questions and avenues for future work. To begin with, not all of the case studies of variable linking have provided a comprehensive analysis of the alternations in question: I have remained agnostic about certain aspects of the agent-affectee alternation (Chapter 4.4), I have discussed various outstanding problems for my analysis of pseudo-symmetric predicates (Chapter 4.5, see also Chapters 5.2.4 and 6.1), and my account of statives of ingestives has been preliminary (Chapter 4.6). Analyzing the syntax and semantics of these constructions in detail was not the goal of this study, but I hope that the perspective from variable linking will prove fruitful for investigating these data in the future.

On a theoretical level, several ideas pitched in this study have remained tentative and will need to be developed further. In the discussion of direct causatives of unergatives in Chapter 5.2.1, I have pointed out that in the same way that an agent is always an agent *of* a particular eventuality, a causing event is always a causing event *of* a particular eventuality, and that as a result, the proposal developed here for argument interpretation might apply to event semantics as well. This might require revising the formal semantic framework I have relied on here, as also suggested by the data from entailments with pseudo-symmetric predicates discussed in Chapter 6.1. I have left all of this to further research. Moreover, in Chapter 6.2, I have laid out an answer to the question of how speakers know in which syntactic environments a given verb can be used, arguing that verbs do not have a particular syntactic distribution because they have a particular semantics, but that they have a particular semantics because they have a particular syntactic distribution. Again, this proposal is so far only a sketch.

The arguably main challenge for the analysis has been, and remains, to clarify the semantics of prethematic verbs. I believe to have made some progress in this matter. For one, I have discussed the question in which sense the events denoted by prethematic verbs do or do not contain arguments. What is more, I have proposed that the meaning of prethematic verbs can be usefully analyzed in terms of kinds, but this point was made only very briefly. Fleshing it out and implementing it formally is an outstanding task. Lastly, further work on this question will need to take more seriously the perspective from polysemy, which the present study has largely steered away from.

As a final limitation, I wish to highlight that the analysis proposed here rests on a standard generative machinery, and it rests in particular on a radically decompositional syntax of the verbal domain where θ -roles are associated with independent heads entering the derivation. If all semantic composition is syntactic composition, then this is unavoidable. But this is not an architectural assumption that is universally adopted, and in this particular case, it is not rooted in a whole lot of evidence from overt morphosyntax. For instance, I have had nothing to say about the extent to which θ -heads are spelled out as independent morphemes cross-linguistically. There might be a way of approaching the research question by retaining a more holistic view on verbs; for instance, by defining what I have called here θ -roles as partial overlaps between the meanings of different lexical items. I would be excited to see such an approach being worked out.

To conclude, I have begun this work by assuming a sharp line between lexical meaning and grammatical meaning, between roots and trees. It seems that this line has sometimes been taken to suggest that roots are pieces of pure world knowledge and therefore outside the area of expertise (or interest, for that matter) of syntacticians and formal semanticists. This is, upon reflection, a strange thought. If roots occupy a position in the tree, then they are syntactic objects, and their meaning must semantically compose with that of other elements in the verbal domain. Generative research on argument structure in the last few decades has been marked by the demise of lexicalist frameworks like projectionism and the rise of syntactiocentric frameworks like DM, and while I believe that this has been a positive development on the whole, it has also made it difficult to get an analytic handle on lexical meaning and the role it plays in argument-structural alternations. Treating lexical meaning as a black box that does not affect theories of syntax and compositional semantics is not fruitful. The interface between grammatical structure and world knowledge remains a major puzzle in linguistic theory, and the present study has hardly scratched its surface.

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